Answering questions with data

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This comprehensive resource offers a free, accessible textbook for environmental science students embarking on introductory statistics. The package includes a practical lab manual and a dedicated course website, all provided under a CC BY-SA 4.0 license.

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Preface

Second Draft (version 0.1 = August 18th, 2025)

Welcome to the second edition of this Open Educational Resource (OER) textbook, specifically adapted for Environmental Science students enrolled in the SPEA E-538 statistics course at Indiana University (IU).

This textbook is an adaptation of a thorough introductory statistics textbook originally developed for undergraduate Psychology students by Matthew Crump and colleagues (refer to Acknowledgements for more details). As part of IU's Course Materials Fellowship Program (CMFP), I've had the opportunity to mold this material, refining it to serve as a specialized resource for students studying Environmental Science.

Online Textbook:https://malloryb.github.io/statistics_E538/

Citation for original textbook: Crump, M. J. C., Navarro, D. J., & Suzuki, J. (2019, June 5). Answering Questions with Data (Textbook): Introductory Statistics for Psychology Students. https://doi.org/10.17605/OSF.IO/JZE52

Acknowledgements

I wish to express my deepest appreciation to the contributors of the original textbook, without whom this adaptation would not have been possible. I am deeply grateful for the expertise and vision of Matthew Crump, Alla Chavarga, Anjali Krishnan, Jeffrey Suzuki, and Stephen Volz. Their exceptional groundwork laid the foundation for this project.

My heartfelt thanks also go to the Course Materials Fellowship Program (CMFP) at Indiana University (IU), which has been instrumental in supporting this adaptation effort. The CMFP is an initiative designed to incentivize the discovery, implementation, and creation of cost-effective course materials. Its aim is to foster the use of Open Educational Resources (OERs)—freely accessible and customizable learning materials that make education more equitable and accessible.

I am particularly grateful to Sarah Hare, the Bloomington lead for the CMFP program, and Adam Mazel, a digital publishing librarian at IU. Their guidance, expertise, and steadfast

support have been crucial to this project's success. Their contributions to the advancement of affordable and accessible education are truly commendable.

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Copying the textbook

This textbook was written in R-Studio, using R Markdown, and compiled into a web-book format using the bookdown package. In general, I thank the larger R community for all of the amazing tools they made, and for making those tools open, so that I could use them to make this thing.

All of the source code for compiling the book is available from the GitHub repository for the original textbook:

https://github.com/CrumpLab/statistics

and my github repository:

https://github.com/malloryb/statistics_E538

In principle, anyone can fork or download the E-538 textbook, or the original textbook, which is what I did. You can load the Rproj file in RStudio, compile the entire book, and then edit the individual .Rmd files for content and style to fit your needs.

If you'd like to contribute to this version, you can submit pull requests on GitHub or use the issues tab to share suggestions.

The vision behind this textbook

The aim of this textbook is twofold: to make core statistical concepts accessible to Environmental Science students, and to promote the use of open-source tools like R as flexible, transparent resources for learning and research.

1 Measuring and Describing Data

Adapted to environmental science by Mallory Barnes. Portions adapted from Chapters 1 and 2 in Navarro, D. J. "Learning Statistics with R." https://compcogscisydney.org/learning-statistics-with-r/

> *To call in statisticians after the experiment is done may be no more than asking them to perform a post-mortem examination: They may be able to say what the experiment died of.* – Sir Ronald Fisher

1.1 The Role of Statistics in Environmental Science

Environmental questions are almost always questions about change. Is this stream warming? Are these bees declining? Did the restoration work? What makes them hard is that the systems involved never sit still to begin with. Streams warm and cool over a single day, bee counts swing with the weather, and vegetation recovers in fits and starts. Something is always changing, so finding a real effect means separating it from all the change that would have happened anyway.

That is the problem statistics exists to solve. It gives you a way to say how much of what you observed is the thing you were looking for, how much is ordinary variation, and how much confidence you should have in telling the two apart. Every environmental claim you will read, make, or have to defend rests on that distinction, whether or not the person making it says so out loud.

1.1.1 Why Numbers Matter

Common sense is useful, but it is prone to bias. When we already believe something to be true, we are more likely to interpret new evidence as supporting it, even if it does not. This tendency can lead us astray.

For example, if you're convinced that industrial farming is the main driver of bee declines, you might see every new drop in bee abundance as confirmation of that belief. Maybe you're right, but maybe disease or weather are stronger contributors. Statistics gives us tools to test these ideas systematically. It helps us separate patterns from noise, avoid being misled by our own expectations, and draw conclusions that are more likely to hold up under scrutiny. It's not magic, but it makes our conclusions far more reliable.

Another challenge is that the same data can tell very different stories depending on how they are aggregated. **Simpson's paradox** occurs when a trend seen in combined data reverses after you split the data into meaningful groups.

Here's a recent example. During the COVID-19 pandemic, early data showed that Italy's overall case fatality rate was higher than China's. But, once researchers *disaggregated by age group,* the trend flipped: within every age group, the fatality rate was actually higher in China. The apparent contradiction was explained by differences in the age distribution of cases (von Kügelgen, Gresele, and Schölkopf 2021).

Using case fatality rate (CFR), here are some illustrative numbers (made up for teaching).

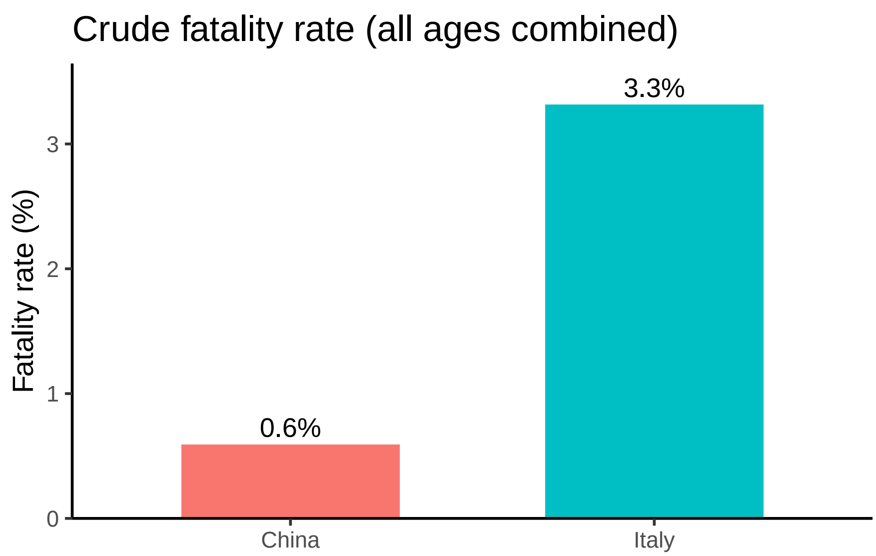

Figure 1.1: Crude case fatality rates. Italy appears to have a higher fatality rate than China when all cases are combined.

At face value, Italy looks worse. But what happens when we stratify by age group? Because age strongly affects COVID fatality, splitting cases into 10-year bands reveals a different picture. When we do, China's fatality rate is higher in every band.

Why the reversal? Italy's population skews older while China's skews younger, and baseline risk rises steeply with age. It is not that people in China fared better, it is that Italy had a greater proportion of older people getting sick, which pulls Italy's overall average up and China's down. Aggregating without adjusting for age hid the within-group pattern entirely.

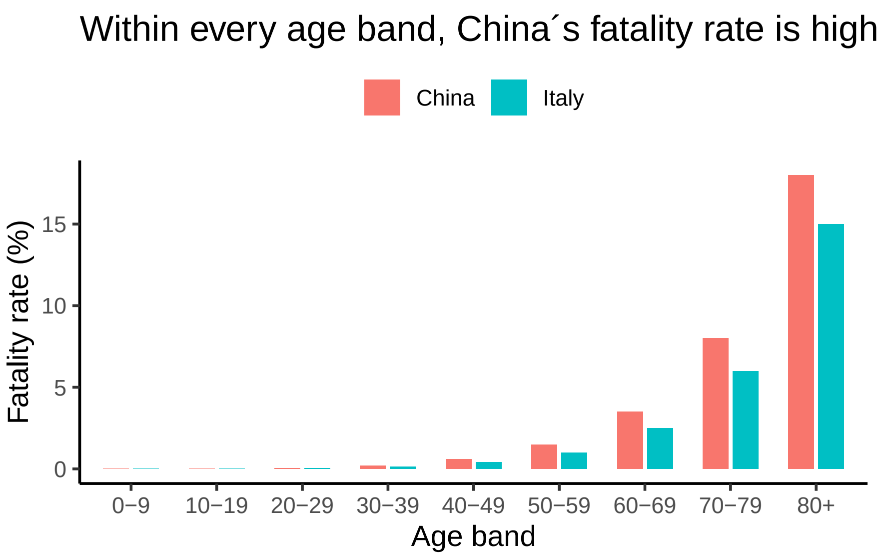

Figure 1.2: Fatality rates by age group (same made-up numbers). Within each age band, China's CFR exceeds Italy's.

That is the disciplined check statistics offers: the answer can reverse once you ask the data a more specific question.

Sheer volume is the other reason. Satellites measure surface temperature, field stations log rainfall, and sensors track air quality, so a single study can generate thousands of rows. Meanwhile river flow changes by the hour, forests grow unevenly, and human activity layers on more variability still. Statistics is how we compress all of that into something we can reason about, and how we ask whether an apparent change is real or just the fluctuation we would expect anyway.

It would be reasonable to ask why you need to learn this yourself rather than handing the analysis to someone who already knows it. Three reasons:

1. **Design and analysis go together.** A good study starts long before you run a statistical test. If you want to study how fertilizer affects crop yields, your sampling plan and your analysis are inseparable. Poor design, like measuring only in unusually wet fields, can't be rescued by sophisticated analysis.
2. **Understanding the Science:** Scientific papers on climate change, biodiversity, or pollution are built on statistical results. To interpret them, you need to know what the numbers mean.
3. **Practicality:** Hiring a specialist for every question isn't realistic. A working knowledge of statistics makes you more self-sufficient, whether you end up working as a scientist, an environmental manager, or a decision-maker who has to evaluate someone else's analysis.

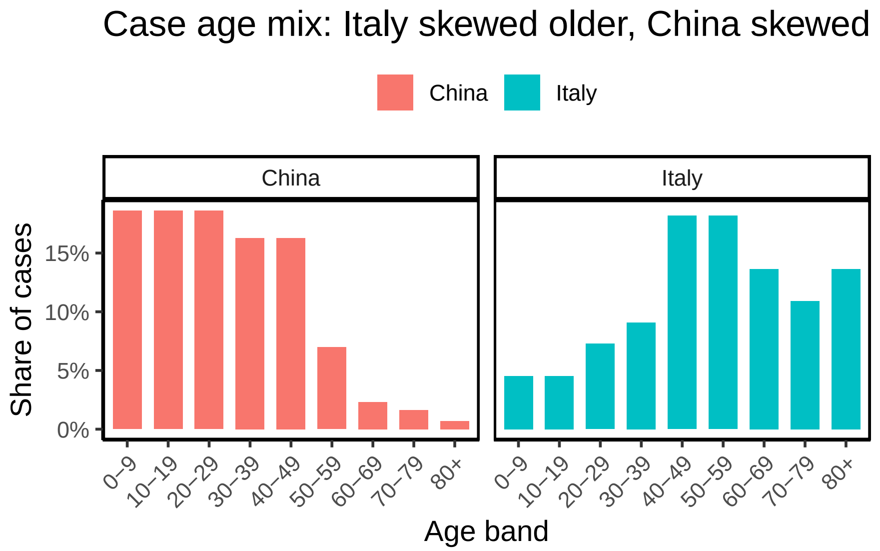

Figure 1.3: Case age distribution by country. Italy's cases are older on average; China's are younger.

> *"We are drowning in information, but we are starved for knowledge"*
>
> – Various authors, original probably John Naisbitt

Finally, none of this is only for researchers, or only for environmental science. Weather forecasts, air quality alerts, and wildlife population trends are all communicated through numbers, and so are the statistical claims that turn up in news coverage, policy debates, and podcasts. Once you can recognize the difference between a correlation and a cause, or spot a claim resting on a sample too small to support it, you start noticing those moves everywhere. A basic knowledge of statistics is what lets you tell whether the numbers support the claim being made, or whether someone is stretching them. In that sense, it is part of being an informed citizen in a data-saturated world.

1.2 Introduction to measurements

Every dataset begins with measurement: assigning numbers, labels, or categories to aspects of the world so they can be recorded and analyzed. Science often deals with broad ideas that need to be pinned down before they can be studied. For example, *soil health* or *forest change* are meaningful, but vague. To analyze them, we have to decide exactly what we mean and how to capture it. That process is called **operationalization**: turning a general concept into something measurable.

Before moving on, let's clarify how operationalization fits with three related terms:

- **Operationalization**: the step where you define exactly how a broad idea will be captured with a measure
- **Measure**: the tool or method used to make observations
- **Variable**: the actual values you record once the measure is applied. Variables are the actual data that end up in our dataset.

Example:

- Concept of interest: *Soil health*
    - **Operationalization:** concentration of organic carbon in the top 30 cm
    - **Measure:** laboratory analysis of soil samples
    - **Variable:** recorded carbon concentration values for each plot

Soil health is the idea we care about, but you cannot go outside and measure an idea. Organic carbon is a defensible stand-in, since healthy soils generally store more of it, but soil health also involves compaction, pH, microbial activity, and how well the soil holds water. So the first decision is which facet will stand in for the whole:

- **Which facet?** pH compaction ***organic carbon*** microbial activity water-holding capacity

Choosing carbon means the other facets go unmeasured. And the choice is not finished, it has only narrowed. How deep will you sample? Thirty centimeters is conventional, but 10 cm or a full meter would give different answers:

- **Sampling depth?** 10 cm ***30 cm*** 1 m

Then come a dozen more, each one a place where another reasonable person might go the other way:

- **Which carbon counts?** ***organic*** carbonate minerals total
- **How many cores per plot?** 1 3 ***6*** 9
- **Where in the plot?** wherever is easiest ***stratified random*** along a fixed transect
- **When in the season?** spring ***after harvest*** same calendar date each year

Every checked box narrows what "soil health" now means in your dataset, and every unchecked box is a part of soil health your data can no longer speak to. Another researcher could work down the same list, check different boxes, report a different number, and both of you would be right, because you measured different things and gave them the same name.

Operationalization is rarely straightforward, and there is no single correct way to do it. The best choice depends on the question you are asking and how the data will be used. In many fields, scientists have developed common practices, but every project still requires case-by-case judgment. Even so, some principles of good operationalization apply across studies: be precise about what you mean, how you will measure it, and what values are possible.

1.3 Scales of measurement

As the previous section indicates, the outcome of a measurement is called a **variable**. Not all variables are the same type, and knowing the type matters because it determines which statistical tools make sense. The four classic scales are **nominal, ordinal, interval,** and **ratio**.

1.3.1 Nominal scale

A **nominal scale** variable (also referred to as a **categorical** variable) is one in which the values are just names. They do not have an inherent order, and it makes no sense to average them.

Example: land cover class. A pixel on a satellite map might be classified as forest, grassland, wetland, or urban. None of those is "greater" than another, they cannot be put in a meaningful order, and there is no such thing as an average land cover.

It is worth being precise about *why*, because the arithmetic itself does not fail. Map classes are drawn in colors, and colors have numeric codes, so a computer will happily average forest, grassland, wetland, and urban and hand you a result. Figure 1.4 shows what comes back.

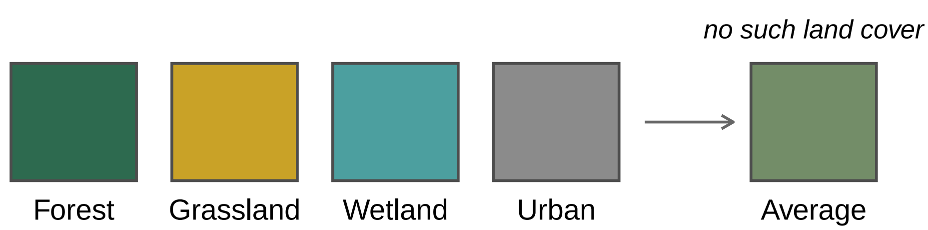

Figure 1.4: Averaging the color codes of four land cover classes returns a color, but not a land cover class. Nothing on the map is that color.

You get a color back. Nothing on the map is that color, and more importantly there is no class it could represent, because forest and urban have no midpoint. That is what makes a variable nominal: the values are **labels, not quantities**, so arithmetic on them runs perfectly well and tells you nothing. Land cover is unusual in even having numbers attached that *could* be averaged, and only because someone chose colors for a map. Most nominal variables, like species identity or watershed name, have nothing there to average in the first place.

1.3.2 Ordinal scale

Ordinal scale variables have a meaningful order, but the spacing between values is not defined mathematically. You can rank the values, but you can't assume equal steps between them, nor calculate a meaningful average.

Example: finishing position in a competition. First place comes before second, and second before third, but the ranks themselves tell you nothing about how far apart the competitors actually were.

An Olympic weightlifting meet makes this concrete, because the thing being ranked is a weight, so we can see both scales at once. Figure 1.5 shows a podium alongside the weights the three athletes actually lifted.

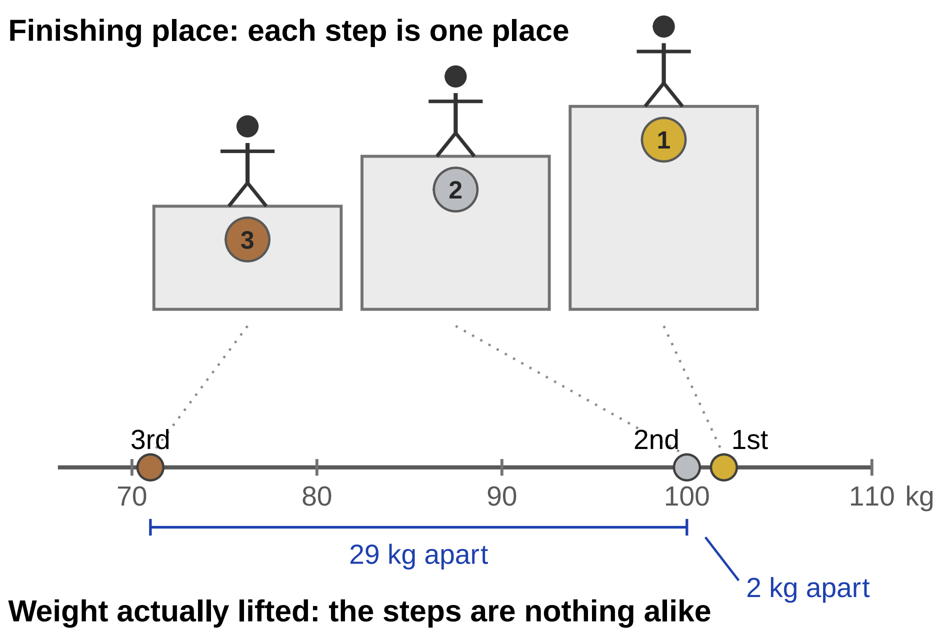

Figure 1.5: The same three athletes shown two ways. On the podium the steps are one place apart and one place apart again. On the scale below, the weights they lifted are 29 kg and 2 kg apart.

On the podium, second place looks exactly as far from first as it does from third. On the scale below, second place is nearly tied with first and nowhere near third. That is the cost of ranking: it keeps the order and throws away the distances. Notice also that a single event gives us both kinds of variable. The finishing place is ordinal, the weight lifted is ratio, and only one of them can be averaged.

1.3.3 Interval scale

Interval variables have equal intervals between values, so differences are meaningful. However, zero is arbitrary, so multiplication and division are not valid.

Example: temperature in degrees Celsius. A difference of 3° means the same regardless of whether it is $7 \to 10$ or $15 \to 18$. But 0°C does not mean "no temperature." That zero is defined by the freezing point of water, which is a useful convention rather than a fact about heat. So you can say today is 3° warmer than yesterday, but you cannot say 20°C is "twice as hot" as 10°C.

Alternate example: calendar years. The ten years from 1990 to 2000 are the same length as the ten years from 2010 to 2020, so differences work fine. But year 0 is a convention we agreed on, not the beginning of time, and other calendars put their zero somewhere else and are equally correct. That is why the year 2000 is not "twice as late" as the year 1000.

1.3.4 Ratio scale

Ratio scale variables have all the properties of interval variables, plus a true zero, meaning zero really does indicate none of the thing. That is what makes multiplication and division valid.

Example: age in years. Zero means no age at all, and someone who is 20 really is twice as old as someone who is 10. Differences ($20 - 10 = 10$ years) and ratios ($20 \div 10 = 2$) are both meaningful.

Temperature makes an unusually clean comparison here, because the same physical quantity can be recorded on either kind of scale depending on where you put the zero. Figure 1.6 shows both.

The consequence is easy to check. Going from 10°C to 20°C doubles the Celsius number, so it is tempting to call it twice as hot. On the Kelvin scale those same two temperatures are 283 K and 293 K, an increase of about 3.5%. The physical change is small; the doubling was an artifact of where Celsius happens to put its zero. Kelvin, with a true zero, gives ratios you can trust: 200 K really does carry twice the thermal energy of 100 K.

	can rank	can subtract/add	can multiply/divide	example
nominal				land cover class
ordinal	x			finishing place
interval	x	x		temperature (°C)
ratio	x	x	x	age

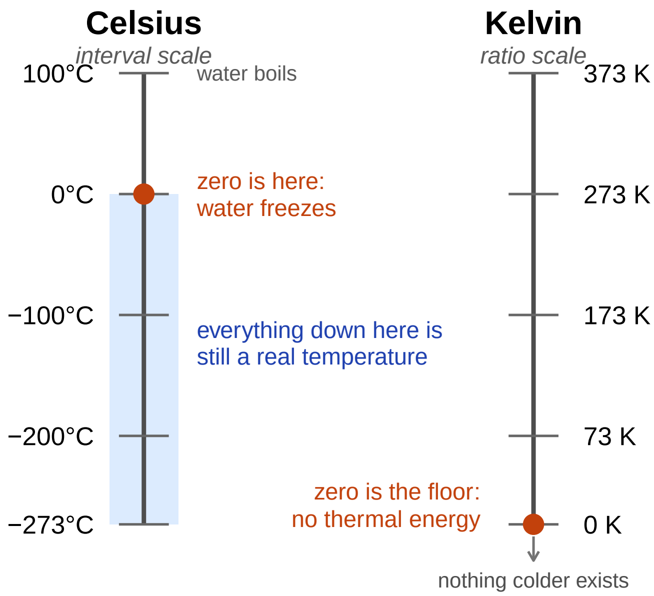

Figure 1.6: The same physical range of temperatures on two scales. Both rails cover identical ground; only the placement of zero differs. Celsius anchors its zero to the freezing point of water, Kelvin to the absence of thermal energy.

1.3.5 Continuous versus discrete variables

Another useful distinction is whether a variable can take on values in between others.

- A **continuous variable** can, in principle, take on any value within a range. For example, consider height. If you are 72 inches tall and your friend Cameron is 71 inches tall, Alan could be 71.4 inches and David 71.49 inches. Because we can always imagine a new value in between two others, height is continuous.
- A **discrete variable** is, in effect, a variable that isn't continuous. Discrete variables have separate, distinct values with nothing in between. Nominal variables are always discrete: there is no land cover class that falls "between" wetland and urban in the same way that 71.4 falls between 71 and 72. Ordinal variables are also discrete: although second place falls between first and third, nothing can logically fall between first and second.

Interval and ratio variables can be either. Height (a ratio variable) is continuous. But the number of people living in a household (a ratio variable) is discrete: you cannot have 4.2 people. Temperature in degrees Celsius (an interval variable) is also continuous. But the year you started college (an interval variable) is discrete: there is no year between 2022 and 2023.

The table below shows how the scales of measurement relate to this distinction. Cells with an "x" mark what is possible.

	continuous	discrete
nominal		x
ordinal		x
interval	x	x
ratio	x	x

1.3.6 A note on real data

These categories are guides, not hard rules. For example, survey responses on a 1–5 "strongly disagree" to "strongly agree" scale are technically **ordinal**, but researchers often treat them as "quasi-interval" because the spacing is assumed to be roughly equal.

1.4 The role of variables: predictors and outcomes

One last piece of terminology before we leave variables behind. In most studies we have many variables, but when we analyze them, we usually split them into two roles: the **thing we're**

trying to explain and the **thing doing the explaining**. To keep it straight, we use Y for the variable being explained, and a X_1, X_2, etc. for the variables used to explain it.

Traditionally, X is called the **indepdendent variable (IV)** and Y is the **dependent variable (DV).** The logic is that if there's a relationship, Y depends on X. These terms can be clunky and can be confusing because: (a) IVs are rarely actually "independent of everything else" and (b) if there's no relationship, then the DV doesn't actually "depend" on the IV at all.

Two alternative vocabularies are often clearer. In experiments, IVs are **manipulations** and DVs are **measurements**. When we are using X to predict Y rather than manipulating anything, the pair is **predictor** and **outcome**.

role of the variable	classical name	in experiments	in prediction
"to be explained" (Y)	dependent variable (DV)	measurement	outcome
"to do the explaining" (X)	independent variable (IV)	manipulation	predictor

Note

Which terms this course uses. You should recognize all three pairs, since all three appear in the literature. This book uses **IV and DV** when comparing groups (t-tests, ANOVA) and **predictor and outcome** for regression, where nothing is manipulated and "independent variable" would mislead. A later chapter switching vocabulary is following the convention of the method, not introducing a new idea.

1.5 Experimental and non-experimental research

A central distinction in research is between experimental and non-experimental studies. What matters here is the degree of control the researcher has.

In **experimental research**, the researcher deliberately manipulates something (the predictor/IV) and measures its effect on outcomes (the outcome/DV). The goal is to isolate causal effects. To avoid the problem of "something else" influencing the outcome, researchers try to hold other factors constant. In practice, it's almost impossible to identify *everything* that might matter, much less keep it constant. The standard solution is **randomization**. Randomization doesn't eliminate confounds, but it makes them less likely to systematically bias results.

For instance, suppose we wanted to know if smoking causes lung cancer. Observing smokers and non-smokers can only get us so far, because those groups differ in many ways besides smoking, like occupation, income, and diet. A true experiment would require randomly assigning people to smoke or not. You can see how that would be deeply unethical. The same problem comes up in medicine: we know surprisingly little about how certain drugs or exposures affect pregnant people, precisely because we cannot ethically assign them to risky conditions.

In **environmental science** the limitation is usually not ethics but feasibility. There is no control planet to set aside as a baseline, and nobody can dial precipitation or temperature up and down. So the field leans heavily on **non-experimental research**, which mostly takes three forms.

Quasi-experiments compare conditions the researcher did not create. You cannot control what a factory discharges into a river, but you can compare reaches above and below the outfall and use statistical tools to account for the confounding variables you could not hold constant.

Time series follow one system over a long stretch, which matters because many environmental processes unfold across decades or centuries. Tracking global temperature or sea level takes years of consistent measurement, and statistics is what separates the trend from the natural year-to-year noise.

Case studies go deep on a single event or place: the aftermath of a wildfire, the ecology of a threatened habitat, the consequences of a policy. One case is tied to its context, but a well-designed one can still expose a mechanism that travels. A drought in one forest can reveal how water potential and transpiration respond to stress, which helps anticipate what other forests will do.

1.6 From measurement to description

The first half of this chapter was about what data *are*: how a broad concept gets operationalized into a variable, what scale that variable lives on, and what kind of study produced it. The rest of the chapter is about what to do with a dataset once you have one.

> Far better an approximate answer to the right question, which is often vague, than an exact answer to the wrong question, which can always be made precise.
> John W. Tukey

Statistics often begins with the problem of **too much data**. Rainfall at hundreds of stations, thousands of tree measurements in a forest survey, millions of satellite pixels: looking at raw numbers alone is overwhelming and uninformative. We need tools to compress, summarize, and visualize, and those tools are **descriptive statistics**.

There is more than one useful way to describe a dataset, and which one you reach for depends on the pattern you are trying to reveal. Every method in this half of the chapter was invented for that reason.

Table 1.4: The first six days of the airquality dataset.

Ozone	Solar.R	Wind	Temp	Month	Day
41	190	7.4	67	5	1
36	118	8.0	72	5	2
12	149	12.6	74	5	3
18	313	11.5	62	5	4
NA	NA	14.3	56	5	5
28	NA	14.9	66	5	6

1.7 Looking at the data

Everything up to now has been about where numbers come from. From here on we work with a real dataset, so the tools have something concrete to act on.

R ships with `airquality`, a record of daily air quality measurements taken in New York from May through September of 1973. Each row is one day, with ozone concentration (parts per billion), solar radiation, wind speed, and maximum temperature.

Look at the `NA` entries. Ozone and solar radiation were not recorded every day, and real environmental datasets almost always have gaps like this. That has an immediate practical consequence:

```
mean(airquality$Ozone)
#> [1] NA
```

R returns `NA` rather than a number, because it has no way of knowing what the missing days would have been. You have to tell it explicitly to set those days aside:

```
mean(airquality$Ozone, na.rm = TRUE)
#> [1] 42.12931
```

That `na.rm = TRUE` will follow you through the rest of the course. Now, to the data itself.

1.7.1 Scatter of raw values

Plot every value. Before summarizing anything, it is worth simply looking at all of it.

Figure 1.7 shows all 153 daily temperatures at once, with day number across the bottom and that day's maximum temperature up the side. Already this tells us more than a table of 153 numbers would: there is a clear seasonal arc, cooler in May, warmest through July and August, cooling again by September.

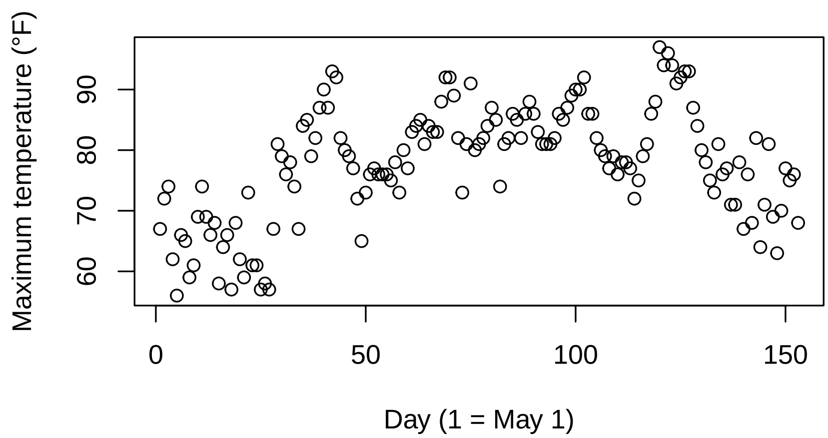

Figure 1.7: Daily maximum temperature at a New York station, May through September 1973.

But notice what the plot does *not* answer well. If you want to know whether most days were warm or cool, you end up squinting at a cloud of dots, because the seasonal ordering gets in the way. For that we need a different view, a **histogram**.

1.7.2 Histograms

A histogram summarizes data by grouping values into bins rather than showing each point separately. It throws away the day-to-day ordering on purpose, and in exchange it shows you the shape of the data.

Each bar counts the days falling in a temperature range, and the pattern is now easy to read: most days land in the 70s and 80s, with a tail of cooler days near 56°F and relatively few above 90°F.

Bin width is a choice you make. Wide bins look smooth but hide detail; narrow bins show more variation but can dissolve into noise. Figure 1.9 compares the same temperatures at four bin widths.

All four show the same general shape, but the extremes are unhelpful. Five bins is so coarse that the peak becomes one wide block; fifty spreads 153 days so thinly that bars jump around at random. The middle two are the ones you would actually use. A histogram trades away the individual values and the day-to-day ordering, and in exchange it shows you the shape of the data, but only if you pick a sensible bin width.

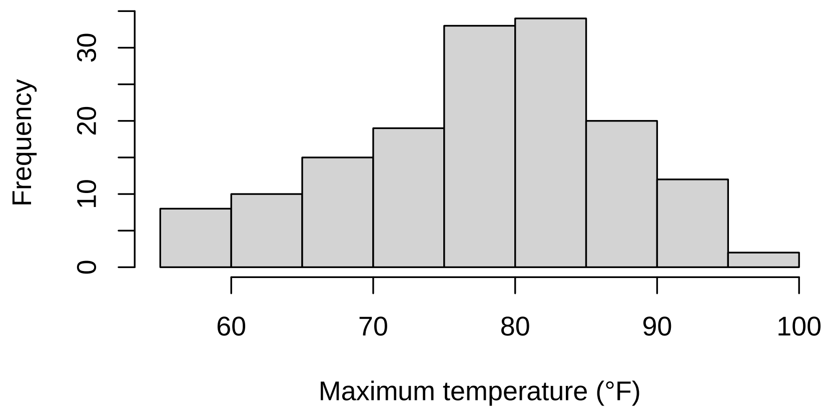

Figure 1.8: A histogram of the same daily temperatures.

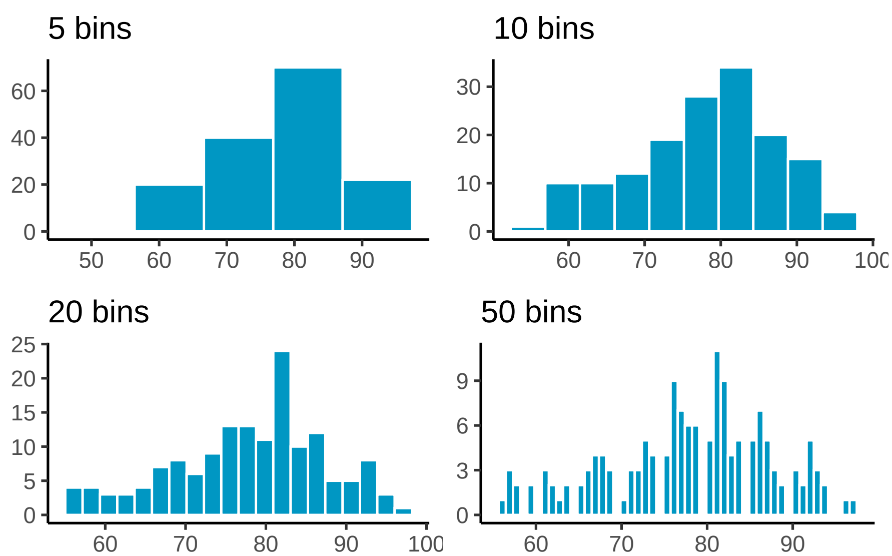

Figure 1.9: Four histograms of the same temperature data using different numbers of bins.

1.8 Important Ideas: Distribution, Central Tendency, and Variance

Three terms will keep coming up: **distribution, central tendency, and variance.** Their everyday meanings aren't far from the statistical ones.

- **Distribution**: how values are spread across their range. A histogram is one way of showing a distribution. The shape of the distribution tells us where values are concentrated, where they're rare, and how far the extremes reach. Later we'll see that distributions don't just describe data, we also use them to **generate theoretical data**, which is the basis for sampling distributions and tools like t-tests.
- **Central tendency** is about *sameness*: where values cluster. In the temperature histogram, most days sit in the high 70s and low 80s, so we say the data have a central tendency around 80°F. Central tendency doesn't mean "everything is the same", just that many values gather in a common area. Some datasets even have more than one cluster, and therefore more than one central tendency.
- **Variance** is about *differentness*: how spread out the values are. If every day had been 78°F, variance would be zero. Instead the days run from 56°F to 97°F, so there is real variance to describe. In practice, variance is a family of measures we use to quantify differences, and we'll introduce those next.

1.9 Measures of Central Tendency (Sameness)

Plots and histograms show where values fall, but lack precision. We can summarize data with a single number representing its "center" - these **measures of central tendency** show what values are generally like.

1.9.1 Mode

The **mode** is the most frequently occurring value: count how often each value appears, and whichever wins is the mode. In `1 1 1 2 3 4 5 6` the mode is 1, since it appears three times.

Ties are allowed. In `1 1 1 2 2 2 3 4 5 6` both 1 and 2 appear three times, so the data have two modes.

The mode does not always land where you would call the center. In `1 1 2 3 4 5 6 7 8 9` the mode is 1, but nearly every value is larger. It is most useful when a dataset has genuinely repeating values, and like any tool it needs to suit the data you have.

1.9.2 Median

The **median** is the exact middle of the data once they are ordered from smallest to largest. For example:

1 5 4 3 6 7 9

Before we can compute the median, we need to order the numbers from smallest to largest. Ordered:

1 3 4 **5** 6 7 9

The median is 5, with three numbers on each side.

With an even count there is no single middle value, so the median is the midpoint of the two central numbers. In `1 2 3 4 5 6` that is halfway between 3 and 4, or 3.5.

The median holds still when extreme values do not. Consider `1 2 3 4 4 4 5 6 6 6 7 7 1000`. Most values are modest and one is wildly out of line, yet the median is still 5, sitting right where the bulk of the data is. That value of 1000 is an **outlier**, a value much farther out than the rest. Outliers can drag the mean a long way while leaving the median untouched, and how to handle them is a question we return to later in the course.

1.9.3 Mean

The **mean**, also called the average, is the sum of a set of numbers divided by how many numbers there are.

$$Mean = \bar{X} = \frac{\sum_{i=1}^{n} x_i}{N}$$

The $\sum$ symbol is **sigma**, and it means "sum these up." The x_i are the individual numbers, i runs from the first to the last, N is how many there are, and $\bar{X}$ is the mean. In plainer terms, it is the sum of your numbers divided by the count of your numbers.

For the five numbers 3, 7, 9, 2, and 6, that gives

$$\bar{X} = \frac{3+7+9+2+6}{5} = \frac{27}{5} = 5.4$$

Whether the mean is a *good* summary, as with the mode, depends on the data.

1.9.4 What does the mean mean?

Knowing how to calculate a mean is not the same as knowing what it represents. The formula divides a sum by a count, but what does that operation actually *do*?

Think about plain division. When we compute $\frac{12}{3} = 4$, we are splitting 12 into three equal parts of 4 each. Division equalizes the numerator into identical pieces.

The same logic applies to the mean. Suppose you add up every daily temperature across the whole summer into one large total. If that total were redistributed equally across all the days, each day would get the same value. That value is the mean. Individual days were originally warmer or cooler, but the mean is the balance point created by spreading the total evenly.

This is why the mean is more than just arithmetic. It is the one number that can replace every observation so that, when all those replacements are added together, you get back the original total.

That is what makes the mean unique. It is the only single value that can replace every observation and still leave the total unchanged.

1.9.5 Comparing the mean, median, and mode

So far we have three summaries of the "middle" of a dataset. It is worth seeing what happens when they disagree, because the disagreement itself is informative.

Temperature was a well-behaved variable: its mean (77.9°F) and median (79°F) sit almost on top of each other, which is what you expect when a distribution is roughly symmetric. Ozone is a different story.

All three land in different places, and the order they fall in is not an accident. For a right-skewed distribution you will nearly always find **mode < median < mean**, which is exactly what Figure 1.10 shows: the mode at 23 ppb, the median at 31.5 ppb, and the mean at 42.1 ppb.

They differ because each one is sensitive to something different.

- The **mean** uses every value, so it feels the full weight of those bad-air days out past 150 ppb. Add one extreme day and the mean moves.
- The **median** only cares about which value sits in the middle. Those same extreme days push it over by one position at most, so it stays down where the bulk of the days actually are.
- The **mode** ignores position entirely and reports whichever value occurred most often. It answers a genuinely different question: not "what is the center" but "what happened most".

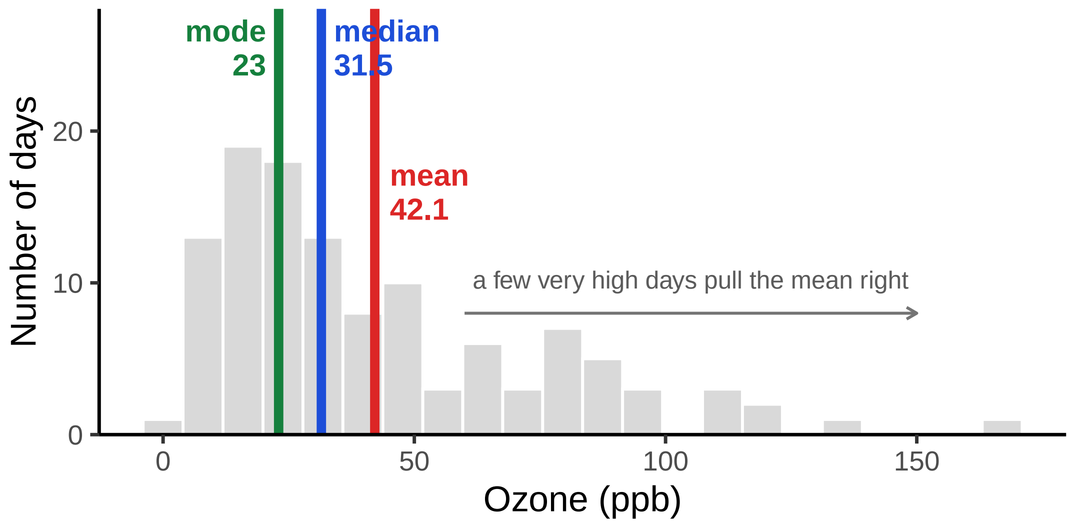

Figure 1.10: Daily ozone concentration with all three measures of central tendency marked. They fall in the order mode, median, mean, which is the standard signature of a right-skewed distribution.

Neither the mean nor the median is wrong here. They answer different questions. If you want to know what a typical day looked like, the median is the better guide. If you care about the total ozone burden across the summer, the mean is the number that relates to that total.

> **Note**
>
> **When the mode gets fragile.** Ozone is a continuous measurement, so exact repeats are largely an accident of rounding. The value 23 ppb occurs 6 times out of 116 days, barely more often than several neighbors. Round differently and the mode moves. It is at its best for categorical variables, where "most common land cover class" is exactly the question, and for values that genuinely repeat.

1.10 Measures of Variation (Different*ness*)

Central tendency tells us what values have in common. Measures of **variation** tell us how they differ. Any dataset with more than one value will show some variation, and summarizing that spread is as important as finding the center.

1.10.1 The Range

Consider these 10 ordered numbers:

1 3 4 5 5 6 7 8 9 24

The smallest is 1, the largest is 24. Together they define the **range**. The range quickly shows the boundaries of the data and can flag possible outliers. For instance, if you expected values between 1 and 7 but found one at 340,500, you'd know something unusual happened and might investigate.

1.10.2 Difference Scores

It would be nice to summarize the amount of different*ness* in the data. Think about these 10 ordered numbers:

1 3 4 5 5 6 7 8 9 24

One idea is to compare every number to every other number. It works, but it scales badly: 10 numbers give 100 pairwise differences, and 500 numbers would give 250,000. If all the values were identical every difference would be 0 and the lack of variation would be obvious, but with real data we just get a large pile of differences. The natural way to summarize a pile of numbers is the one we just learned, so let's take their **average**.

Let's try it on a set small enough to see all at once:

1 2 3

	1	2	3
1	0	1	2
2	-1	0	1
3	-2	-1	0

The mean of these nine difference scores is:

mean of difference scores $= \frac{0+1+2-1+0+1-2-1+0}{9} = \frac{0}{9} = 0$

This will always happen: the positives and negatives cancel. So the mean of raw difference scores is not a useful measure of variation.

Notice also that the matrix is redundant: the diagonal is always zero, and values above and below the diagonal are the same except for sign.

These problems motivate why we compute **variance** and **standard deviation**. They solve the cancellation issue and give us a practical summary of spread.

1.10.3 The Variance

We have used the words variability, variation, and variance a lot, and they all point at the same idea: numbers differ. The word *variance* does double duty, though. Loosely it means that general idea. Precisely it names a specific statistic, the **mean of the squared deviations from the mean**.

To calculate it, subtract the mean from each score to get its **difference score**, square those differences, then average them. Squaring is the trick that stops positive and negative deviations from cancelling. In short:

$variance = \frac{\text{Sum of squared difference scores}}{\text{Number of Scores}}$

A little further on we'll distinguish between dividing by N (when you have the entire population) and $n-1$ (when you only have a sample, which is almost always the case). For now, just use N.

1.10.3.1 Deviations from the mean

Earlier we compared every number to every other number, which got unmanageable fast. The simpler approach is to compare each score to the mean instead: find the mean, then subtract it from every score. What comes back tells us both how well the mean represents the data and how much spread there is around it.

scores	values	mean	Difference_from_Mean
1	1	4.5	-3.5
2	6	4.5	1.5
3	4	4.5	-0.5
4	2	4.5	-2.5
5	6	4.5	1.5
6	8	4.5	3.5
Sums	27	27	0
Means	4.5	4.5	0

The mean here is $\frac{1+6+4+2+6+8}{6} = \frac{27}{6} = 4.5$. The third column just repeats it on every row, which looks redundant but makes the earlier point concrete: six copies of 4.5 add back to the original total of 27. The fourth column is the **deviation from the mean**, $X_i - \bar{X}$, so the score of 1 sits 3.5 below and the score of 6 sits 1.5 above.

And now the problem. Those deviations sum to zero. The positives and negatives cancel exactly, which would suggest there is no variation at all, and that is obviously false. Squaring will fix it, but first it is worth seeing *why* the cancellation happens.

1.10.3.2 Mean as the Balancing Point

The mean is the balancing point of the data. Imagine laying a ruler across your finger. The place where it balances is the point where the weight on each side is equal. Data works the same way: if we treat each value like a weight, the mean is where the total "mass" to the left and right cancel out.

This balancing property explains why the deviations always sum to zero. The values below the mean contribute negative deviations, the values above contribute positive ones, and together they cancel:

$-x + x = 0$

To see this more concretely, consider the numbers

$X = (1, 2, 6, 7, 9)$

Figure 1.11: Values on a number line.

Now imagine the number line as a teeter-totter. Only when the fulcrum is placed at 5 does the board balance:

Formally, this means the signed distances from the mean always cancel out:

$(1 - 5) + (2 - 5) + (6 - 5) + (7 - 5) + (9 - 5) = 0$

This makes the mean unique. It is the **only value** for which the sum of deviations equals zero:

$\sum_{i=1}^{n} x_{i-a} = 0 \quad \Rightarrow \quad a = \bar{x}$

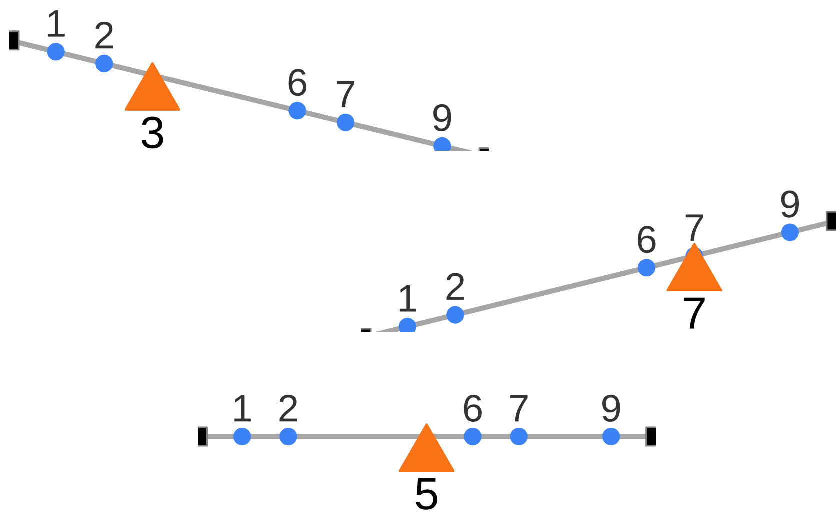

Figure 1.12: Mean as the unique balancing point. Only at the mean do signed deviations cancel.

The mean is not just a "typical" value, it is the one point where the data balance. And that balancing property is exactly why the raw deviations always sum to zero. To summarize variation, we need a way around this problem.

1.10.3.3 The squared deviations

The standard trick is to square the deviations. Since $2^2 = 4$ and $(-2)^2 = 4$, squaring turns every negative into a positive, so the deviations can no longer cancel. We call these **squared deviations**. Here is the same table with a column for them:

scores	values	mean	Difference_from_Mean	Squared_Deviations
1	1	4.5	-3.5	12.25
2	6	4.5	1.5	2.25
3	4	4.5	-0.5	0.25
4	2	4.5	-2.5	6.25
5	6	4.5	1.5	2.25
6	8	4.5	3.5	12.25
Sums	27	27	0	35.5
Means	4.5	4.5	0	5.91666666666667

The first score of 1 sits 3.5 below the mean, so its deviation is -3.5 and its squared deviation is $(-3.5)^2 = 12.25$. Now that nothing is negative we can add them up, and that total is the

sum of squares (SS). You will meet this quantity again in the ANOVA chapter, but there is nothing more to it than the sum of those squared deviations.

1.10.3.4 Finally, the variance

We have already computed the variance, one step at a time, without naming it. The goal was a single number summarizing spread, but deviations give us as many numbers as we have data points, so we want their *average*. Sum the squared deviations, divide by how many there are, and that is **the variance**:

$$variance = \frac{SS}{N}$$

For our data it comes to 5.916. On its own that number is hard to interpret, because squaring the deviations put it on a *squared* scale, so it is no longer in the same units as the data. The fix is to take the square root, $\sqrt{5.916} \approx 2.43$.

1.10.4 The Standard Deviation

That square root is the **standard deviation (SD)**, so we had already computed it a step earlier without naming it:

$$\text{standard deviation} = \sqrt{Variance} = \sqrt{\frac{SS}{N}} = \sqrt{\frac{\sum_i^n (x_i - \bar{x})^2}{N}}$$

The square root "unsquares" the variance, putting the measure of spread back on the original scale of the data. Here is the table one more time:

scores	values	mean	Difference_from_Mean	Squared_Deviations
1	1	4.5	-3.5	12.25
2	6	4.5	1.5	2.25
3	4	4.5	-0.5	0.25
4	2	4.5	-2.5	6.25
5	6	4.5	1.5	2.25
6	8	4.5	3.5	12.25
Sums	27	27	0	35.5
Means	4.5	4.5	0	5.91666666666667

Our standard deviation, $\sqrt{5.916} \approx 2.43$, makes far more sense than the variance did. A variance of 5.916 feels too large for this data because it is on the squared scale, but an SD of 2.43 lines up with the actual deviations, since most scores fall within about 2.4 of the mean of 4.5.

That is what makes the SD so useful. Told only that a dataset has a mean of 4.5 and an SD of 2.4, you already have a picture of it: values cluster around 4.5, not exactly at it, with a typical spread of two to three units either way. It gives you the *typical deviation* from the mean while staying in the original units.

1.10.5 Sample versus population: a note on $n-1$

One detail we glossed over: is N the size of the whole population, or just the size of our sample? For the mean it does not matter. The sample mean $\bar{x}$ is **unbiased**, meaning that averaged across many samples it lands right on the true population mean.

The standard deviation is not so well behaved. A sample standard deviation computed by dividing by N comes out systematically **too small**. The reason is that $\bar{x}$ is calculated *from* the same points used to measure spread, which pulls those points artificially close to it, so the distances we are averaging are shorter than they should be.

The fix is to divide by $n-1$ instead of n:

$$s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

This is what R's `sd()` function already does, and it is what "the sample standard deviation" means unless you are told otherwise. Chapter 3 returns to this and shows the bias directly with a simulation, once we have the vocabulary of populations, samples, and estimation to explain *why* the correction is the right size.

1.10.6 Mean Absolute Deviation

So far we've handled the "differences cancel out" problem by squaring deviations. But there's another simple option: take the **absolute value** of each deviation from the mean. That way, every difference becomes positive, and when we add them up we don't end up at zero.

Here's what that looks like for our data:

scores	values	mean	Difference_from_Mean	Absolute_Deviations
1	1	4.5	-3.5	3.5
2	6	4.5	1.5	1.5
3	4	4.5	-0.5	0.5
4	2	4.5	-2.5	2.5
5	6	4.5	1.5	1.5
6	8	4.5	3.5	3.5
Sums	27	27	0	13
Means	4.5	4.5	0	2.16666666666667

The last column shows absolute deviations. If we take their mean, we get the mean absolute deviation (MAD). Notice this acts a lot like the standard deviation: it's the average size of a deviation from the mean, only without squaring. Both approaches, squaring and taking absolute values, solve the same problem in slightly different ways. Squaring is more common because it leads to nice mathematical properties (like in regression and ANOVA), but the MAD can be more intuitive to interpret.

1.11 Remember to look at your data

Descriptive statistics are useful, but they are also compressed summaries. They reduce a dataset to a few numbers, which means they always lose detail. That's fine if the summary captures the important features, but sometimes it doesn't.

The safest habit is to **always combine descriptive statistics with graphs**. Compute the summaries, then plot the data, and check that the two tell the same story. We just did this with ozone: the gap between the mean and the median said "skewed," and the histogram confirmed it.

When the numbers and the graph disagree, believe the graph. A summary can only report what it was designed to measure, while a plot shows you everything, including whatever you failed to anticipate. The next two examples show just how badly summaries can mislead when you skip that step.

1.11.1 Anscombe's Quartet

Francis Anscombe (Anscombe (1973)) made this point with a famous example, now called *Anscombe's Quartet*. Each panel in Figure Figure 1.13 shows pairs of x and y values as a scatterplot.

Visually, the four panels are very different: one looks linear, one curved, one is mostly linear except for an outlier, and one forms a near-vertical line except for one point.

Yet if we summarize each quartet with just its descriptive statistics:

quartet	mean_x	var_x	mean_y	var_y
1	9	11	7.500909	4.127269
2	9	11	7.500909	4.127629
3	9	11	7.500000	4.122620
4	9	11	7.500909	4.123249

All four datasets have the **exact same descriptive statistics**:

same mean and variance for x

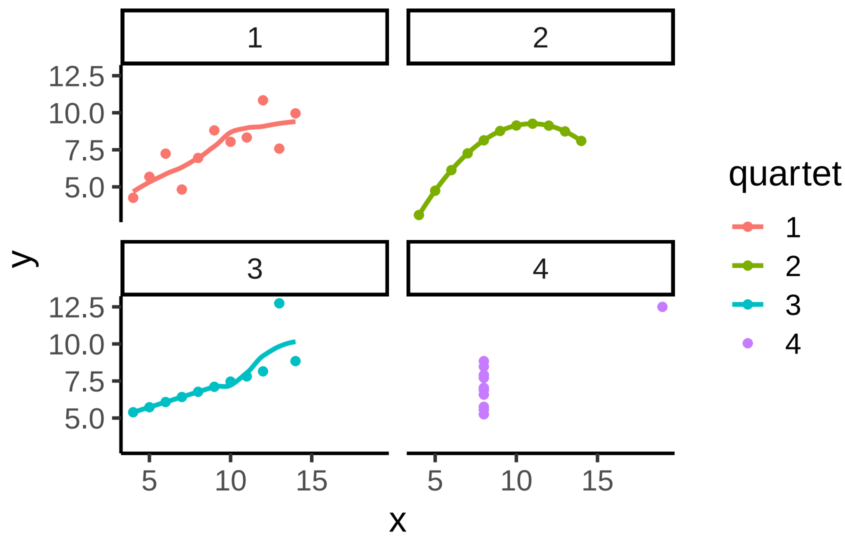

Figure 1.13: Anscombe's Quartet

same mean and variance for y

same correlation between x and y

Yet the scatterplots could not look more different.

1.11.2 Datasaurus Dozen

The Datasaurus Dozen (Matejka and Fitzmaurice 2017) takes Anscombe's point further: 13 datasets with nearly identical means, variances, and correlations, but when plotted, the points form shapes like a star, a circle, or even a dinosaur. A reminder that your numbers might look like a dinosaur if you don't plot them.

1.12 Chapter Summary

Why it matters. Every analysis later in this course assumes you already have numbers that mean something and a sense of what they look like. This chapter covers both halves of that: where data come from, and how to describe them once you have them.

On measurement and study design:

- **Why statistics matters.** It will not answer every scientific question, but it provides a disciplined way to separate signal from bias, and to avoid being fooled by aggregated data, as Simpson's paradox shows.

- **Operationalization and measurement.** Turning a broad concept like *soil health* into something recordable is a decision you have to make and justify, not something the data hands you.
- **Scales of measurement.** Nominal, ordinal, interval, and ratio, plus the continuous-versus-discrete distinction. These determine which statistical tools are available to you, which is why we return to them all semester.
- **Predictors and outcomes.** The roles variables play in an analysis, and the several vocabularies used to describe them.
- **Experimental and non-experimental research.** Environmental science rarely permits full experimental control, so the field leans on quasi-experiments, long-term time series, and case studies.

On describing data:

- **Plot first.** Scatter plots and histograms show the shape of a distribution, and bin width is a choice that changes what you see.
- **Central tendency.** The mode, median, and mean each summarize the "middle" differently, and they diverge when data are skewed.
- **Variation.** The range, then deviations from the mean, then squared deviations, the sum of squares, the variance, and finally the standard deviation, which returns the measure of spread to the original units.
- **Always look at the graph.** Anscombe's Quartet and the Datasaurus Dozen make the point that identical summary statistics can describe radically different data.

1.13 Videos

1.13.1 Terms of Statistics

1.13.2 Measures of center: Mode

1.13.3 Measures of center: Median and Mean

1.13.4 Standard deviation part I

1.13.5 Standard deviation part II

2 Correlation

> Correlation does not equal causation.*
> Every Statistics Instructor Ever

In the last chapter, we worked with data that felt overwhelming at first. We used plots and histograms to make the data visible, and descriptive statistics to summarize their central tendency and variability.

The reason we learned those tools is simple: we want to use data to answer questions. That theme runs through this course. As you read each section, keep asking yourself: how does this concept help me answer questions with data?

In Chapter 1, we described a season of daily air quality measurements. The data were just numbers, but they already raised useful questions. How spread out were the ozone readings? Did most days cluster near the middle, or were there clusters of especially clean and especially polluted days?

We also saw that numbers are imperfect stand-ins for the thing we actually care about. Ozone concentration at one station is one way of capturing air quality, and it may not capture the whole idea, but it is meaningful enough to work with.

Every one of those questions was about **one variable at a time**. That is the limit we run into now. Describing a single variable well tells you nothing about whether it travels with anything else, and most interesting questions are about exactly that. Does ozone rise on hotter days? Do wetter winters produce taller trees? Does income track education?

This sets up the next step: moving from description to explanation. Rather than asking only how much of something we observed, we start asking what it moves along with. Questions like these push us beyond describing data toward identifying relationships, the core of correlation analysis.

2.1 What data look like when one thing changes another

To use data for explanation, we need tools for recognizing when two things move together. Sometimes that reflects a real cause-and-effect relationship, and sometimes it doesn't. The challenge is learning to tell the difference. In this chapter we'll build intuition for what data look like when two variables are unrelated, when they move together, and when the pattern might mislead us.

2.1.1 More Chocolate = More Happiness?

Suppose a person's chocolate supply has a causal influence on their happiness: more chocolate means more happiness, less chocolate means less. Because happiness is also caused by plenty of other things in a person's life, we'd expect the relationship between chocolate and happiness to be real but imperfect, not a perfect one-to-one match.

Our first step is to collect some imaginary data from 100 people. We walk around and ask the first 100 people we meet to answer two questions:

1. how much chocolate do you have? and
2. how happy are you?

For convenience, both the scales will go from 0 to 100. For the chocolate scale, 0 means no chocolate, 100 means lifetime supply of chocolate. Any other number is somewhere in between. For the happiness scale, 0 means no happiness, 100 means all of the happiness, and in between means some amount in between.

Here is some sample data from the first 10 imaginary subjects.

subject	chocolate	happiness
1	1	1
2	2	2
3	3	3
4	2	3
5	5	4
6	4	4
7	4	6
8	5	8
9	9	7
10	10	8

We asked each subject two questions so there are two scores for each subject, one for their chocolate supply, and one for their level of happiness. You might already notice some relationships between amount of chocolate and level of happiness in the table. To make those relationships even more clear, let's plot all of the data in a graph.

2.1.2 Scatter plots

When you have two measurements worth of data, you can always turn them into dots and plot them in a scatter plot. A scatter plot has a horizontal x-axis, and a vertical y-axis. You get to choose which measurement goes on which axis. Let's put chocolate supply on the x-axis, and happiness level on the y-axis. Figure 2.1 shows 100 dots. Each dot is for one subject and represents both their happiness level and their chocolate supply.

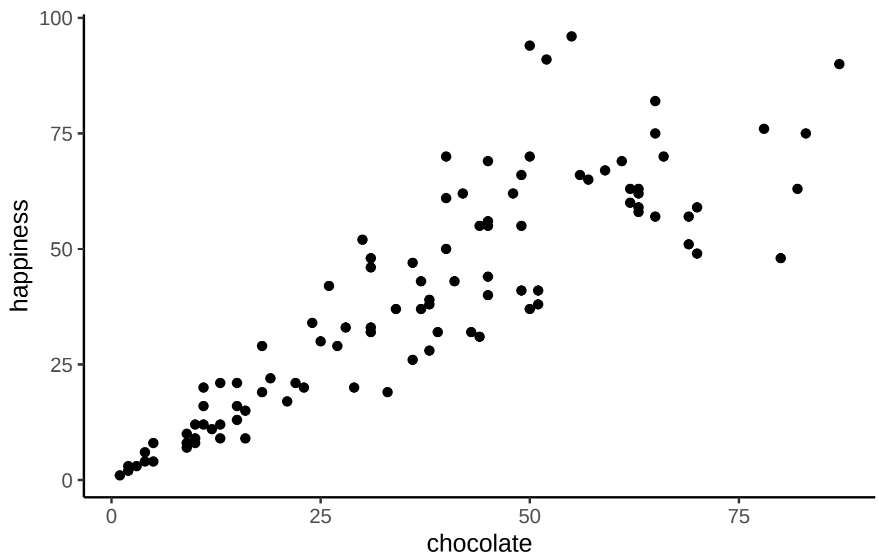

Figure 2.1: Imaginary data showing a positive correlation between amount of chocolate and amount happiness

Each dot has two coordinates, an x-coordinate for chocolate, and a y-coordinate for happiness. Looking at the **scatter plot**, we can see that the dots show a pattern. People with less chocolate tend to report lower happiness, while those with more chocolate tend to report higher happiness. It looks like the more chocolate you have the happier you will be, and vice-versa. This kind of relationship is called a **positive correlation**.

2.1.3 Positive, Negative, and No-Correlation

Two other patterns are worth seeing: a reversed relationship, where more chocolate means less happiness, and no relationship at all, where chocolate supply has nothing to do with happiness. Figure 2.2 shows all three side by side:

The first panel shows a **negative correlation**: happiness goes down as chocolate supply increases, more of X means less of Y. The second panel shows a **positive correlation**: happiness goes up as chocolate supply increases, more of X means more of Y. The third panel shows **no**

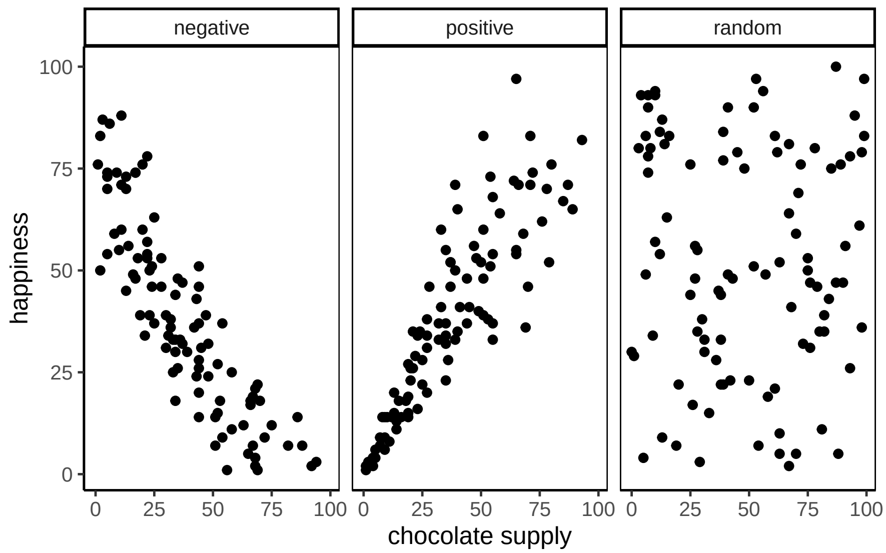

Figure 2.2: Three scatterplots showing negative, positive, and zero correlation

correlation: there's no obvious relationship between chocolate supply and happiness, changes in X don't relate to changes in Y.

> Note
>
> **Always plot first.** If the scatter shows a U-shape or other curve, a single straight-line correlation will mislead. Fit a curved model (e.g., add a quadratic term) or transform variables instead of reporting r alone.

We've already covered how to generate descriptive statistics for individual variables through means, standard deviations, etc. But to summarize how two variables move together in a single descriptive statistic, we need a new one: Pearson's r. The path there starts with understanding covariance.

2.2 Covariance

Let<80><99>s revisit **variance**. For one variable, variance summarizes how much values differ from their mean. **Covariance** extends this to two variables: it tells us whether being above the mean on X tends to come with being above the mean on Y (positive), below the mean on Y (negative), or neither (near zero).

In our chocolate<80><93>happiness example, the scatter plot suggested that higher chocolate tends to come with higher happiness. That means the deviations from each mean often have the same sign, so their products are mostly positive: evidence of *positive* covariance. Put simply, the two variables **vary together**.

Covariance is important because it underlies correlation, regression, and many other statistical tools. It can feel abstract at first. Keep the core idea in mind: look at how each variable<80><99>s deviations from its mean behave together. If both are above or below their means at the same time, the covariance is positive. If one is above while the other is below, it<80><99>s negative. If there<80><99>s no consistent pairing, the covariance is near zero. With practice, this way of thinking becomes intuitive.

> Note
>
> **Covariance intuition.** Think of a three-legged race. When partners move forward together, their steps align (positive covariance). When one moves forward as the other moves back, their steps oppose (negative covariance). If their steps are uncoordinated, there<80><99>s no consistent pattern (covariance near zero).

2.2.1 Turning the numbers into a measure of Covariance

To turn "chocolate and happiness vary together" into an actual number, we need a way to measure that relationship directly. Consider our chocolate and happiness scores.

subject	chocolate	happiness	Chocolate_X_Happiness
1	1	1	1
2	2	2	4
3	3	3	9
4	2	3	6
5	5	4	20
6	4	4	16
7	4	6	24
8	5	8	40
9	9	7	63
10	10	8	80
Sums	45	46	263
Means	4.5	4.6	26.3

We<80><99>ve added a new column called `Chocolate_X_Happiness`, which is simply each person<80><99>s chocolate score multiplied by their happiness score. Why multiply? Because the products reveal how the two values move together.

- Small <97> small <86><92> small product
- Large <97> large <86><92> large product
- Small <97> large <86><92> product in between

Table 2.1: Aligned: sum = 385

X	Y	XY
1	1	1
2	2	4
3	3	9
4	4	16
5	5	25
6	6	36
7	7	49
8	8	64
9	9	81
10	10	100
55	55	385

Table 2.2: Reversed: sum = 220

A	B	AB
1	10	10
2	9	18
3	8	24
4	7	28
5	6	30
6	5	30
7	4	28
8	3	24
9	2	18
10	1	10
55	55	220

Now extend this idea from pairs to the whole dataset. If chocolate and happiness rise together (both above average at the same time), the summed products will be large. If one tends to rise while the other falls, the summed products will shrink. This is the basic logic behind **covariance**.

Suppose we have two variables, X and Y. If X and Y go up together (both small at the same time, both large at the same time), their products will also be large when summed across the dataset. If one goes up while the other goes down (large pairs with small), the summed products will be much smaller.

Here are two tables to illustrate:

X and Y are perfectly aligned: 1 with 1, 2 with 2, up to 10 with 10. Their products add up to **385**, the *largest* possible sum.

A and B are perfectly reversed: 1 with 10, 2 with 9, and so on. Their products add up to **220**, the *smallest* possible sum.

Any other pairing of 1<80><93>10 with 1<80><93>10 will fall between these two extremes. Figure 2.3 shows the sums of products from 1000 random pairings. Every result falls between 220 (perfect negative alignment) and 385 (perfect positive alignment).

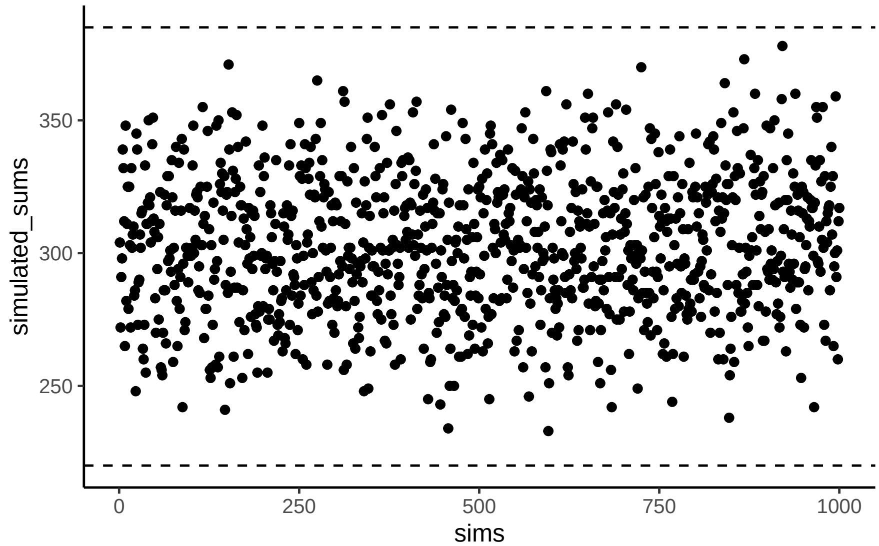

Figure 2.3: Simulated sums of products showing the kinds of values than can be produced by randomly ordering the numbers in X and Y.

The **sum of the products** tells us about the direction of the relationship:

- Close to 385 <86><92> strong positive relationship.
- Close to 220 <86><92> strong negative relationship.
- Near the middle <86><92> little or no relationship.

This is the foundation of *covariance*: products of paired values reveal whether two variables rise together, fall together, or move in opposite directions.

Takeaway:

- Positive correlation <86><92> large values in X pair with large values in Y <86><92> sum of products near the maximum.
- Negative correlation <86><92> large values in X pair with small values in Y <86><92> sum near the minimum.
- No correlation <86><92> pairings are random <86><92> sum lands in the middle.

2.2.2 Covariance, the measure

Earlier we saw that multiplying values from two variables can highlight how they vary together: large-with-large and small-with-small pairs push the product up, while mixed pairs keep it lower. That intuition is useful, but in statistics we define **covariance** more precisely.

Instead of multiplying raw values of X and Y, we multiply their *deviations from the mean*. This centers each variable, so the product reflects how the two variables co-vary around their averages:

$cov(X,Y) = \frac{\sum_i^n (x_i - \bar{X})(y_i - \bar{Y})}{N}$

In practice, the steps are straightforward: subtract the mean of X from each x_i, subtract the mean of Y from each y_i, multiply the deviations, and then average those products. A positive covariance means the two variables tend to move together (above or below their means at the same time). A negative covariance means they move in opposite directions. A covariance near zero suggests no consistent relationship.

The following table shows this process:

subject	chocolate	happiness	C_d	H_d	Cd_x_Hd
1	1	1	-3.5	-3.6	12.6
2	2	2	-2.5	-2.6	6.5
3	3	3	-1.5	-1.6	2.4
4	2	3	-2.5	-1.6	4
5	5	4	0.5	-0.6	-0.3
6	4	4	-0.5	-0.6	0.3
7	4	6	-0.5	1.4	-0.7
8	5	8	0.5	3.4	1.7
9	9	7	4.5	2.4	10.8
10	10	8	5.5	3.4	18.7
Sums	45	46	0	0	56
Means	4.5	4.6	0	0	5.6

Deviations from the mean for chocolate (column `C_d`), deviations from the mean for happiness (column `H_d`), and their products. The mean of those products (in the bottom right corner of the table) is the official **covariance**.

Covariance is a powerful idea, but it has a major drawback: its magnitude is hard to interpret. Just as variance is tied to squared units, covariance is tied to the scales of both variables, so a covariance of 3.22 isn't inherently "large" or "small" the way a correlation is. All we can read directly from it is the sign, not the strength. To compare strength across variables with different units or scales, we need to normalize covariance, which leads to Pearson's r.

2.2.3 Pearson's r

We often want a relationship measure that is **directional**, **unitless**, and **bounded**. **Pearson<80><99>s** r does exactly this by rescaling covariance to lie between -1 and 1:

- $r = 1$: perfect positive linear relationship
- $r = 0$: no linear relationship
- $r = -1$: perfect negative linear relationship

Values closer to ± 1 indicate stronger linear association; values near 0 indicate weak or no linear association.

Let's take a look at a formula for Pearson's r:

$r = \frac{cov(X,Y)}{\sigma_X \sigma_Y} = \frac{cov(X,Y)}{SD_X SD_Y}$

Here, σ is the standard deviation (SD), so r is the covariance of X and Y divided by the product of their standard deviations. Dividing by the standard deviations **normalizes** the covariance, constraining r to fall between -1 and 1, which makes correlation values comparable across variables measured in different units or on different scales.

> **i** Note
>
> **Computing *r*.** You'll see several algebraically equivalent formulas for Pearson's r, some easier for hand calculation, others more compact. They all produce the same result. Since we rely on software for calculations, our focus is on what r means and how to interpret it, not manual computation, you'll run the calculation in R during lab.

To confirm r really does stay between -1 and 1, Figure 2.4 shows a simulation: we randomly ordered the numbers 1 to 10 for both an X measure and a Y measure, computed Pearson's r between them, and repeated the process 1000 times. Every dot falls between -1 and 1.

2.3 Regression: A mini intro

This section adds one more piece to our understanding of correlation: **linear regression**, at an introductory level.

Figure 2.5 shows the same three scatter plots as before, now with a fitted line added to each.

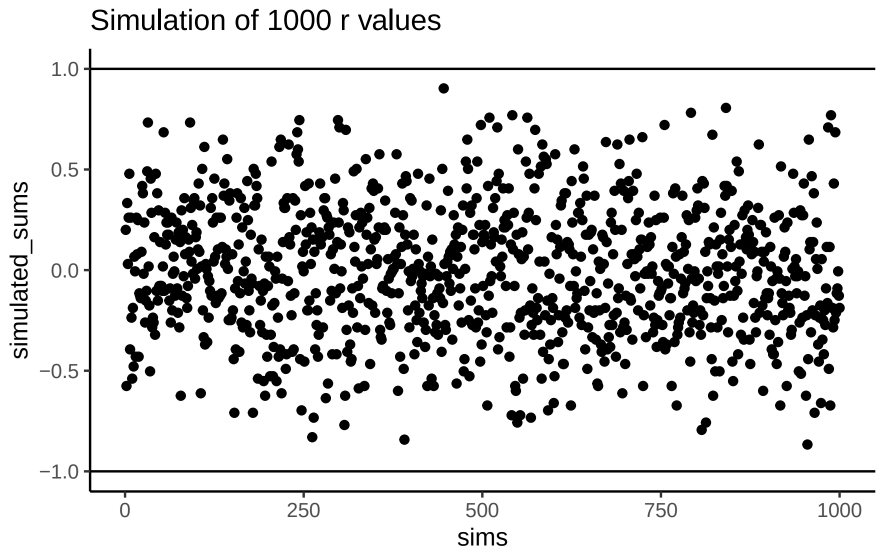

Figure 2.4: A simulation of correlations. Each dot is the r-value between an X and Y variable that each contain the numbers 1 to 10 in random orders.

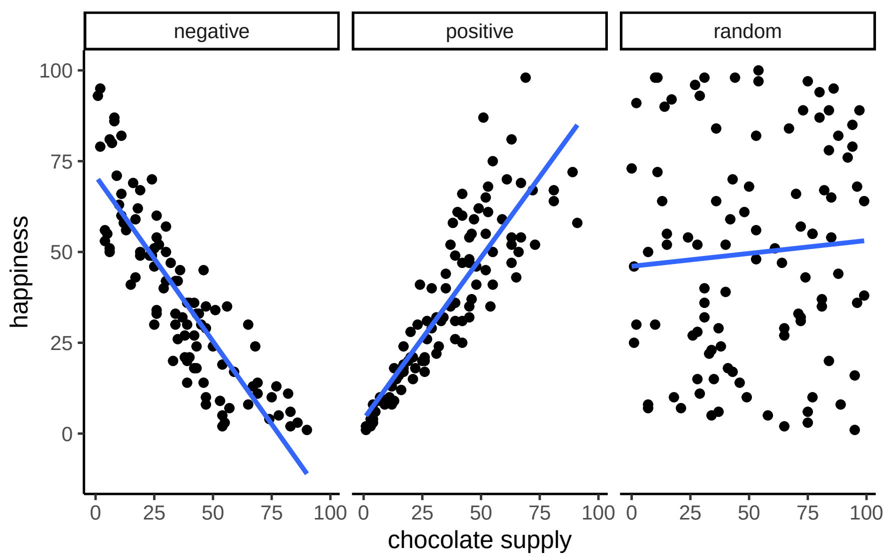

Figure 2.5: Three scatterplots showing negative, positive, and a random correlation (where the r-value is expected to be 0), along with the best fit regression line

2.3.1 The best fit line

Notice the blue lines in these panels. Each is a **best fit** line: the single straight line that best summarizes the overall pattern of dots.

With one variable, we use the mean to describe central tendency in one dimension. With two variables plotted together, we need a two-dimensional equivalent. A line serves that purpose, capturing the average relationship across the cloud of points.

Many lines could be drawn through a scatter of points, but most would miss the pattern. The best fit line is the one with the least error, defined precisely below.

Figure 2.6 shows what that error looks like: a line runs through the dots, and short vertical lines drop from each dot to the line. These are **residuals**, the vertical differences between the observed data and the fitted line, showing how far off the line's prediction is for each point. Since not all dots fall exactly on the line, some error is inevitable; the best-fit line is the one that minimizes it overall.

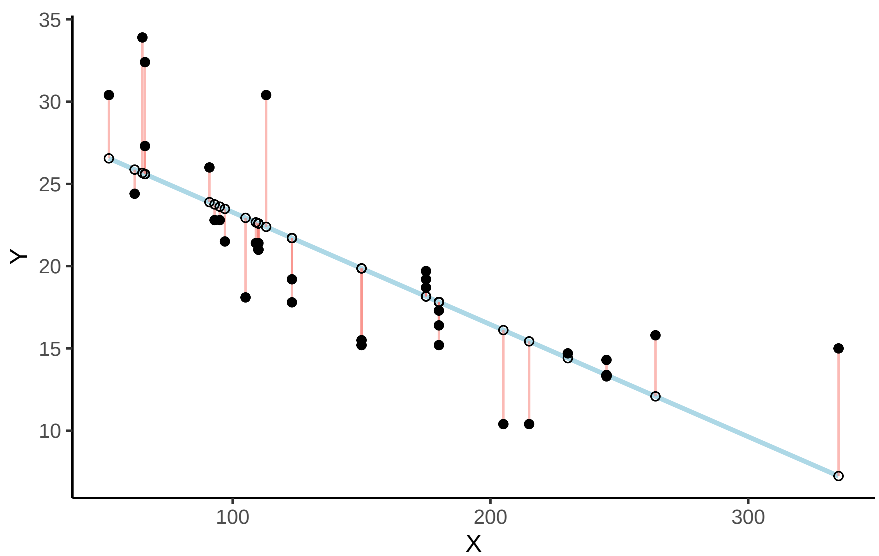

Figure 2.6: Black dots are the data, white dots are the values the regression line predicts, and the red lines are the errors between them. R code for residual visualization adapted from Simon Jackson: https://drsimonj.svbtle.com/visualising-residuals

Figure 2.6 shows a scatter plot of two variables X and Y. Each black dot represents observed data; the blue line is the regression line, and the white dots show its predicted values. The red lines are **residuals**: they drop vertically from each observed value to its predicted value on the line, showing how far the prediction differs from what was actually observed.

The regression line is chosen to minimize the total size of these residuals. In practice, we compute the **sum of squared deviations**: square each residual, then add them together.

The blue line produces the smallest possible total squared error; any other line would yield a larger one.

Figure 2.7 animates this directly, comparing the best fit line (blue) to other possible lines (black) as the black line moves up and down. The red lines show the error between the black line and the data points at each position, and the shaded grey area shrinks to its minimum exactly when the black line reaches the best fit line, then expands again as it moves away.

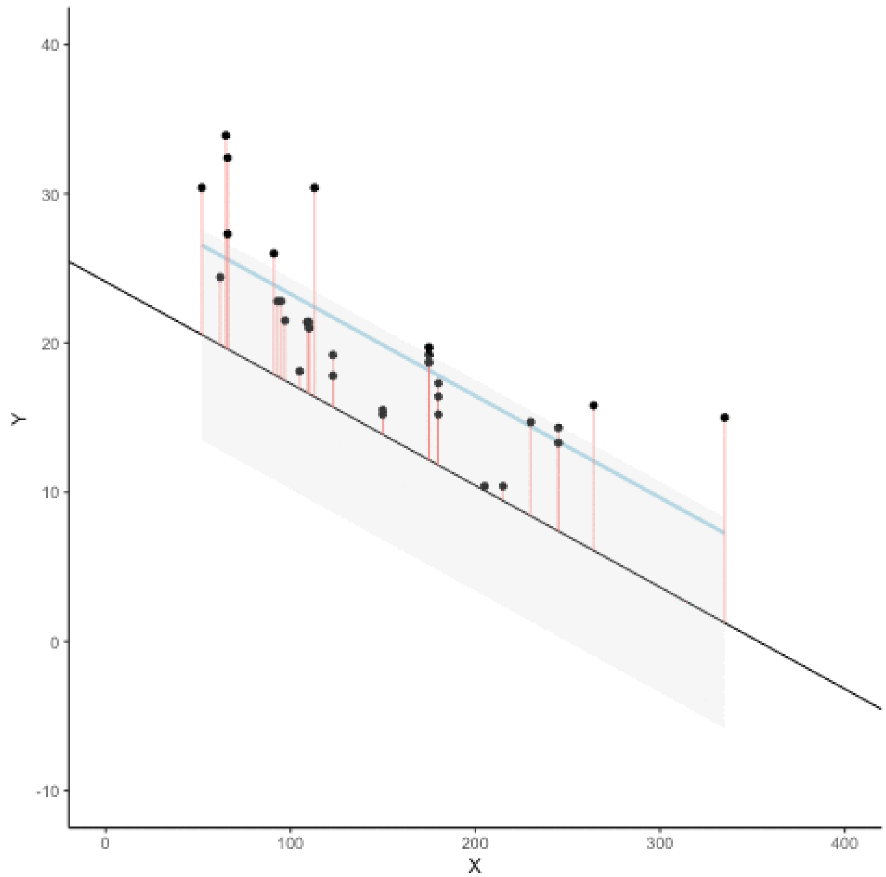

Figure 2.7: The black line moves through alternative positions while the blue best-fit line stays put. Red lines show each point's error, and the shaded grey area is their total, which shrinks to a minimum exactly at the best-fit line. (Final frame; the animated version is in the web edition of this book.)

Whenever the black line does not overlap with the blue line, it fits the data less well: the blue regression line is the one that minimizes the overall error, not too high, not too low, but the line with the least total deviation.

Figure 2.8 makes that error concrete: for each position of the black line, we square the length of every red line from the animation above and add them up, giving the total sum of squared deviations at that position. The dots start high on the left, swoop down to a minimum in the middle, and rise again on the right. That minimum point is the best fit regression line.

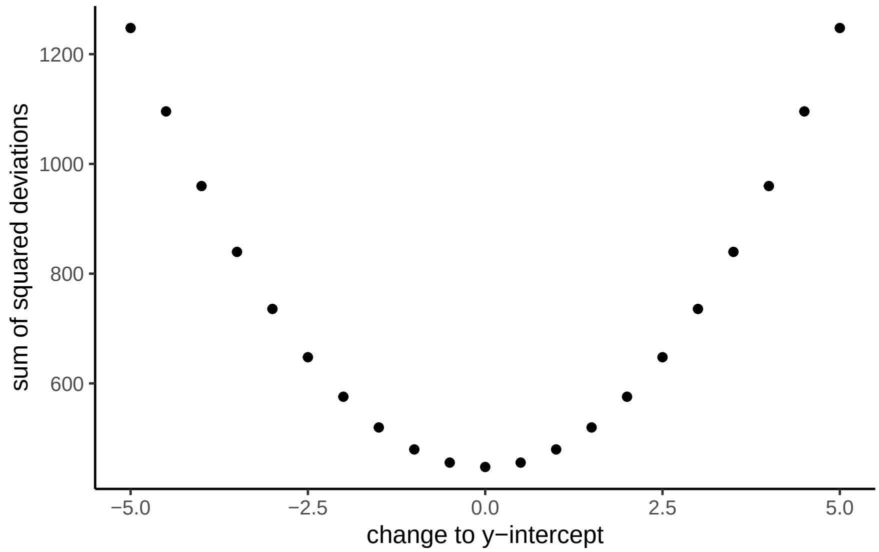

Figure 2.8: A plot of the sum of the squared deviations for different lines moving up and down, through the best fit line. The best fit line occurs at the position that minimizes the sum of the squared deviations.

This graph shows how the total error behaves as the line's y-intercept moves up and down: the error rises whether we shift the line down (0 to -5) or up (0 to +5), and is smallest right in the middle, at the original best fit line.

2.3.2 Computing the best fit line

How do we find the line that minimizes the sum of squared deviations? In principle, you could try many possible lines and compare their errors, but that's not efficient. Instead, we use formulas (or, more realistically, software) to identify the line with the smallest total error.

Next we<80><99>ll look at the formulas for calculating the slope and intercept of a regression line, and work through one example by hand. Many statistical formulas were designed to simplify hand calculations before computers were common. Today we rely on software, but working through examples by hand helps illustrate the logic behind the formulas.

Here are two formulas we can use to calculate the slope and the intercept, straight from the data. We won't go into why these formulas do what they do. These ones are for "easy" calculation.

$intercept = b = \frac{\sum y \sum x^2 - \sum x \sum xy}{n \sum x^2 - (\sum x)^2}$

$slope = m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$

In these formulas, the x and the y refer to the individual scores. Here's a table showing you how everything fits together.

scores	x	y	x_squared	y_squared	xy
1	1	2	1	4	2
2	4	5	16	25	20
3	3	1	9	1	3
4	6	8	36	64	48
5	5	6	25	36	30
6	7	8	49	64	56
7	8	9	64	81	72
Sums	34	39	200	275	231

The table has 7 pairs of x and y scores, plus columns for x^2, y^2, and xy (each x score times its matching y score), with column sums at the bottom. These sums are everything the formulas need. Here's what they look like with numbers plugged in:

$intercept = b = \frac{\sum y \sum x^2 - \sum x \sum xy}{n\sum x^2 - (\sum x)^2} = \frac{39*200-34*231}{7*200-34^2} = -.221$

$slope = m = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2} = \frac{7*231-34*39}{7*275-34^2} = 1.19$

To check the work, plot the scores and draw a line through them with slope = 1.19 and y-intercept = -.221. As shown in Figure 2.9, the line runs through the middle of the dots, as it should.

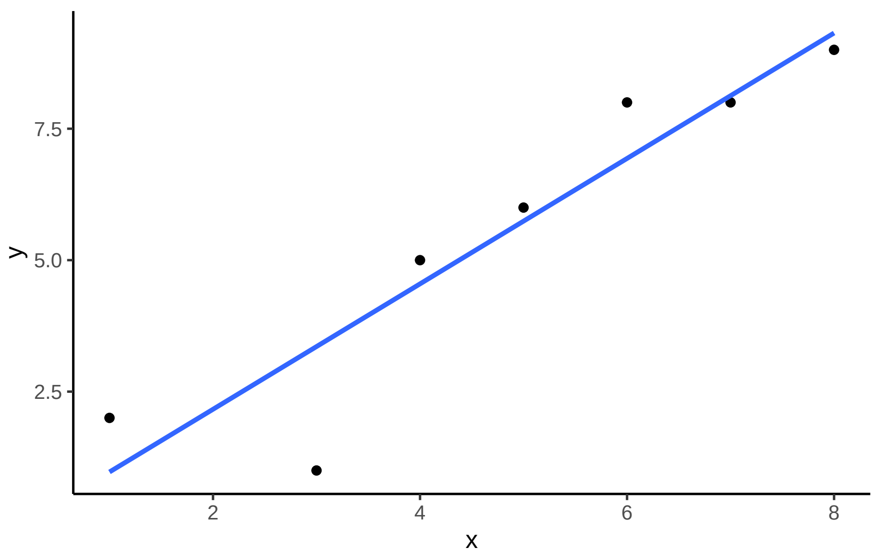

Figure 2.9: An example regression line with confidence bands going through a few data points in a scatterplot

2.4 Interpreting Correlations: why caution is warranted

You’ve heard the warning: **correlation does not equal causation**. But what does a correlation really tell us, and what can it mislead us about? Interpreting correlations requires care, because the same number can arise from very different situations: random chance, confounding influences, or nonlinear patterns. In this section, we<80><99>ll look at why correlations can be unreliable and how to recognize their limits.

2.4.1 Spurious correlation: chance alone can look convincing

Another very important aspect of correlations is the fact that they can be produced by random chance. This means that you can find a positive or negative correlation between two measures, even when they have absolutely nothing to do with one another. You might have hoped to find zero correlation when two measures are totally unrelated to each other. Although this certainly happens, unrelated measures can accidentally produce **spurious** correlations, just by chance alone.

Let<80><99>s demonstrate how correlations can appear by chance when there is no causal connection between two measures. Imagine two independent participants, one at the North Pole and one at the South Pole. Each has a lottery machine containing balls numbered 1 through 10. With replacement, each participant randomly draws 10 balls and records the numbers. Because the draws occur independently and half a world apart, there is no possible causal link between the two sets of numbers. The outcomes are determined by chance alone.

Here is what the numbers on each ball could look like for each participant:

Ball	North_pole	South_pole
1	2	10
2	3	1
3	7	3
4	6	6
5	5	6
6	6	5
7	3	10
8	8	6
9	8	9
10	7	7

In this one case, if we computed Pearson’s r, we would find that r = -0.1318786. But, we already know that this value does not tell us anything about the relationship between the balls chosen in the north and south pole. We know that relationship should be completely random, because that is how we set up the game.

The better question is what random chance alone can do: if we ran this game thousands of times, each time drawing new balls and computing the correlation, the r value would fluctuate, sometimes positive, sometimes negative, sometimes large, sometimes small. Looking at the shape of that fluctuation gives us a window into the kinds of correlations chance alone can produce.

2.4.1.1 Monte-carlo: what chance produces

We can use a computer to repeat the lottery game as many times as we like. This process, repeated random sampling to explore possible outcomes, is called a **Monte Carlo simulation**.

Running the game 1,000 times, each time generating 10 random numbers for both the North Pole and South Pole participants and computing the correlation between them, gives 1,000 different values of Pearson's r, each produced under conditions where we know no real relationship exists.

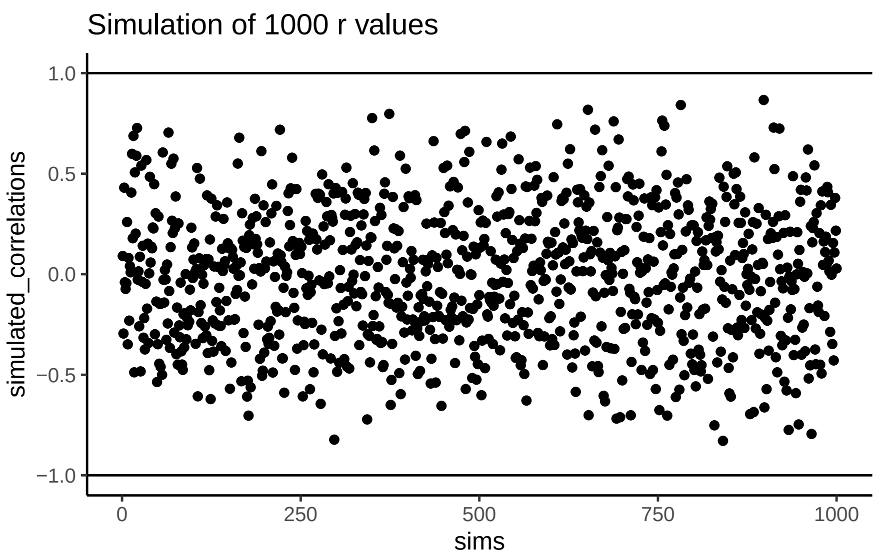

Figure 2.10: Another figure showing a range of r-values that can be obtained by chance.

Figure 2.10 shows the outcome. Each dot is one simulated correlation. All of them fall between <80><93>1 and 1, but the values scatter widely: sometimes strongly positive, sometimes strongly negative, often near zero. The point is that **chance alone can generate correlations of any size**. Finding a correlation in your data does not by itself guarantee a meaningful relationship.

To visualize this process more concretely, we can animate a single run. In Figure 2.11, each frame shows two sets of 10 random values plotted against each other, along with the fitted regression line. Because the data are random, we expect no consistent pattern. Yet across repetitions, the line sometimes slopes upward, sometimes downward, and occasionally appears flat. This is what spurious correlation looks like.

The animation makes clear how quickly random samples can mimic each of these patterns.

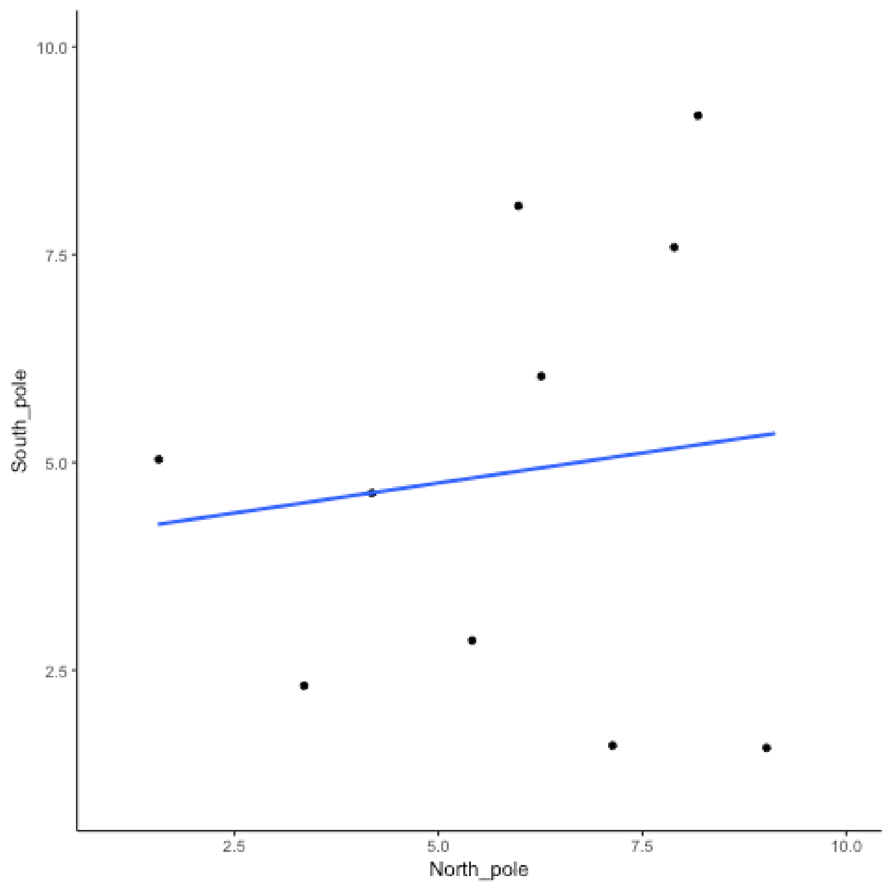

Figure 2.11: Completely random data points drawn from a uniform distribution with a small sampl-size of 10. The blue line twirls around sometimes showing large correlations that are produced by chance (Final frame; the animated version is in the web edition of this book.)

You might find this unsettling: if two random variables with no relationship can still produce correlations, how can we trust any correlation at all? This raises a key question: **how do we know whether the correlation we observed is meaningful or just chance?** One way forward is to look not at a single outcome, but at the distribution of outcomes produced by chance. By simulating 1,000 random correlations and plotting them in a histogram, we can see which values occur frequently and which are rare.

Figure 2.12 shows that most chance correlations cluster near zero. Moderate correlations

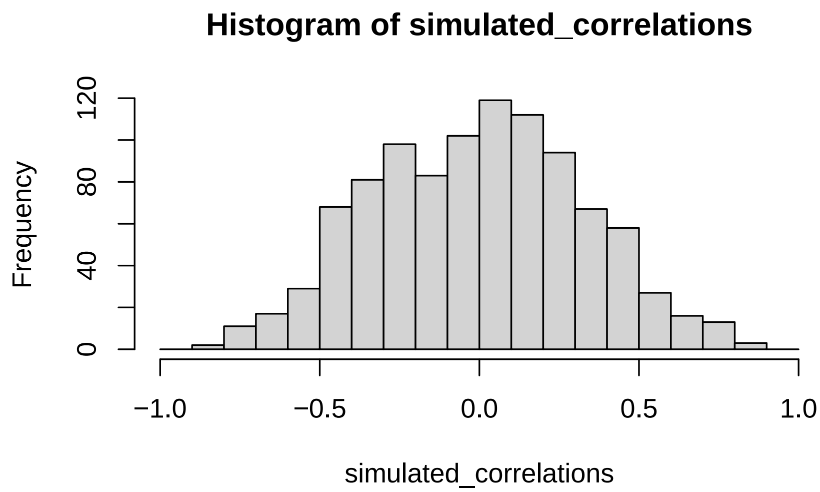

Figure 2.12: A histogram showing the frequency distribution of r-values for completely random values between an X and Y variable (sample-size=10). A rull range of r-values can be obtained by chance alone. Larger r-values are less common than smaller r-values

(around ± 0.5) occur less frequently, and near-perfect correlations (± 0.95) are almost absent. This distribution defines a <80><9c>window of chance.<80><9d> When we compute a correlation in real data, we can ask: is our observed r consistent with what chance alone could plausibly generate?

For example, if you observed $r = 0.1$, the histogram shows that such values are common in random data, so chance could easily explain it. An observed $r = 0.5$ is less common but still possible by chance. An observed $r = 0.95$, however, falls outside what chance typically produces, suggesting that something more systematic is driving the relationship.

2.4.2 Sample size and stability

So far, our lottery game has used 10 draws per participant. With such small samples, chance produces a wide range of correlations: sometimes strongly positive, sometimes strongly negative. But what happens if we increase the sample size?

We can repeat the simulation under four conditions: each participant draws 10, 50, 100, or 1000 numbers. In each case, we run 1000 simulations and record the correlation values.

Figure 2.13 shows the resulting four histograms of Pearson r values.

The pattern is clear:

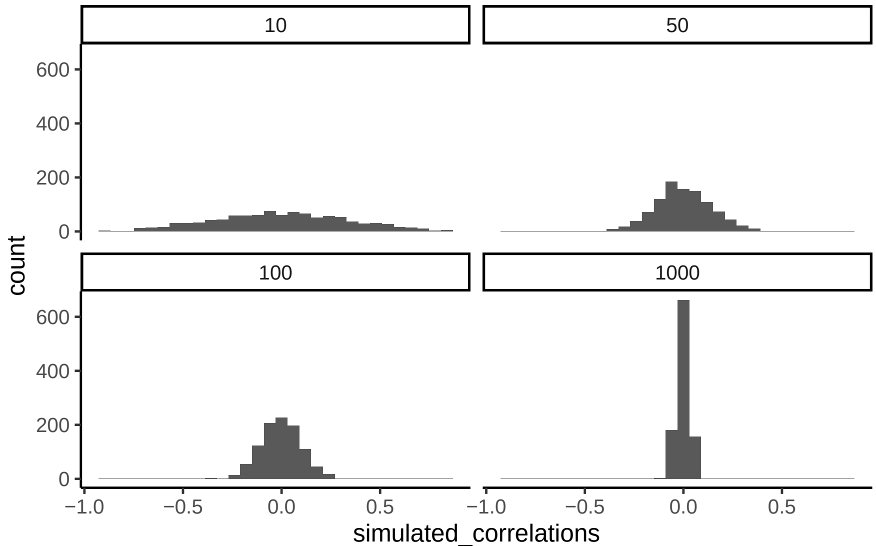

Figure 2.13: Distributions of r-values between completely random X and Y variables, at four sample sizes. The distributions narrow as sample size grows.

- With **n=10**, correlations spread widely across the range <80><93>1 to 1. Chance alone frequently produces values that would look <80><9c>strong<80><9d> in real data.
- With **n=50 or 100**, the spread narrows, but moderate correlations still appear.
- With **n=1000**, almost all simulated correlations fall close to zero.

The takeaway is that **larger samples reduce the influence of chance**. Small samples are highly unstable, making it difficult to distinguish signal from noise. As sample size increases, the distribution of chance correlations collapses toward zero, so observing a large r becomes far less likely under randomness alone.

This illustrates why researchers place so much emphasis on collecting sufficient data. With too few observations, even strong-looking correlations may be spurious. With larger samples, a correlation of the same size is much harder to attribute to chance, and therefore more likely to reflect a real underlying relationship.

> **i** Note
>
> **Experimental design tip: sample size.** Bigger n narrows the <80><9c>window of chance.<80><9d> If you expect a modest linear effect (e.g., |r| <89><88> 0.2<80><93>0.3), plan for larger samples so random swings are less likely to mimic or mask the effect.

2.4.2.1 Visualizing the effect of sample size

Animations make it easier to see how sample size influences the stability of correlation estimates. When the sample is very small, chance variation can create apparent patterns in any direction. As the sample size increases, those chance fluctuations diminish.

When there is no true correlation

In Figure 2.14, each panel shows two random variables sampled with no real relationship. The panels differ only in sample size: 10, 50, 100, or 1000 observations. Each frame of the animation re-samples the data, plots them, and fits a line.

With $n = 10$, the line swings dramatically from positive to negative slopes. At $n = 50$ and $n = 100$, the line still shifts but is more restrained. At $n = 1000$, the line is consistently flat, as expected when no correlation exists.

Which line should you trust? The line from the sample with 1000 observations is the most reliable. It stays nearly flat across repetitions, as we would expect when no true correlation exists. The line from a sample of only 10 observations swings wildly, sometimes positive, sometimes negative. The lesson is straightforward: claims about correlation are only as trustworthy as the sample size behind them. Small samples can produce misleading patterns by chance, while large samples rarely do.

We can also repeat this simulation with data drawn from a normal distribution rather than a uniform one. The result is the same: small samples fluctuate dramatically, but with large samples the estimated correlation stays close to zero. Figure 2.15 illustrates this.

When there is a true correlation

In Figure 2.16, the simulation is repeated with a real positive correlation built into the data. Again, we vary the sample size from 10 to 1000.

With $n = 10$, the regression line jumps around and occasionally even points downward, despite the underlying positive relationship. As the sample size grows, the line becomes more stable and consistently reflects the true positive slope. Sampling error still introduces some noise, but with larger samples the real correlation is much easier to detect.

Both the histograms and the animations show that **small samples make correlations unreliable, while larger samples are more likely to reveal the true pattern**. Even when a real positive correlation exists, a small sample might miss it, or even suggest the opposite, simply due to chance. Because we usually only have one sample, we can never know for certain whether we were <80><9c>lucky<80><9d> or <80><9c>unlucky.<80><9d> The best protection is to collect more data: larger samples reduce the influence of chance and make genuine relationships easier to detect.

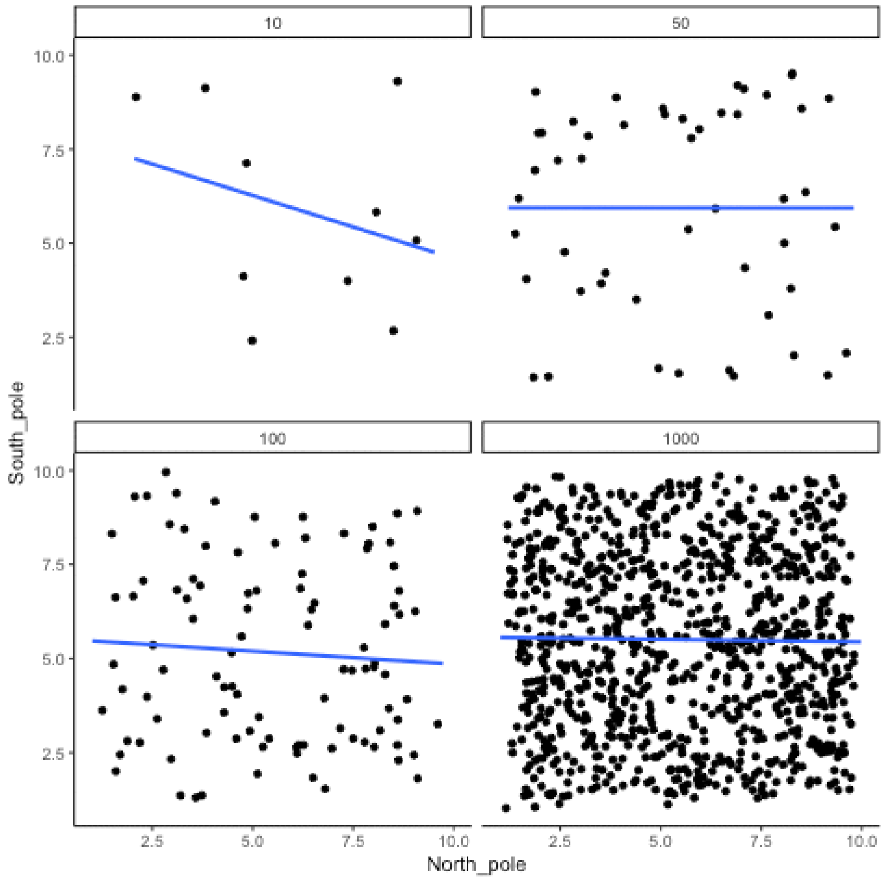

Figure 2.14: Animation of how correlation behaves for completely random X and Y variables as a function of sample size. The best fit line is not very stable for small sample-sizes, but becomes more reliably flat as sample-size increases. (Final frame; the animated version is in the web edition of this book.)

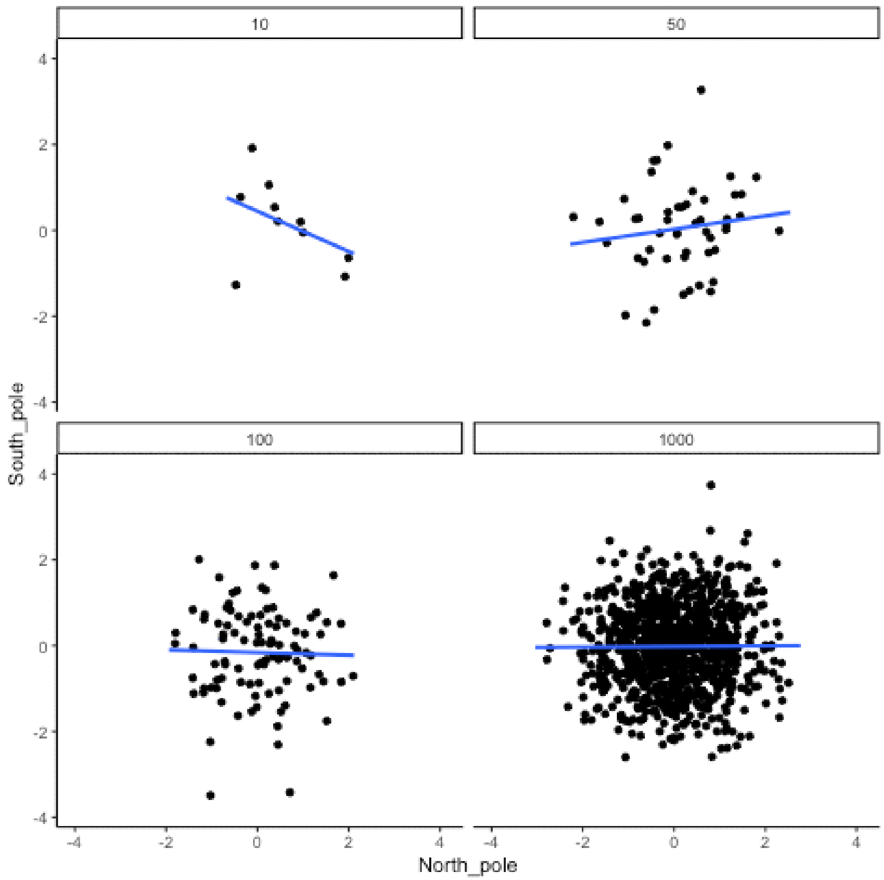

Figure 2.15: Animation of correlation for random values sampled from a normal distribution, rather than a uniform distribution. (Final frame; the animated version is in the web edition of this book.)

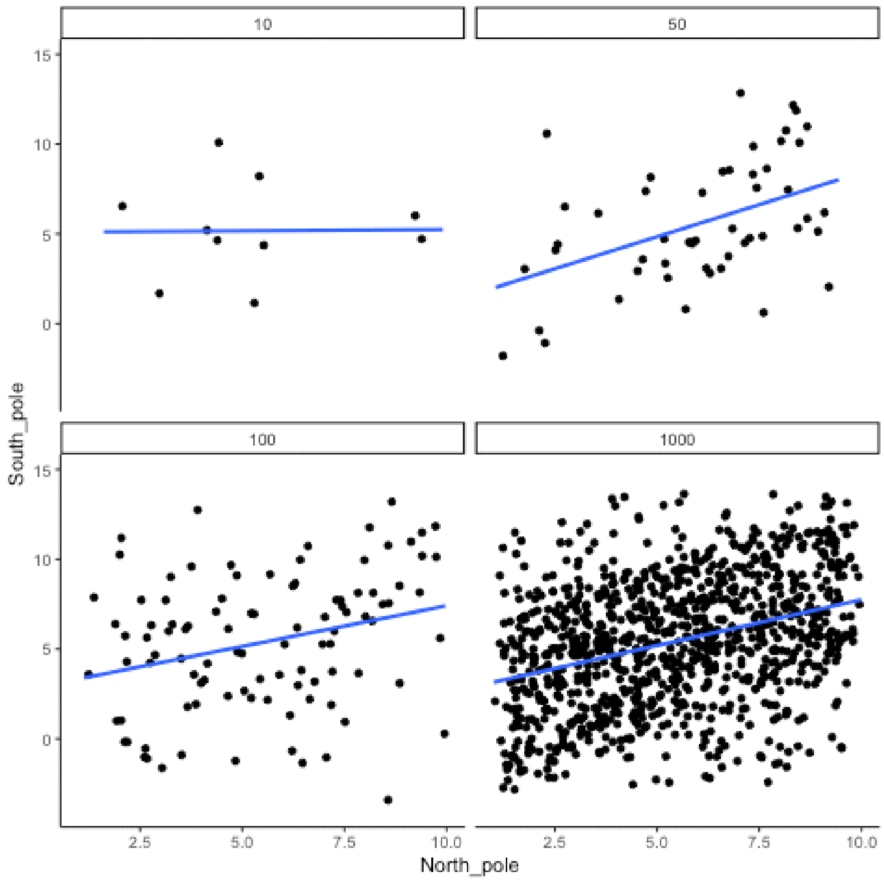

Figure 2.16: How correlation behaves as a function of sample-size when there is a true correlation between X and Y variables. (Final frame; the animated version is in the web edition of this book.)

2.4.3 Nonlinearity: when r misses real relationships

Consider a snake plant. Like most plants, it needs water to stay alive, but also needs *the right amount* of water. Imagine we grow 1000 snake plants, each receiving a different amount of water per day, ranging from *no water* to *extreme overwatering* (say 1000 teaspoons a day), and measure their weekly growth. Water is clearly causally related to plant growth, so we'd expect to see a correlation.

Plants given no water at all will eventually die, showing little or no weekly growth. With only a few teaspoons per day, plants may survive but grow only modestly. In a scatter plot, those cases would appear near the bottom left (low water, low growth) and rise upward as water increases. Growth continues to improve until a threshold is reached, beyond which excess water harms the plants. Severely overwatered plants also fail to grow and eventually die, just as those given no water do.

The scatter plot would form an inverted U or V: as water increases, growth rises to a peak and then declines with overwatering. A relationship like this can produce a Pearson's r close to zero, even though water clearly affects growth. Figure 2.17 illustrates this pattern.

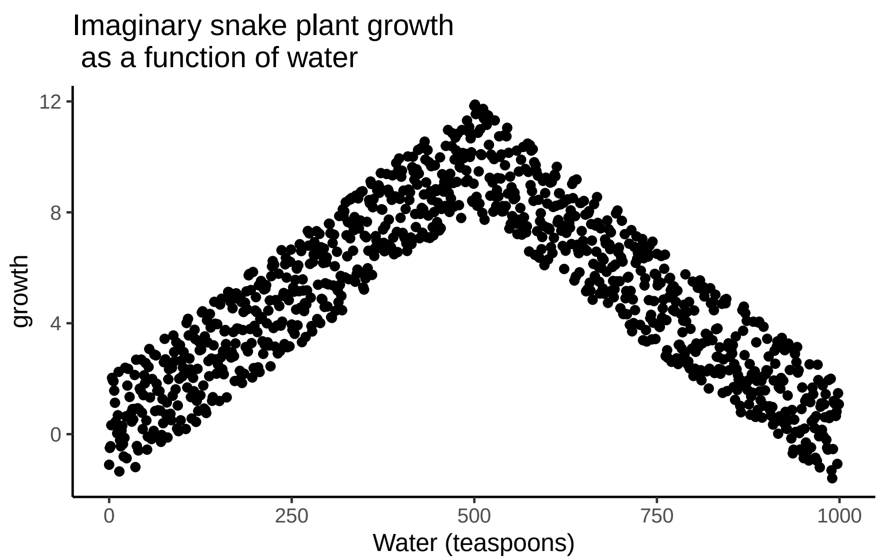

Figure 2.17: Illustration of a possible relationship between amount of water and snake plant growth. Growth goes up with water, but eventually goes back down as too much water makes snake plants die.

There is clearly a relationship between watering and snake plant growth. But, the correlation isn't in one direction. As a result, when we compute the correlation in terms of Pearson's r, we get a value very close to zero, suggesting no relationship.

```
#> [1] -0.02108469
```

What this really means is there is no relationship that can be described by a *single straight line*. When we need lines or curves going in more than one direction, we have a nonlinear relationship.

This example shows why correlations can be tricky to interpret. Water is clearly necessary for plant growth, yet the overall correlation can be misleading. The first half of the data suggests a positive relationship, the second half a negative one, and the full dataset little or none. Even with a real causal link, a single correlation coefficient may fail to capture it.

> Pro Tip: This is one reason why plotting your data is so important. If you see an upside U shape pattern, then a correlation analysis is probably not the best analysis for your data.

2.4.4 Confounding variables (third-variable problem)

Correlations can arise because of a third variable: something not directly measured that influences both variables of interest. Returning to the snake plant example: imagine we measure water and plant growth, but ignore sunlight. If plants in sunnier windows both receive more water and grow faster, we might attribute the effect to water alone, when in fact sunlight is doing much of the work.

The same issue appears in human data. Suppose we find that income and happiness are positively correlated. Money itself might be the cause, but other factors, such as housing quality, healthcare access, or social opportunities, may be the true drivers, correlated with income and independently influencing happiness.

2.5 Chapter Summary

Why it matters. Correlation is how we quantify whether two variables move together. It<80><99>s essential for describing patterns (e.g., temperature and canopy cover, chocolate and happiness) and for deciding whether a relationship is worth modeling or acting on.

Core ideas

- **Scatter first, always.** A scatter plot shows direction (up, down, none), form (straight vs curved), and outliers.
- **Covariance <86><92> correlation.** Covariance measures whether deviations from each mean tend to align. Pearson<80><99>s r rescales covariance by the standard deviations so the result is unitless and bounded ([-1, 1]).
- **Interpreting r.** Sign indicates direction; magnitude indicates strength of a *linear* relationship. r near 0 can still hide a strong *nonlinear* pattern.

- **Regression link.** The best-fit line summarizes the average relationship; residuals are the vertical gaps between data and the line. The line is chosen to minimize the **sum of squared residuals**.

Common pitfalls

- **Chance (spurious) correlations.** Small samples can produce large $|r|$ by accident. Larger samples shrink the <80><9c>chance window<80><9d> toward 0.
- **Nonlinearity.** Curved relationships (e.g., too little or too much water harming plant growth) can yield r <89><88> 0 despite a real effect.
- **Confounding.** A third variable can create or inflate a correlation (e.g., sunlight affecting both watering patterns and plant growth).

Practice checklist

1. Plot the scatter; look for linearness, curvature, and outliers.

2. Report r with **sample size** (n) and, when possible, a **confidence interval**.
3. Avoid causal language: say <80><9c>associated with,<80><9d> not <80><9c>caused by.<80><9d>

4. If the pattern is curved, don<80><99>t stop at r; fit a suitable model (e.g., add a quadratic term) or transform.

5. If results hinge on a small sample, say so and treat conclusions as tentative.

Takeaway. Correlation is a precise way to summarize linear association, but it<80><99>s only as trustworthy as your plot, your sample size, and your awareness of curvature and confounding. Plot first, quantify carefully, and interpret cautiously.

3 Probability and Distributions

> I have studied many languages-French, Spanish and a little Italian, but no one told me that Statistics was a foreign language. – Charmaine J. Forde

3.1 From description to inference

Up to this point, we've focused on experimental design and descriptive statistics: how to collect data and how to summarize it with graphs and averages. But statistics is more than description. Its power is in inference, using limited data to say something about a broader population. Inference relies on two foundations: probability theory and the behavior of samples from distributions.

3.2 Probability vs. Statistics

Probability and statistics are closely related but not the same.

Probability starts with a known model of the world and asks: what kinds of data are likely?

- Example: What is the chance of flipping 10 heads in a row with a fair coin?
- Example: What is the chance of drawing five hearts from a shuffled deck?

Here, the **model is known** (fair coin, fair deck). Data are imagined.

Statistics starts with data and asks: what model of the world generated these data?

- Example: If a coin comes up heads 10 times in a row, is it fair?
- Example: If five cards in a row are hearts, was the deck shuffled?

Here, the **data are known**. The model is what we want to learn about.

> **Note**
>
> **Shortcut.** Probability goes model <86><92> data. Statistics goes data <86><92> model.

3.3 What does probability mean?

Statisticians agree on the rules of probability but not always on what the word itself means. In everyday life we're comfortable saying something is "likely" or "unlikely," but the formal meaning varies.

Imagine a soccer game between Arsenal and West Ham. You say Arsenal has an 80% chance of winning. That could mean:

1. Long-run frequency: if the teams played many times, Arsenal would win ~8 out of 10.
2. Betting odds: you'd only take a wager if payoffs reflected an 80/20 split.
3. Subjective belief: your confidence in Arsenal is four times stronger than in West Ham.

All three interpretations follow the same rules but reflect different philosophies.

For our purposes, the math works the same either way. In this course we'll use the **frequentist** view, so you can think of probability as the long-run proportion of times an event would occur if you could repeat the process many times (that's #1 above).

3.4 Basic probability theory

At the core of probability theory is the idea of a distribution: a set of possible outcomes and their probabilities.

Suppose we track which ice cream flavor a customer chooses: chocolate, vanilla, or strawberry. Each choice is an *elementary event*, and the set of all possible events is the *sample space*, S:

$S = \text{chocolate}, \text{vanilla}, \text{strawberry}$

If we record how often each flavor is chosen, we might estimate the following probabilities:

Flavor Probability:

- Chocolate $P(\text{chocolate}) = 0.5$
- Vanilla $P(\text{vanilla}) = 0.3$
- Strawberry $P(\text{strawberry}) = 0.2$

These probabilities form a distribution. Each value lies between 0 and 1, and together they sum to 1.

Non-elementary events

We can also define events that group multiple outcomes.

For example, let $A =$ "not chocolate." Then:

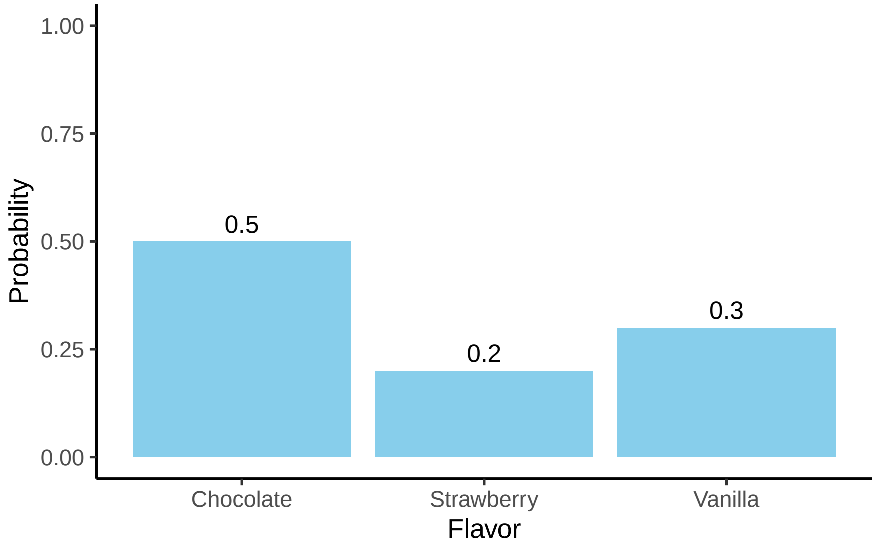

Figure 3.1: A simple probability distribution: the probabilities for three flavors must add to 1.

$$P(A) = P(vanilla) + P(strawberry) = 0.5$$

3.5 Rules of Probability

Some basic rules follow from this framework. Let's consider a sample space X that consists of elementary events x, and two non-elementary events, which we'll call A and B.

Here are our rules:

English	Notation	Formula
not A	$\neg A$	$P(\neg A) = 1 - P(A)$
A and B	$A \cap B$	$P(A \cap B) = P(A \mid B)\,P(B) = P(B \mid A)\,P(A)$
A or B	$A \cup B$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
A given B	$A \mid B$	$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$

These rules look abstract at first, but they are just formal ways of describing common-sense situations. Let's work through them step by step with our ice cream example. We'll assign probabilities to each flavor and then show how the rules play out with real numbers. That way, the notation is always tied to something you can picture.

Let's add a fourth flavor and assign some probabilities:

Flavor Probability:

- Chocolate $P(\text{chocolate}) = 0.4$
- Vanilla $P(\text{vanilla}) = 0.3$
- Strawberry $P(\text{strawberry}) = 0.2$
- Rocky Road $P(\text{rocky road}) = 0.1$

Notice these add to 1, as they must.

We'll have two non-elementary events, A and B, defined as:

$A = (\text{chocolate, vanilla, strawberry})$

$B = (\text{chocolate, rocky road})$

3.5.1 Not A

$A = (\text{chocolate, vanilla, strawberry})$

$P(A) = 0.4 + 0.3 + 0.2 = 0.9$

So, $P(\neg A) = 1 - P(A) = 1 - 0.9 = 0.1$

Explanation: This adds up logically. The chance of *not* choosing chocolate, vanilla, or strawberry is equal to the chance of Rocky Road.

3.5.2 A and B (the intersection)

Think of *and* as meaning <80><9c>both at once.<80><9d> For probabilities, that translates into the **overlap** of the two events.

- The only overlap between A and B is chocolate: $A \cap B = (\text{chocolate})$
- The intersection $A \cap B$ contains only chocolate, and its probability is 0.4.
- So $P(A \cap B) = 0.4$

To reiterate: *And* means overlap, not <80><9c>add the probabilities together.<80><9d>

3.5.3 A or B (the union)

Or in probability is inclusive: it means <80><9c>in A, or in B, or in both.<80><9d>

- If we combine everything in either set, we get all four flavors.
- $A \cup B =$ (chocolate, vanilla, strawberry, rocky road)
- That<80><99>s the whole sample space, so $P(A \cup B) = 1$.
- So $P(A \cup B) = 1$.

To reiterate: *or* covers everything in either group. We subtract the overlap ($A B$) so we don<80><99>t double-count it.

3.5.4 Conditional probability

Suppose we want $P(B \mid A)$: the probability of choosing B (chocolate or rocky road) given that we already know the choice was from A (chocolate, vanilla, strawberry).

By the rule,

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{0.4}{0.9} \approx 0.44$$

Explanation: Given the choice is from the A group, the only overlap with B is chocolate. So there's about a 44% chance the flavor is chocolate.

3.6 Probability Distributions

A **probability distribution** is a rule that assigns probabilities to all possible outcomes of a random variable. Many distributions exist, but this book mostly uses five: Binomial, Normal, t, <87> ("chi-square"), and F. We'll focus on Binomial and Normal here; the others appear when we do inference.

The binomial and the normal represent two kinds of distributions: **discrete** and **continuous**.

- **Discrete distributions** assign probabilities to distinct outcomes.
- **Continuous distributions** describe ranges of values, where probabilities are measured as *areas* under a curve.

3.6.1 The binomial distribution (discrete)

The binomial distribution models the number of "successes" in a fixed number of independent trials.

- A trial is a single attempt with two possible outcomes (success or failure).
- The success probability is the chance of success on any one trial, usually written as θ.
- The size parameter N is the number of trials.
- The random variable X is the number of successes observed in those N trials.

We write this as:

$X \sim \text{Binomial}(N, \theta)$

For example, if you flip a fair coin 20 times ($N = 20$, $\theta = 0.5$), the binomial distribution tells you the probability of getting 0 heads, 1 head, 2 heads … all the way up to 20 heads. The bars of the distribution in Figure 3.2 show the probability of each outcome, and the total always sums to 1.

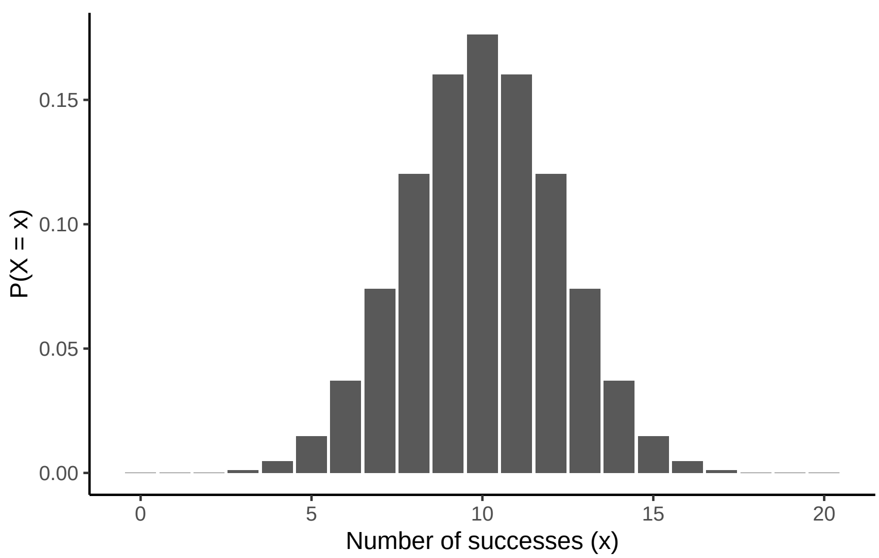

Figure 3.2: Binomial probabilities for X ~ Binomial(N = 20, = 0.5). Bars show P(X = x).

Two common queries

1. Exact probability. For example, the chance of exactly 4 heads: $P(X = 4)$

```
dbinom(x = 4, size = 20, prob = 0.5)
#> [1] 0.004620552
```

The probability is about 0.5%.

2. Cumulative probability. For example, $P(X \leq 4)$:

```
pbinom(q = 4, size = 20, prob = 0.5)
#> [1] 0.005908966
```

The probability is about 0.6%

The probability is slightly higher here, because it also includes the probability we could get 3, 2, or 1 heads as well.

> **Note**
>
> **R's distribution helpers.** Every distribution in R comes with the same four:
>
> - d* = exact probability (PDF)
> - p* = cumulative probability $P(X \leq q)$
> - q* = quantile (inverse of p*)
> - r* = random draws
>
> For binomial: dbinom, pbinom, qbinom, rbinom. For discrete distributions, some percentiles don't exist exactly; qbinom returns the smallest x with P(X x) p.

3.6.2 The normal distribution (continuous)

The normal distribution describes variables that can take on a wide range of values, clustered around a central point. The normal distribution is our most important distribution. It's also sometimes called a Gaussian distribution or a bell curve.

- The mean μ is the center of the distribution.
- The standard deviation σ measures spread: how tightly or widely values are clustered around the mean.
- The random variable X represents the outcome, which can be any real number.

We write this as:

$X \sim \text{Normal}(\mu, \sigma)$

It looks like Figure 3.3.

> **Note**
>
> **For continuous distributions.** The probability of any single exact point is 0, so probability is always an area under the curve.

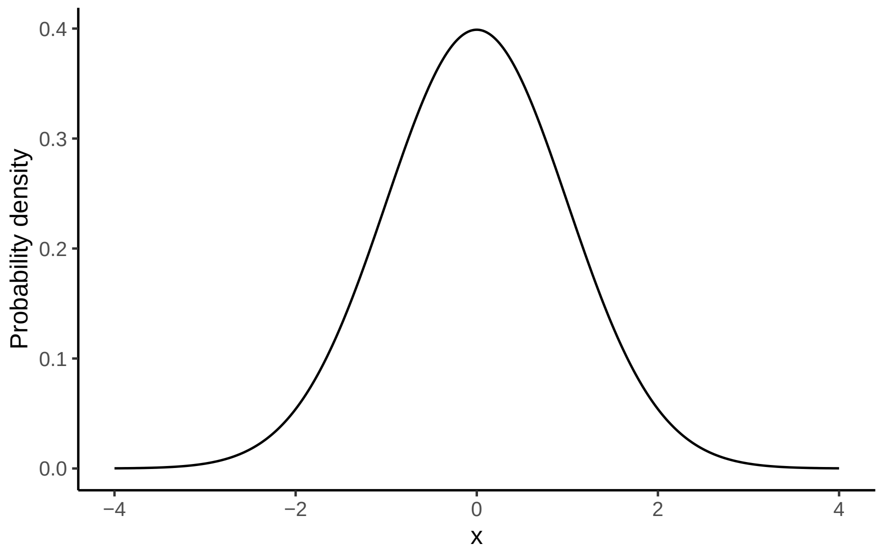

Figure 3.3: Standard normal Probability Distribution Function (PDF), where = 0, <83> = 1. For continuous variables, probabilities are *areas* under the curve.

Changing μ shifts the curve left/right; changing σ widens/narrows it (area always = 1).

For continuous distributions $P(X = x) = 0$; instead, we compute $P(a \leq X \leq b)$ as the area between a and b.

See Figure 3.4 for an illustration.

This leads to a key property of the normal distribution (explored further in the Z-scores section):

68–95–99.7 rule:

- ~68% of values fall within 1<83> of the mean
- ~95% fall within 2<83>
- ~99.7% fall within 3<83>

Two common queries

1. Probability of being below a value, e.g. $P(X \leq 1.64)$ for $X \sim \text{Normal}(0, 1)$

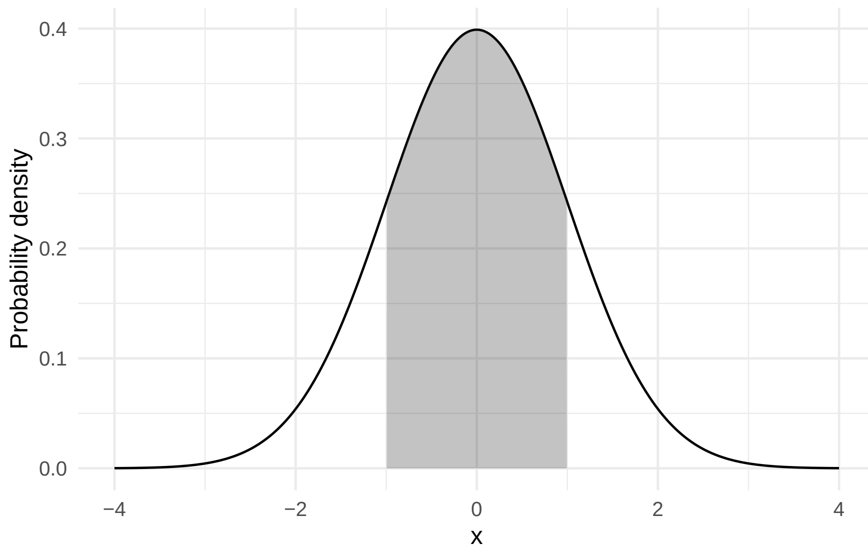

Figure 3.4: Shaded area shows P(<88><92>1 <89> X <89> 1) <89><88> 0.68 for X ~ Normal(0,1).

```
pnorm(q = 1.64, mean = 0, sd = 1)
#> [1] 0.9494974
```

In a normal distribution with a mean of 0 and standard deviation of 1, the probability the value is less than or equal to 1.64 is about 95%.

2. Value for a percentile, e.g. the 97.5th percentile:

```
qnorm(p = 0.975, mean = 0, sd = 1)
#> [1] 1.959964
```

Takeaway: For continuous variables, use areas (via pnorm/qnorm), not bar heights.

3.6.3 What's probability density?

For continuous distributions like the normal, probabilities work differently than in the discrete case. You cannot ask for the probability of one exact value (e.g., $P(X = 23)$), because the answer is essentially zero. Instead, we calculate the probability of being in a **range**, such as $22.5 \leq X \leq 23.5$.

This is why the y-axis of a normal curve is labelled **probability density**. The height of the curve at a point, $p(x)$, is not itself a probability. Instead, probabilities come from the **area**

under the curve between two values. For example, about 68% of values from a standard normal fall between <88><92>1 and 1, because the area between <88><92>1 and 1 is 0.68.

In R, functions like `dnorm()` return the density (the curve's height), while `pnorm()` and `qnorm()` work with cumulative probabilities and quantiles, which are usually more useful for applied problems.

3.7 Summary of Probability

In this section we introduced the building blocks of probability, with an emphasis on the parts most relevant for statistics.

- What probability means: We use the frequentist view, in which probability is the long-run proportion of times an event occurs.
- Distributions: A probability distribution assigns probabilities to all possible outcomes of a random variable. For discrete variables (like coin flips), probabilities are attached to each outcome; for continuous variables (like exam scores), probabilities come from areas under a curve.
- Rules of probability: Complements, intersections (and), unions (or), and conditional probabilities all follow from the basic distribution framework.
- Key distributions:
  - Binomial: counts successes across N independent trials.
  - Normal: describes continuous outcomes centered at a mean, with spread determined by the standard deviation.
  - Others (t, χ^2, F) appear later when we turn to inference.

Takeaway: probability provides the language and tools that make statistical inference possible. Once we understand how outcomes behave under these distributions, we can ask questions about samples, uncertainty, and whether results are surprising enough to change what we believe about the world.

3.8 Samples, populations and sampling

Descriptive statistics summarize what we **do** know. Inferential statistics aim to "learn what we do not know from what we do." The central questions are: what do we want to learn about, and how do we learn it? Introductory statistics typically divides inference into two big ideas: estimation and hypothesis testing. This chapter introduces estimation, but first we need to cover sampling. Estimation only makes sense once you understand how samples relate to populations.

Sampling theory specifies the assumptions on which statistical inferences rest. To talk about making inferences, we must be clear about what we are drawing inferences **from** (the sample) and what we are drawing inferences **about** (the population).

In almost every study, what we actually have is a **sample**, a finite set of data points. We cannot survey every voter in a country or measure every tree in a forest. Earlier, our goal was just to describe the sample. Now we turn to how samples can be used to generalize.

3.8.1 Defining a population

A sample is tangible: it is the dataset you can open on your computer. A **population**, by contrast, is the (usually much larger) set of all possible individuals or observations you want to draw conclusions about. In an ideal world, researchers would start with a clear definition of the population of interest, since that shapes how a study is designed. In practice, the population is often only loosely defined. Still, the basic idea is simple: the sample should represent some broader population we care about.

3.8.2 Simple random samples

The sample–population relationship depends on the **procedure** used to select the sample. This is the **sampling method**, and it matters.

Suppose we have a bag with 10 chips, each uniquely labeled, some black and some white. This bag is the population. If we shake the bag and draw 4 chips without replacement, each chip has the same chance of being selected. This is a **simple random sample**.

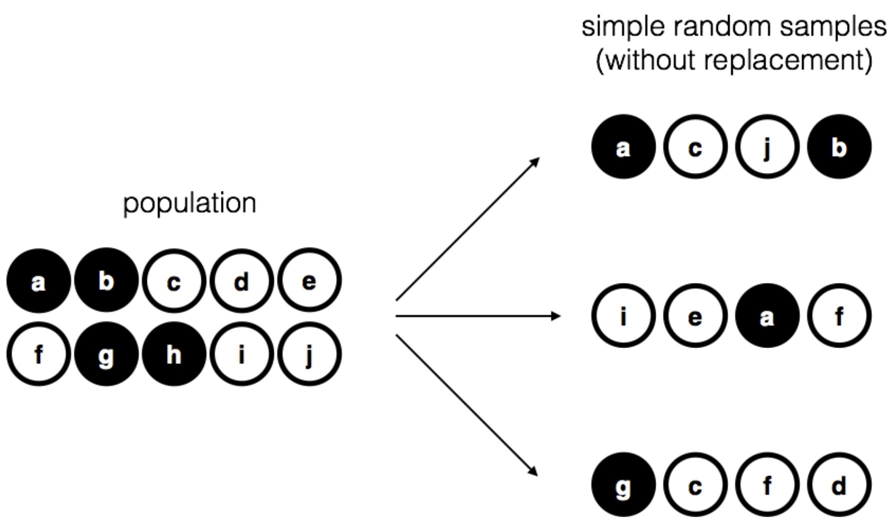

Figure 3.5: Simple random sampling without replacement from a finite population

Now imagine a different procedure: someone peeks inside and deliberately picks only black chips. That is a **biased sample**. The difference is critical. With a random sample, we can use statistical tools to generalize from the sample to the population; with a biased sample, we cannot.

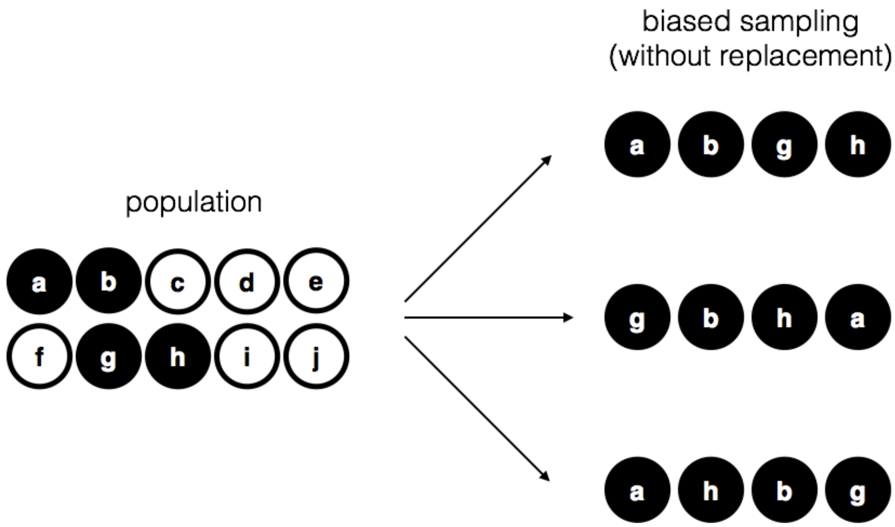

Figure 3.6: Biased sampling without replacement from a finite populations.

Another variation is to draw chips **with replacement**, meaning each chip is returned to the bag before the next draw. In practice, most studies are "without replacement," but much of statistical theory assumes replacement. When the population is large, the difference is negligible.

3.8.3 Random and non-random samples in practice

Random samples are the gold standard, and in many areas of environmental science they are common: inventory plots, transects, and monitoring networks are usually designed with probability-based methods. Still, many studies rely on modified approaches:

- **Stratified sampling** divides the population into subgroups (strata) and samples within each. This ensures representation of small but important groups, such as rare habitat types.
- **Opportunistic or logistically constrained sampling** selects cases that are accessible or already monitored. These are not random, but they are common in practice when budgets, terrain, or historical networks limit where data can be collected. .

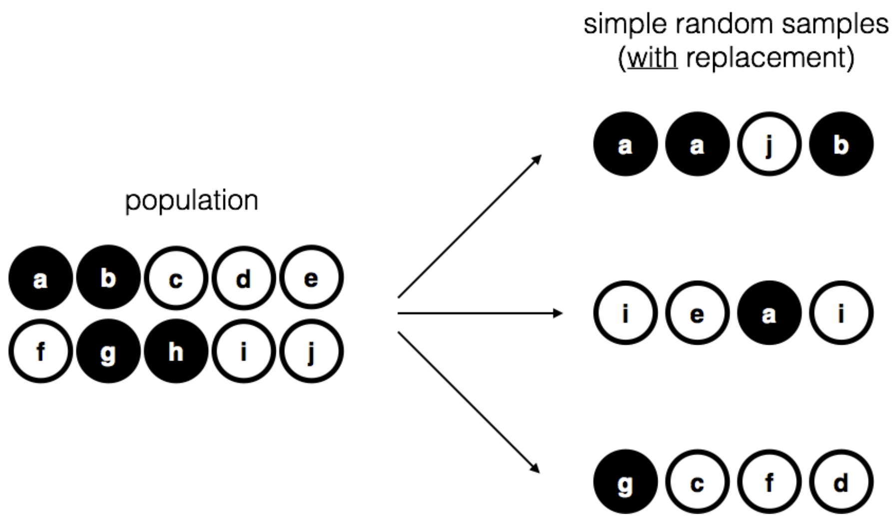

Figure 3.7: Simple random sampling with replacement from a finite population.

3.8.4 How much does it matter if you don't have a random sample?

The short answer: it depends. Stratified sampling, while biased by design, is often useful and correctable with statistical adjustments. Opportunistic samples can be acceptable if the bias is unlikely to affect the phenomenon being studied. For example, if you want to study tree growth rates, sampling only trees on one side of a valley may or may not be problematic, depending on whether the valley side influences growth.

The key is to consider whether the way you sampled could plausibly distort the conclusions. When designing your own studies, aim for sampling methods that are as appropriate as possible for the population of interest. And when critiquing others, be specific about how a sampling choice might bias results, rather than simply noting that it is not random.

3.8.5 Population parameters and sample statistics

So far we have talked about populations the way a scientist might: a group of people, a stand of trees, a set of rivers. Statisticians formalize this idea by treating a population as a **probability distribution**, the abstract source from which data are drawn.

For example, suppose daily coffee consumption in adults follow an approximately normal distribution with mean $\mu - 2$ cups/day and standard deviation $\sigma = 1$ cup/day. These are the **population parameters**.

Now suppose we take a random sample of 100 adults. The histogram in panel (b) of Figure 3.8 shows that the sample resembles the population distribution, but not perfectly. The sample

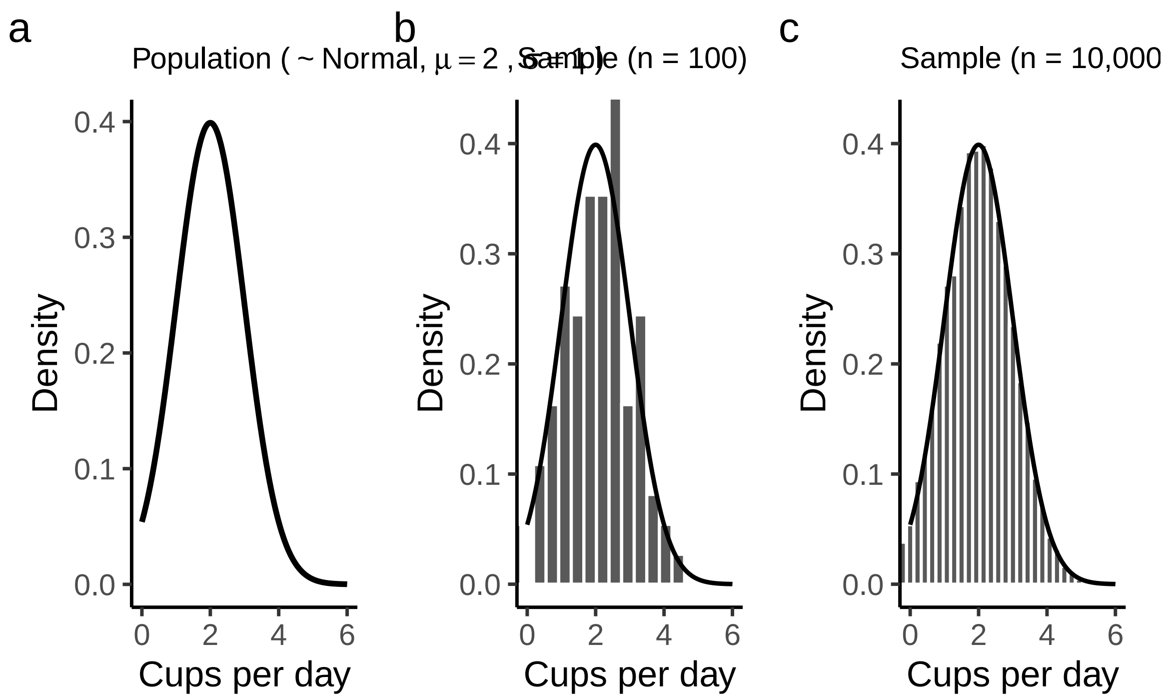

Figure 3.8: The population distribution of daily coffee consumption (a) and two random samples: one of size 100 (b) and one of size 10,000 (c).

mean might be 1.9 cups and the sample standard deviation 1.1, close to the population values but not exactly equal to them.

These are **sample statistics**: numbers we calculate from the data in hand. They approximate the population parameters, which describe the entire distribution. The central problem of inference is how to use sample statistics to estimate population parameters, and how much confidence we can place in those estimates.

3.9 The law of large numbers

In the last section we compared a small sample of coffee drinkers ($n = 100$) to a much larger sample ($n = 10{,}000$). The difference was clear: the larger sample produced a histogram that looked much closer to the true population distribution. The sample mean and standard deviation were also much closer to the population values ($\mu = 2$, $\sigma = 1$ cups/day).

This illustrates the **law of large numbers**. Informally: larger samples give better information. More precisely, as the sample size grows, the sample mean $\bar{X}$ converges toward the population mean μ. Symbolically:

$$N \to \infty \quad \Rightarrow \quad \bar{X} \to \mu$$

Although easiest to see with the mean, the law applies to many sample statistics: proportions, variances, correlations, and more. The guarantee is that, with enough data, the statistics you calculate from a sample will hover ever closer to their population counterparts.

The idea is obvious enough that Jacob Bernoulli, who first formalized it in 1713, remarked that "even the most stupid of men" already know it by instinct. His phrasing hasn't aged well, but the insight remains correct: averaging over more observations reduces the "danger of wandering from one's goal."

In practice, this means any single sample statistic will be off the mark, but if we keep collecting data, those statistics will tend to get closer and closer to the true population parameters. The law of large numbers is one of the fundamental guarantees that makes statistical inference possible.

3.10 Sampling distributions and the central limit theorem

The law of large numbers tells us that with enough data, our sample mean will get close to the population mean. But in practice, we never have infinite data. What we need is a way to understand how the sample mean behaves with *finite* samples. This is where the **sampling distribution of the sample means** comes in.

3.10.1 Sampling distribution of the sample means

The phrase is clunky, but the idea is straightforward.

- A **sample mean** is just the average of one sample.
- A **sampling distribution** is what you get when you take many samples, compute the mean for each, and then look at the distribution of those means.

So the **sampling distribution of the sample means** is simply the distribution formed by repeating your study many times and collecting all those sample averages.

Why is this useful? Because it tells us how much the sample mean varies from sample to sample, and how close we can expect it to be to the population mean.

3.10.2 Seeing the pieces

We can build this step by step:

1. Start with a distribution to sample from.
2. Draw repeated random samples of the same size n.
3. Compute the mean of each sample.
4. Plot those means in a histogram.

For example, suppose we draw numbers from a uniform distribution on 1 to 10: Figure 3.9.

Individual samples of size 20 look very different from each other, but their means cluster near 5.5 (the population mean).

If we repeat this thousands of times and plot all the sample means, we get a new distribution: the sampling distribution of the sample means. It is centered on 5.5, but spread out because of sampling variation.

We had a uniform distribution originally, but Figure 3.10 does not look like a uniform distribution. This is key takeaway: the histogram of sample means does **not** look like the original uniform distribution. Instead, it has its own shape and variability. That fact leads directly to the **central limit theorem**, which we'll introduce next.

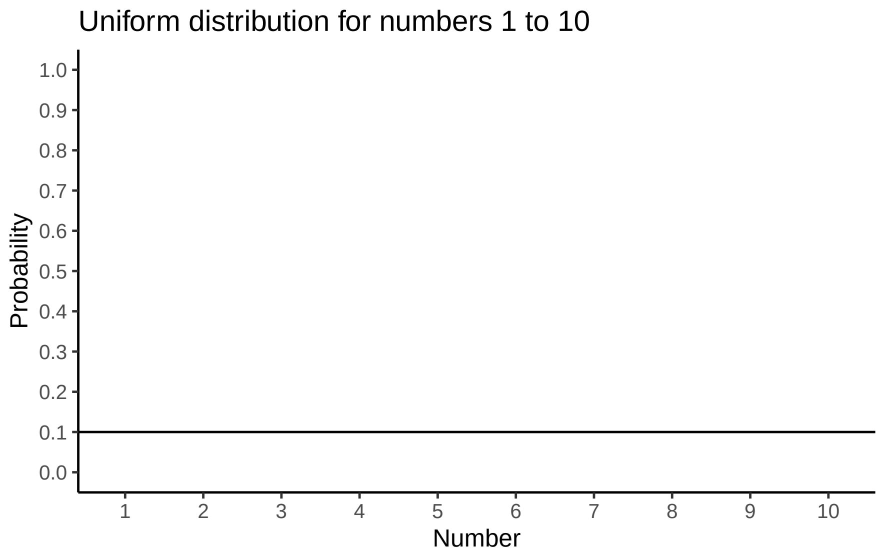

Figure 3.9: A uniform distribution illustrating the probabilites of sampling the numbers 1 to 10. In a uniform distribution, all numbers have an equal probability of being sampled, so the line is flat indicating all numbers have the same probability

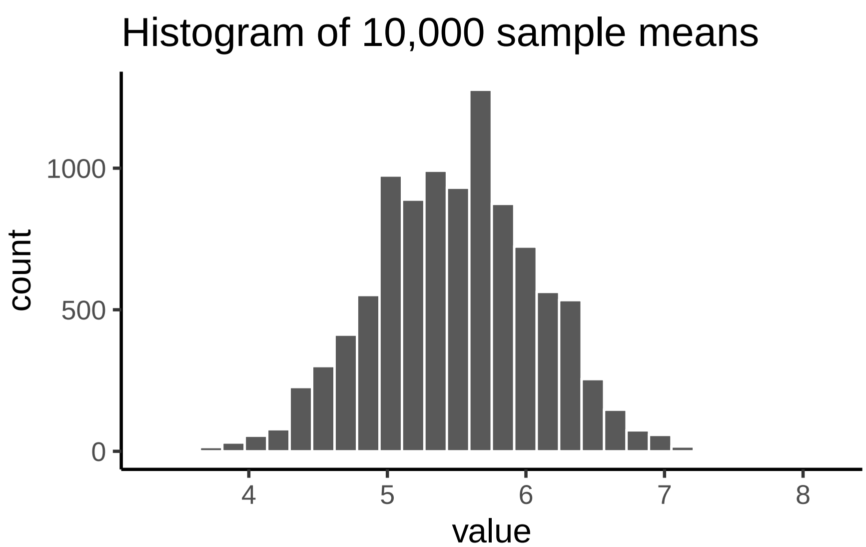

Figure 3.10: Sample means from 10,000 samples of size 20, drawn from the uniform distribution of numbers 1 to 10. The histogram centers on the expected value of 5.5, but individual samples scatter around it.

3.10.3 Beyond the mean

Although we've focused on means, the same idea applies to any statistic: medians, standard deviations, maximum values, and so on. Each has its own sampling distribution, which describes how that statistic behaves across repeated samples. To illustrate, suppose we draw many samples of size $n = 50$ from a normal distribution with mean 100 and standard deviation 20. For each sample we compute the mean, standard deviation, maximum, and median, then plot the histograms of these values (Figure 3.11).

The sample mean is just the most important case, because it underpins much of classical inference.

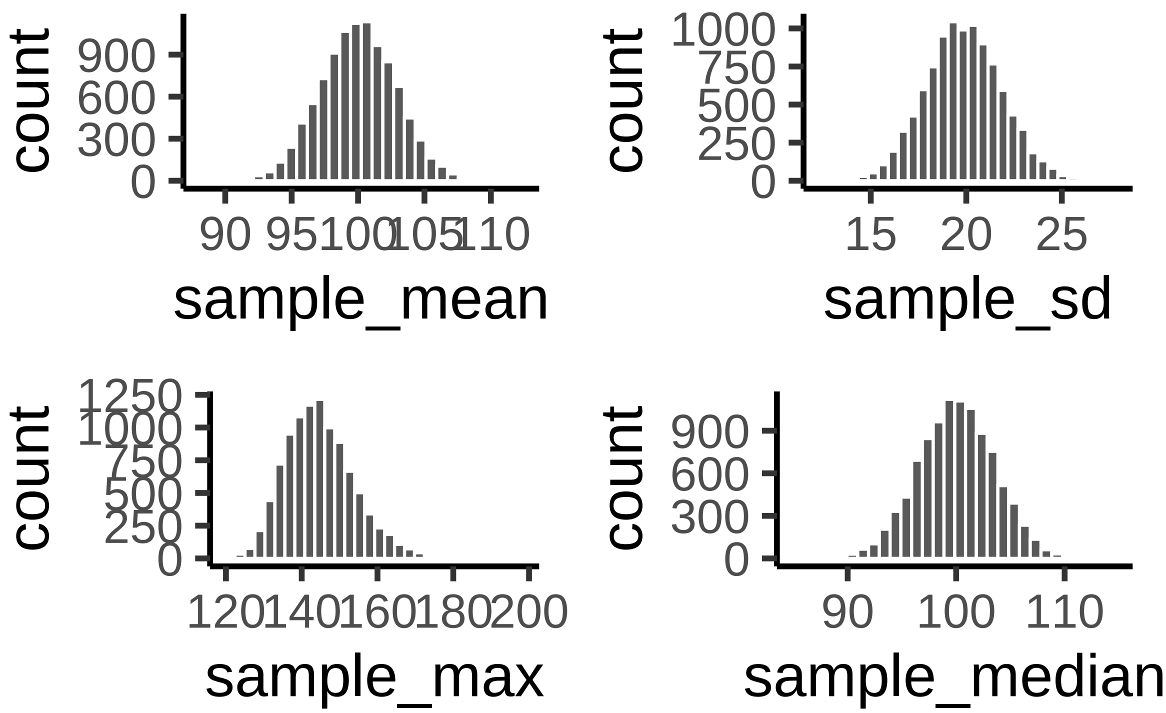

Figure 3.11: Each panel shows a histogram of a different sampling statistic

3.11 The central limit theorem

So far, you've seen that sample means vary from one sample to the next, and that their variability shrinks as sample size grows. The **sampling distribution of the mean** is just the distribution you get if you compute the mean from many different samples of the same size.

Intuitively: - Small samples <86><92> sample means bounce around more <86><92> wide sampling distribution. - Large samples <86><92> sample means are more stable <86><92> narrow sampling distribution.

Figure 3.12 animates this idea. Each panel shows samples of size 10, 50, 100, or 1000 drawn from a normal distribution (red line). Grey bars show the sample itself, the red line marks that sample's mean, and the blue histogram shows the distribution of sample means across many repetitions. Notice how the blue distribution shrinks around the population mean as n increases.

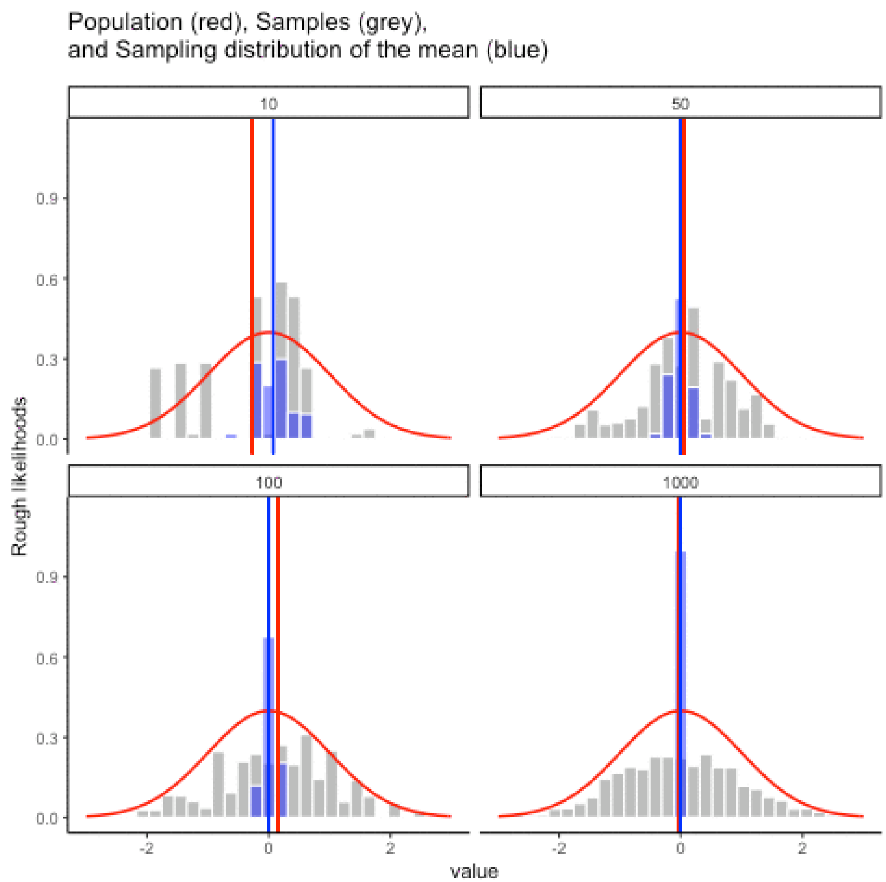

Figure 3.12: Simulation of samples (grey), their means (red line), and the sampling distribution of the mean (blue). As sample size grows, the distribution of sample means narrows and centers on the population mean. (Final frame; the animated version is in the web edition of this book.)

So, the sampling distribution of the mean is itself a distribution, with its own variability. When the sample size is small, the distribution of sample means is wide; when the sample size is large, it is narrow. The standard deviation of this sampling distribution has a special name:

the standard error (SE). In this context, it is the standard error of the mean. As you can see in the animation, as the sample size N increases, the SE decreases.

But there's more. The **central limit theorem** tells us three powerful facts about the sampling distribution of the mean:

1. Its mean equals the population mean, μ.
2. Its standard deviation equals the population σ divided by $\sqrt{N}$:

$$\text{SEM} = \frac{\sigma}{\sqrt{N}}$$

3. Its shape approaches normal as N grows, no matter what the shape of the population is.

Together, these three facts form our practical version of the **central limit theorem (CLT)**. The first fact, that sample means converge to the population mean, comes from the law of large numbers. The second formalizes how the SE shrinks with larger N. And the third shows why the normal curve appears so often: the distribution of sample means tends toward normality, even when the underlying data are skewed or flat.

The full CLT is mathematically more general and has a complicated proof, but for our purposes these three results capture what we need. To see the third point in action, let's look at histograms of **sample means** drawn from different underlying distributions. Remember: these are distributions of *sample means*, not of individual observations.

In Figure 3.13, we are smapling from a *normal* distribution (red line). The sampling distribution of the mean (blue bars) is also normal.

Let's try sampling from a uniform population: the parent distribution is flat (red), but the sampling distribution of the mean is bell-shaped (Figure 3.14).

Sampling from an exponential population: the parent distribution is skewed, but the sampling distribution of the mean still looks normal (Figure 3.15).

Together, these results are the essence of the central limit theorem (CLT). Formally, if the population has mean μ and standard deviation σ, then the sampling distribution of the mean also has mean μ and standard error

$$SE = \frac{\sigma}{\sqrt{N}}.$$

This formula shows how sample size controls reliability. As N increases, the denominator grows and the SE falls. The drop is not linear: doubling N cuts the SE by only about 30%, which is why the benefits of ever-larger samples taper off. Still, larger samples consistently produce more stable and trustworthy estimates of μ.

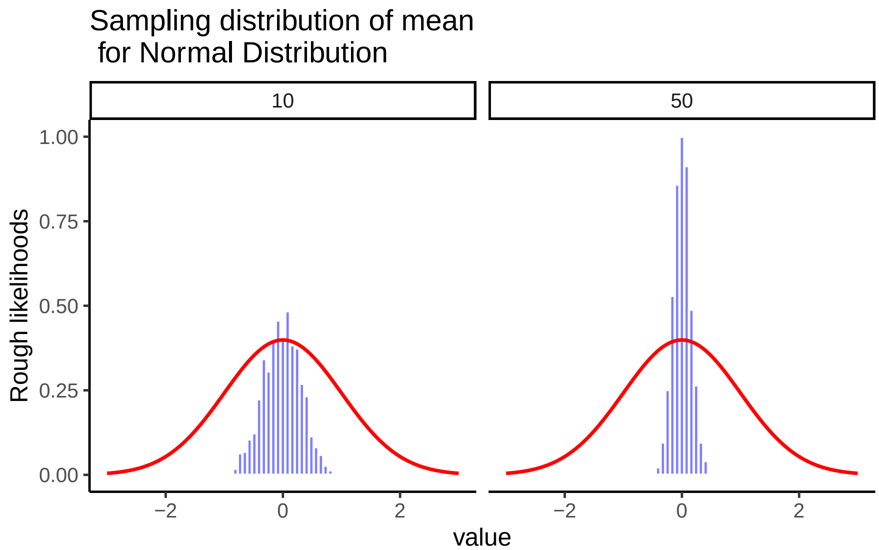

Figure 3.13: Sampling distributions of the mean from a normal population. As sample size increases, the distribution tightens around the population mean.

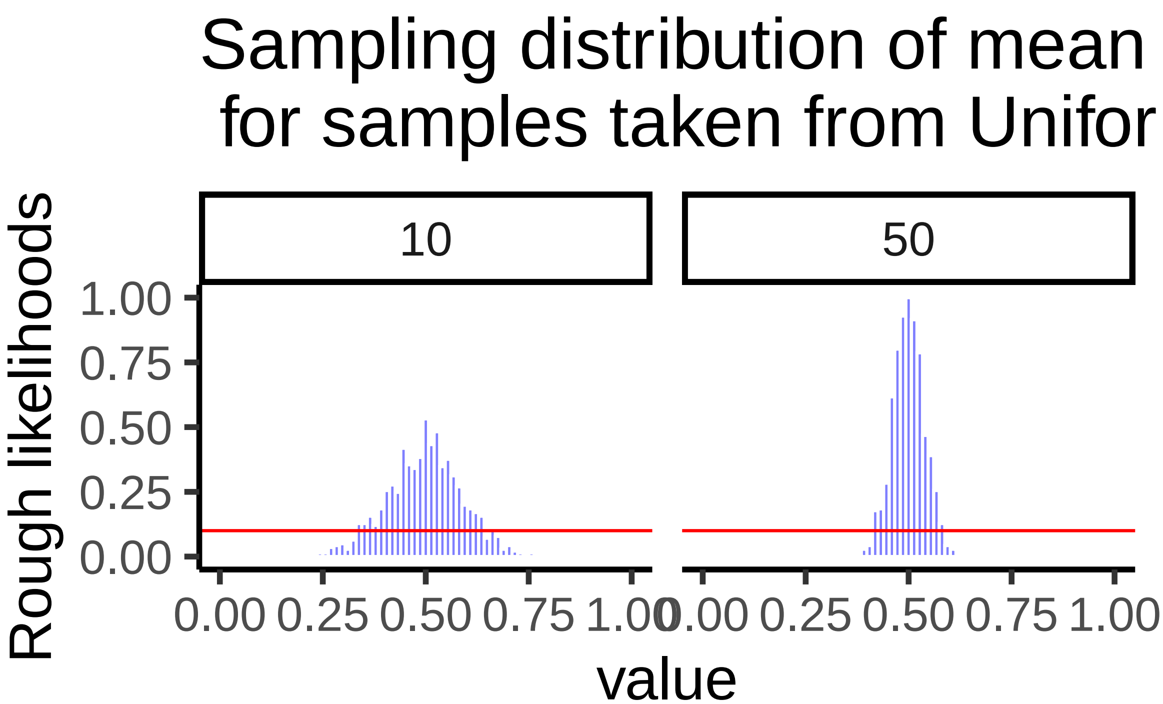

Figure 3.14: Even when samples come from a flat uniform distribution, the distribution of sample means is approximately normal.

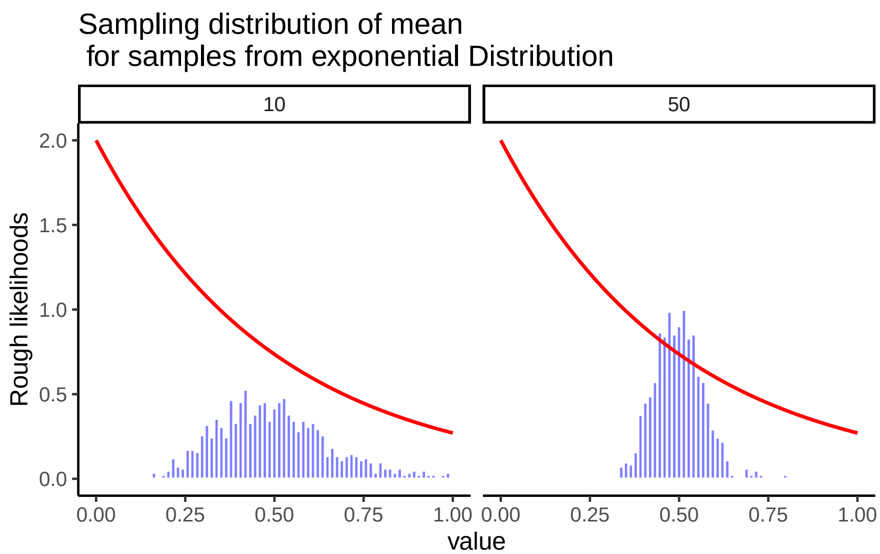

Figure 3.15: Even when samples come from a skewed exponential distribution, the distribution of sample means is approximately normal.

The CLT is one of the cornerstones of statistics. It explains why sample means are useful, why bigger studies are more reliable, and why the normal curve appears so often in practice. Whenever you average across many influences, exam scores, crop yields, daily temperatures, the distribution of those averages tends to be normal.

3.12 z-scores

A **z-score** is a standardized way to express how far a value is from its mean, measured in units of standard deviation. Z-scores are useful because (i) areas under the normal curve translate directly into probabilities, and (ii) by the CLT, many averages (e.g., sample means) are approximately normal, so z-scores let us read off probabilities for those, too.

First, recall the geometry of the normal curve. In Figure 3.16 we draw vertical lines at 0, 1, 2, and 3 standard deviations for a standard normal ($\mu = 0, \sigma = 1$). The labels show familiar areas: about 34.1% between 0 and 1, 13.6% between 1 and 2, and so on. Altogether, about 68.2% of values lie between <88><92>1 and +1 SD.

Those same proportions hold for any normal distribution, regardless of its mean or SD. In Figure 3.17 (with $\mu = 100$, $\sigma = 25$), the region from 100 to 125 (one SD above the mean) still contains about 34.1% of values.

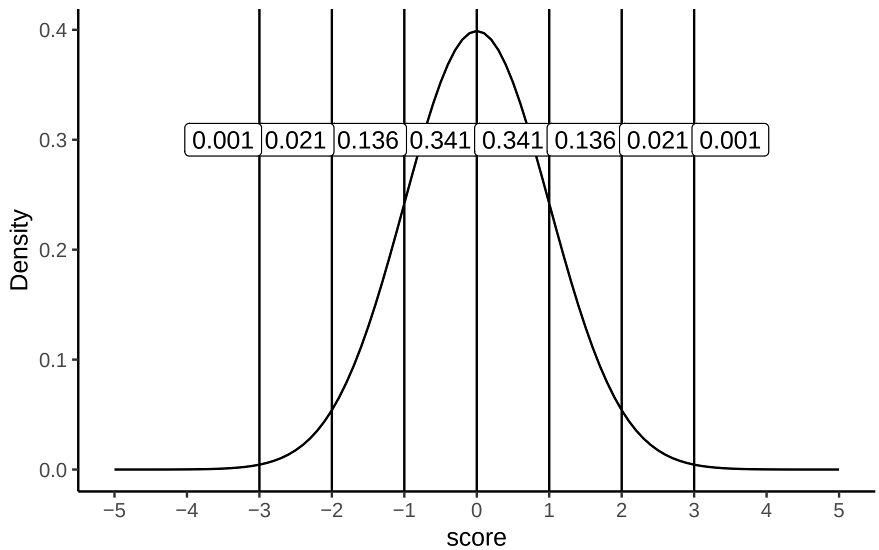

Figure 3.16: A normal distribution. Each line marks a standard deviation from the mean; labels show the proportion between adjacent lines.

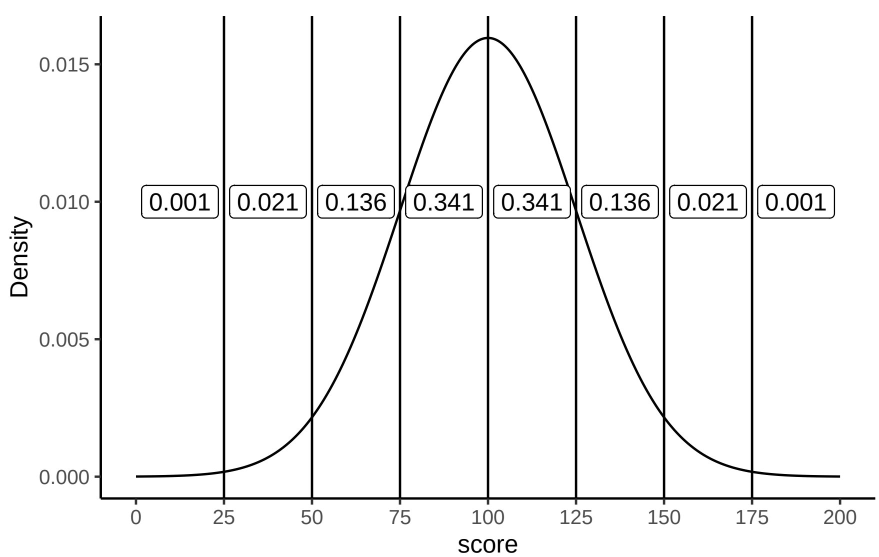

Figure 3.17: The same area rules apply for any normal: one SD on either side covers ~68.2%, etc.

3.12.1 What "standardized" means

Standardizing converts raw scores to z-scores by subtracting the mean and dividing by the standard deviation:

$z = \frac{x-\mu}{\sigma}$

Interpretation is immediate: - z=0 is the mean, - z=1 is one SD above the mean, - z=-2 is two SDs below, etc.

Because z-scores put everything on the same scale, you can compare values across different normal variables (or read probabilities from the standard normal). And by the CLT, you can do the same with *sample means* when N is modestly large.

3.12.2 Calculating z-scores

If the population mean and SD are known, use them in the formula. In practice they're often unknown, so we standardize with the sample mean $\bar{x}$ and sample SD s as estimates.

Example: generate ten observations from a $N(100, 25^2)$ distribution, then standardize:

```
#> # A tibble: 10 x 2
#>    Observation Score
#>          <int> <dbl>
#>  1           1 122.
#>  2           2 102.
#>  3           3 113.
#>  4           4  92.4
#>  5           5 103.
#>  6           6 160.
#>  7           7  86.4
#>  8           8  92.8
#>  9           9 115.
#> 10          10  53.9
```

Using the known population values:

$(scores - 100)/25$

```
#> # A tibble: 10 x 3
#>    Observation Score Zscore
#>          <int> <dbl>  <dbl>
#>  1           1 122.    0.88
```

```
#>  2             2 102.    0.1
#>  3             3 113.    0.51
#>  4             4  92.4  -0.3
#>  5             5 103.    0.13
#>  6             6 160.    2.41
#>  7             7  86.4  -0.54
#>  8             8  92.8  -0.29
#>  9             9 115.    0.6
#> 10            10  53.9  -1.84
```

Once standardized, areas under the normal curve tell you how unusual a score is. z between <88><92>1 and 1 is common (~68% of cases), between 1 and 2 is less common (~13.6% on each side), and beyond 2 is relatively rare (~2.1% in each tail segment between 2 and 3, even less beyond 3).

Finally, z-scores are also called **standardized scores** because they express distance from the mean in SD units.

3.13 The chi-square distribution

Not every named distribution is symmetric like the normal. The **chi-square (χ^2) distribution** is another shape you'll see repeatedly once we get to hypothesis testing, particularly for categorical data, where the question isn't "how far is this mean from that one?" but "how far are these observed counts from what we'd expect?"

Two things make chi-square look different from the normal shapes above. First, a chi-square statistic is built from *squared* deviations between observed and expected counts, so it can never be negative: the distribution starts at 0 and has no left tail at all. Second, its exact shape depends on degrees of freedom: with few df it's sharply right-skewed (most of the probability mass sits near 0, with a long tail to the right), and as df grows it stretches rightward and becomes more symmetric.

This also means chi-square tests are naturally **one-tailed**: because deviations in *either* direction from expected get squared before being added up, only large positive values of the statistic are ever "surprising." There's no equivalent of a negative z-value to worry about.

You'll put this distribution to work later for the chi-square goodness-of-fit and test-of-independence, once you have the hypothesis-testing framework (Chapter 5) to build a decision around it.

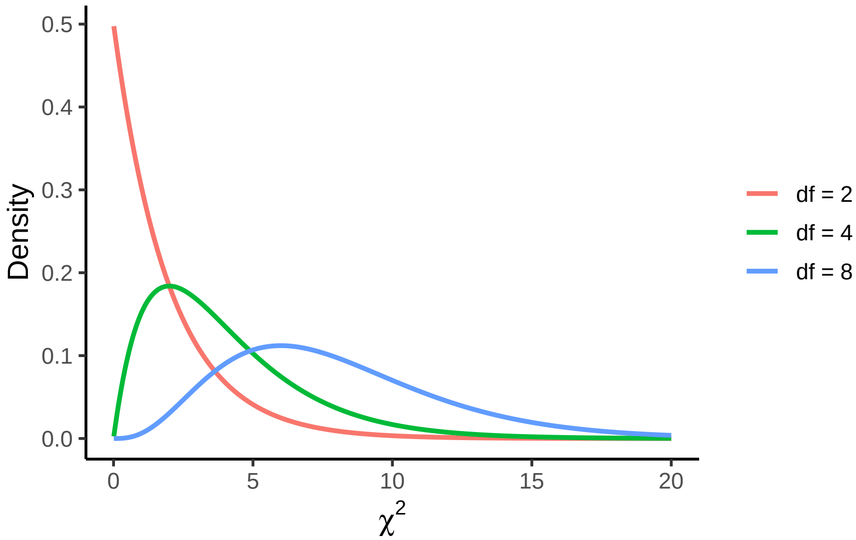

Figure 3.18: Chi-square distributions for three degrees of freedom. Unlike the (symmetric) normal distribution, chi-square is always non-negative and right-skewed; as df increases, the distribution shifts right and becomes more symmetric.

3.14 Estimating population parameters

3.14.1 Why estimate? (two quick motivations)

- **Concrete**: A bootmaker wants the mean and spread of adult foot lengths to plan inventory. Measuring everyone is impossible; a well-designed sample can estimate those parameters closely enough to act on.
- **Abstract**: In experiments, we often ask whether a manipulation changes a response. Even if the broader "population" is fuzzy, we still estimate the **mean response** and its **variability** under each condition to compare them.

3.14.2 Sample statistic vs. estimate

A statistic describes your data; an estimate is your best guess about the corresponding population parameter. For the mean, these coincide:

Symbol	What is it?	Do we know it?
$\bar{X}$	Sample mean	Yes, calculated from the raw data
μ	True population mean	Almost never known for sure

Symbol	What is it?	Do we know it?
$\hat{\mu}$	Estimate of the population mean	Yes, for simple random samples $\hat{\mu} = \bar{X}$

3.14.3 Estimating the population standard deviation

Estimating the population mean was straightforward: the sample mean $\bar{X}$ is an **unbiased estimator** of μ, meaning that across many samples it lands on the true mean on average. Standard deviation is trickier.

Here we encounter **bias**. An estimator is biased if its expected value systematically differs from the true population parameter. The sample standard deviation *sd* is such a case: it tends to underestimate σ, especially when N is small.

That is why back in Chapter 1 we saw two formulas for the standard deviation, one dividing by N and one by $N-1$. We can now explain why.

3.14.4 Simulating bias

We can see this by creating a **sampling distribution of the standard deviation**. Suppose the true population IQ distribution has mean 100 and standard deviation 15. If we repeatedly draw small samples (say, $N = 2$) and compute s each time, the histogram of those s values looks like Figure 3.19.

Even though the true σ is 15, the average **sample SD** is only about 8.5. This is very different from the sampling distribution of the mean, which is centered on μ.

If we repeat the simulation for a range of sample sizes, the contrast is clear:

- Figure 3.20 shows that the sample mean stays centered on μ (unbiased).
- Figure 3.21 shows that the sample SD underestimates σ when N is small, though the bias shrinks as N grows.

Figure 3.21 shows the sample standard deviation as a function of sample size. Unlike the mean, the sample standard deviation does not center on the population value. Small samples especially understate variability.

3.14.5 Why variance needs a small correction

We just saw *that* the sample standard deviation is biased. Here is *why*, and why the fix takes the particular form it does.

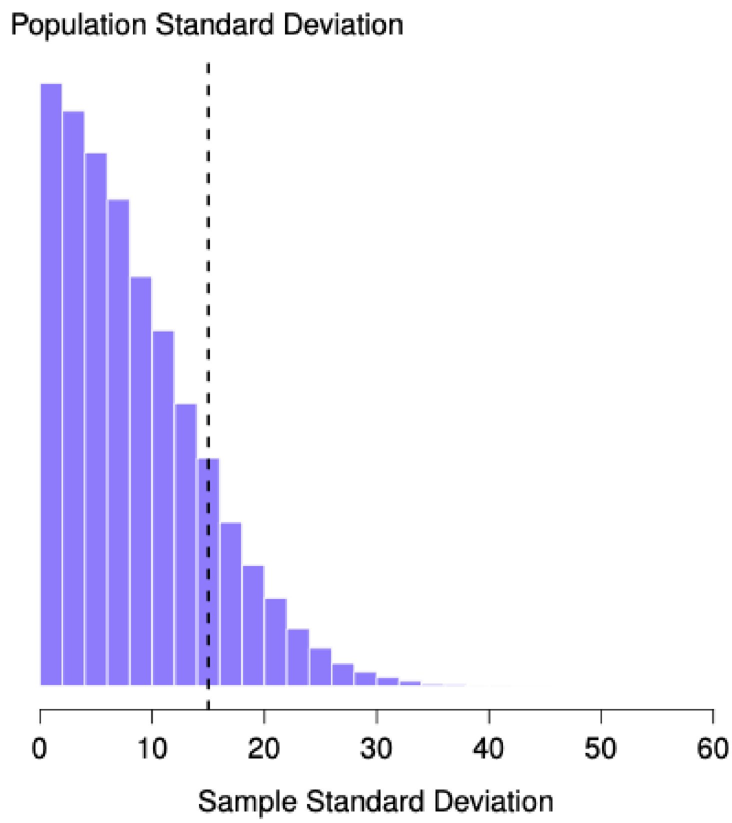

Figure 3.19: Sampling distribution of the sample standard deviation for N = 2. The true population SD is 15 (dashed line), but most experiments produce smaller values. The average sample SD is only 8.5, showing that s underestimates *sigma*.

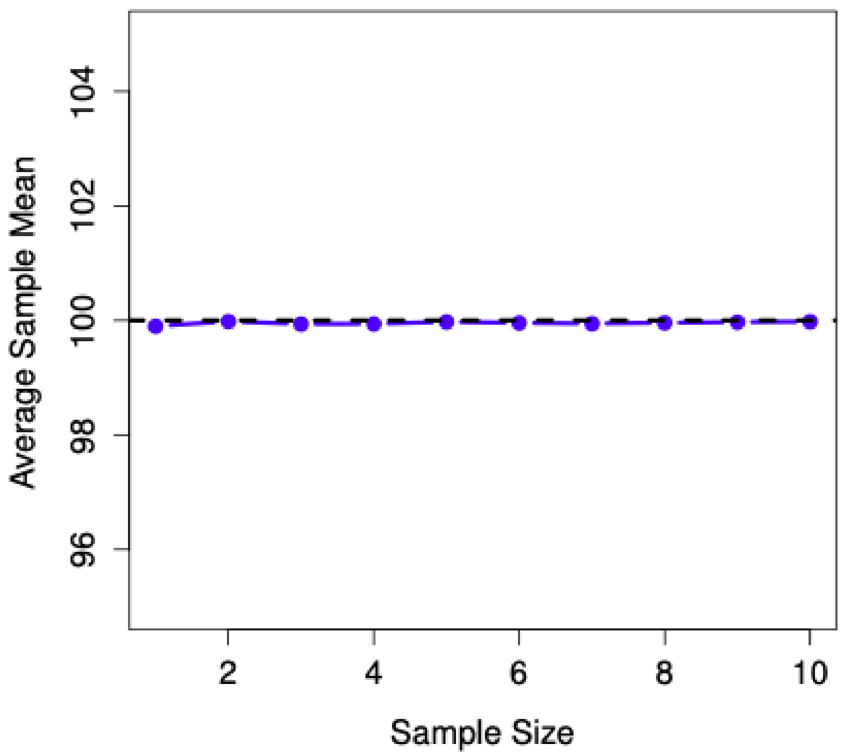

Figure 3.20: The sample mean is an unbiased estimator of the population mean.

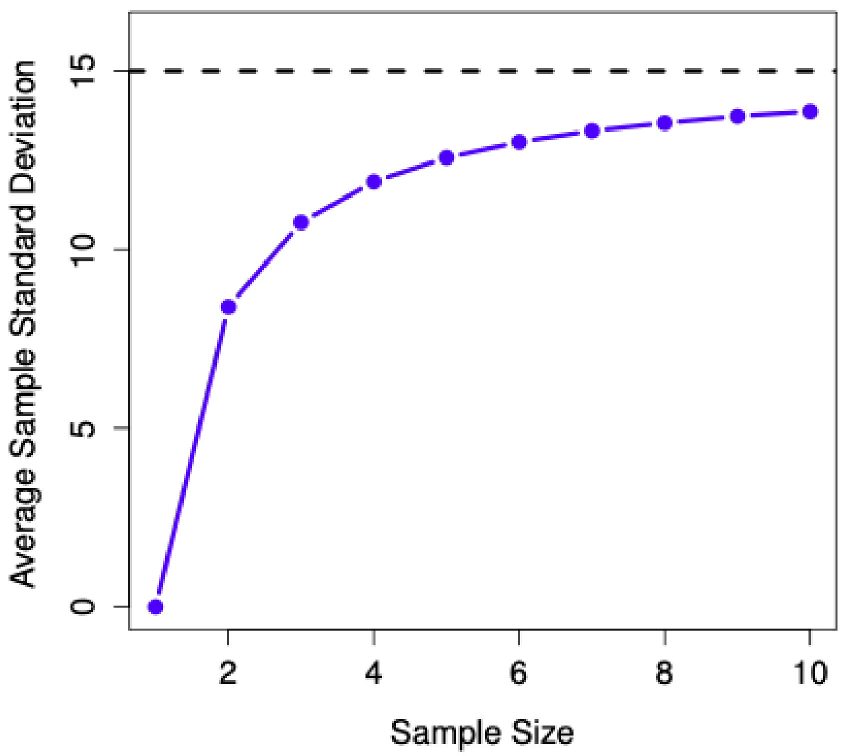

Figure 3.21: The sample standard deviation is a biased estimator of the population standard deviation.

3.14.5.1 Why does this happen?

Picture a tiny sample of $n = 3$ values. The sample mean $\bar{x}$ might land close to the true population mean μ, or it might land far away, and we don<80><99>t know in advance which. But $\bar{x}$ is always computed **from** those three sample points, which pulls it toward the middle of *those specific* points. As a result, distances from the sample points to $\bar{x}$ tend to be a little **shorter**, on average, than distances from those same points to the true μ would be.

Since variance and SD are built from squared distances to the mean, this shrinkage carries through: the sample variance computed the <80><9c>obvious<80><9d> way (divide by n) is biased **small**, and the sample SD inherits that bias.

The fix is to divide by $n - 1$ instead of n. Writing it out, the naive sample variance is

$$s_n^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2,$$

and it systematically underestimates σ^2, the population variance. The corrected version divides by $n - 1$ instead:

$$s_{n-1}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

This s_{n-1}^2 is an **unbiased** estimator of σ^2, so averaged across many samples it lands right on the true population variance. Take the square root of either formula to get back to an SD. One small wrinkle: because square roots aren<80><99>t linear, even the corrected s_{n-1} is *technically* still very slightly biased for σ. In practice that residual bias is tiny and shrinks as n grows, which is why s_{n-1} is the standard choice for a sample SD.

3.14.5.2 Seeing the bias

Let<80><99>s make this concrete. Suppose our full population is every even number from 80 to 120, with true mean $\mu = 100$, marked in red in Figure 3.22.

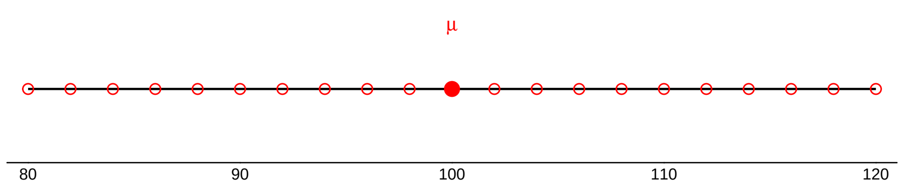

Figure 3.22: Our imaginary population: every even number from 80 to 120.

Now let<80><99>s draw three separate samples of $n = 3$ from that population, each on its own number line. The light blue point is the sample mean $\bar{x}$; the three filled blue points are the sampled values. On the right of each panel:

- **uncorrected** s (divide by n): the naive sample SD,
- **corrected** s (divide by $n - 1$): the conventional sample SD,
- σ: the true population SD, about 12.1.

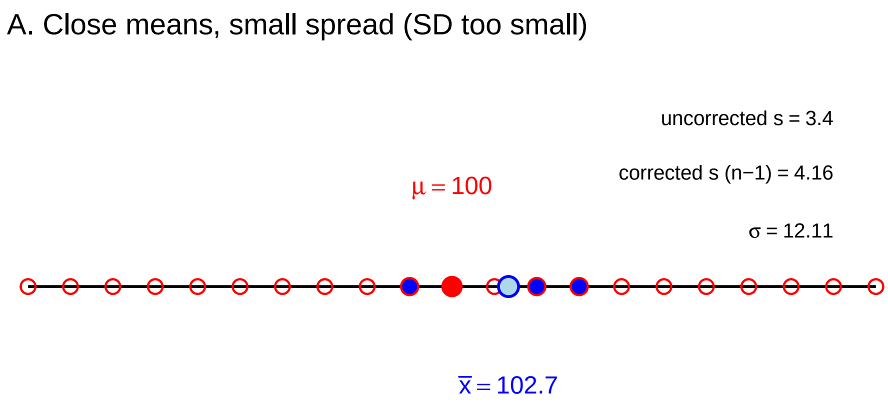

Figure 3.23: Example A: close sample mean, small spread.

In Example A, the sample mean lands close to μ, but the three points sit close together. σ is about 12.1, uncorrected s is 3.4, corrected s is 4.2. Both versions undershoot σ badly, but the correction helps.

In Example B, the sample mean lands farther from μ, but the points are still tight together. σ is about 12.1, uncorrected s is 2.5, corrected s is 3.1, the same story.

In Example C, the sample mean lands close to μ again, but this time the three points happen to be spread wide. σ is about 12.1, uncorrected s is 14.1, corrected s is 17.2. Here *s overshoots* σ instead.

So sometimes s undershoots and sometimes it overshoots. Why call it <80><9c>biased small,<80><9d> then? Because outcomes like A and B (tight, unrepresentative samples) happen more often than outcomes like C. Average over enough samples, and the sample SD comes in low more often than it comes in high. Let<80><99>s check that directly by simulating 8,000 samples, shown in Figure 3.26.

Each panel is a histogram of 8,000 sample SDs. The red dashed line marks the true $\sigma = 12.11$; the blue solid line marks the average sample SD across all 8,000 simulations.

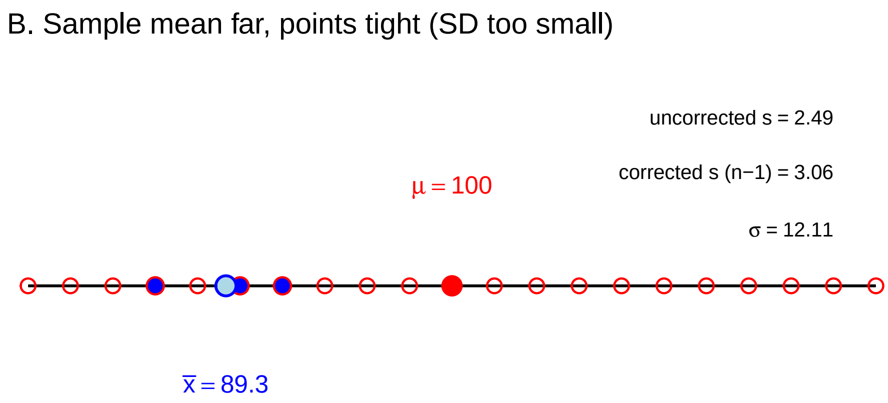

Figure 3.24: Example B: sample mean far from mu, points tight together.

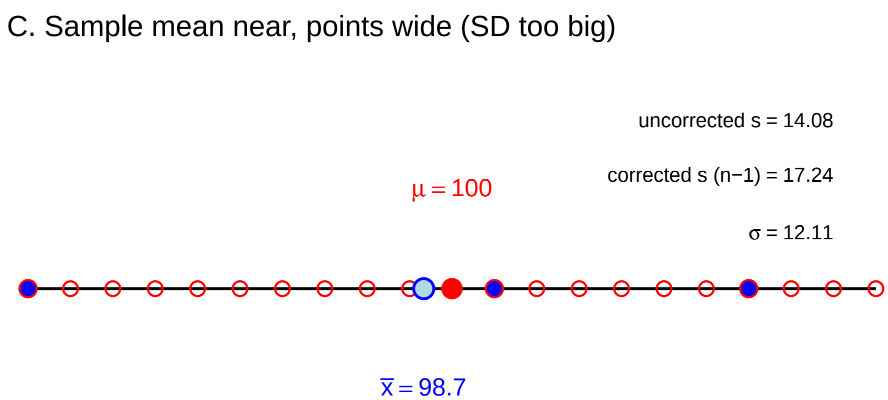

Figure 3.25: Example C: close sample mean, points spread wide.

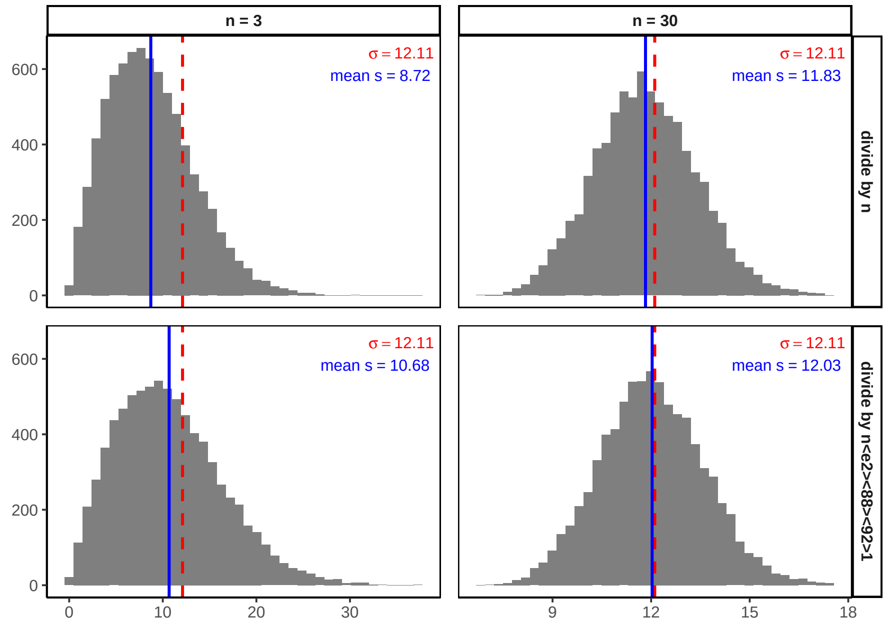

Figure 3.26: Sample SD bias, simulation results for n = 3 vs. n = 30.

Two things to notice. First, compare the top row (divide by n) to the bottom row (divide by $n-1$): the blue line sits closer to the red line in the bottom row, confirming the correction helps. Second, compare the columns: at $n = 30$ the bias is much smaller than at $n = 3$, in both rows. The $n-1$ correction matters most for small samples, and becomes negligible as sample size grows, exactly the pattern we'd expect if bigger samples give us better estimates.

3.14.5.3 Key points to remember

- The sample mean is unbiased for μ, so no correction is needed.
- The naive sample SD (divide by n) tends to underestimate σ.
- Dividing by $n-1$ instead corrects this bias, and is the standard choice for sample data.
- The correction matters most for small n, and fades as n grows.
- On an exam, if you<80><99>re asked for <80><9c>the sample standard deviation,<80><9d> use the $n-1$ version unless told otherwise. R<80><99>s `sd()` function already does this for you.

> **ℹ Note**
>
> **Connecting back to the law of large numbers.** Both of the ideas we met earlier in this chapter bear directly on what we just saw:

- **Law of large numbers:** as n grows, both $\bar{x}$ and s_{n-1} converge toward their true population values. That is exactly the pattern in Figure 3.26: bigger samples, tighter histograms, less bias.
- **Central limit theorem:** the CLT describes the sampling distribution of the **mean**, not the standard deviation. So we cannot assume the distribution of sample SDs is normal no matter how large n gets, even though the histograms above happen to look reasonably bell-shaped.

To summarize:

Symbol	Definition	Known?
s^2	Sample variance (divide by N)	Yes, calculated directly from data
σ^2	Population variance	No, almost never known exactly
$\hat{\sigma}^2$	Estimated population variance (divide by $N-1$)	Yes, but distinct from s^2

3.15 Estimating a confidence interval

> Statistics means never having to say you're certain – Unknown origin

So far, we've focused on using sample data to estimate population parameters, but every estimate comes with some uncertainty. A single number, like "the mean IQ is 115," doesn't capture how confident we are. What we want is a range that likely contains the true population mean. That range is called a **confidence interval (CI)**.

You'll use this directly in Chapter 5, where a confidence interval becomes one of three equivalent ways to decide whether to reject H_0. For now, let's see where it comes from.

The Idea

Thanks to the central limit theorem, we know the sampling distribution of the mean is approximately normal. In a normal distribution, 95% of values fall within about 1.96 standard deviations of the mean.

If we **knew** the population mean and SD (μ, σ)), the central limit theorem tells us where **sample means** tend to fall: a sample mean ($\bar{X}$) is usually within about (1.96) standard errors of (μ), where

SEM = $\frac{\sigma}{\sqrt{N}}$

That's the *forward* view (model <86><92> data).

In practice we want the reverse (data <86><92> model): given an observed ($\bar{X}$), what can we say about (μ)? Rearranging the same inequality gives

$$\bar{X} - 1.96\,\frac{\sigma}{\sqrt{N}} \leq \mu \leq \bar{X} + 1.96\,\frac{\sigma}{\sqrt{N}}$$

so the **95% confidence interval** is

$$CI_{95} = \bar{X} \pm 1.96\,\frac{\sigma}{\sqrt{N}}$$

The constant (1.96) is the normal cutoff for 95%; for other levels use the corresponding normal quantile (e.g., (1.04) for 70%).

The catch

There's one snag: this formula assumes we know the population standard deviation σ. But in practice, σ is almost never known. We estimate it with $\hat{\sigma}$, which introduces extra uncertainty.

To account for this, we swap the normal cutoffs for those from the t-distribution, which depends on sample size. With small N, the t values are larger, so the CI widens. As N grows, t values shrink toward 1.96, and the t-based CI becomes indistinguishable from the normal version.

> Tip
>
> **Rule of thumb.** Larger N <86><92> smaller SE <86><92> narrower CI.

Why it matters

Confidence intervals tell us how precise our guess is, and they force us to acknowledge uncertainty. A wide interval tells us we don't have much precision; a narrow one suggests greater precision.

3.16 Chapter Summary

Why it matters. Every inference you make for the rest of the course rests on one move: reasoning from a sample you have to a population you don't. This chapter builds the machinery that makes that move legitimate, and explains why it works even when your data are not normal.

Core ideas

- **Probability runs model <86><92> data; statistics runs data <86><92> model.** Probability starts from a known process and predicts what data it produces. Statistics starts from observed data and reasons backward to the process. The rest of the book is the second direction.
- **A population is a distribution, not just a group.** Treating the population as a probability distribution is what lets us say anything precise about how samples behave.
- **Law of large numbers.** As N grows, a sample statistic settles onto the population value. Bigger samples give more reliable estimates, and this is why.
- **Sampling distributions are the key abstraction.** Any statistic computed over repeated samples has its own distribution. The sampling distribution of the mean is the one that matters most, and its spread is the **standard error**, $\sigma/\sqrt{N}$.
- **The central limit theorem.** The sampling distribution of the mean centers on μ, narrows as N grows, and approaches normal *regardless of the population's shape*. That last clause is what lets normal-based methods work on data that are not themselves normal.
- **Not every estimator is unbiased.** The sample mean lands on μ on average, but the sample SD systematically underestimates σ. Dividing by $N-1$ corrects it, and the correction matters most when N is small.
- **Confidence intervals express uncertainty.** Rather than reporting a single estimate, a CI reports the range of population values the data are consistent with.

3.17 Videos

3.17.1 Introduction to Probability

Jeff has several more videos on probability that you can view on his statistics playlist.

3.17.2 Chebychev's Theorem

3.17.3 Z-scores

3.17.4 Normal Distribution I

3.17.5 Normal Distribution II

4 Foundations for inference

> Data and data sets are not objective; they are creations of human design. We give numbers their voice, draw inferences from them, and define their meaning through our interpretations.
> Katie Crawford

Inferential statistics is the branch of statistics concerned with drawing conclusions about populations or causal relationships from sample data. It provides the tools to answer questions like: did our manipulation actually cause a change, or could the pattern we observed have arisen by random chance alone?

This chapter builds the conceptual foundations for inference using simulation, no formulas yet. The goal is to make the underlying logic explicit and visible before it gets formalized in later chapters.

4.1 Brief review of Experiments

A controlled experiment is a structured method of data collection designed to support inferences about causality. Experiments involve deliberately manipulating one condition while holding others constant, then measuring the response.

Every experiment has two core components: the **independent variable**, the condition under experimental control, and the **dependent variable**, what is measured.

For example, a researcher studying the effects of wastewater discharge on stream health might compare dissolved oxygen (DO) levels at sites downstream of a discharge point against levels at unaffected reference sites, expecting DO to be lower downstream if the discharge is having an effect (wastewater typically consumes oxygen as it decomposes). Distance from discharge (affected vs. reference) is the independent variable; DO is the dependent variable.

Causal forces produce detectable change. If the discharge is truly reducing dissolved oxygen, then measurements from affected sites should differ systematically from measurements at reference sites. The experiment is designed to detect that change.

The central question becomes: how do we know if a measured difference between conditions reflects a real causal effect? Data always contain variability: measurements fluctuate even when nothing has changed. To decide whether a difference is real, we need tools that can distinguish meaningful change from background noise.

4.2 The data came from a distribution

Understanding where data come from is foundational to inference. This section returns to sampling and distributions, but now to ask what they reveal about the process that generated our data, not just to describe them.

A **distribution** specifies what numbers are likely to occur and how often. It sets the constraints on what we can expect to see when we take measurements.

4.2.1 Uniform distribution

As a reminder from last chapter, Figure 4.1 shows that the shape of a uniform distribution is completely flat.

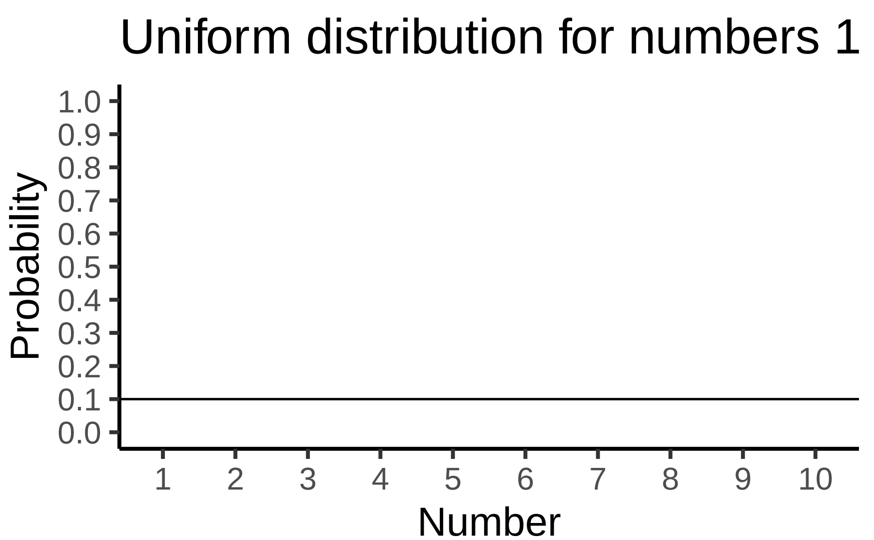

Figure 4.1: Uniform distribution showing that the numbers from 1 to 10 have an equal probability of being sampled

The y-axis shows probability (0 to 1) and the x-axis shows the values 1 through 10. The flat horizontal line indicates that every value from 1 to 10 has an equal probability of occurring: $1/10 = 0.1$. This is the defining property of a uniform distribution: no value is more likely than any other.

Think of the uniform distribution as a number-generating process. If it generates numbers repeatedly, it produces each value with equal frequency: roughly 10% 1s, 10% 2s, and so on. Let's sample 100 numbers from this distribution.

We used the uniform distribution to generate these numbers. Formally, this is called **sampling** from a **distribution**. Drawing a subset of observations from a larger process or population is sampling; if you could observe every possible value the process could produce, you would have the **population**. We will return to the distinction between samples and populations throughout the course.

Because we used the uniform distribution to create these numbers, we already know where they came from. But suppose someone handed you this same list without telling you its source: could you tell, just by looking, that it came from a uniform distribution? You'd need to check is whether all the numbers occur with roughly equal frequency, since that's what a uniform distribution predicts: if each number had the same chance of occurring, each should show up roughly the same number of times.

Putting the sample of 100 numbers into a histogram makes the counts easy to compare. If all values from 1 to 10 occur with equal frequency, each individual number should occur about 10 times. Figure 4.2 shows the histogram:

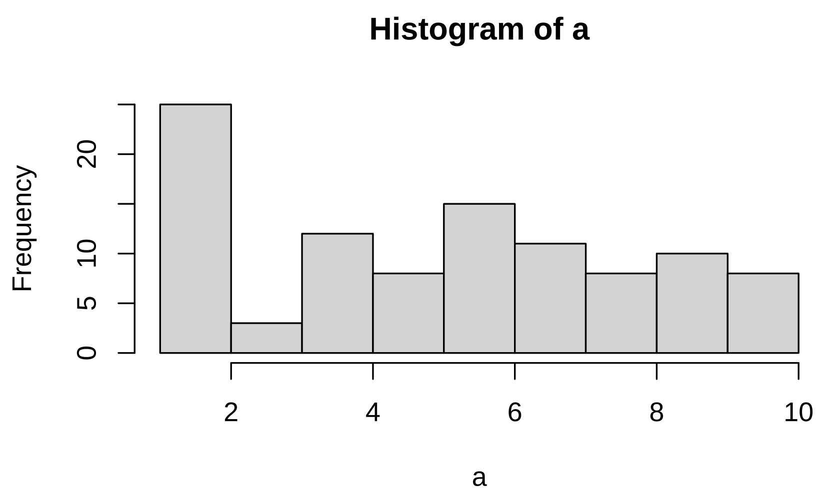

Figure 4.2: Histogram of 100 numbers randomly sampled from the uniform distribution containing the integers from 1 to 10

As the histogram shows, not all numbers occurred exactly 10 times. The bars vary in height even though all values had equal probability. This is an illustration of **sampling variability**: random samples do not perfectly reproduce the distribution they came from. The discrepancy between what we observe and what the distribution predicts is called **sampling error**.

4.2.2 Not all samples are the same, they are usually quite different

Let's look at sampling error more closely. We will sample 20 numbers from the uniform distribution. We should expect that each number between 1 and 10 occurs about two times each. As before, this expectation can be visualized in a histogram. To get a better sense of sampling error, let's repeat the above process ten times. Figure 4.3 has 10 histograms, each showing what 10 different samples of twenty numbers looks like:

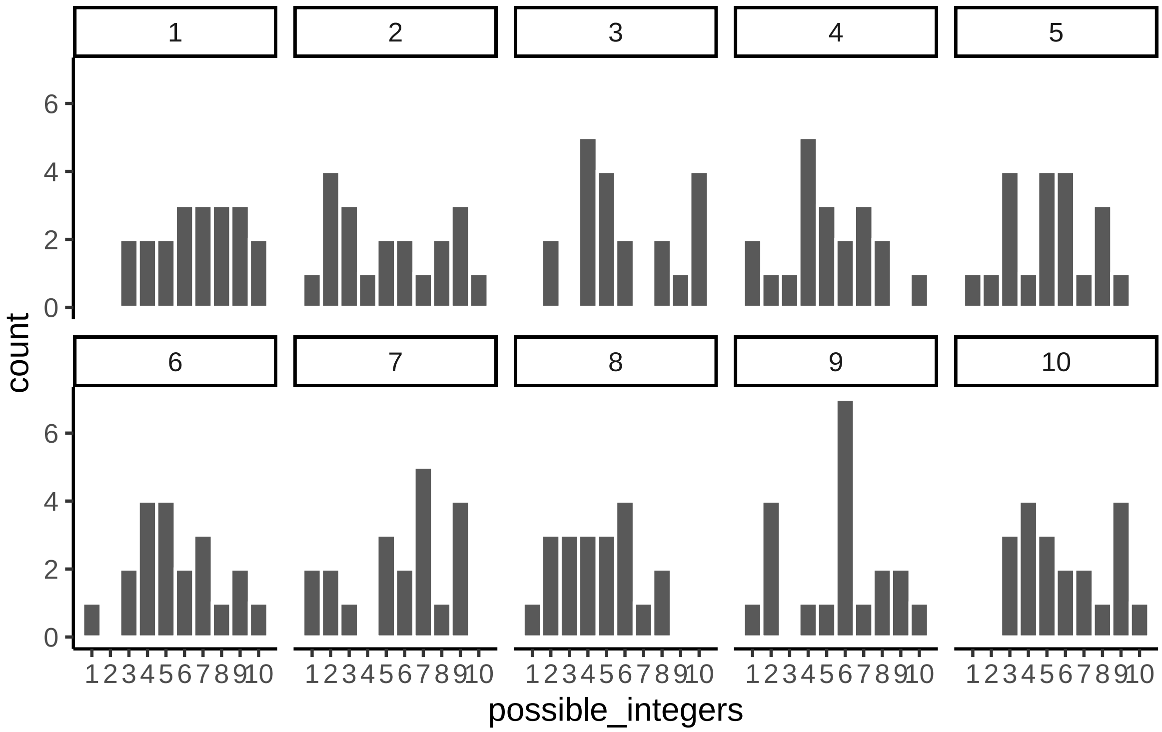

Figure 4.3: Histograms for 10 different samples from the uniform distribution. Each sample contains 20 numbers. The histograms all look quite different. The differences between the samples are due to sampling error or random chance.

You might notice right away that none of the histograms are the same. Even though we are randomly taking 20 numbers from the very same uniform distribution, each sample of 20 numbers comes out different. That's our sampling variability.

Figure 4.4 shows an animated version of the process of repeatedly choosing 20 new random numbers and plotting a histogram. The horizontal line shows the flat-line shape of the uniform distribution. The line crosses the y-axis at 2; and, we expect that each number (from 1 to 10) should occur about 2 times each in a sample of 20. However, each sample bounces around quite a bit, due to random chance.

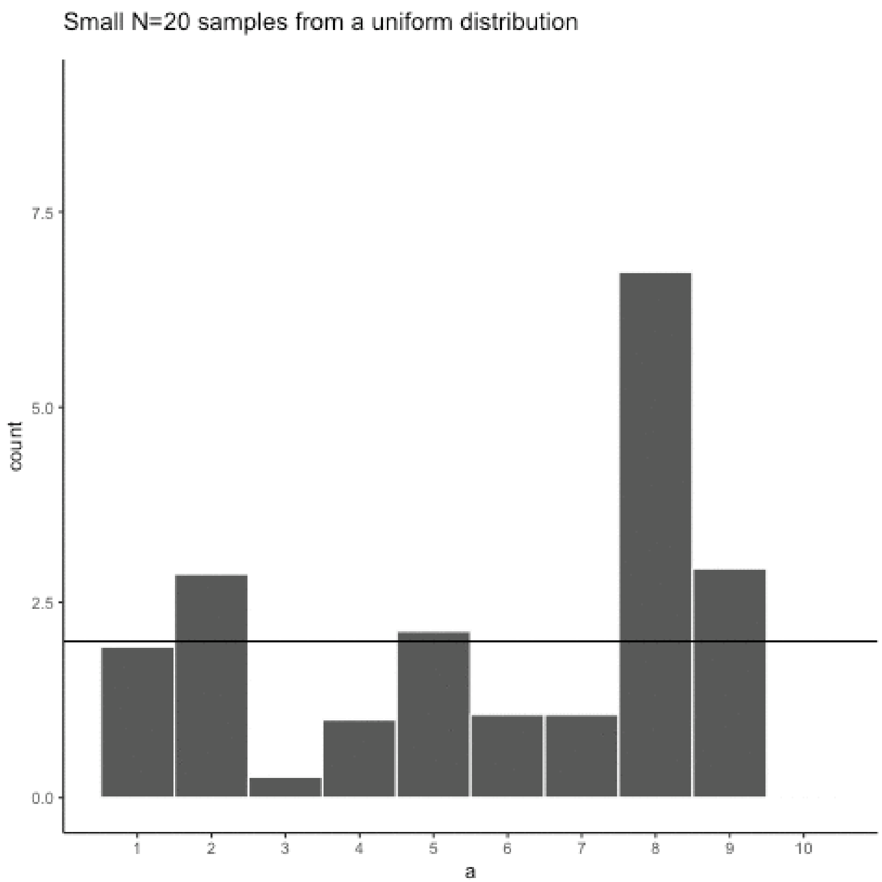

Figure 4.4: Animated histograms of repeated samples of 20 from a uniform distribution. The black line marks the expected count of 2 per number; the bars rarely match it. (Final frame; the animated version is in the web edition of this book.)

Small samples show high variability because random chance can produce very uneven counts even when every value is equally likely. This variability is sampling error, and it makes it harder to identify the true distribution from a single sample.

4.2.3 Large samples are more like the distribution they came from

Let's refresh the question. Which of the two samples in Figure 4.5 do you think came from a uniform distribution?

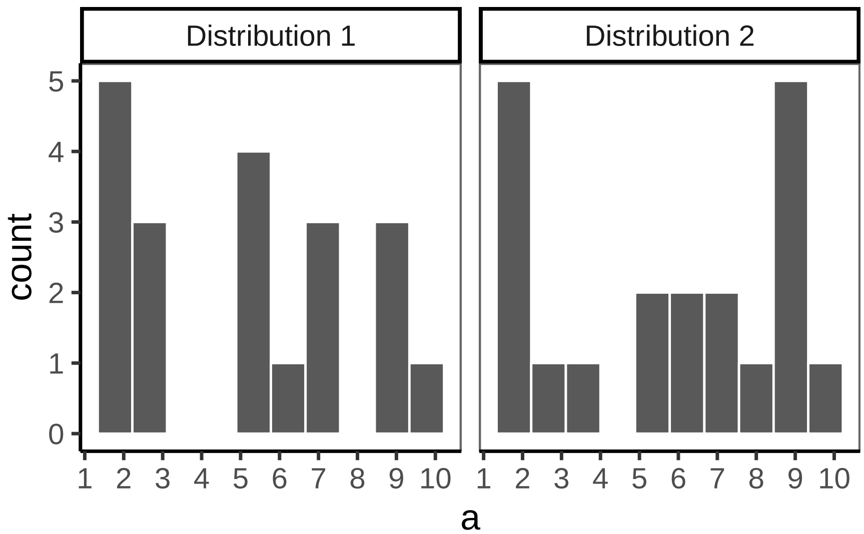

Figure 4.5: Which of these two samples came from a uniform distribution?

Trick question. Both samples came from the uniform distribution, yet neither histogram looks flat. Sampling error can obscure the true shape of a distribution, even with a sample of this size.

Increasing the **sample size** makes it easier to see whether a sample came from a uniform distribution. We'll often use the letter `n` for sample size: the number of observations in a sample.

Increasing each sample from 20 to 100 observations, and again creating 10 samples drawn from the same uniform distribution, we'd now expect each number from 1 to 10 to occur about 10 times per sample. The histograms are shown in Figure 4.6.

Again, most of these histograms don't look very flat: the bars still vary up and down instead of sitting exactly at 10 each. Sampling error is still there, and it always will be, at any sample size.

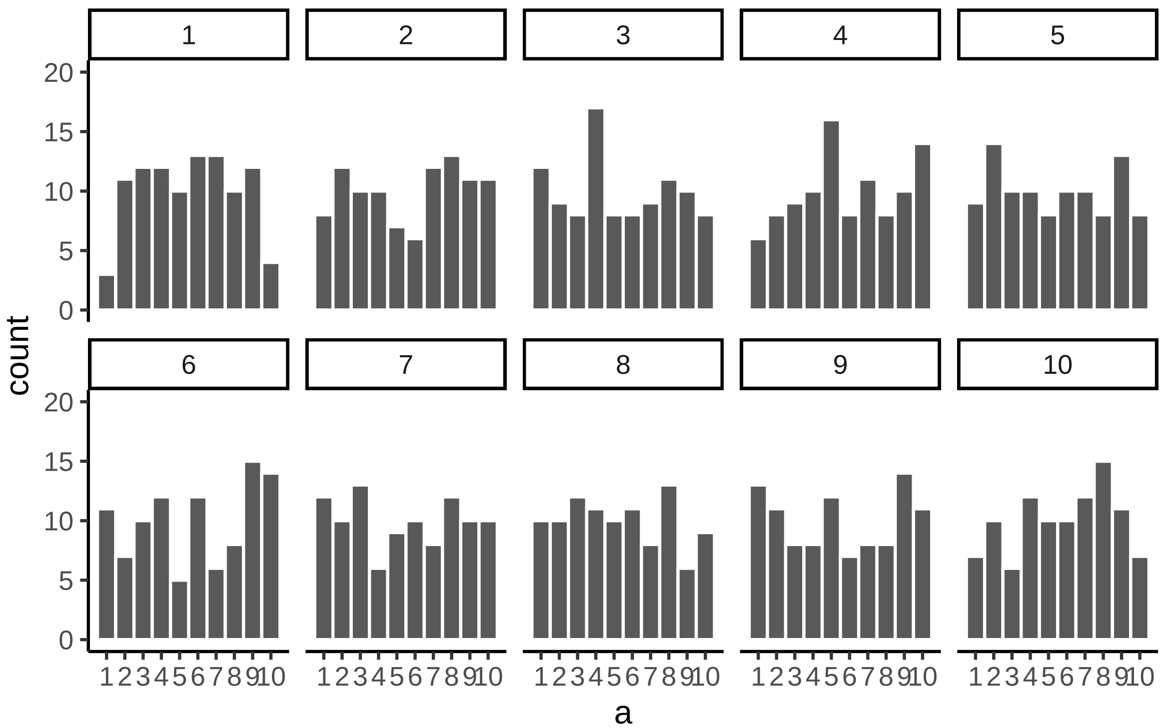

Figure 4.6: Histograms for different samples from a uniform distribution. N = 100 for each sample.

Increasing N from 100 to 1000 observations per sample, we'd now expect every number to appear about 100 times each.

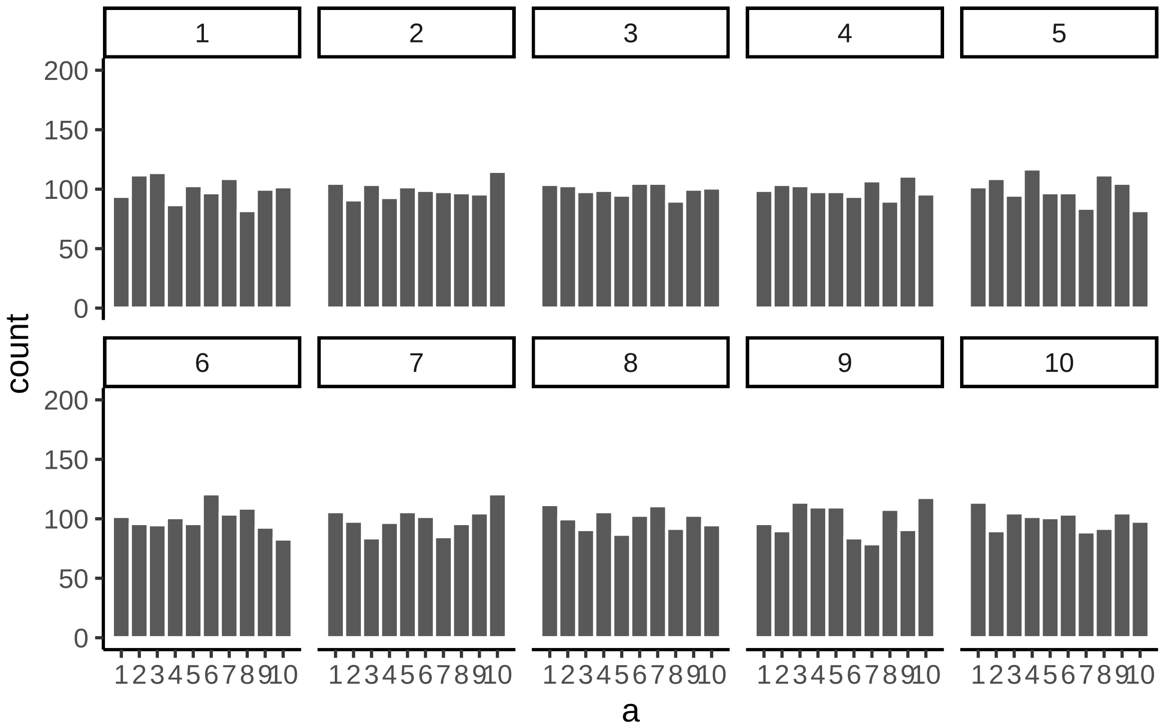

Figure 4.7: Histograms for different samples from a uniform distribution. N = 1000 for each sample.

Figure 4.7 shows the histograms starting to flatten out. The bars are still not perfectly at 100, because there is still sampling error, but a flat-looking histogram from a large sample gives more confidence that the numbers came from a uniform distribution.

Pushing sample size further, to 100,000 observations per sample, we'd expect each number to occur about 10,000 times each.

Figure 4.8 shows the histograms for each sample starting to look the same. With 100,000 observations each, chance has enough opportunity to distribute the numbers evenly, so the bars all land very close to 10,000, right where they should be for a uniform distribution.

Tip

Key pattern. The pattern behind a sample will tend to stabilize as sample size increases. Small samples show a wider variety of patterns because of sampling error (chance).

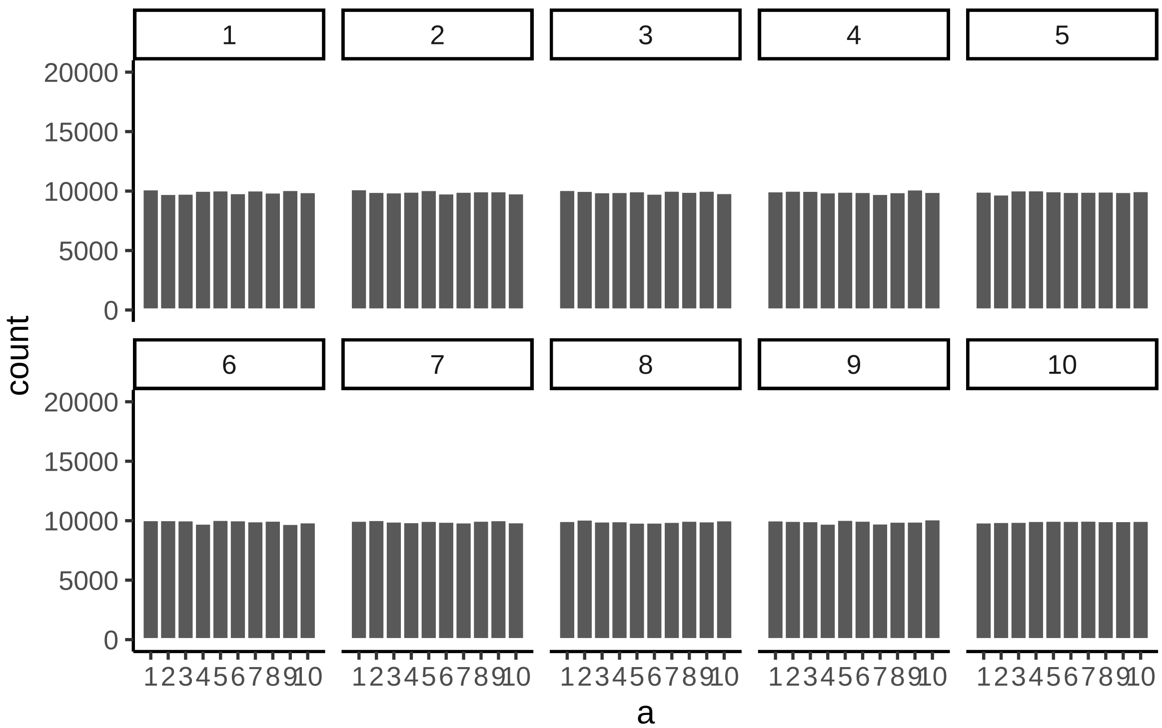

Figure 4.8: Histograms for different samples from a uniform distribution. N = 100,000 for each sample.

Before returning to experiments, one more comparison, and this time it isn't a trick question: in Figure 4.9, one of the two 20-observation samples really did come from a different distribution than the other.

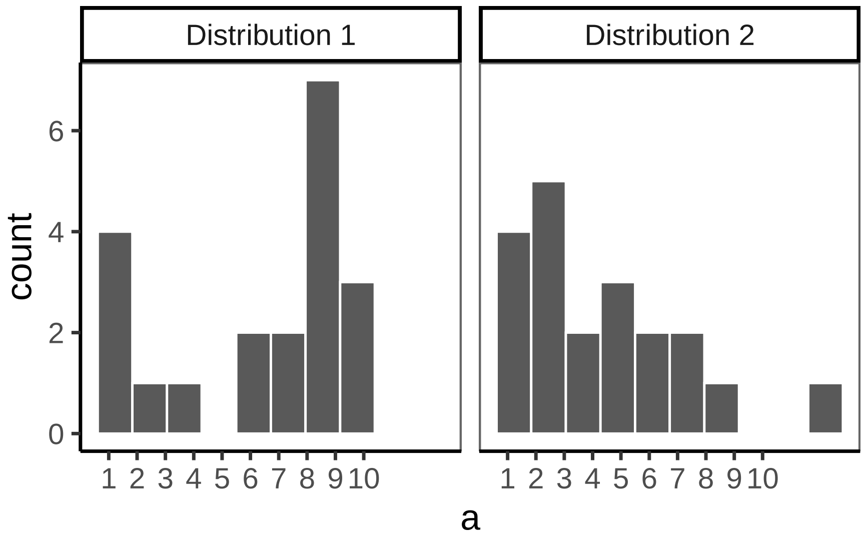

Figure 4.9: Which of these samples came from a uniform distribution?

If you're not confident in the answer, that's because **sampling error** (randomness) is obscuring the histograms, even though the two samples really do come from different distributions this time.

Here's the same question again, this time with 1,000 observations per sample. Which histogram in Figure 4.10 do you think came from a uniform distribution, and which did not?

Now that we have increased N, we can see the pattern in each sample becomes more obvious. Distribution 1 has bars near 100, not perfectly flat, but it resembles a uniform distribution. Distribution 2 is not flat looking at all.

By comparing the two histograms, we've already performed a basic statistical inference: Distribution 2 did not come from a uniform distribution. This is the same logic underlying the formal tests covered in later chapters: arrange the data to make the source distribution visible, then judge whether the observed pattern is consistent with a hypothesized one.

4.3 Is there a difference?

Back to experiments. We want to know if an independent variable (our manipulation) causes a change in a dependent variable (measurement); if it does, we should see differences in the measurement as a function of the manipulation.

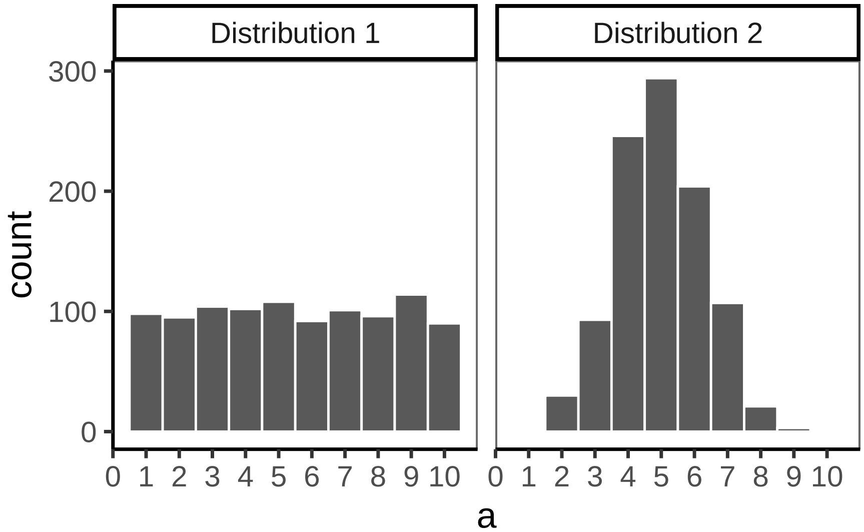

Figure 4.10: Which of these samples came from a uniform distribution?

Some manipulations produce effects too large and consistent to need any statistics at all:

> **Note**
>
> **Light switch experiment.** Flip the switch up (condition 1) and the light goes on (measurement). Flip it down (condition 2) and the light goes off (measurement). The light changes as a direct function of the switch position, every time. When an effect is this large and this consistent, there's no ambiguity: you don't need a statistical test to tell you the manipulation is working.

Most environmental effects aren't like the light switch. They're a real signal buried in noisy, variable measurements, which is exactly the kind of case inference is built for.

4.3.1 Chance can produce differences

Random chance can produce apparent differences between groups even when no real difference exists. We have already seen this with sampling: two samples from the same distribution come out differently. The same principle applies when we compare group means. This is a fundamental problem for inference.

To make this concrete, consider a scenario where we expect to find no real difference. A researcher samples pH at 10 randomly selected sites from a stream network. All 10 sites are drawn from the same population: no treatment, no environmental gradient, no systematic difference between them. The sites are arbitrarily divided into two groups of 5 (Group A and

Group B). Even though there is no real difference, we might still observe a difference in mean pH between the two groups simply because of sampling error.

Here is data from one run of this scenario:

group	ph
A	7.66
A	6.84
A	6.48
A	7.34
A	7.36
B	6.66
B	6.60
B	7.45
B	6.90
B	7.26

This is a long-format table. Each row is one site. The first column shows the group assignment, and the second shows the pH reading. Did Group A have a higher mean pH than Group B?

It is easier to see the comparison in a bar graph (Figure 4.11), which shows the mean pH for each group.

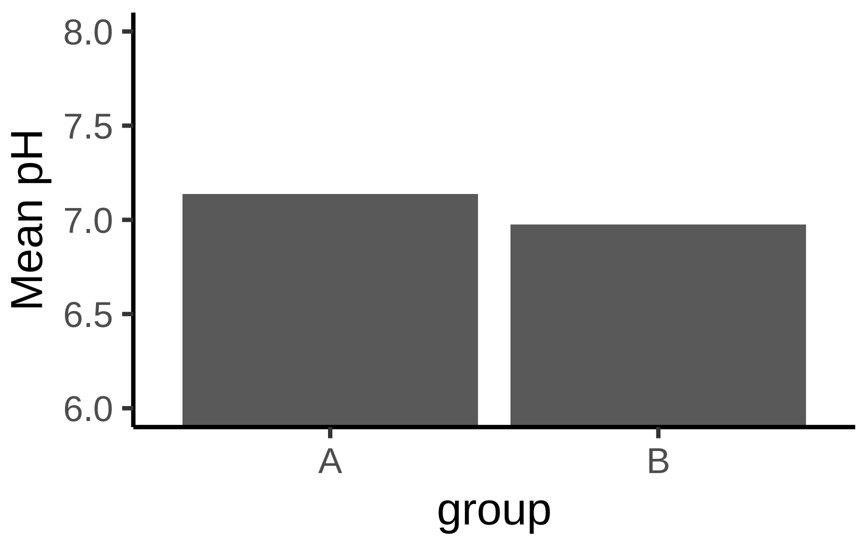

Figure 4.11: Mean pH for two groups drawn from the same stream population. Note the zoomed y-axis, which does not start at 0.

The bars are not the same height: Group A and Group B have different mean pH values. But both groups were drawn from the very same stream population, so this isn't a real environmental difference, it's sampling error.

This is the core inference problem: differences appear even when nothing real caused them. **How can we tell whether an observed difference is real or just the result of chance?**

4.3.2 Differences due to chance can be simulated

Chapter 2 showed that chance alone can produce spurious correlations. The same is true for group means: we can characterize exactly what chance is capable of producing in a comparison of group means. Once we know what chance typically does, and what it cannot do, we have a basis for judging whether our observed difference falls within or outside its range.

The first step is to replicate the sampling scenario many times. (ah I see, this used to be gumballs, but now it's PH for "environmental science" reasons. Tbh the PH isn't working becuase again it's log scale. I think something that's at least a discrete # of things. Like maybe, how many bats in your sample have white nose fungus, or need to think of some other idea that is nicer than Ph for this example) Figure 4.12 shows bar graphs of mean pH for Group A and Group B across 10 replications of the same experiment: all 10 sites sampled from the same population, split 5+5, and means computed.

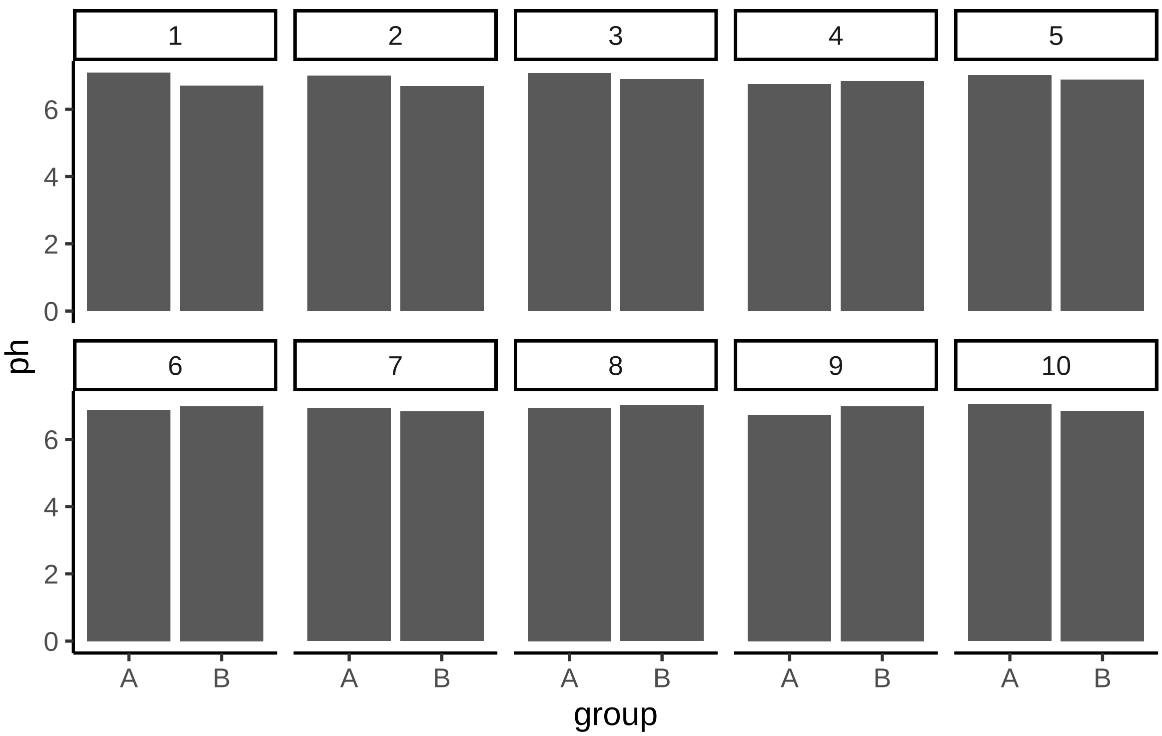

Figure 4.12: Ten replications of sampling pH at 10 stream sites and splitting them arbitrarily into two groups of 5. Every difference here is chance alone.

These 10 replications show that chance produces different mean differences each time. Sometimes Group A is higher, sometimes Group B is higher, sometimes the means are nearly equal. All of this variation arises from sampling error alone: there is no real difference in the population.

4.4 Chance makes some differences more likely than others

We have seen that chance can produce group mean differences. But we still need to characterize what chance usually does and what it rarely does. If we can establish the range of differences that chance produces, we can build a window: differences inside the window could plausibly have arisen by chance; differences outside the window could not.

We summarize each replication as a single **difference score**: mean pH for Group A minus mean pH for Group B. Figure 4.13 shows those difference scores for the 10 replications above.

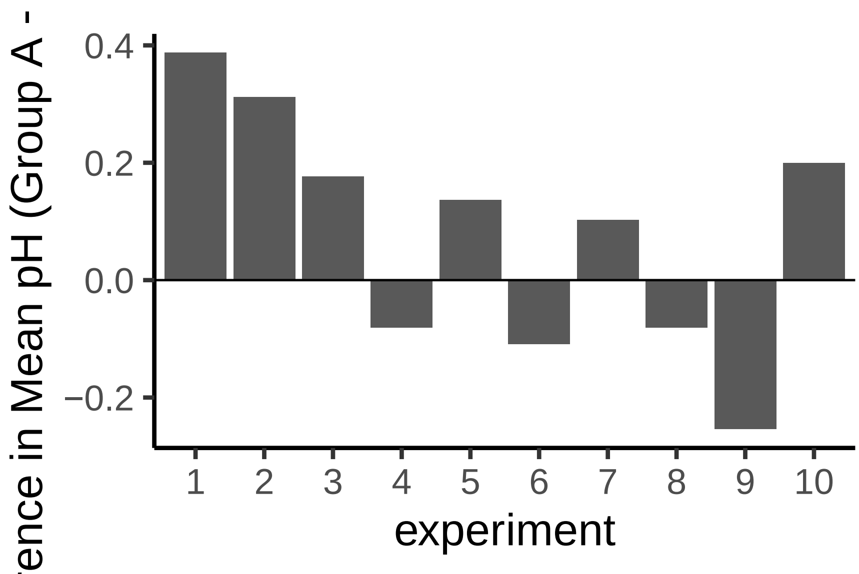

Figure 4.13: Difference in mean pH between Group A and Group B across 10 replications. The true difference is zero, but sampling error produces non-zero differences each time.

A bar at zero means the two groups had the same mean pH in that replication. Positive values indicate Group A was higher; negative values indicate Group B was higher. The signs of the differences would flip if we reversed the subtraction order.

The differences in these 10 replications mostly fall in a narrow range. To better characterize what chance can do, we run 100 replications. The results are shown in Figure 4.14.(The #s are all smushed in the below figure's x-axis)

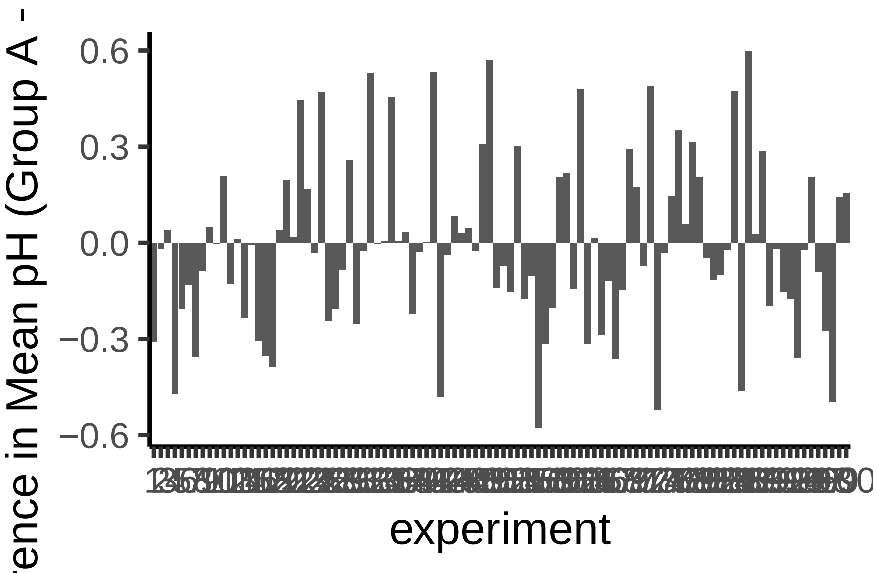

Figure 4.14: Difference in mean pH between Group A and Group B across 100 replications. There are many possible differences that chance can produce, but the y-axis shows the range is bounded.

The x-axis spans 100 experiments and is difficult to read, but the important information is on the y-axis. Many different difference sizes occur, but the range is bounded: very large differences (say, greater than 0.5 pH units in either direction) do not appear. This begins to define the window of what chance can produce.

To get a clearer picture, we now run 10,000 replications and plot the distribution of difference scores as a histogram. This gives a precise view of which differences happen often and which are extremely rare.

This histogram is the **chance window** for this study design. It shows that chance most often produces a difference near zero, the tallest bar in the center. Larger differences in either direction occur less and less frequently. The distribution has a clear boundary: extreme differences are possible but very rare.

We can use this window to evaluate any observed difference. If a study found a mean pH difference of 0.1 between two groups, that falls well inside the chance window: differences of that size occur frequently by random sampling alone. If the study found a difference of 0.6, that would fall far outside the window; chance produced a difference that large 0 times out of 10,000. When an observed result almost never happens by chance, we have grounds to conclude that something other than chance produced it.

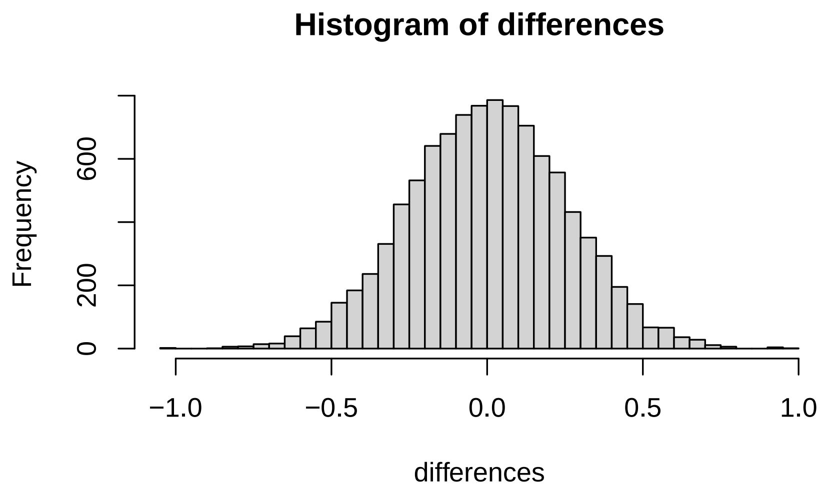

Figure 4.15: Histogram of mean pH differences between Group A and Group B over 10,000 replications. All differences arise from chance alone. The most frequent difference is near zero; larger differences occur less often. Chance cannot produce everything.

4.5 Simulation-Based Inference

Most of the inference done throughout the rest of this course comes down to one question: did chance produce the difference in the data? In an experiment, we want to know if our manipulation caused a difference in our measurement. But everything we measure has natural variability, so we'll always find *some* difference between conditions, whether the manipulation matters or not. What we want to know is whether that difference could plausibly have been produced by chance alone. If chance is a very unlikely explanation, we infer that something about the manipulation, not chance, produced the difference.

> **Note**
>
> **Beyond ruling out chance.** Statistics is not only about determining whether chance could have produced a pattern in the observed data. The same tools we are talking about here can be generalized to ask whether any kind of distribution could have produced the differences. This allows comparisons between different models of the data, to see which one was the most likely, rather than just rejecting the unlikely ones (e.g., chance). But, we'll leave those advanced topics for another textbook.

The simulation-based approach introduced here answers the same question as every formal test in later chapters: **could chance alone have produced the observed difference?** The advantage of starting here is transparency: every step is explicit, and the logic is visible before any formulas are introduced.

4.5.1 Intuitive methods

The approach in this section uses only simple arithmetic operations that you already understand to build a tool for inference. Specifically, we will use:

1. Sampling numbers randomly from a distribution
2. Adding and subtracting
3. Division, to find the mean
4. Counting
5. Graphing and drawing lines
6. NO FORMULAS

4.5.2 Part 1: Frequency-based intuition about occurrence

Question: How many times does something need to happen for it to happen "a lot"? How rare does something need to be before you stop worrying about it?

Environmental scientists already have a working vocabulary for this: the **100-year flood**. A "100-year flood" doesn't mean floods that size happen once a century on a schedule; it means that in any given year, there's about a 1-in-100 chance of a flood that severe. Would you build a house in that floodplain? Many people do, and often regret it, because a 1% annual chance still adds up over a 30-year mortgage (roughly a 26% chance of at least one such flood in that span). A location with a 1-in-10,000 annual flood risk is very different: over the same 30 years, the cumulative chance is under 0.3%. Same underlying logic, wildly different practical decisions.

That's the intuition we need going forward: **rare events still happen, but how rare something is changes what you're willing to conclude from seeing it.** Later in this chapter, we'll use 1-in-10,000 as our own working definition of "rare enough to call surprising."

4.5.3 Part 2: Judgment and Decision-making

We already have everything we need to make this concrete: the pH simulation above gave us 10,000 mean differences that chance alone can produce when there is no real difference between groups. That histogram (Figure 4.15) *is* our reference distribution. Now we use it to make a decision.

Let's draw some lines on that same histogram and label two kinds of regions:

1. **Region of chance.** Chance did it. Chance could have done it.
2. **Region of not chance.** Chance didn't do it. Chance couldn't have done it.

The boundaries are the minimum and maximum values chance actually produced across the 10,000 replications. Figure 4.16 shows this applied to our pH differences.

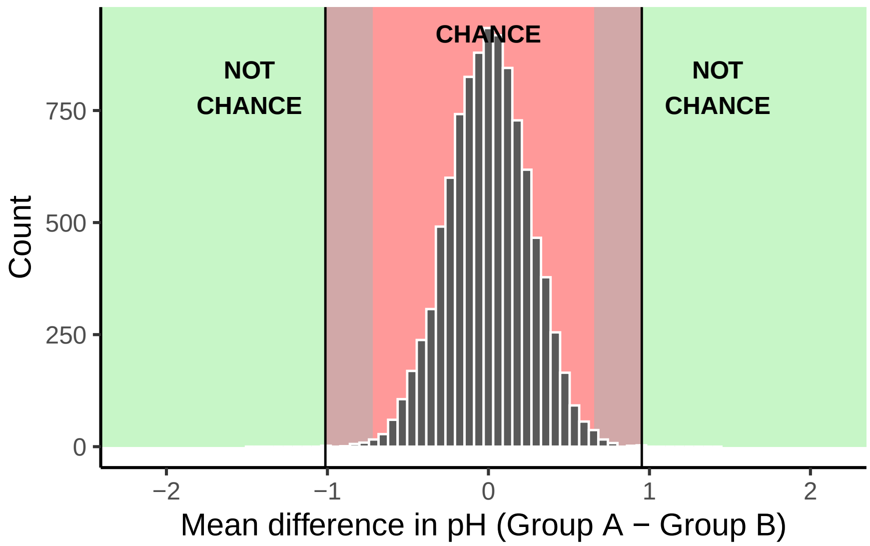

Figure 4.16: Decision boundaries applied to the histogram of pH mean differences from Figure 4.15. Green marks differences chance never produced in 10,000 replications; red marks the range chance did produce; grey marks a zone chance produced only rarely.

Here's how the decision works. A pH difference beyond the green boundary (roughly -1.01 to 0.95, from our 10,000 simulated replications) never happened by chance in this simulation, so observing something out there would give real grounds to suspect it wasn't just sampling error. A difference well inside the red zone, close to 0, is exactly what chance produces routinely: no grounds for suspicion.

The grey bands are the honest part: differences near the edge of what chance can do, but not quite past it. Chance *could* produce something in the grey zone, it's technically still inside the red window, but it does so rarely. That's not a clean yes/no; it's a "probably not chance, but I can't rule it out," and how you handle that ambiguity is a judgment call, not a fact. More conservative researchers shrink the grey zone (fewer false alarms, more missed real effects); more permissive researchers expand it (the reverse trade-off). Chapter 5 formalizes this trade-off with a standard convention ($\alpha = .05$) so that researchers aren't each inventing their own boundary from scratch.

4.5.3.1 Making decisions and being wrong

No matter how you plan to make decisions about your data, you will always be prone to making some mistakes. You might call one finding real, when in fact it was caused by chance. This

Table 4.1: Smallest and largest pH mean differences produced by chance, by sites per group.

sample_size	smallest	largest
5	-0.80	0.83
10	-0.82	0.63
20	-0.43	0.42
40	-0.31	0.40

is called a **type I** error, or a false positive. You might ignore one finding, calling it chance, when in fact it wasn't chance (even though it was in the window). This is called a **type II** error, or a false negative.

How you make decisions influences how often you make errors over time. A researcher who only accepts results as real when they fall in the strict "no chance" zone will make few type I errors, but will also make more type II errors, missing real effects that the decision criterion happened to call chance. A researcher who accepts results in the grey zone as real will make more type I errors but fewer type II errors. Using cutoff regions for decision-making forces this trade-off. The Bayesian view sidesteps it somewhat: rather than a binary real/not-real decision, it treats confidence as continuous, updated by each new experiment, which avoids committing to a fixed trade-off between the two error types.

There's also a way to reduce both error types, and increase confidence in your results, before you even run the experiment: designing the study well in the first place.

4.5.4 Part 3: Study Design and Sensitivity

Study design directly controls how sensitive an analysis is. It's often possible, and important, to choose a design that can actually detect the size of effect you care about.

Two forces govern whether a study can detect a real effect: the size of the effect, and the amount of noise in the data. The main design choice under your control is sample size, which determines how much that noise obscures the signal.

Recall what happens to the sampling distribution of the mean as sample size increases: it narrows. The same is true for the distribution of mean differences. As n increases, the range of differences that chance alone produces shrinks. This means that with enough sites or plots, even a small real effect can be reliably distinguished from noise, while with too few, even a fairly large real effect can get lost in it.

Let's return to the stream pH scenario, but now vary the number of sites sampled per group: 5 (what we started with), 10, 20, and 40.

The pattern is clear: the chance window is wide with only 5 sites per group and narrows steadily as sites are added. Consider what this means for a real study. Suppose restoration

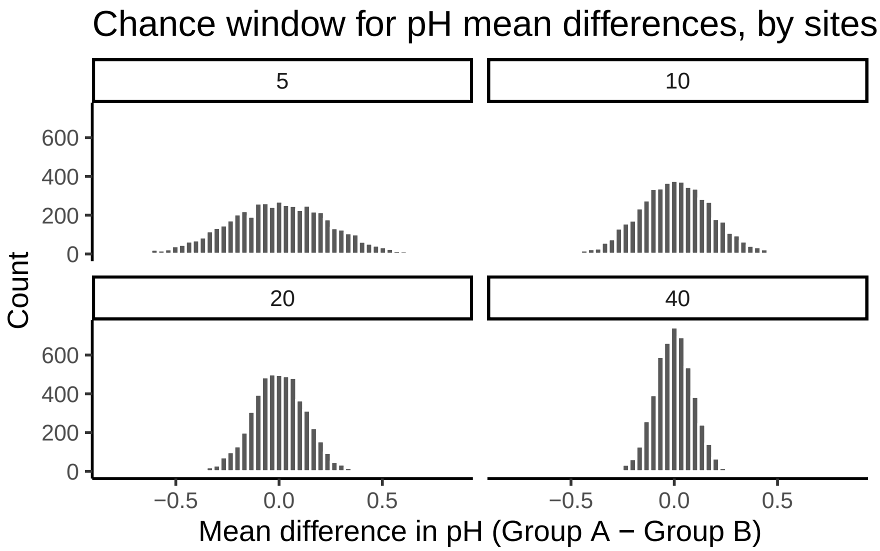

Figure 4.17: The width of the chance window for mean pH differences shrinks as the number of sites per group increases.

of a wetland is expected to raise pH by about 0.3 units. With only 5 sites per group, that effect size sits well inside the chance window in the table above, so a real 0.3-unit effect could easily be mistaken for sampling noise, or a sampling-noise result could be mistaken for a real effect. With 40 sites per group, the same 0.3-unit difference sits outside the window chance can produce, and you could detect it with real confidence.

The design of a study determines the size of effect it can reliably detect. Sample size is the main lever, but not the only one: taking repeated measurements at each site (reducing measurement noise) and choosing conditions with a genuinely strong contrast both help too. This idea, how sensitive a given design is to a real effect of a given size, has a name, **statistical power**, which Chapter 5 develops formally alongside the standard $\alpha = .05$ convention.

4.5.5 Summary of Simulation-Based Inference

We built the core logic of inference using nothing but sampling, arithmetic, and histograms: simulate what chance alone can produce, compare an observed result to that simulated distribution, and use where it falls to judge whether chance is a plausible explanation. This comparison isn't perfectly clean; there's a grey zone near the edges where judgment, not a formula, has to do the work, and the width of the chance window itself depends on how much data you collect.

4.6 The randomization test (permutation test)

This is the first formal inferential statistic in this textbook. Everything so far has built the intuition; now we formalize it into a **randomization test**, which uses random chance to judge whether an experimental finding was likely due to chance or not. The ideas behind it are exactly the ideas already developed in this chapter, now assembled and given a name.

Here's the big idea. When you run an experiment and collect some data you get to find out what happened that one time. But, because you ran the experiment only once, you don't get to find out what **could have happened**. The randomization test is a way of finding out what **could have happened**. And, once you know that, you can compare **what did happen** in your experiment, with **what could have happened**.

4.6.1 Pretend example: does distance from a highway affect NO<82><82> levels?

Let's say you are monitoring nitrogen dioxide (NO<82><82>) concentrations at air quality sites near a major highway. You assign 20 monitoring sites to a "near" group (within 50 m of the highway) and 20 different sites to a "far" group (500 m or more from the highway). If highway traffic is a meaningful source of NO<82><82>, then the near group should have higher concentrations on average than the far group.

Let's say the data looked like this:

site	near_ppb	far_ppb
1	47	22
2	57	13
3	45	38
4	55	27
5	36	38
6	39	34
7	61	32
8	61	32
9	54	28
10	50	23
11	47	13
12	36	36
13	54	14
14	48	13
15	45	30
16	50	31
17	42	32
18	59	19
19	45	11
20	37	11
Sums	968	497
Means	48.4	24.85

So, did the near-highway sites have higher NO<82><82> than the far sites? The mean for near sites was 48.4 ppb, and for far sites was 24.85 ppb. Just looking at the means, it looks like proximity to the highway matters.

But could this just be chance? Maybe the near sites happened to be in a windier or more industrialized area, so their higher readings reflect those conditions rather than the highway itself, or maybe the difference arose simply from random sampling variation. That's a legitimate concern worth taking seriously. We already know how the data came out; what we want to know is how they *could have* come out, across all the possibilities.

The data would have come out a bit different if some sites from the near group had been placed in the far group, and vice versa. There are many ways the 40 sites could have been assigned to two groups, and the group means would come out differently depending on the assignment.

It's not practical to run the experiment every possible way, that would take too long, but we can estimate how those experiments might have turned out using simulation.

We take all 40 NO<82><82> readings from both groups, pool them together, then randomly assign 20 to a "near" group and 20 to a "far" group, as if the original assignment had been done differently. We compute the new group means and their difference, and repeat this process many times to build up a picture of what the difference could have been under random assignment.

4.6.1.1 Doing the randomization

Before we do that, let's show how the randomization part works. We'll use fewer numbers to make the process easier to look at. Here are the first 5 NO<82><82> readings from each group.

site	near_ppb	far_ppb
1	47	22
2	57	13
3	45	38
4	55	27
5	36	38
Sums	240	138
Means	48	27.6

Things could have turned out differently if some of the near-highway sites had been placed in the far group instead, and vice versa. Here's how we can do some random switching using R.

```
all_readings      <- c(near[1:5], far[1:5])
randomize_scores  <- sample(all_readings)
new_near          <- randomize_scores[1:5]
new_far           <- randomize_scores[6:10]
print(new_near)
#> [1] 27 13 22 36 55
print(new_far)
#> [1] 57 38 38 47 45
```

We have taken the first 5 readings from each group and put them all into a variable called `all_readings`. Then we use the `sample` function in R to shuffle the readings. Finally, we take the first 5 readings from the shuffled numbers and put them into `new_near`, and the last five into `new_far`.

Repeating this a couple of times and putting the results in a table shows the group means shifting each time the sites are shuffled around:

site	near_ppb	far_ppb	near2	far2	near3	far3
1	47	22	45	22	57	47
2	57	13	47	27	13	22
3	45	38	36	38	38	45
4	55	27	13	55	55	36
5	36	38	57	38	27	38
Sums	240	138	198	180	190	188
Means	48	27.6	39.6	36	38	37.6

4.6.1.2 Simulating the mean differences across the different randomizations

In our pretend study we found that the mean NO<82><82> for near-highway sites was 48.4 ppb, and for far sites was 24.85 ppb. The mean difference (near <88><92> far) was 23.55 ppb. This is a pretty big difference. This is **what did happen**. But, **what could have happened**? If we tried out all the possible ways to assign 40 sites to two groups, what does the distribution of the possible mean differences look like? Let's find out. This is what the randomization test is all about.

When we do our randomization test we will measure the mean difference in NO<82><82> between the near and far groups. Every time we randomize we will save the mean difference.

Figure 4.18 animates what's happening in a randomization test, using data from a different fake study (we'll return to the NO<82><82> example after). The purple dots show the original mean of each group (lower-exposure vs. higher-exposure sites); since one sits below the other, we want to know whether chance could produce a difference that size. The light green and red dots are the individual readings from each of 10 sites per group, the raw data the purple means are built from. As the animation randomizes, those readings shuffle between groups, and the moving yellow dots show the resulting new group means after each shuffle. The spread of the yellow dots shows the range of differences chance alone could produce.

We are engaging in some visual statistical inference. By looking at the range of motion of the yellow dots, we are watching what kind of differences chance can produce. In this animation, the purple dots, representing the original difference, are generally outside of the range of chance. The yellow dots don't move past the purple dots, as a result chance is an unlikely explanation of the difference.

If the purple dots were inside the range of the yellow dots, we would know that chance is capable of producing the difference we observed, and that it does so fairly often. As a result, we should not conclude the manipulation caused the difference, because it could have easily occurred by chance.

Let's return to the NO<82><82> example. After we randomize our readings many times and compute the new means and mean differences, we will have loads of mean differences to look

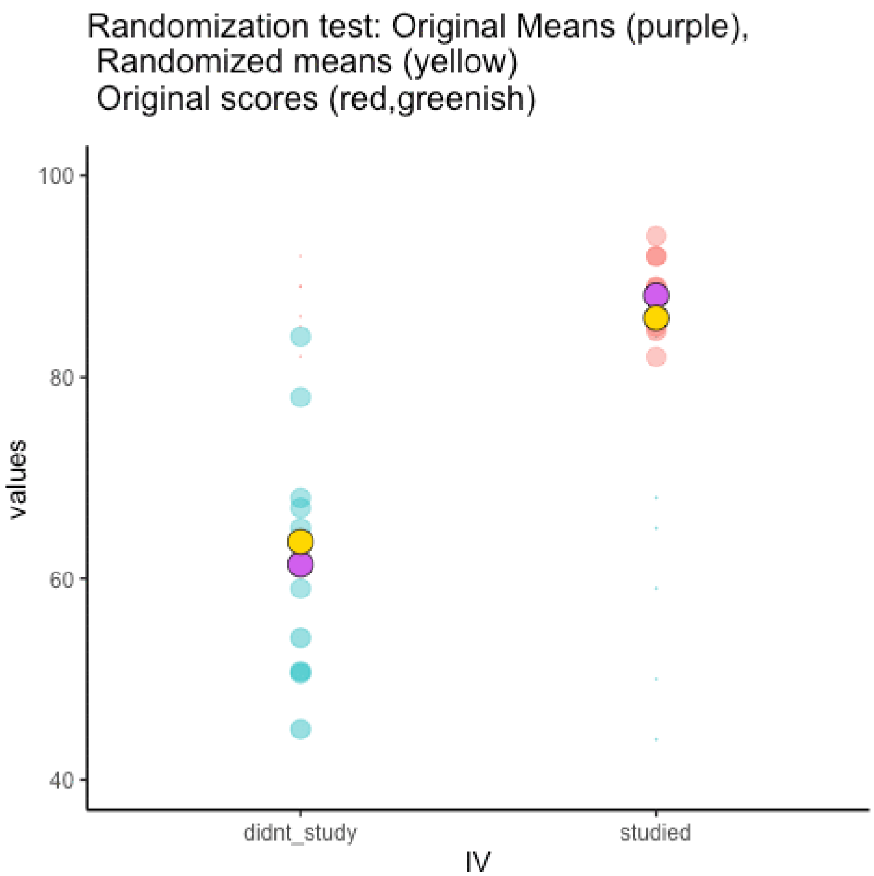

Figure 4.18: Animation of a randomization test. Purple dots are the original condition means, yellow dots the means after each reshuffle, and blue and red dots track how individual scores move. (Final frame; the animated version is in the web edition of this book.)

at, which we can plot in a histogram. The histogram gives a picture of **what could have happened**. Then, we can compare **what did happen** with **what could have happened**.

Here's the histogram of the mean differences from the randomization test. For this simulation, we randomized the results from the original study 10,000 times. This is what could have happened. The blue line in Figure 4.19 shows where the observed difference lies on the x-axis.

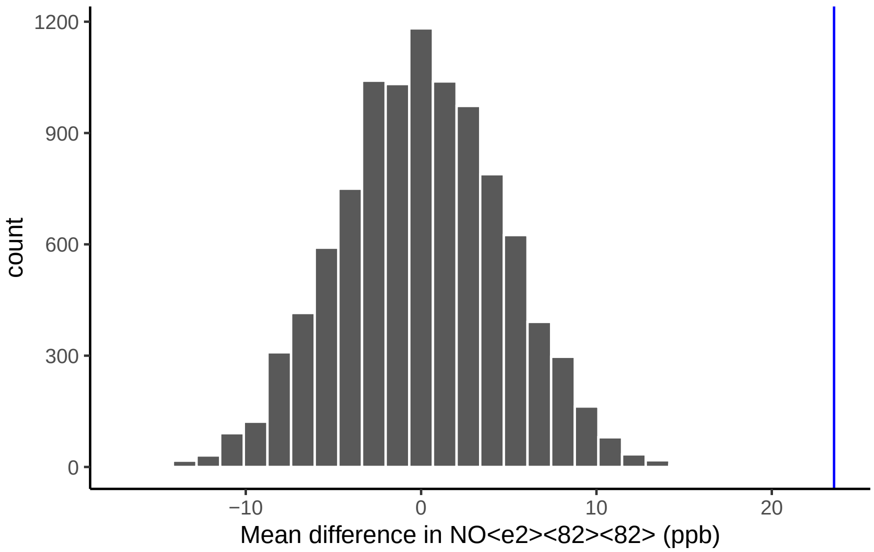

Figure 4.19: A histogram of simulated mean differences in NO<82><82> (ppb) for a randomization test. The blue vertical line marks the observed difference from our study.

Could the difference represented by the blue line have been caused by chance? Probably not. The histogram shows us the window of chance, the range of differences that random assignment alone can produce, and the blue line falls outside it. This means we can be pretty confident that the difference we observed was not due to chance alone, and is more consistent with a real environmental effect of highway proximity on NO<82><82>.

This is another window of chance: a histogram of the mean differences that could have occurred if the 40 sites had been randomly assigned to the near and far groups differently. The differences range from negative to positive, symmetric around a most-frequent value of 0, and grow less frequent the larger they get in either direction. The blue line, **the observed difference** from our study, sits well out to the right, outside the histogram entirely. When we compare **what could have happened** by chance to **what did happen**, what did happen turns out to be rare under chance alone.

Important: what could have happened always means what could have happened **by chance**. Comparing what did happen to what chance could have done is how we assess whether a result is likely due to chance or something real.

Now suppose we'd gotten a much smaller mean difference when we first ran the study. New lines (blue and red) represent smaller differences we might have found instead.

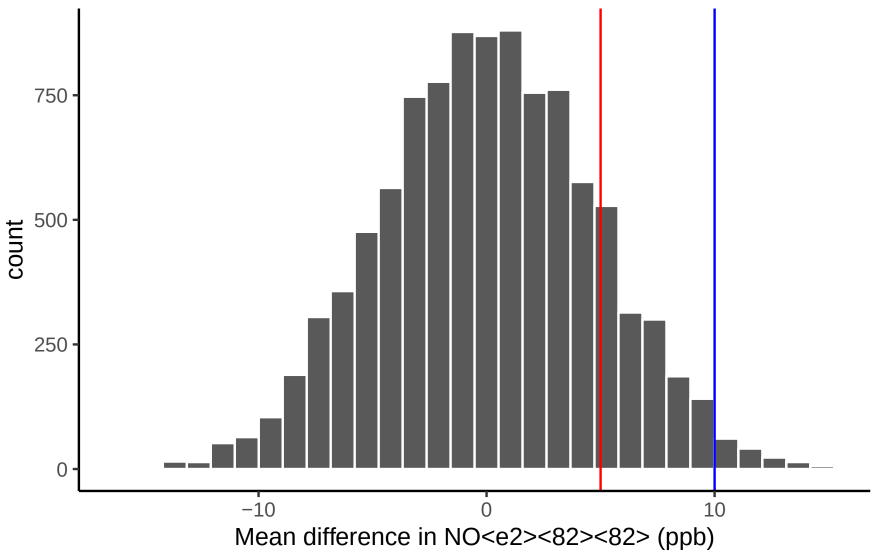

Figure 4.20: Would you expect a mean difference represented by the blue line to occur more or less often by chance compared to the mean difference represented by the red line?

Look at the blue line in Figure 4.20, a mean difference of 10 ppb. It sits inside the chance window: differences that size do happen sometimes by chance. You might lean toward "probably not chance" since it's near the edge, but with some real skepticism. The red line, a difference of +5 ppb, sits well inside the chance window, and differences that size happen fairly often, so it would take more evidence before concluding highway proximity was driving it.

4.7 Chapter Summary

Why it matters. This chapter is the conceptual hinge of the course. Every formal test that follows is a shortcut for what we did here by hand: work out what chance alone could produce, then check whether your result sits inside that range.

Core ideas

- **The question inference actually asks.** Not "is there a difference," but "where did this difference come from?" Chance alone produces differences between groups all the time, so the real question is whether chance is a sufficient explanation for yours.

- **Simulate what chance can do.** Randomly reshuffling your data breaks any real relationship while keeping everything else intact. Doing that many times builds a picture of the differences chance alone can generate for a dataset your size.
- **Then compare.** If your observed difference sits comfortably inside that chance window, chance remains a plausible explanation. If it sits far outside, chance becomes hard to defend as the whole story.
- **The grey zone is real.** Results near the edge of the chance window are genuinely ambiguous, and where you draw the line is a judgment call rather than a fact. The next chapter adopts a convention ($\alpha = .05$) so researchers aren't each inventing their own boundary.
- **Design determines what you can detect.** Sample size is the main lever, but repeated measurements and a stronger contrast between conditions matter too. A study can fail to find a real effect simply because it was never built to see one.
- **The picture becomes a number.** From here on we replace the histogram with a single summary, the probability that chance could have produced your result. That number is the p-value, and the next few chapters build toward it.

4.8 Videos

4.8.1 Null and Alternate Hypotheses

4.8.2 Types of Errors

5 Hypothesis Testing

5.1 What you'll learn

Hypothesis testing is the backbone of statistical inference: it lets us decide whether the patterns we see in a sample are strong enough to draw conclusions about the population. By the end of this chapter, you should be able to:

- **Define the key players** in hypothesis testing (null, alternative, true/estimated, and population/sampling distributions).
- **Explain and use decision rules**: decide whether to reject H_0.
- **Connect confidence intervals and hypothesis tests** for common models.
- **Identify and interpret errors** (Type I and Type II) that can arise in testing.
- **Recognize the role of effect size and power** in planning a study, even if your main focus is running tests.
- **Write a clear results statement** that communicates both the statistical and substantive meaning of your test.

5.2 Core Players (distributions)

To make sense of hypothesis testing, it helps to picture four kinds of distributions. Later in the chapter we'll put them together, but first we define them one at a time:

1. **Null Distribution** - The hypothesized population distribution under the assumption that the null hypothesis H_0 is true.

2. **Estimated Population Distribution** - The distribution we infer from our sample. This represents what the data suggest about H_a. (Sometimes called the *inferred parent distribution.*)

3. **True Population Distribution** - The actual distribution from which our sample was drawn. We never know this perfectly, but it's the reality we're trying to approximate.

4. **Sampling Distribution of the Sample Mean** - The distribution we'd get if we took many samples and calculated $\bar{X}$ each time. This is key for connecting sample evidence back to the population.

Null value notation. We'll write the hypothesized population mean as μ_0. Hypotheses compare the population mean μ to this constant μ_0. The sample mean $\bar{X}$ is never part of H_0 or H_a; it's the evidence we bring to test them.

Note on terms. In this chapter, you'll sometimes see the phrase **parent distribution** in the prose. This is just a metaphor: a sample is like a "child" drawn from a "parent" distribution. The standard statistical term is **population distribution**. Both terms mean the same thing here. For clarity, we'll use *population distribution* in figures and definitions, and you can think of *parent* as a memory aid.

5.3 Terms for Decision Rules

When running a hypothesis test, three terms guide our decision:

Significance level (α). Chosen *before* the test. α is the long-run probability of a Type I error (rejecting a true H_0). A common choice is $\alpha = 0.05$. Smaller α makes tests more cautious and produces wider confidence intervals.

p-value. Assuming H_0 is true, the p-value is the probability of seeing a test statistic at least as extreme as what we observed. Small p means our data look unusual under H_0.

Confidence level. Confidence level is the long-run proportion of confidence intervals that contain the true parameter when we repeat the entire data collection and interval procedure.

$$\text{Confidence Level} = 1 - \alpha$$

With $\alpha = 0.05$, about 95% of intervals constructed this way will capture the true value.

It's tempting to misinterpret confidence intervals. Do **not** say:

> "There is a 95% probability this interval contains μ."

Instead say:

> "We used a procedure that captures μ about 95% of the time in the long run."

5.4 Three equivalent ways to test H_0

1. **Test statistic / critical values.** Choose α, find the corresponding **critical value(s)** for your test statistic, and compute your observed statistic. Reject H_0 if the statistic falls in the **rejection region**.
2. **Confidence interval (two-sided).** Build a $(1-\alpha)$ CI for the parameter. Reject H_0 if μ_0 lies **outside** the CI; fail to reject if it lies **inside**.
3. p-value (probability). Compute the p-value for your observed statistic under H_0 and compare to α. Reject H_0 if $p < \alpha$.

Note. Methods 1–2 yield a yes/no at α; method 3 yields a yes/no plus a graded measure of evidence (exact p).

5.4.1 Test statistic/critical values

Figure 5.1 shows how these ideas come together in a two-sided test. The blue curve is the sampling distribution of the test statistic if the null hypothesis H_0 is true. The dashed magenta line marks the expected center at μ_0. The red vertical lines are the **critical values** for $\alpha = 0.05$. They cut off the most extreme 5% of the distribution: 2.5% in each tail. Those shaded tails are the **rejection regions**.

Here's the decision rule:

- If the observed test statistic falls inside the central region, we **fail to reject** H_0 because the result is consistent with chance variation.
- If it falls in either rejection region, we **reject** H_0 because the result is too extreme to attribute to chance alone at the 5% level.

This picture makes clear why α and the p-value are linked. α sets the cutoff for how extreme is "too extreme," while the p-value tells us how far into the tails our actual result lies.

5.4.2 Confidence intervals

Figure 5.2 shows another way to make the same decision. The **blue curve** is the null distribution (H_0 true), and the **dark green curve** is the sampling distribution of the mean based on our sample. The **dashed green curve** is the estimated population distribution we infer from the sample.

The **thick dark green bar** at the bottom is the 95% confidence interval around the sample mean $\bar{X}$. The **magenta dashed line** marks μ_0, the hypothesized mean under H_0.

Decision rule:

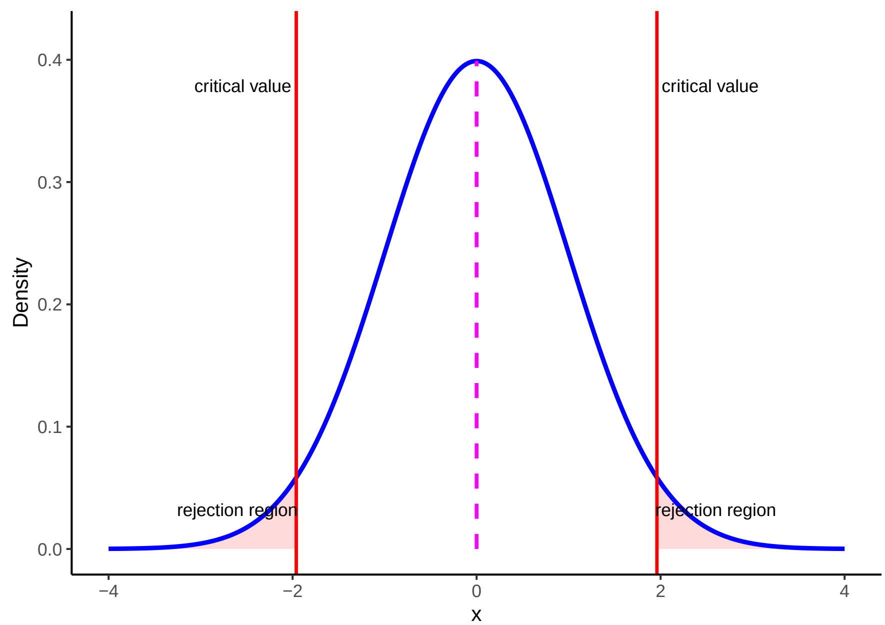

Figure 5.1: Two-sided test: blue curve is the sampling distribution under H_0 (centered at μ_0). Magenta dashed line marks μ_0. Red vertical lines mark the critical values for $\alpha = 0.05$; shaded tails are the rejection regions.

- If μ_0 lies **inside** the CI bar, we **fail to reject** H_0.
- If μ_0 lies **outside** the CI bar, we **reject** H_0 at $\alpha = 0.05$.

For common models, a two-sided level-α test of $H_0: \mu = \mu_0$ is equivalent to checking whether μ_0 lies inside the $100(1-\alpha)$ confidence interval. The **CI** is often the most intuitive way to see this: it shows both the decision (reject or not) *and* the range of parameter values still consistent with the data.

5.4.3 p-values

The p-value is the probability, **under** H_0, of obtaining a test statistic **at least as extreme** as the one you observed. In practice, you compute it from your observed statistic using software or a table. Reject if $p < \alpha$.

5.5 One- and Two-Tailed Tests

Two-tailed tests allow for differences in *either* direction:

One-tailed tests only look in *one* direction (smaller or larger than μ_0):

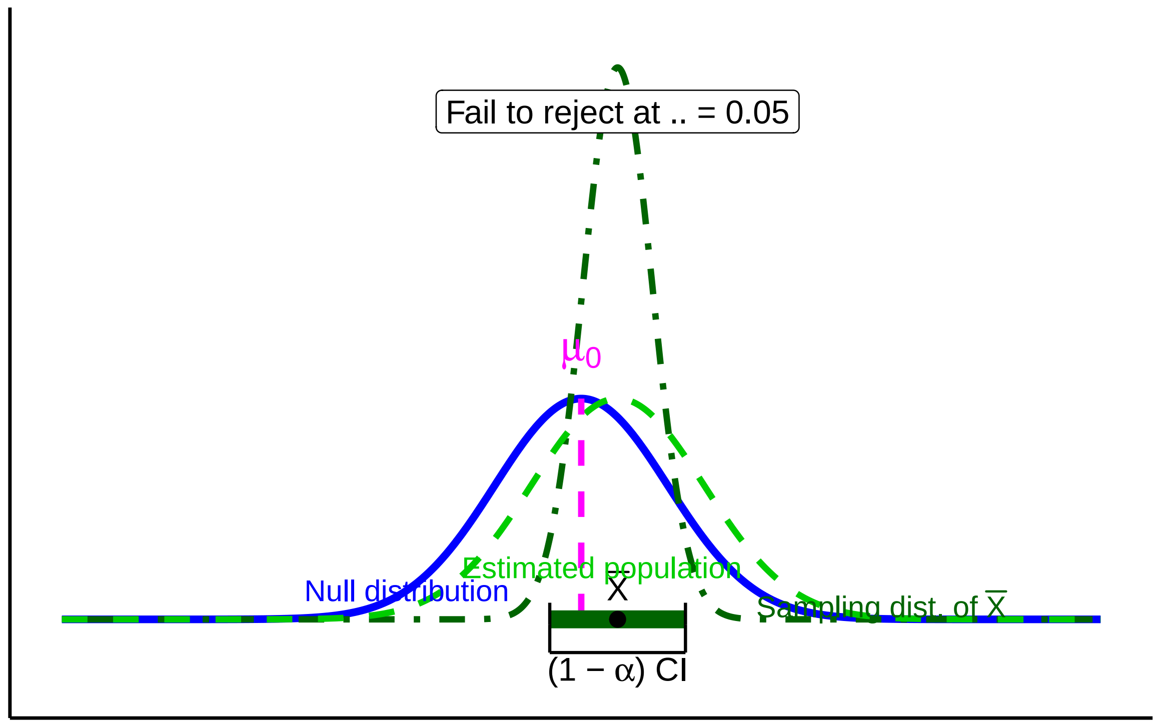

Figure 5.2: Confidence-interval lens: null (blue), sampling distribution of $\bar{X}$ (dark green), estimated population (light-green dashed). The thick bar is the $(1-\alpha)$ CI around $\bar{X}$; the magenta dashed line marks μ_0. Decision uses whether μ_0 is inside or outside the CI.

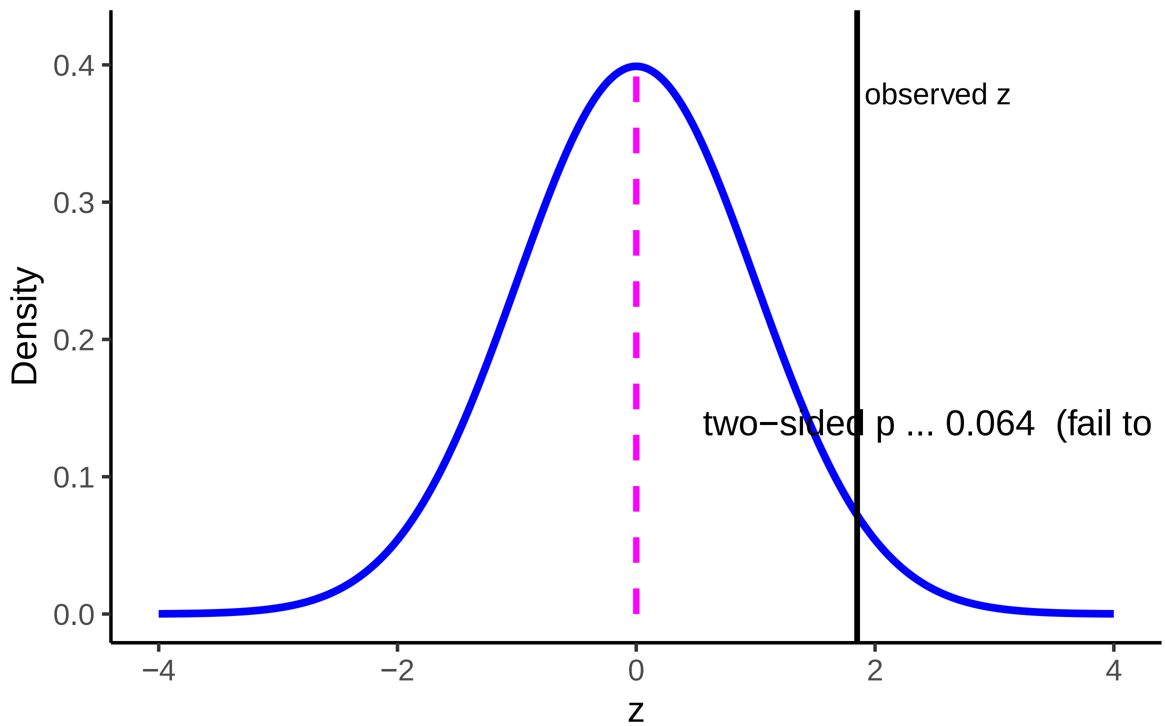

Figure 5.3: p-value lens: compute the p-value from the observed statistic (vertical line) under the null.

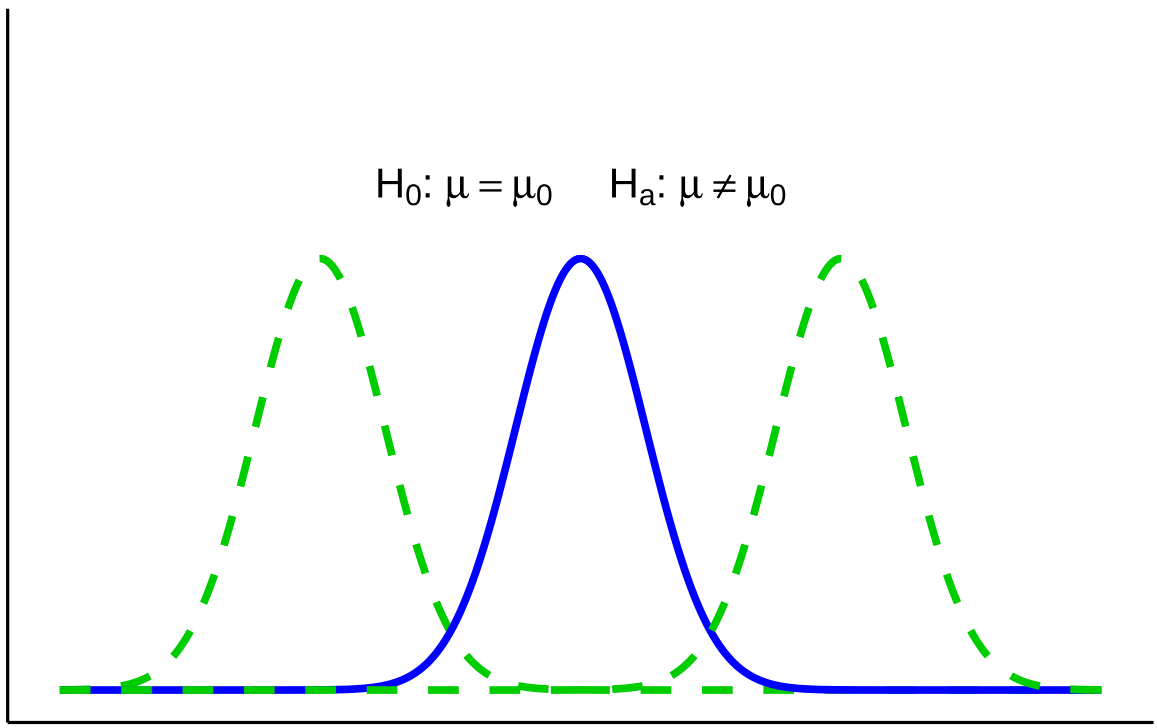

Figure 5.4: Two-sided test: $H_0:\ \mu = \mu_0$ vs $H_a:\ \mu \neq \mu_0$; estimated population distributions shifted left and right of the null.

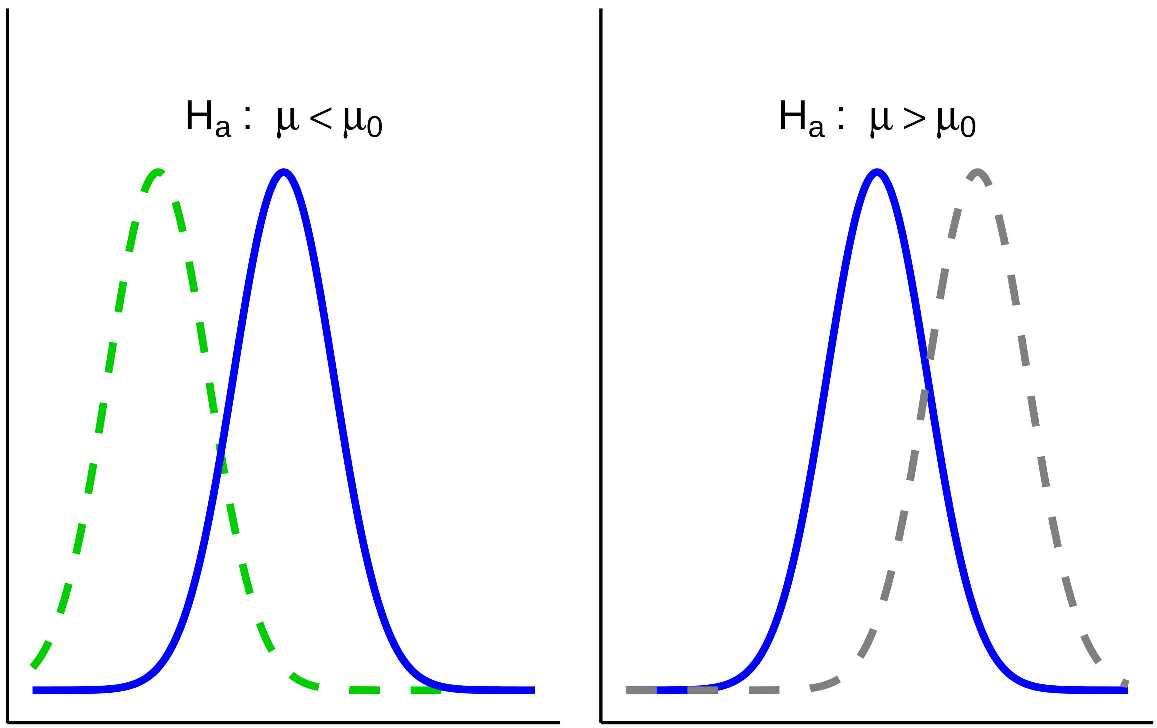

Figure 5.5: One-sided tests: $H_a: \mu < \mu_0$ (left) and $H_a: \mu > \mu_0$ (right), compared to the null.

5.5.1 One- vs. Two-Sided: which and why?

Use this rule of thumb:

- Two-sided (default): You care whether μ differs from μ_0 **in either direction**.
- One-sided (only if justified): Differences in the opposite direction are **scientifically irrelevant** and **would not change your decision or action**.

5.6 Putting it all together

Let's bring the key players together. The figure below shows the elements we'll use throughout the chapter:

- **Blue solid line** = Null distribution (H_0).
- **Orange solid line** = True population distribution (what actually generated the data).
- **Light green dashed line** = Estimated population distribution (inferred from the sample; our stand-in for H_a).
- **Dark green dot-dashed line** = Sampling distribution of the sample mean, $\bar{X}$.
- **Magenta dashed line** = Hypothesized mean μ_0 (under H_0).

- **Thick dark green bar** = 95% confidence interval (CI) around the sample mean $\bar{X}$.

Here they are layered together: this figure serves as your legend for the rest of the section.

Up to now we've introduced each distribution separately. Let's layer them to see how they connect.

1. Null vs. true population.

In an idealized world where H_0 is exactly true, the true population (orange) and the null distribution (blue) are identical. In reality, we never know the orange curve. It's shown here for illustration.

2. Adding the sampling distribution. We never know the true parent distribution directly. What we do have is a sample, and from that we can construct the **sampling distribution of the mean** (dark-green dot-dash). This distribution underlies our ability to estimate a CI (dark green horizontal line).

3. Adding the estimated population distribution.

From our sample we also infer an **estimated population distribution** (light-green dashed). In the best case, this lines up closely with the true population (orange). The CI bar shows that the sample mean is consistent with μ_0.

Putting everything together, you'll see graphs moving forward that are all variations on the figure below. By now you should recognize each element.

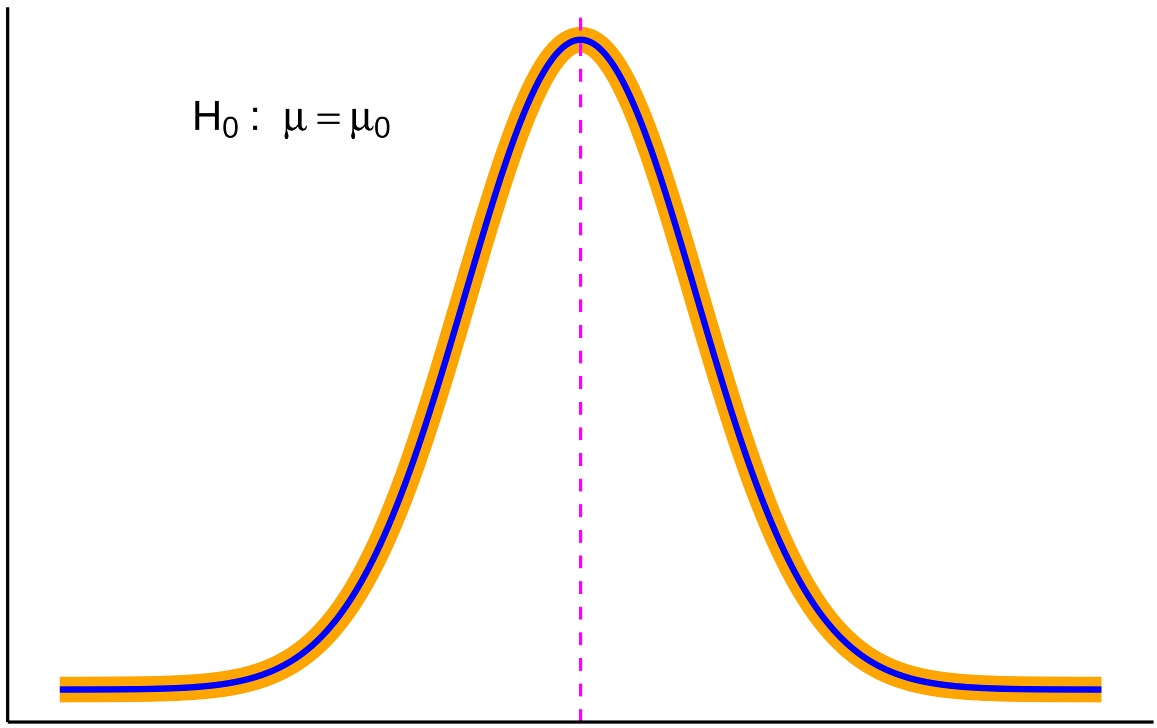

Figure 5.6: True population distribution equals the null distribution (H_0 true): $\mu = \mu_0$.

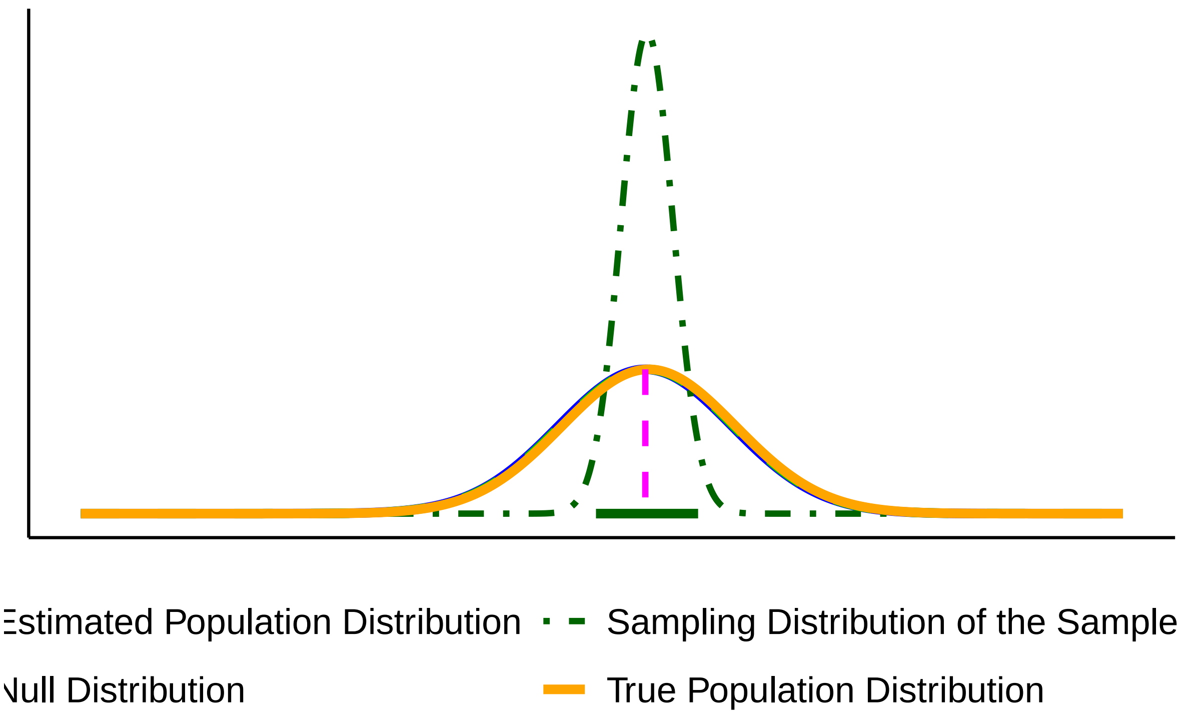

Figure 5.7: Perfectly representative sample: estimated population, null, and true population distributions align; the CI for $\bar{X}$ includes μ_0.

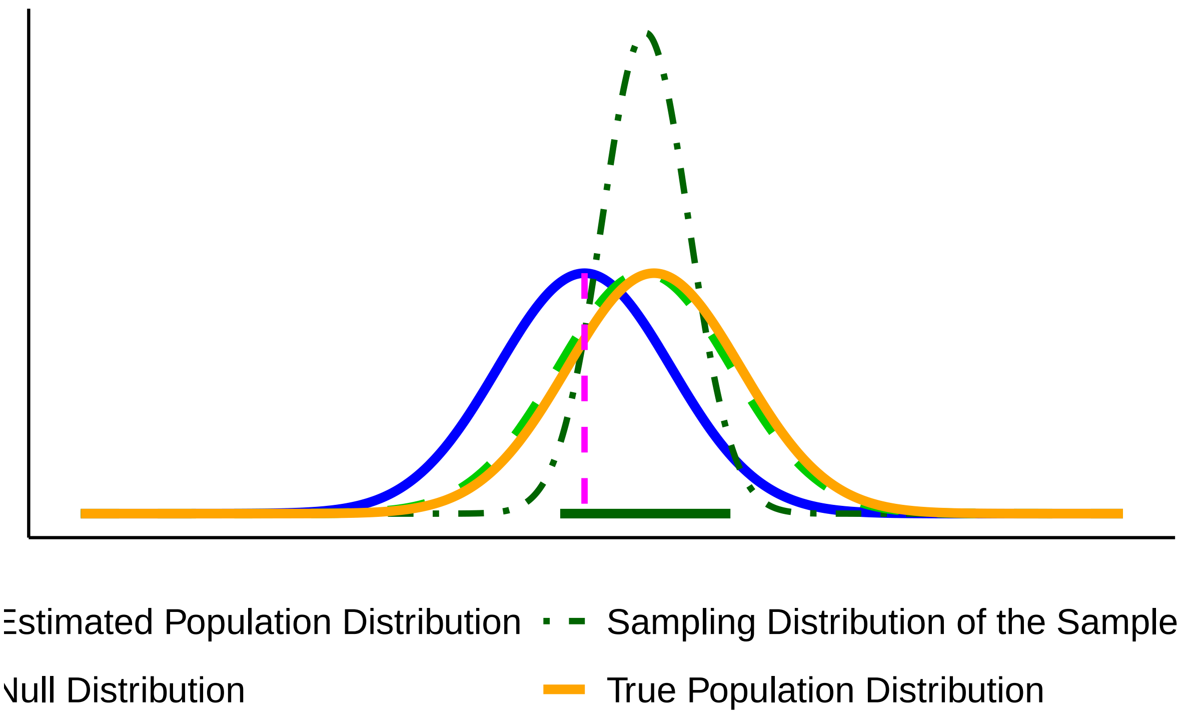

Figure 5.8: Imperfect but convenient sample: estimated population close to the null; the two-sided CI for $\bar{X}$ includes μ_0 (fail to reject H_0).

Key reminder: μ_0 (pink dashed line) comes from the null hypothesis. The unknown μ is what we're trying to learn about, and we approximate it from the sample through the estimated population distribution and the sampling distribution of $\bar{X}$.

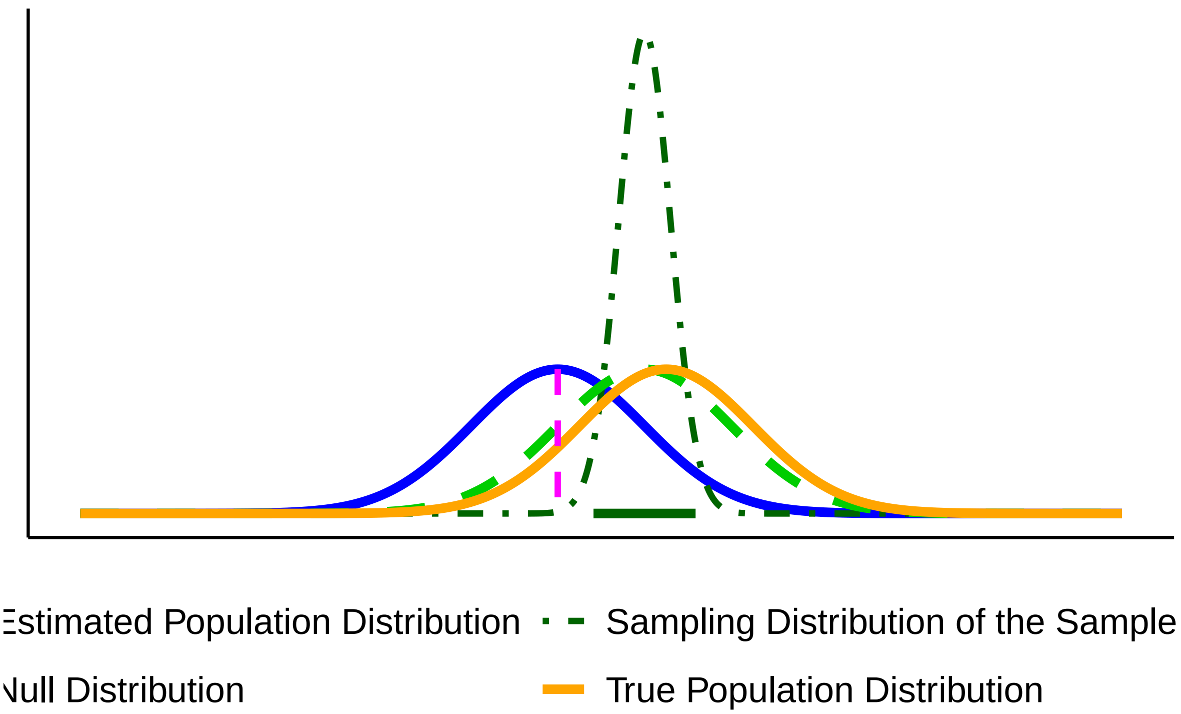

Figure 5.9: Key elements of hypothesis testing graphs: null (blue), true population (orange), estimated population (light-green dashed), sampling distribution of $\bar{X}$ (dark-green dot-dash), null mean μ_0 (magenta dashed), and CI for $\bar{X}$ (thick dark-green bar).

5.6.1 Decision Outcomes

Test outcomes are **binary**: we either **reject** H_0 or **fail to reject** H_0. We never *accept* H_0, because sampling variability means we can't prove it's true, only that the data are consistent with it. Similarly, we don't formally *accept* H_a, but we can say the results provide **support for** H_a when H_0 is rejected.

If the observed statistic falls in either tail beyond the critical value(s), reject H_0. Otherwise, fail to reject H_0. Report the exact p and an effect size or CI..

Here are our options (for a two tailed test).

1. **Reject the Null Hypothesis (Two-Tailed Test).** The sample mean falls beyond the critical values, in the rejection region. The confidence interval also excludes the null

mean (magenta line). In this example, the mean is more than 3 standard errors from μ_0, which is typically considered significant for common choices of α.

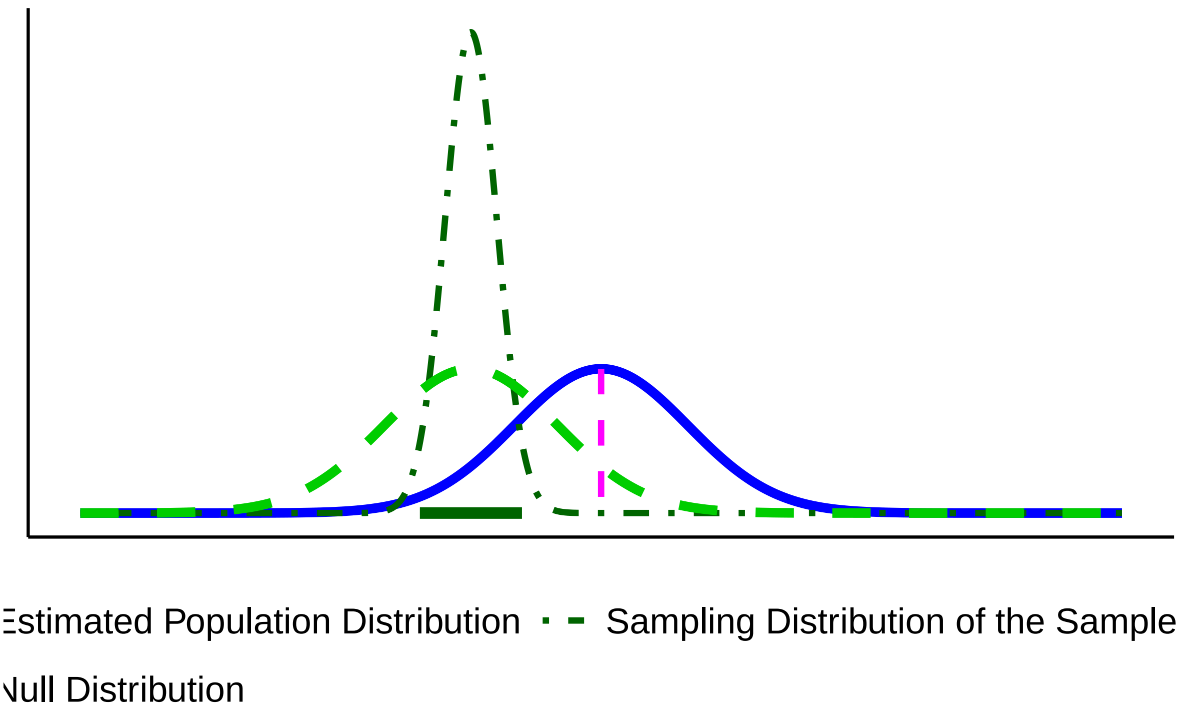

Figure 5.10: Two-sided test, reject H_0: sample mean far from μ_0 in either direction.

2. **Fail to Reject the Null Hypothesis (Two-Tailed Test).** The sample mean falls in the non-rejection region, and the confidence interval includes the null mean. In this case, the evidence is consistent with chance variation, so we fail to reject H_0.

5.7 Errors in Hypothesis Testing

Sometimes, even when we apply the test correctly, sampling variability can lead us to the wrong conclusion.

- **Type I Error (α)**: Rejecting H_0 when it is actually true. This is like a smoke alarm going off when there's no fire: a false alarm. By convention, α is often set at 0.05 (5%).
- **Type II Error (β)**: Failing to reject H_0 when it is actually false. This is like a smoke alarm staying silent when there *is* a fire: a missed signal.

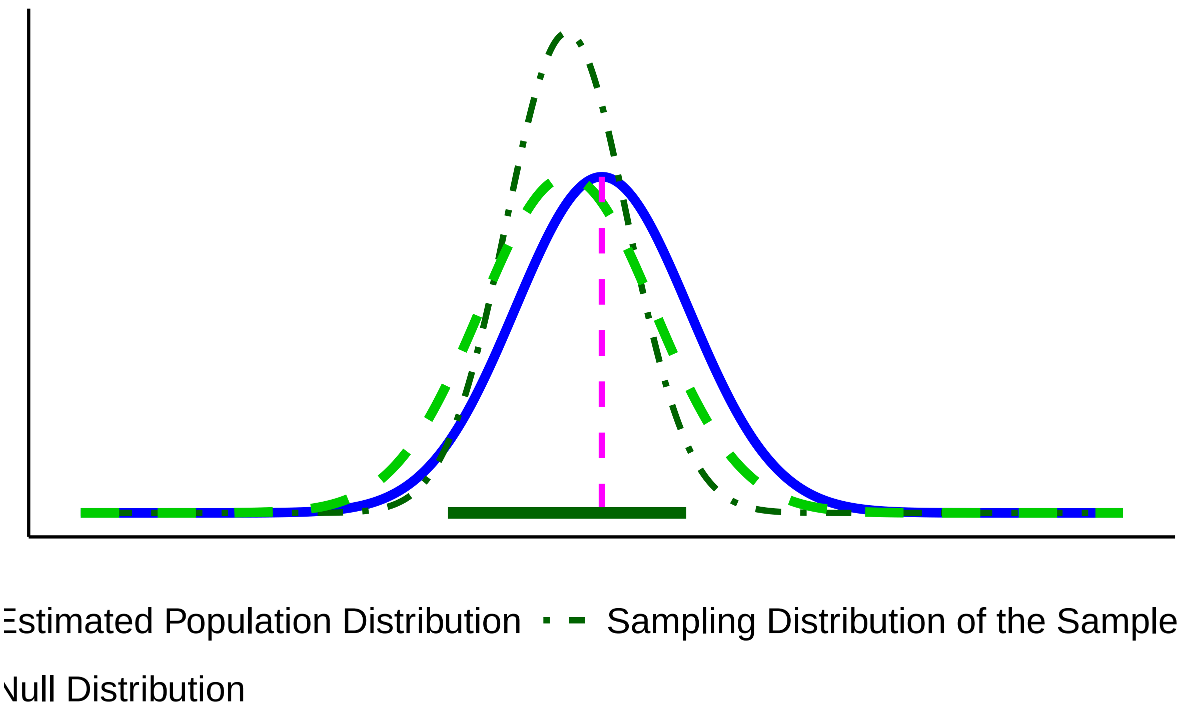

Figure 5.11: Two-sided test, fail to reject H_0: sample mean not greater than critical value (within $3\,SE$ from μ_0).

Decision	If the null hypothesis is True	If the null hypothesis is False
Reject H_0	Type I error (prob = α)	Correct (prob = 1 - β)
Fail to reject H_0	Correct (prob = 1 - α)	Type II error (prob = β)

> **Key Takeaway**: As α gets smaller, β gets bigger, and vice-versa.

Errors happen because samples are imperfect.

- In Figure 5.12, the true population (orange) is almost identical to the null (blue). By chance, however, the sample mean lands far away, and the CI excludes μ_0. The test rejects H_0 even though it is true: a **Type I error**.

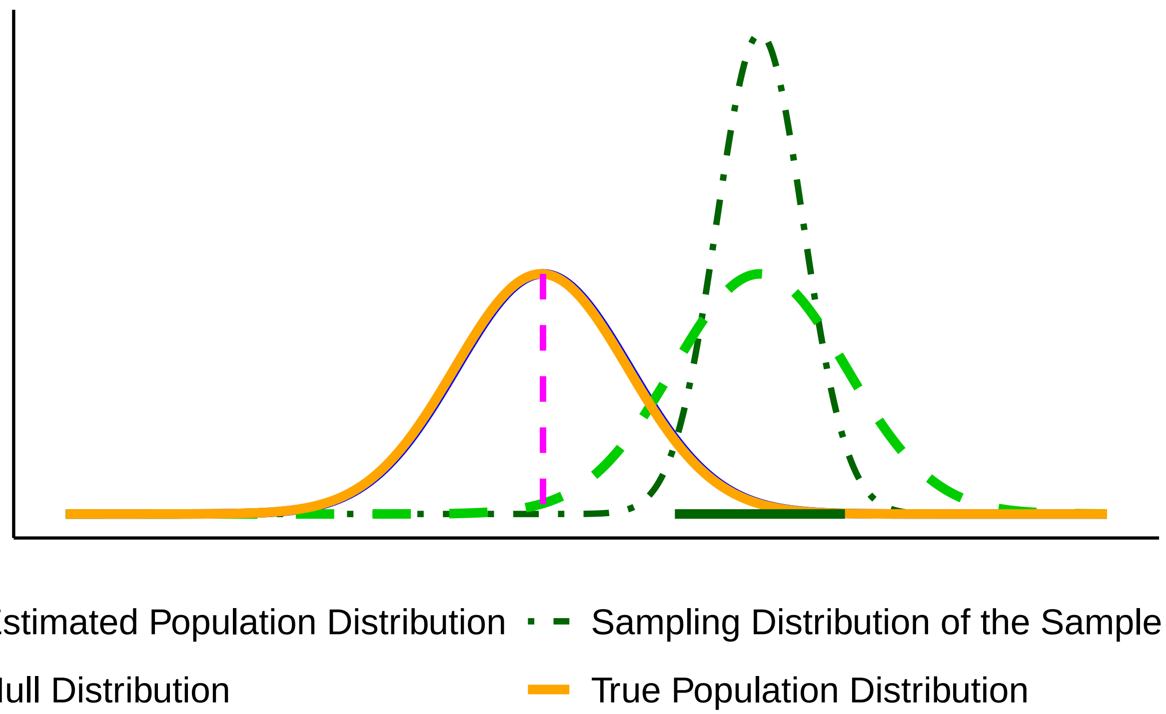

Figure 5.12: Type I error: H_0 is true ($\mu = \mu_0$), but the sample leads us to reject H_0.

- In Figure 5.13, the true population mean (orange) is far from μ_0, but the particular sample looks close to the null. The CI includes μ_0, so the test fails to reject when it should reject the null: a **Type II error**.

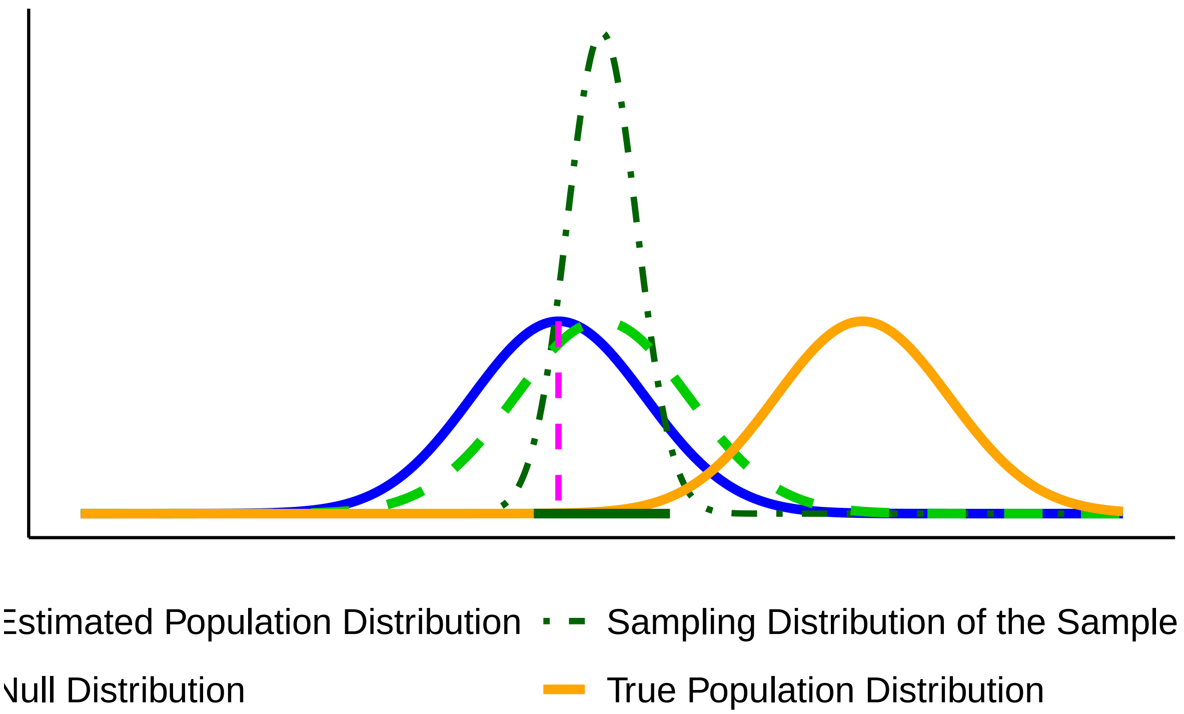

Figure 5.13: Type II error: H_0 is false (true population mean differs from μ_0), but the sample leads us to fail to reject H_0.

5.7.1 Effect of Sample Size on Errors

With fixed α and correct assumptions, the long-run Type I error rate is α for any n. **Sample size mainly affects Type II error:** with small n, the sampling distribution is wide, true effects are harder to detect, and β increases (power decreases). As n grows, the sampling distribution tightens, β falls, and power rises. Apparent changes in the Type I rate at small n usually reflect **assumption violations** (e.g., non-normality, poor variance estimates, multiple looks), not n itself.

Takeaway: Increasing n improves estimation precision and **power** (reduces β); **Type I error remains at** α when assumptions hold.

5.7.2 How to interpret p-values

A p-value is the probability, **assuming** H_0 is true, of obtaining a test statistic **at least as extreme** as the one we observed. We **reject** at level α when $p < \alpha$.

Tails. Match the tail(s) to H_a:

- Two-sided ($H_a: \ \mu \neq \mu_0$): $p = 2 \cdot P(\text{tail beyond } |z_{\text{obs}}|)$
- One-sided ($H_a: \ \mu > \mu_0$): $p = P(Z \geq z_{\text{obs}})$
- One-sided ($H_a: \ \mu < \mu_0$): $p = P(Z \leq z_{\text{obs}})$

Common mistakes to avoid.

1. p is **not** the probability that H_0 is true.
2. A smaller p does **not** imply a larger effect; report effect size and a CI.
3. With very **large** n, tiny effects can yield tiny p; assess practical importance.
4. With small n, true effects can yield large p (low power).

Takeaway. Use p for the decision, but interpret results alongside **effect size** and a **95% CI**.

5.7.3 Reporting results

A results statement gives the plain-language conclusion and the key statistics in a fixed order. Aim for one clean sentence plus context.

Template (two-sided test): *Plain-language finding. (test, statistic, df or n, value of statistic, p-value, effect size/CI if relevant).*

What to include (in order):

1. Test name (e.g., one-sample z, two-sample t, ^{2}, regression).
2. Test statistic and its degrees of freedom (or sample size for z).
3. Exact p (report $p < 0.001$ when very small).
4. An effect size and/or a 95% CI when available.

Examples

- **One-sample z:** The average mercury level within 10 km of the smelter was higher than the legal limit ($z = 2.85$, $n = 32$, p = 0.004).
- **One-sample t with CI:** Mean nitrate exceeded the target by 0.41 mg/L ($t(31) = 2.6$, p = 0.013, 95% CI [0.09, 0.73] mg/L).
- **Two-sample t with effect size:** Downstream sites had higher turbidity than upstream sites ($t(58) = 3.1$, p = 0.003, Cohen's (d = 0.80)).
- **Not significant (be explicit):** Mean canopy temperature did not differ from the baseline ($t(47) = 1.2$, p = 0.24); we fail to reject H_0.

Rounding & style

- Round statistics to 2 decimals; p to 3 decimals (use $p < 0.001$ when needed).
- Use "fail to reject H_0" (not "accept").
- Match tail to hypothesis in prose (e.g., "higher than," "lower than," or "different from").

Do / Don't

- **Do** pair the sentence with a short, substantive takeaway (what it means for the question).
- **Do** include a CI or effect size when it aids interpretation.
- **Don't** restate methods; **don't** claim proof; **don't** omit units.

> **Takeaway:** One sentence is enough: state the direction in words, then report (*test*, statistic, df/n, (p), and effect size/CI).

5.7.4 Beyond the 0.05 cutoff: Effect-size and power

Why go beyond a yes/no at $\alpha = 0.05$? Because scientists, managers, policymakers, and communities need to know **how large** an effect is and **how likely** a study is to detect it. That's what **effect size** and **power** provide.

Working definitions

- **Effect size**: how big the difference or association is (in natural units or standardized units like Cohen's d).
- **Power** ($1 - \beta$): the probability your design will detect a real effect of a specified size at your chosen α.

Use them before you sample. Decide what effect is *meaningful* for your context (e.g., a 0.3 mg/L nitrate reduction) and size your study to have adequate power to detect that effect.

5.7.5 Effect size: practical vs. standardized

What is effect size? It's how big the difference or association is. You can express it in **original units** (practical meaning) or in **standardized units** (comparability across scales).

Practical (original units). Use the scale stakeholders recognize.

- Exams: "Treatment improved scores by **5 percentage points**" (a letter-grade–sized change).
- Time-on-task: "New training reduces practice time by **1,000 hours** out of 10,000" (a 10% reduction).

These statements are easy to interpret, but they don't travel well across contexts or measures.

Standardized (unitless). When units are unfamiliar or scales differ, standardize by typical variability. The most common standardized effect size metric is *Cohen's d*. For two means with a common SD σ,

$$d = \frac{\mu_1 - \mu_2}{\sigma}$$

Now "how big?" is expressed in **SD units**. If you notice, this is just a kind of z-score. It is a way to standardize the mean difference in terms of the population standard deviation.

Environmental example (biodiversity index). Suppose an index increases from 3 to 4 after reforestation.

- In **raw units**: +1 index point.
- In **standardized terms**:
  - If the SD across sites is 1.0, then $d = 1.0$: a large shift.
  - If the SD is 10, then $d = 0.10$: a small shift.

  Takeaway: Report both when possible: original units for meaning (policy/action) and a standardized effect (e.g., $Cohen's d$) for comparison across studies.

Notes for practice

- In real data, σ is unknown; estimate it (pooled SD for two groups, with small-sample corrections as needed).
- Heuristics (context matters): $d \approx 0.2$ (small), $d \approx 0.5$ (medium), $d \approx 0.8$ (large).

- A "small" d can still be important (e.g., small reductions in lead or $PM_{2.5}$ can matter for health).
- Pair effect sizes with **95% CIs** to show precision, not just magnitude.

Let's ground effect size in a concrete example. Suppose we're measuring **final exam performance** (percent correct). The class mean is 65% with a standard deviation of 5%. Group A is a control, while Group B receives some instructional treatment.

Figure 5.14 shows four possible scenarios:

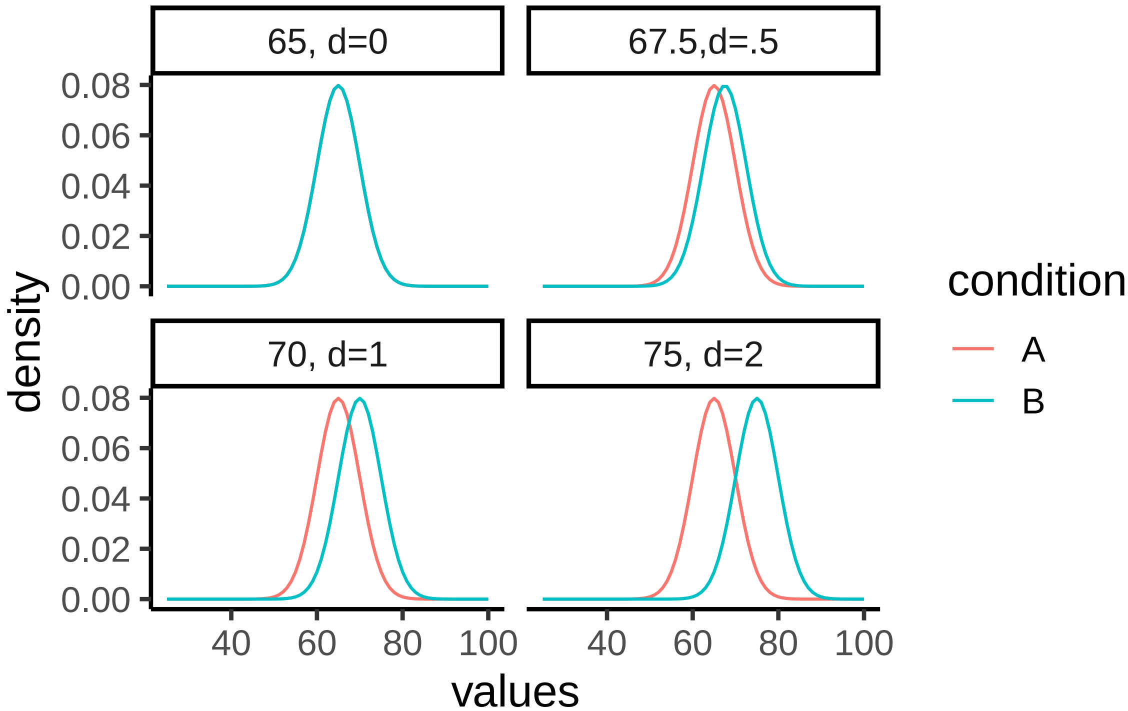

Figure 5.14: Same idea with concrete scores (mean 65, $sd = 5$): treatment shifts correspond to $d = 0, 0.5, 1, 2$.

- **Panel 1** **(**$d = 0$**)**: No effect. Both groups sample from the same distribution (mean = 65).
- **Panel 2** **(**$d = 0.5$**)**: Group B's mean shifts upward by 2.5 points (67.5).
- **Panel 3** **(**$d = 1$**)**: A 5-point shift (70).
- **Panel 4** **(**$d = 2$**)**: A large 10-point shift (75).

The takeaway is that effect size puts raw differences into a standard-deviation scale. A half–SD shift ($d = 0.5$) is noticeable but modest; a two–SD shift ($d = 2$) is very large. In practice, we rarely know beforehand how big a shift to expect, that's why we conduct the study.

5.8 Power

When a real effect exists, your design needs to be sensitive enough to detect it, otherwise the test has little value. We've already seen that effects can vary in size. Now we add a second idea: study designs differ in their ability to reliably detect those effects. This sensitivity is called **statistical power**.

Power rises when:

1. **Effect size is larger.**
2. **Sample size (n)** is larger (smaller SE).
3. **The test is less conservative** (larger α).
4. **Measurement noise is lower** (smaller SD).

> **Question:** "With $\alpha = 0.05$, how large must n be to achieve ≥ 80 power for the smallest effect that matters?"

5.8.1 Simulation walkthrough

Let's see how design choices affect power with a simple simulation. Suppose we have two groups, each with $n = 10$. Group A (control) is sampled from a normal distribution with mean $= 10$ and $SD = 5$. Group B (treatment) has a mean $= 12.5$, a shift of 2.5 points (Cohen's $d = 0.5$), considered a medium effect.

If we simulate this experiment 1,000 times and run an independent-samples t-test each time, how often do we reject H_0 at $\alpha = 0.05$?

```
set.seed(2025)
p <- numeric(1000)
for(i in 1:1000){
  A <- rnorm(10, 10, 5)
  B <- rnorm(10, 12.5, 5)
  p[i] <- t.test(A,B,var.equal=TRUE)$p.value
}
```

```
#> [1] 0.207
```

This proportion is the **power** of the design: the probability of detecting the true effect in repeated experiments. With $n = 10$, it's only about 20%. Even though we know a medium-sized effect exists, the test usually fails to pick it up with such a small sample.

5.8.2 How power changes

1. **Larger** n. Doubling to $n = 20$ per group reduces sampling error and roughly doubles power:

```
#> [1] 0.328
```

2. ***Smaller** α. Making the test more conservative (e.g., $\alpha = 0.01$) lowers power, because fewer outcomes count as "significant":

```
#> [1] 0.123
```

3. **Bigger effects are easier to detect.** If the treatment produces a big shift (say 2 standard deviations), the same design has near-perfect power:

```
p <- numeric(1000)
for(i in 1:1000){
  A <- rnorm(20, 10, 5)
  B <- rnorm(20, 20, 5)   # 2SD effect
  p[i] <- t.test(A,B,var.equal=TRUE)$p.value
}
```

```
#> [1] 1
```

5.8.3 Power curves

It's best to think of power as a **profile**, not a single number. A power curve shows how sensitive a design is across a range of possible effects.

In Figure 5.15, we fix the design to a two-sample t-test with $n = 10$ per group and $\alpha = 0.05$. Power is plotted on the y-axis, and the true effect size (Cohen's d) is on the x-axis.

Walkthrough.

- With $n = 10$, the design only detects a medium effect ($d = 0.5$) about 20% of the time.
- A larger effect ($d = 0.8$) is caught about half the time.
- Very large effects ($d = 2$) are detected almost every time.

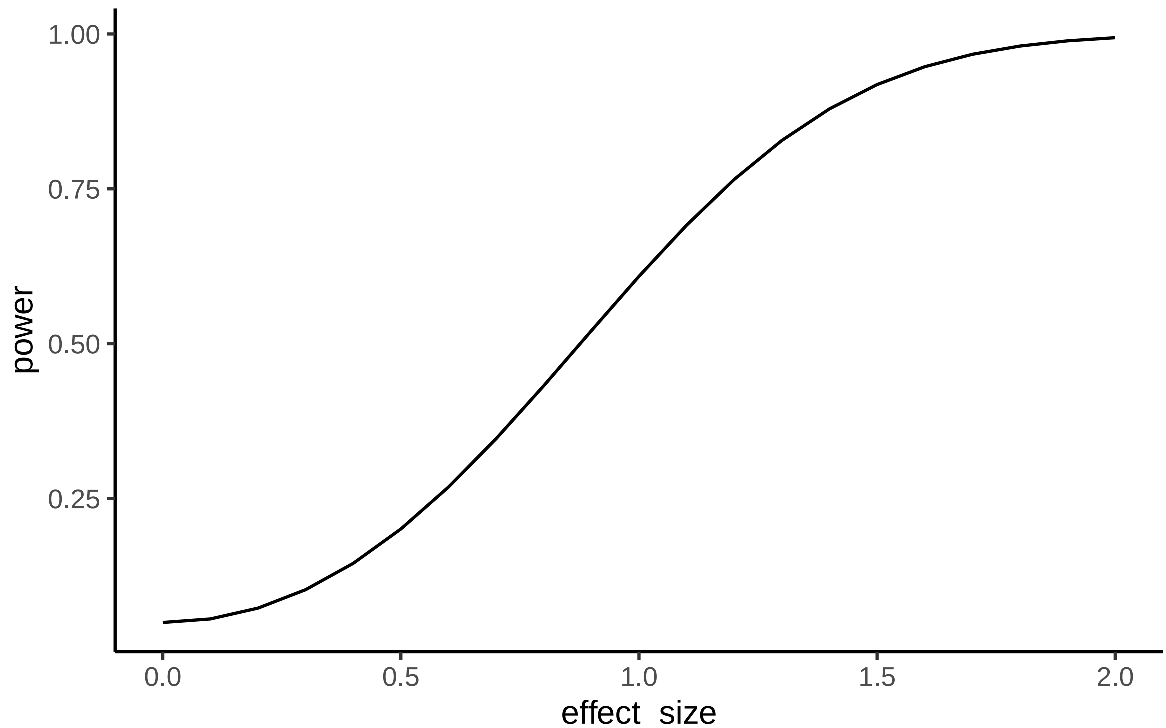

Figure 5.15: Power vs. effect size (Cohen's d) for a two-sample t-test with $n = 10$ per group and $\alpha = 0.05$.

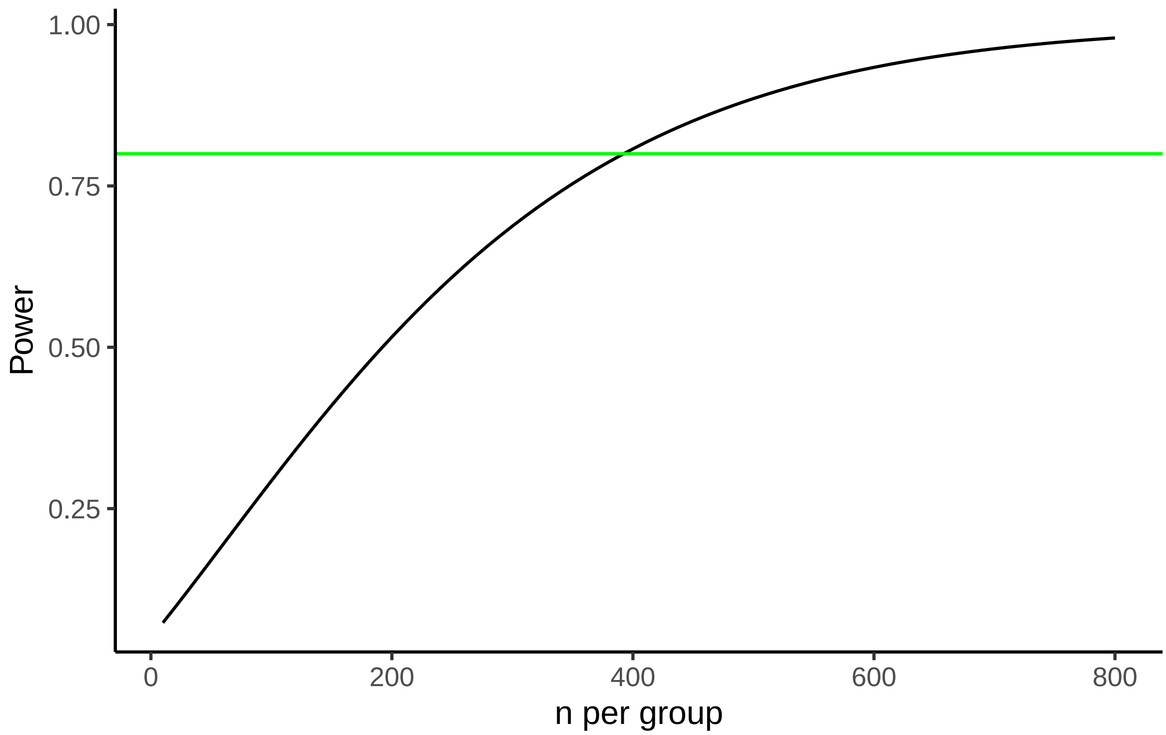

Figure 5.16: Power vs. sample size for detecting $d = 0.2$ with a two-sample t-test at $\alpha = 0.05$; green line marks 0.80 power.

This shows that small samples are only sensitive to big, obvious effects.

Now suppose prior research suggests the effect size is small ($d = 0.2$). Instead of varying the effect, we can ask: *how many subjects per group do we need to have decent power?*

Figure 5.16 plots power against sample size for detecting $d = 0.2$. The green line marks 80% power, a common planning target.

Walkthrough.

- At $n = 10$ per group, power to detect $d = 0.2$ is near zero.
- To reach 80% power, we need about **380 per group**.
- Even then, 20% of studies would still fail to reject H_0 just by chance.

Takeaway:

- A power curve shows, for a fixed design, how power rises as the true effect grows.
- A power-vs-n curve shows how many samples you need to hit a target power for a specific effect.
- Power analysis is mainly a **planning tool**, not a results-reporting tool. When you design a study, don't just aim for "power = 0.8." Always state the **assumed effect size** and the **standard deviation** you used for that calculation. Otherwise, "0.8" has no meaning: power depends entirely on those assumptions.

5.8.4 Planning your design

Our discussion of effect size and power highlight the importance of the understanding the statistical limitations of an experimental design. In particular, we have seen the relationship between:

- **Define a meaningful effect** (environmentally or policy-relevant minimum).
- **Choose** α (often 0.05; justify if different).
- **Estimate variability** (pilot data, literature, or historical monitoring).
- **Compute** n for ≥ 80 power (or explain tradeoffs if you can't reach it).
- **Pre-specify** one- vs two-tailed before data collection.

As a general rule of thumb, small-n designs can only reliably detect very large effects, whereas large-n designs can reliably detect much smaller effects. As a researcher, it is your responsibility to plan your design accordingly so that it is capable of reliably detecting the kinds of effects it is intended to measure.

5.8.5 Two cautions (common pitfalls)

- **Low power inflates doubt, even when (p<0.05).** Underpowered studies rarely replicate. A design with only 30% power to detect an expected effect might, by luck, produce a "significant" result, but most replications will not. Worse, false positives from small samples tend to look exaggerated: if a spurious finding reaches significance, the estimated effect size is often large enough to seem convincing.
- **Huge n finds tiny effects.** With very large samples, trivial differences can register as statistically significant. For example, a billion-person trial might show a drug reduces pain by 0.01%. That effect is "real" but practically meaningless. This is why effect size and confidence intervals should be reported alongside p-values: significance alone is not enough.

Figure 5.17 shows the first pitfall graphically. Even when the null hypothesis is true, 5% of tests are expected to fall below $\alpha = 0.05$. In small samples, those false positives tend to be paired with large apparent effect sizes. This can give the illusion of a strong, replicable finding when in reality it is just sampling noise.

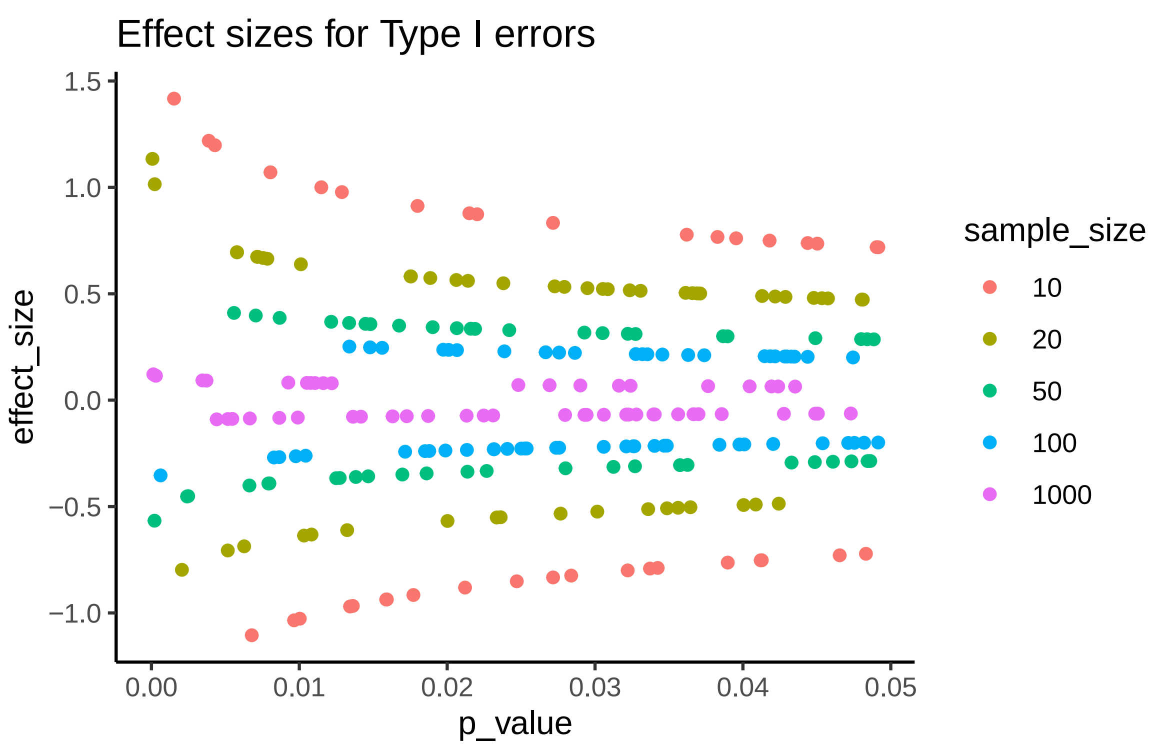

Figure 5.17: Effect size estimates among Type I errors ($p < 0.05$) under the null, by sample size, for a one-sample t-test. Small n inflates apparent effects.

Takeaway: statistical significance does not guarantee scientific significance. Always interpret p in the context of effect size, confidence intervals, and study design.

Key point

Power is not a fixed property of a test; it depends entirely on assumptions about n, effect size, α, and variability. Think of it as a **family of numbers**, not a single value. That's why power analysis is most useful **before** you collect data: it tells you whether your design is capable of detecting the kind of effect you care about

5.9 Chapter Summary

Why it matters. Hypothesis testing is the machinery behind most quantitative claims you will read, make, or have to defend. It gives you a disciplined way to decide whether a pattern in your sample says something real about the population, and, just as importantly, an honest account of how often that decision will be wrong.

Core ideas

- **Three routes, one decision.** A test statistic against a critical value, a confidence interval, and a p-value are three ways of asking the same question, and for a given α they always agree. Which you report is a matter of audience and convention, not correctness.
- **What a p-value is.** The probability of data at least this extreme *if* H_0 *were true*. It is not the probability that H_0 is true, and on its own it tells you nothing about how large the effect is.
- **Two ways to be wrong.** A Type I error rejects a true H_0, at a rate you choose by setting α. A Type II error fails to reject a false one. Tightening one loosens the other, which is why $\alpha = .05$ is a convention balancing the two rather than a fact about nature.
- **Significance is not size.** With a large enough n, a trivially small difference will clear any threshold. Report an effect size and a confidence interval next to every p-value so a reader can judge whether the result matters, not just whether it cleared a line.
- **Power is a design decision.** Power is your chance of detecting a real effect when one exists, and it depends on n, effect size, α, and variability together. An underpowered study fails in both directions: it usually misses real effects, and on the occasions it does reach significance, the effect it reports tends to be badly inflated. That is why power analysis belongs *before* data collection.
- **Commit to one- or two-tailed in advance.** A one-tailed test is only justified when theory genuinely rules out the other direction. Deciding after seeing the data quietly raises your real Type I error rate above the α you claimed.

6 t-tests

6.1 Intro

In the early 1900s, William Sealy Gosset joined **Guinness**, then the world's largest brewery. Guinness hired scientifically trained brewers and gave them latitude to pursue research that improved the product. As production scaled, quality control mattered: a pint should taste the same in Dublin as in Detroit. Until then, hop quality was judged mostly by appearance and smell, which was unreliable. Chemical assays could measure ingredients, but the core problem was statistical. Inference relied on the normal (Z) distribution, which works well when samples are large enough for a normal approximation. In practice, Guinness could not destroy large volumes of beer or ingredients just to test them.

Gosset set out to quantify the extra uncertainty introduced by **small** samples. How much wider is the error around an estimate when $n = 10$ instead of $n = 1000$? He derived a new sampling distribution that accounts for small-sample variability, now known as the **Student's (t) distribution**. Because Guinness restricted external publications, he published the 1908 paper under the pseudonym **"Student"** (Student 1908). That is why we still say **Student's** *t*-test. The core idea you will use today is the same: measure a signal, scale it by its standard error, and compare the resulting t to the appropriate t distribution for your $n - 1$ degrees of freedom.

Why this matters. We often compare a sample mean to a reference (drinking-water standard, historical average) or compare two conditions (before/after, site A vs site B). A t-test asks whether the observed difference is larger than you would expect from sampling variation alone, given your n and variability.

Three common t-tests you will use.

1. **One-sample t-test.** Compare a sample mean to a target (μ_0).
2. **Paired t-test.** Compare before/after (or matched) measurements on the same units by running a one-sample t on the *differences*.
3. **Independent-samples t-test (also called two-sample t-test)** Compare means from two groups measured on different units. Use Welch's test by default when variances differ.

What you will report.

Always report:

- the effect size (mean difference)
- a 95% CI, the test statistic with df
- and the p-value.

Example: "Mean change was (-0.42) units (95% CI [-0.73,-0.11] ;t(11)=-2.92; p=0.014)."

6.2 From mean & SD to "signal over noise"

A mean tells you where the center is; the SD tells you how spread out the data are. A mean of 50 could be very representative (if SD is tiny) or nearly meaningless (if SD is huge). You need both.

A simple way to combine them is a ratio:

$$\frac{\text{mean}}{\text{SD}}$$

- Big numerator, small denominator $\rightarrow$ big ratio (more confidence that the mean reflects the data).
- Small numerator or big denominator $\rightarrow$ small ratio (less confidence).

For instance, mean = 50 with SD = 1 gives 50; mean = 50 with SD = 100 gives 0.5. Same mean, very different stories.

This "signal over noise" pattern shows up everywhere in inference. In general,

$$\text{statistic} = \frac{\text{effect}}{\text{error}}.$$

For t-tests, we won't actually use SD in the denominator, we use the **standard error** SE, which is the SD of the sample mean. But the logic is the same: measure a signal, scale it by its uncertainty, and see whether the resulting ratio is unusually large or small under the right t distribution.

$$t = \frac{\text{effect}}{\text{standard error}} = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

We now turn that ratio into a test statistic for a *single mean.*

6.3 One-sample t-test

Use a one-sample t when you want to test whether a sample mean differs from a reference value (μ_0). The question: is the observed difference bigger than you'd expect from sampling variability, given n and your sample SD s?

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1$$

After calculating t from our data, we compare the observed t to a t distribution with $n-1$ degrees of freedom to get a p-value and a decision at α.

6.3.1 Comparing z and t

When the population SD is known (or we have a large enough sample to assume convergence), the ratio is called z; when it's estimated from the data, it's t.

Under a known population SD (σ), the standardized mean uses the z statistic:

$$z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

When σ is **unknown** (the usual case), we estimate it with the sample SD (s) and use the t statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \qquad df = n - 1$$

Because s varies from sample to sample, the t distribution has **heavier tails** than the standard normal. As n grows, s stabilizes and t **converges** to z.

> **i** Note
>
> **When is** z appropriate? If the population SD σ is known *and* the sampling distribution of the mean is (approximately) normal, use z. In practice, σ is rarely known, so we use t with s and $df = n - 1$.

Rule of thumb. For two-tailed tests at $\alpha = 0.05$:

- z uses critical values ± 1.96.
- t is a bit wider for small n (e.g., $df = 9 \Rightarrow \pm 2.262$; $df = 29 \Rightarrow \pm 2.045$), and approaches ± 1.96 as n increases.

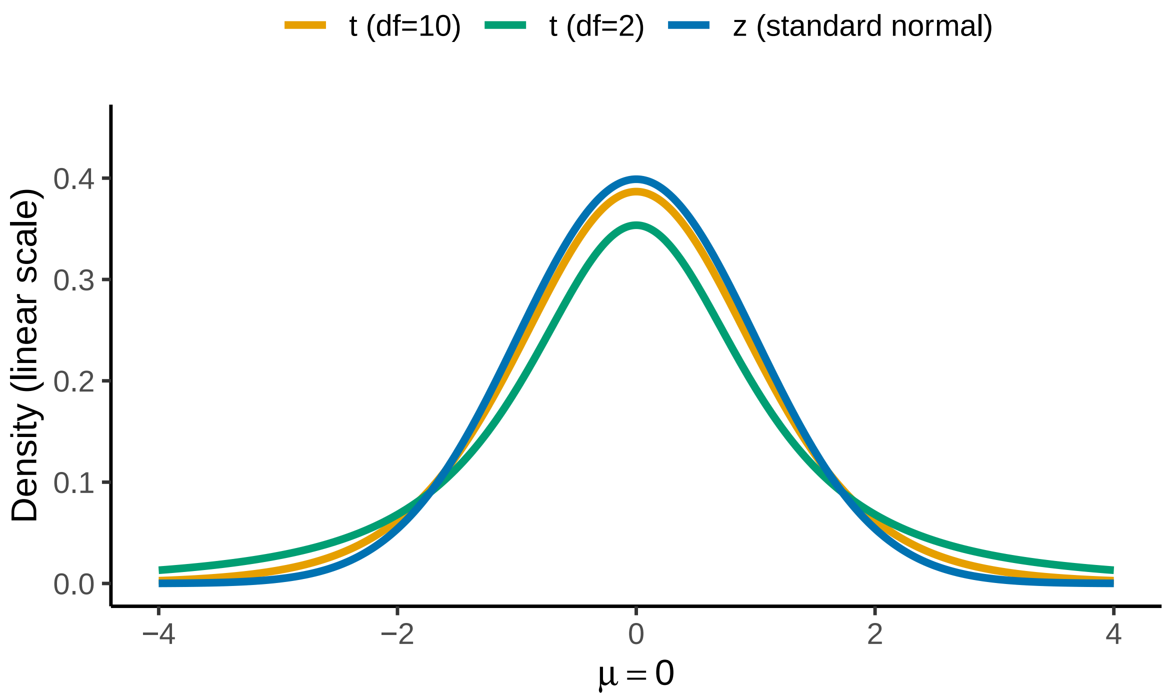

Figure 6.1: Normal (z) vs t distributions: on a linear scale (top), tail differences look small; on a log scale (bottom), the heavy tails of small-n t are clear.

6.3.2 What t represents

Either way, the ratio asks the same question: how far is the mean from the reference in SE units?

t expresses how far a sample mean is from the hypothesized mean, in **standard error units** (signal over noise):

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

t can be positive or negative depending on direction. Under H_0, $t \sim t_{(df=n-1)}$.
Rule of thumb: at $\alpha = .05$ two-tailed, $|t| \gtrsim 2$ is uncommon when $df \approx 20$.

> **Note**
>
> **Standard error, again.** Remember that $s/\sqrt{n}$ is the **standard error (SE)**, the standard deviation of the sampling distribution of the mean.

Let's see how t values behave when samples really come from the null hypothesis. Under the null hypothesis, t follows a t distribution with $df = n - 1$.
Large $|t|$ values are rare; with $df \approx 20$, $|t| \gtrsim 2$ is uncommon at $\alpha = .05$ (two-tailed).

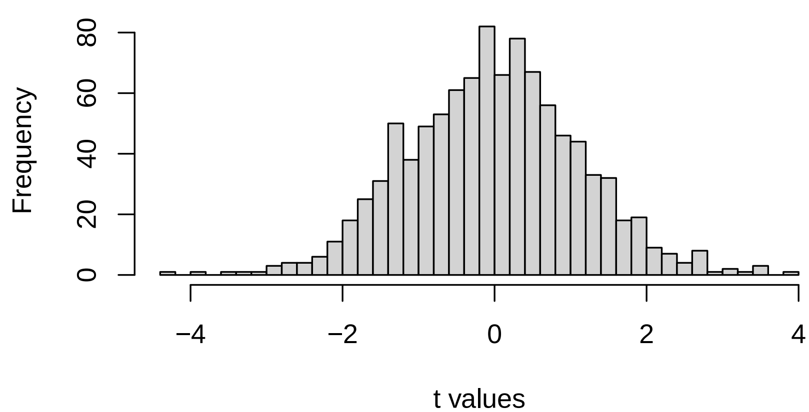

Figure 6.2: Histogram of 1,000 simulated t values ($n = 10$) drawn from a population with $\mu_0 = 0$. Most ts cluster near 0; large positive or negative values are rare.

Takeaway: t measures how far the sample mean is from the null expectation, scaled by its uncertainty. It behaves predictably under H_0, allowing us to decide whether a difference is likely due to chance.

6.3.3 Calculating t from data

Let's calculate a one-sample t statistic step by step using a short example. Ten students each took a 5-question true/false quiz (50% expected by chance). Their scores (percent correct) are:

Scores: 50, 70, 60, 40, 80, 30, 90, 60, 70, 60

By hand, the standard error of the mean is the sample SD divided by the square root of n:

$$\text{SEM} = \frac{s}{\sqrt{n}} = \frac{17.92}{\sqrt{10}} = 5.67$$

And t is the difference between the sample mean (61) and the reference value (50, chance level), divided by that SEM:

$$t = \frac{\bar{X} - \mu_0}{SEM} = \frac{61 - 50}{5.67} = 1.94$$

We can confirm this with a single R command, `t.test(scores, mu = 50)`:

Results: $t(9) = 1.94$, $p = 0.0842$; mean = 61 (95% CI [48.18, 73.82]).

Interpretation: The mean (61%) is above chance (50%), but the difference is not statistically significant at $\alpha = .05$.

Assumptions check: Observations are independent, and with $n = 10$ the data are approximately normal, sufficient for a one-sample t-test.

6.3.4 Assumptions, and what to do when they don't hold

Every test in this book rests on assumptions. Assumptions are conditions that need to be roughly true for the math behind the test to be trustworthy. There are two different kinds of assumptions.

The universal assumption. Every single test we cover in this class (one-sample t, paired t, independent t, ANOVA, regression) assumes your observations are **independent** and came from a **sound study design** (a fair sample, not one that systematically favors certain values). This assumption is so constant that from here on, we won't re-list it test by test. Keep it in the back of your mind, but if an exam or homework question asks you to "list the assumptions of this test," this isn't the answer they're looking for.

Test-specific assumptions. These come from how a particular test's math actually works, and they differ test to test. For the one-sample and paired t-tests, the specific assumption is **approximate normality** of the outcome itself (one-sample) or of the difference scores (paired). Why does this matter? The t-distribution we've been using to get p-values is *derived* assuming the sampling distribution of the mean is normal. If your data are badly skewed and your sample is small, that derivation is on shaky ground, and the p-value it hands you may not mean what it claims to mean.

How worried should you be? It depends enormously on n. With a large sample, the Central Limit Theorem does a lot of the work for you. The sampling distribution of the mean tends toward normal even when individual observations are skewed, so the t-test is fairly **robust** to non-normality once n is reasonably large. A lot of environmental data, particularly field data, instead is from **small samples that are also skewed**. Examples include a handful of expensive water samples, a short list of monitoring wells, five plots you had time to visit before the storm rolled in. Small n and non-normality reinforcing each other is when a t-test's answer becomes questionable.

What to do about it. First, look at your data. A histogram or a QQ-plot will usually tell you whether "approximately normal" is a reasonable description or a stretch. You can also run a formal normality test, the **Shapiro-Wilk test** (`shapiro.test()` in R), which tests H_0: the data came from a normal distribution. Treat it as one more piece of evidence rather than a strict gate though: with a small sample, exactly when a violation matters most, it has low power and will often say "no problem" even when there is one. If your data look like a stretch and your sample is small, you have another option: a **nonparametric test** that doesn't lean on the normality assumption at all. You'll meet these nonparametric alternatives in the next chapter, once we've covered all three t-test forms below.

6.3.5 How does t behave?

Under the null hypothesis (H_0 true), t follows a t distribution with degrees of freedom $df = n - 1$. Large $|t|$ values are rare; extreme sample means are unlikely if H_0 is true.

For moderate sample sizes ($df \approx 20$), a simple rule of thumb is:

> In a two-tailed test at $\alpha = .05$, $|t| \gtrsim 2$ is uncommon.

This means values of t beyond roughly -2 or $+2$ occur less than 5% of the time under the null. That's the foundation of the decision rule for t tests: if your observed t lies in those rare tails, you reject H_0.

Remember, if we obtained a single t from one sample we collected, we could consult the chance window in Figure 6.3 below to find out whether the t we obtained from the sample was likely or unlikely to occur by chance.

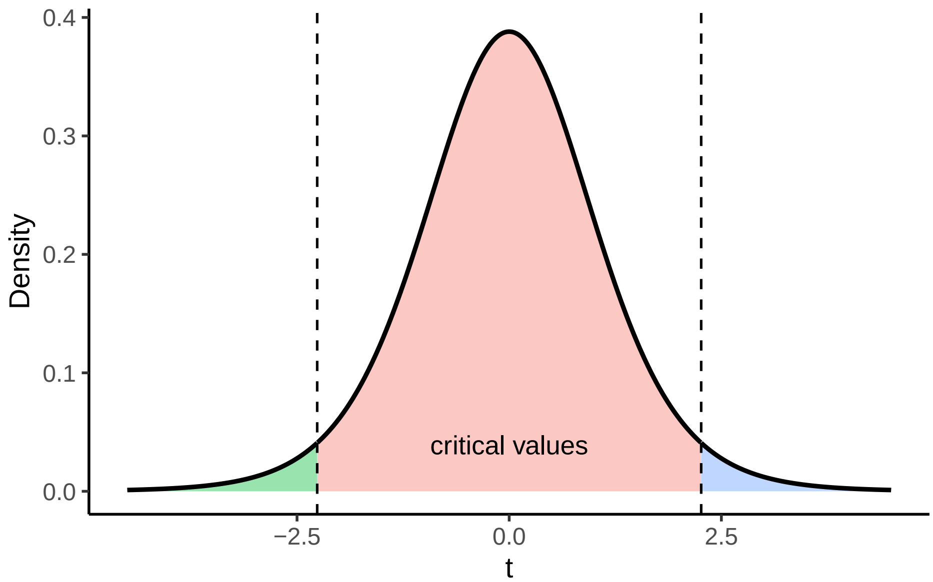

Figure 6.3: Two-tailed rejection regions at $\alpha = .05$ for $t_{(df=9)}$. Shaded tails mark unlikely values under H_0.

6.3.6 Interpreting t

We're asking whether our sample could plausibly have come from the null distribution: the distribution of t-values we'd see across many samples of size 10, if the true mean were exactly 50 (chance level). Figure 6.4 shows what that distribution looks like, at $df = 9$.

t is most likely to land near zero, which makes sense: this is the distribution of *no* differences, so most of the time chance alone won't push it far from 0. Because the distribution is symmetric, a t-value from the null distribution is just as likely to be positive as negative.

That symmetry gives us a few basic probabilities for free:

1. t is always zero, positive, or negative, so $p(t = 0 \text{ or } t > 0 \text{ or } t < 0) = 1$.
2. $p(t \geq 0) = .5$: half of all t-values are zero or greater.
3. $p(t \leq 0) = .5$: half are zero or smaller.

We can get more precise than a 50/50 split. Figure 6.5 divides the same distribution into ten regions, each containing 10% of the distribution. Notice the regions get wider out in the

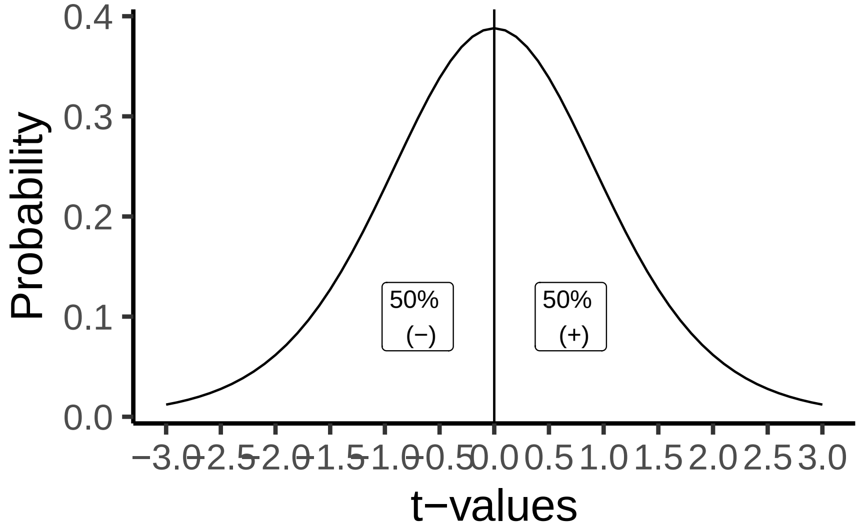

Figure 6.4: A distribution of t-values that can occur by chance alone, when there is no difference between the sample and a population

tails: since most t-values cluster near 0, it takes a wider interval out there to capture the same 10%.

For example:

1. $t \leq -1.38$: $p = 10\%$
2. $-1.38 \leq t \leq -0.88$: $p = 10\%$
3. $-0.88 \leq t \leq -0.54$: $p = 10\%$
4. $t \geq 1.38$: $p = 10\%$

Each range captures the same 10% of the distribution, even though the ranges themselves are different widths.

6.3.7 Getting the p-values for t-values

Where do values like these actually come from? Older textbooks put a table in the back of the book; this one doesn't need one, because software gives you exact t and p values directly. R, Excel, and online calculators all work; you'll get hands-on practice with this in lab, both with software and by simulation.

Software does need a couple of things from you to compute a p-value correctly: the degrees of freedom, and whether the test is one-tailed or two-tailed. We haven't covered either yet, so that's next, starting with one-tailed tests, then two-tailed, then degrees of freedom.

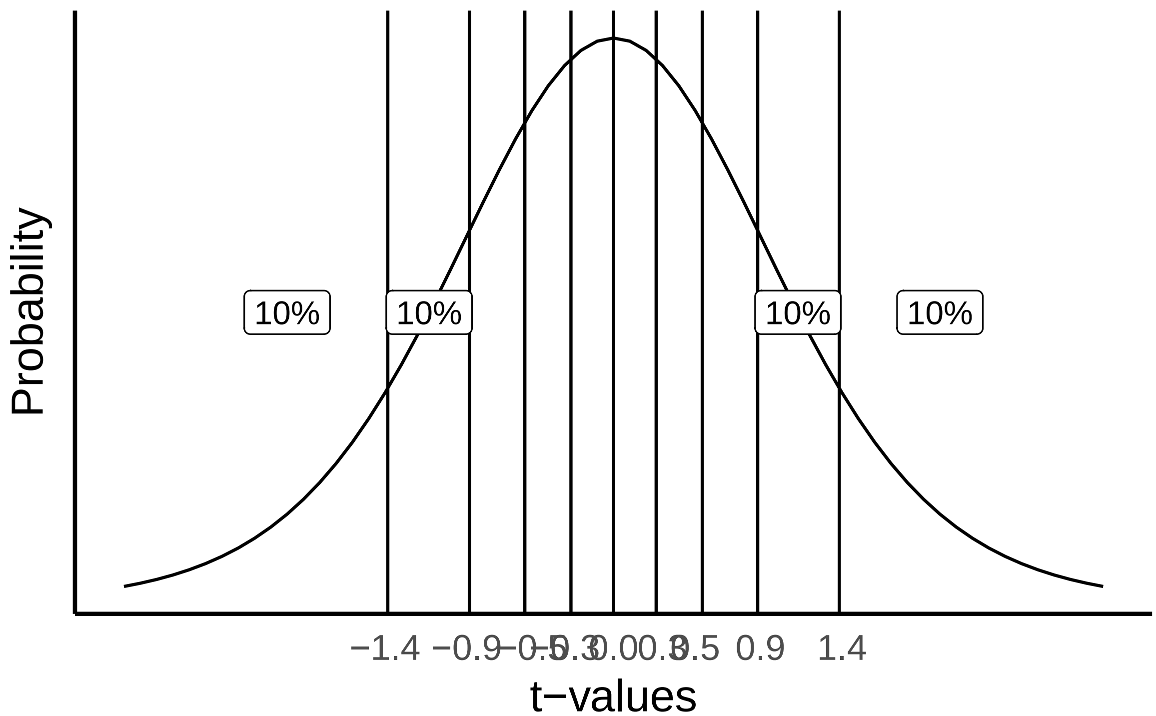

Figure 6.5: Splitting the t distribution up into regions each containing 10% of the t-values. The width between the bars narrows as they approach the center of the distribution, where there are more t-values.

6.3.8 One-tailed vs. two-tailed, applied

Chapter 5 introduced the concept: a two-tailed test asks whether μ differs from μ_0 in either direction, while a one-tailed test commits to one direction in advance, and is only justified when your theory already rules out the other direction. Here's how that choice plays out mechanically for a t-test: it changes the critical value, and therefore the p-value, for the exact same data.

For our quiz-score example ($df = 9$), a one-tailed test only needs to clear 5% of the distribution on one side. A two-tailed test has to clear 2.5% on each side, which pushes the critical value farther out:

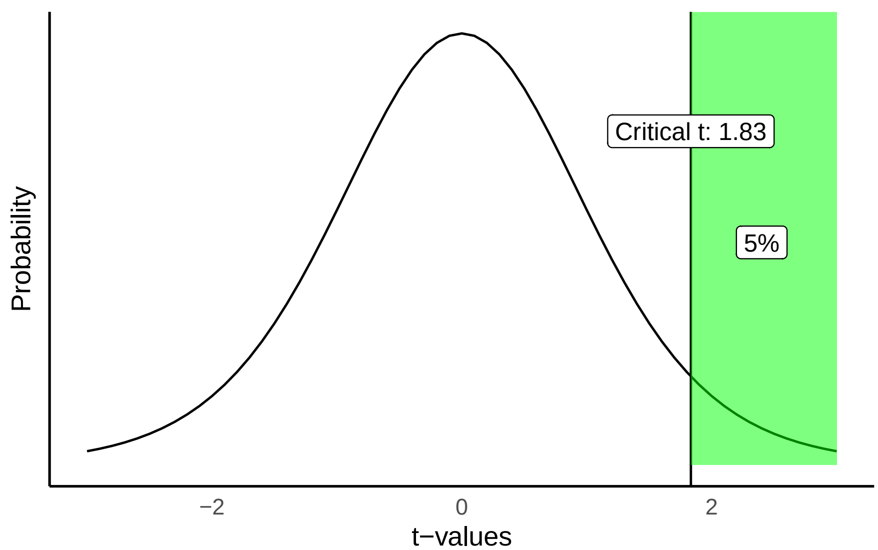

Figure 6.6: One-tailed critical value at df=9: t-values of 1.83 or greater occur 5% of the time.

Applied to our actual result, $t(9) = 1.94$: the one-tailed p is .042, the two-tailed p is .084, exactly double, since a two-tailed test counts the matching extreme region on both sides instead of just one. Here the choice actually changes the decision: one-tailed, this result is significant at $\alpha = .05$; two-tailed, it isn't. That's exactly why a one-tailed test has to be justified by your theory *before* you see the data, not chosen afterward because it gives the answer you wanted.

6.3.9 Which one should you use?

Use a one-tailed test only when your theory already rules out the opposite direction; default to two-tailed otherwise (Chapter 5's rule). There's a second reason two-tailed is the safer default: a one-tailed critical value is always closer to 0 than a two-tailed one, so routinely running one-tailed tests inflates your long-run Type I error rate above your stated α (recall

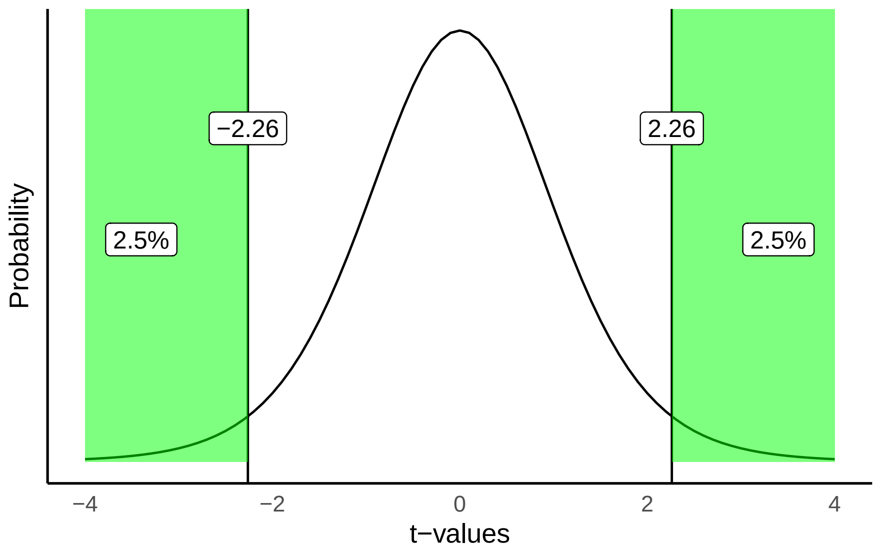

Figure 6.7: Two-tailed critical values at df=9: t-values beyond ± 2.26 occur 5% of the time total, 2.5% in each tail.

from Chapter 5: a Type I error means rejecting H_0 when it's actually true). That's why most researchers default to two-tailed tests even when they do have a directional prediction, and why it's worth being able to justify your choice rather than picking whichever gives the smaller p-value.

6.3.10 Degrees of freedom

You've probably noticed a number in parentheses every time we report a t-test, like $t(9) = 1.94$. That's the degrees of freedom, and it's worth understanding what it means, not just where to find it.

For one-sample and paired t-tests, $df = n - 1$. In our quiz example, $n = 10$ students, so $df = 9$.

Degrees of freedom is both a concept and a correction. The concept: once you've estimated something from your data, like the mean, that estimate constrains how many of the numbers are still free to vary. Consider three numbers with a mean of 2: pick the first two freely, say 51 and -3, and the third number is no longer free. It has to be -42 for the mean to come out to 2. Two of the three numbers had freedom; the third didn't. That's $df = n - 1 = 2$ for three numbers.

The same logic applies to the sample standard deviation. Computing it requires the sample mean first, which uses up one degree of freedom, exactly why the sample SD divides by $n - 1$ rather than n (Chapter 1).

6.3.10.1 Simulating how degrees of freedom affects the t distribution

We can see the effect of degrees of freedom directly by simulating the t distribution at two different sample sizes. Figure 6.8 compares $df = 9$ (our quiz-score example) to $df = 100$ (a much larger sample).

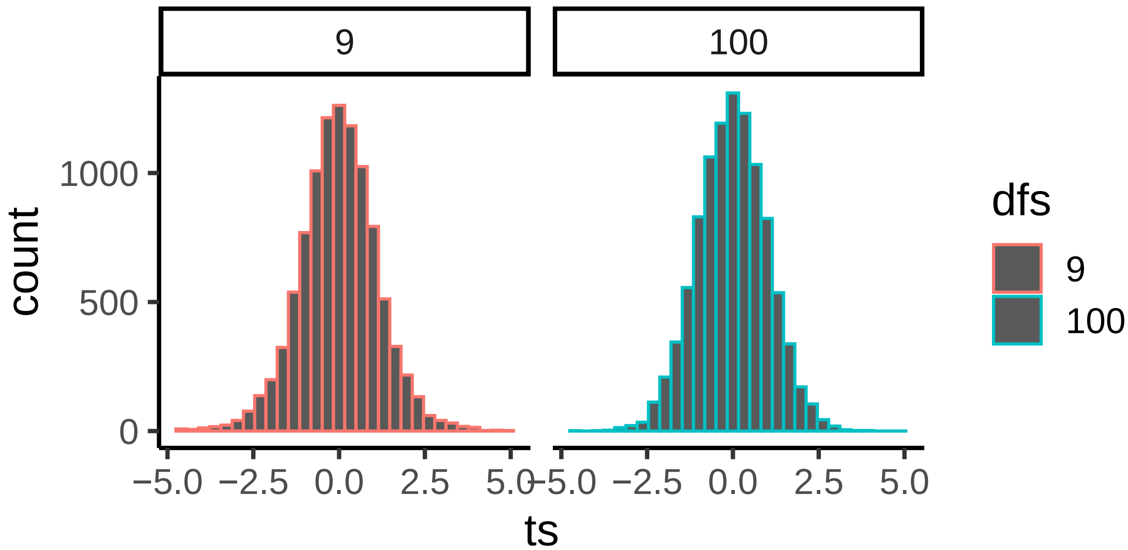

Figure 6.8: The width of the t distribution shrinks as sample size and degrees of freedom (from 9 to 100) increases.

As degrees of freedom increase, the t distribution gets taller in the middle and narrower in the tails, converging toward the standard normal. That's the same pattern from "Comparing z and t" earlier: with more data, the sample SD becomes a more reliable estimate of the population SD, so t behaves more like z.

Why $n - 1$ and not n? Because t uses the sample SD to estimate the standard error, and that SD calculation already spent one degree of freedom estimating the mean, so the t-distribution is built assuming $n - 1$ from the start.

6.4 Paired t: before/after or matched

When the **same units** are measured twice (before/after, matched pairs), the question is whether they **changed**. The natural data unit is the **within-unit difference**.

Design in this worked example. Ecologists measured **soil moisture (%)** at 20 paired field plots before and after wetland restoration. Each plot was measured twice: once during the dry season before restoration work began, and again one year later. We care whether restoration **increased** soil moisture, which would support plant reestablishment and carbon sequestration.

Define differences and their meaning. Choose an order and stick to it:

- $d_i = \text{After}_i - \text{Before}_i$
- $d_i > 0$: plot had **more** soil moisture after restoration (increase)
- $d_i < 0$: plot had **less** soil moisture after restoration (decrease)

This subtraction **cancels between-plot variability** (some plots are naturally wetter than others) and focuses on **within-plot change**.

Test on differences (paired (t)). Compute one difference per plot, then run a **one-sample** t against 0:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \quad s_d = \text{SD}(d_1, \dots, d_n), \quad t = \frac{\bar{d} - 0}{s_d/\sqrt{n}}, \quad df = n - 1.$$

The **sign of** t matches $\bar{d}$ and your difference direction.

Mini walk-through (first 5 plots). Using the first five plots, $\bar{d} = 2.8$ and $s_d = 3.42$. Then

$$t = \frac{2.8}{3.42/\sqrt{5}} \approx 1.83, \qquad df = 4,$$

which (two-tailed) is **not** significant. That's expected with $n = 5$.

Full sample effect. With all 20 plots, $\bar{d}$ is similar but $s_d/\sqrt{n}$ gets smaller, so t increases and the result becomes significant. This is the payoff of pairing: more power from the same number of plots.

What to report. $\bar{d}$ (the mean change), a 95% CI, $t(df)$, and p. Optionally add a standardized effect (Cohen's $d = \bar{d}/s_d$).

R pointers (you'll do this in lab).

- Compute differences: `d <- after - before`
- Paired t: `t.test(d, mu = 0)` or `t.test(after, before, paired = TRUE)`
- CI for $\bar{d}$: from the `t.test` output

Table 6.1: Paired t-test on 5 restoration plots

Mean difference	t	df	p-value
2.8	1.83	4	0.141

6.4.1 Calculate t

How do we calculate t for a paired-samples t-test? We use the exact same one-sample t-test formula from the previous section, just applied to the difference scores instead of raw scores. The paired-samples t-test is really just a variation on the one-sample t-test: there's still only one group of numbers involved, the differences.

Let's find t for the mean difference scores:

plot	Before	After	differences	diff_from_mean	Squared_differences
1	18	21	3	0.2	0.0400000000000001
2	22	26	4	1.2	1.44
3	16	22	6	3.2	10.24
4	25	29	4	1.2	1.44
5	21	18	-3	-5.8	33.64
Sums	102	116	14	0	46.8
Means	20.4	23.2	2.8	0	9.36
				sd	3.421
				SEM	1.53
				t	1.83006535947712

If we did this test using R, we would obtain almost the same numbers (there is a little bit of rounding in the table).

Our t-test results are: t(4) = 1.83, p = .141.

The sign of t matches the sign of the mean difference (ours was +2.8 percentage points), so soil moisture increased on average, exactly what $\bar{d}$ already told us directly. Getting a p-value from this t, and choosing one-tailed vs. two-tailed, works exactly like it did for the one-sample case (see the One-sample t-test section above). Two-tailed, $p = .141$: not significant, this five-plot sample is too small to rule out chance.

6.4.2 Scaling to the full sample

The five plots above are a subset of the full restoration study, which measured all **20** paired plots. With only $n = 5$, $t(4) = 1.83$ wasn't enough to distinguish the observed increase from chance ($p = .141$). Here's what changes with the full sample.

Table 6.2: Paired t-test on all 20 restoration plots

Mean difference	t	df	p-value
1.9	2.5	19	0.022

At $n = 20$, $t(19) = 2.5$, $p = 0.022$: significant. The mean difference (1.9 percentage points) is in the same range as what the 5-plot subset already suggested; what changed is the standard error, which shrank enough with the larger sample to rule out chance. This is the payoff of pairing: same design, more plots, more power to detect a real effect.

6.5 Two independent groups

The independent-samples t-test is the third and last design in this chapter. The logic is the same as the one-sample and paired cases: a mean difference divided by an estimate of its variability. What changes is the research design.

Use an independent-samples t-test for a **between-subjects** design: different people or units in each condition, each measured once. With two groups of 10, that's 20 people total, none of them paired. Without repeated measures on the same unit, there's no meaningful way to subtract scores between conditions the way the paired t-test does.

The numerator works the same way it always has: the mean difference between groups. The denominator is where independent groups differ from the paired case, since now we need a single estimate of variability built from two separate samples instead of one.

A natural first guess is to average the two groups' standard errors:

$$t = \frac{\bar{A} - \bar{B}}{\left(\frac{SEM_A + SEM_B}{2}\right)}$$

That average, however, is a biased estimator of the variability under the null. Instead, the independent-samples t-test **pools** the two groups' variances, weighted by their degrees of freedom, into a single estimate for the denominator:

$$t = \frac{\bar{X}_A - \bar{X}_B}{s_p\sqrt{\frac{1}{n_A} + \frac{1}{n_B}}}$$

where the pooled standard deviation s_p is (note s_A^2 and s_B^2 here are variances, not standard deviations):

$$s_p = \sqrt{\frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}}$$

The example below computes t for two small groups by following that formula step by step in R, then checks it against `t.test()`.

```
## By "hand" using R
a <- c(1,2,3,4,5)
b <- c(3,5,4,7,9)

mean_difference <- mean(a)-mean(b) # compute mean difference

variance_a <- var(a) # compute variance for A
variance_b <- var(b) # compute variance for B

# Compute top part and bottom part of sp formula

sp_numerator <- (4*variance_a + 4* variance_b)
sp_denominator <- 5+5-2
sp <- sqrt(sp_numerator/sp_denominator) # compute sp


# compute t following the formula

t <- mean_difference / ( sp * sqrt( (1/5) +(1/5) ) )

t # print results
#> [1] -2.017991


# using the R function t.test
t.test(a,b, paired=FALSE, var.equal = TRUE)
#>
#>  Two Sample t-test
#>
#> data:  a and b
#> t = -2.018, df = 8, p-value = 0.0783
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#>  -5.5710785  0.3710785
```

```
#> sample estimates:
#> mean of x mean of y
#>       3.0       5.6
```

6.6 Simulating data for t-tests

An "advanced" topic for t-tests is the idea of using R to conduct simulations for t-tests.

If you recall, t is a property of a sample. We calculate t from our sample. The t distribution is the hypothetical behavior of our sample. That is, if we had taken thousands upon thousands of samples, and calculated t for each one, and then looked at the distribution of those t's, we would have the sampling distribution of t!

It can be very useful to get in the habit of using R to simulate data under certain conditions, to see how your sample data, and things like t behave. Why is this useful? It mainly prepares you with some intuitions about how sampling error (random chance) can influence your results, given specific parameters of your design, such as sample-size, the size of the mean difference you expect to find in your data, and the amount of variation you might find. These methods can be used formally to conduct power-analyses. Or more informally for data sense.

6.6.1 Simulating a one-sample t-test

Here are the steps you might follow to simulate data for a one sample t-test.

1. Make some assumptions about what your sample (that you might be planning to collect) might look like. For example, you might be planning to collect 30 subjects worth of data. The scores of those data points might come from a normal distribution (mean = 50, sd = 10).
2. sample simulated numbers from the distribution, then conduct a t-test on the simulated numbers. Save the statistics you want (such as ts and ps), and then see how things behave.

Let's do this a couple different times. First, let's simulate samples with N = 30, taken from a normal (mean= 50, sd = 25). We'll do a simulation with 1000 simulations. For each simulation, we will compare the sample mean with a population mean of 50. There should be no difference on average here. Figure 6.9 is the null distribution that we are simulating.

The simulated t distribution looks like a t distribution should, and it shows us how often we get t values of particular sizes. The p-distribution is flat under the null, which we are simulating here. This means you have the same chance of getting a t with a p-value between 0 and 0.05

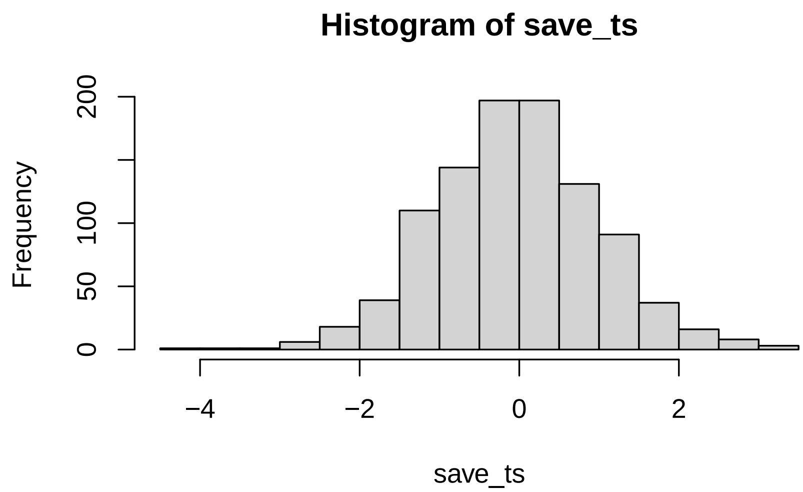

Figure 6.9: The distribution of t-values under the null. These are the t values that are produced by chance alone.

Figure 6.10: The distribution of p-values that are observed is flat under the null.

as you would of getting one between .90 and .95. Those ranges are both ranges of 5%, so there are an equal amount of t values in them by definition.

You can write the same simulation more compactly with `replicate()` instead of a `for` loop; it produces the same two distributions above:

```
simulated_ts <- replicate(1000, t.test(rnorm(30, 50, 25), mu = 50)$statistic)
simulated_ps <- replicate(1000, t.test(rnorm(30, 50, 25), mu = 50)$p.value)
```

6.6.2 Simulating a paired samples t-test

The code below is set up to sample 10 scores for condition A and B from the same normal distribution. The simulation is conducted 1000 times, and the ts and ps are saved and plotted for each.

```
save_ps <- length(1000)
save_ts <- length(1000)
for ( i in 1:1000 ){
  condition_A <- rnorm(10,10,5)
  condition_B <- rnorm(10,10,5)
  differences <- condition_A - condition_B
  t_test <- t.test(differences, mu=0)
  save_ps[i] <- t_test$p.value
  save_ts[i] <- t_test$statistic
}
```

According to the simulation, when there are no differences between the conditions, and the samples are being pulled from the very same distribution, you get these two distributions for t and p. These again show how the null distribution of no differences behaves.

For any of these simulations, if you rejected the null-hypothesis (that your difference was only due to chance), you would be making a type I error. If you set your alpha criteria to $\alpha = .05$, we can ask how many type I errors were made in these 1000 simulations. The answer is:

```
length(save_ps[save_ps<.05])
#> [1] 51
length(save_ps[save_ps<.05])/1000
#> [1] 0.051
```

We happened to make 51 type I errors. The expectation over the long run is a 5% type I error rate (if your alpha is .05).

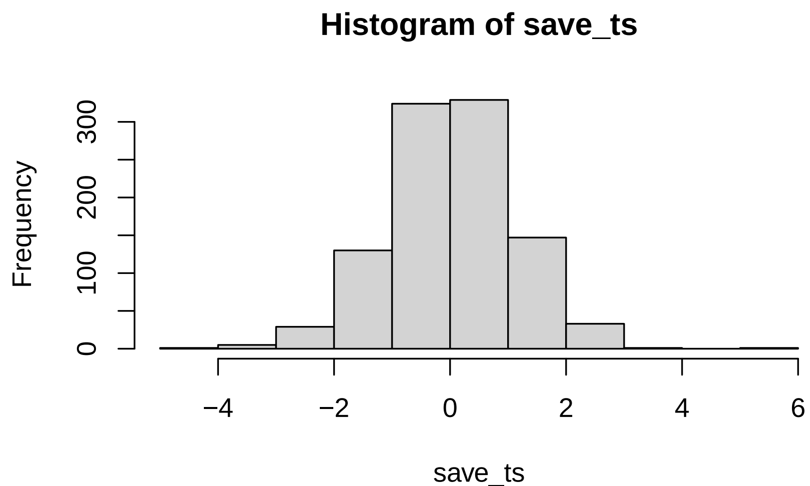

Figure 6.11: 1000 simulated ts from the null distribution

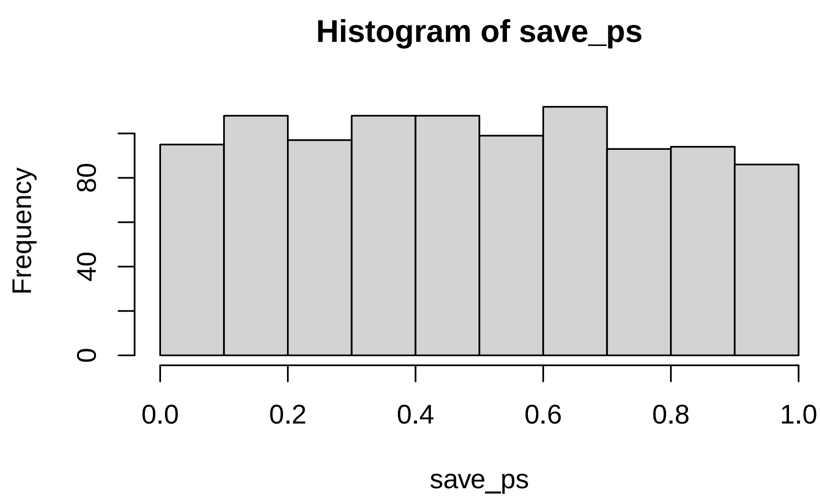

Figure 6.12: 1000 simulated ps from the null distribution

What happens if there actually is a difference in the simulated data, let's set one condition to have a larger mean than the other:

```
save_ps <- length(1000)
save_ts <- length(1000)
for ( i in 1:1000 ){
  condition_A <- rnorm(10,10,5)
  condition_B <- rnorm(10,13,5)
  differences <- condition_A - condition_B
  t_test <- t.test(differences, mu=0)
  save_ps[i] <- t_test$p.value
  save_ts[i] <- t_test$statistic
}
```

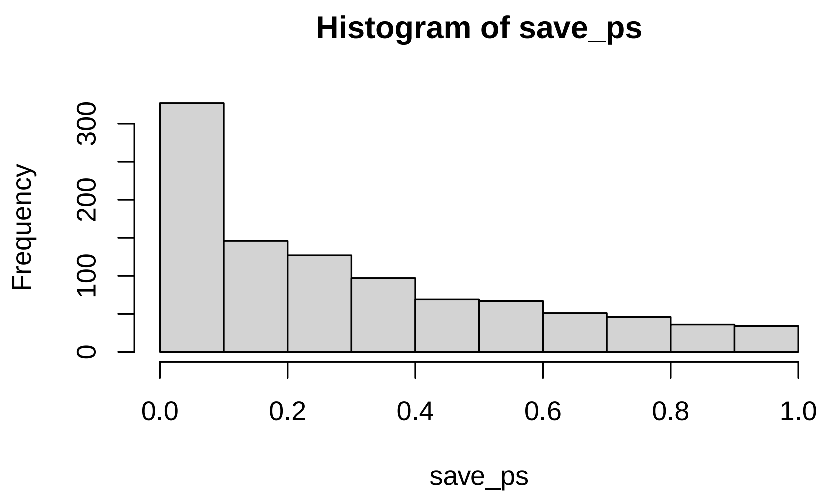

Figure 6.13: 1000 ps when there is a true difference

Now you can see that the p-value distribution is skewed to the left. This is because when there is a true effect, you will get p-values that are less than .05 more often. Or, rather, you get larger t values than you normally would if there were no differences.

In this case, we wouldn't be making a type I error if we rejected the null when p was smaller than .05. How many times would we do that out of our 1000 experiments?

```
length(save_ps[save_ps<.05])
#> [1] 216
```

```
length(save_ps[save_ps<.05])/1000
#> [1] 0.216
```

We happened to get 216 simulations where p was less than .05, that's only 0.216 of experiments. If you were the researcher, would you want to run an experiment that would be successful only 0.216 of the time? Probably not, that's the case for designing a better experiment.

How would you run a better simulated experiment? Well, you could increase n, the number of subjects in the experiment. Let's increase n from 10 to 100, and see what happens to the number of "significant" simulated experiments.

```
save_ps <- length(1000)
save_ts <- length(1000)
for ( i in 1:1000 ){
  condition_A <- rnorm(100,10,5)
  condition_B <- rnorm(100,13,5)
  differences <- condition_A - condition_B
  t_test <- t.test(differences, mu=0)
  save_ps[i] <- t_test$p.value
  save_ts[i] <- t_test$statistic
}
```

```
#> [1] 990
#> [1] 0.99
```

Now almost all of the experiments show a p-value of less than .05 (using a two-tailed test, the default in R). You can use this simulation process to determine how many subjects you need to reliably detect your effect.

6.6.3 Simulating an independent samples t-test

The setup is the same, you just change what goes into `t.test()`. Here's the null case, assuming no difference between two independent groups of $n = 10$:

```
save_ps <- length(1000)
save_ts <- length(1000)
for ( i in 1:1000 ){
  group_A <- rnorm(10,10,5)
  group_B <- rnorm(10,10,5)
  t_test <- t.test(group_A, group_B, paired=FALSE, var.equal=TRUE)
```

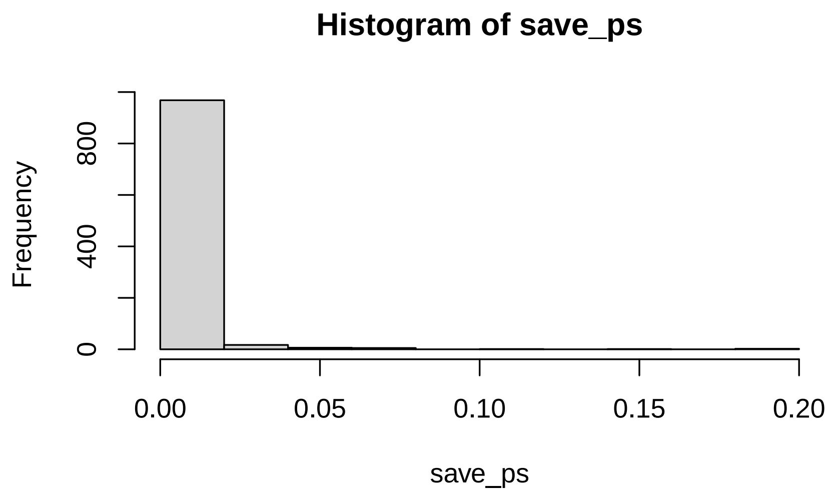

Figure 6.14: 1000 ps for n =100, when there is a true effect

```
  save_ps[i] <- t_test$p.value
  save_ts[i] <- t_test$statistic
}
```

```
#> [1] 51
#> [1] 0.051
```

Same story as before: about 5% of these simulated experiments come out "significant" by chance alone. And the same two levers we saw for the paired test still apply here, giving `group_A` and `group_B` different means will pull power up, and so will increasing n. Try editing the code above to see it for yourself.

6.7 Chapter Summary

Why it matters. The t-test is the first real inferential test in this course, and the logic it runs on, a signal measured against the noise around it, is the same logic behind ANOVA, regression, and everything else that follows. Getting comfortable here pays off for the rest of the semester.

Core ideas

Figure 6.15: 1000 ps for the independent-samples null, n=10 per group

- **Signal over noise.** A difference is only impressive relative to the variability around it. Every t statistic is a ratio: the effect you observed on top, an estimate of how much that effect could bounce around by chance on the bottom.
- **Three forms, one idea.** The one-sample, paired, and independent-samples t-tests differ in what goes into the numerator, but all three ask whether an observed difference is large compared to its own standard error. The paired test is just the one-sample test applied to difference scores.
- **Why t and not z.** When the population SD σ is unknown, we estimate it with s, which adds uncertainty. The t distribution accounts for that by being fatter in the tails than the normal, and it tightens toward the normal as degrees of freedom grow.
- **Assumptions come in two kinds.** *Independence and a sound study design* apply to every test in this book and are assumed from here on. The *test-specific* assumption for these tests is approximate normality, of the outcome for a one-sample test or of the difference scores for a paired test.
- **How worried to be depends on n.** Thanks to the Central Limit Theorem, t-tests are fairly robust to non-normality once samples are reasonably large. The danger zone is small *and* skewed, which describes a lot of environmental field data. Plot your data first; treat a Shapiro-Wilk result as evidence rather than a gate, since it has least power exactly when you need it most.
- **Choose your tails before you look.** Default to two-tailed. Reserve one-tailed tests for cases where theory rules out the opposite direction, and decide before seeing the result.

6.8 Videos

6.8.1 One or Two tailed tests

7 When Assumptions Fail: Nonparametric and Categorical Data

7.1 Parametric vs. nonparametric tests

Chapter 6 covered three flavors of the t-test (one-sample, paired, and independent-samples, also called two-sample). All three need continuous, roughly normal data; the independent-samples test additionally assumes similar variance across the two groups. This chapter covers what to do when those assumptions aren't met. We'll also cover what to do if your data aren't continuous at all.

So far, we've been looking at **parametric** tests, without calling them that. A test is parametric when we can only know the sampling distribution of its test statistic, the shape we compare our result against to get a p-value, because we assumed something specific about the shape of the **population** the data came from. For the t-test, that assumption is normality: the t-distribution isn't the shape of your raw data, it's the shape the t-statistic takes across repeated samples, and we can only derive that shape because we assumed the population itself was normal (or leaned on the Central Limit Theorem to get there for a large enough sample). Nonparametric tests, as you'll see below, still end up with a known distribution for their own test statistics; they just don't need to assume anything about the parent population's shape to get there.

When the distributional assumptions are met, parametric tests are generally the most powerful available. However, when distributional assumptions are not met, parametric tests can become **unreliable**, and it may be difficult to understand how they become unreliable without extensive simulations. The worse the departures from assumed distributions, the worse the errors that may result.

Nonparametric tests make no assumption about the shape of the population's distribution. Nonparametric does **not** mean assumption-free. But they usually have assumptions that are easier to satisfy than those of parametric tests.

Many nonparametric tests are run on the ranks of the data (first, second, third … last), rather than on the raw data. For example, the median - it's a statistic based on ranks. It represents the middle rank in a data set.

We'll talk about two different kinds of tools today:

- **Nonparametric alternatives** (to t-tests). Your outcome is still continuous, but it's skewed, has outliers, or your sample is too small to trust normality. The **Wilcoxon signed-rank test** and **Mann-Whitney-Wilcoxon test** answer the same questions as the one-sample/paired and independent-samples *t*-tests, respectively, without assuming normality.
- **Categorical data.** Your outcome isn't a number at all: a site either has an invasive species or it doesn't, a sample either exceeds a threshold or it doesn't. None of the tests above apply here; you need the **chi-square (χ^2) tests**.

We'll state the assumptions explicitly for each test as we go through them.

A practical bonus of rank-based tests: ordinal data. Because the Wilcoxon and Mann-Whitney-Wilcoxon tests only need a rank ordering, neither is limited to interval- or ratio-scale measurements the way a *t*-test is. A *t*-test needs a mean, which requires numbers where the *distance* between values means something. Plenty of environmental data doesn't work that way: a water-quality rating (poor, fair, good, excellent), a disturbance-severity class, a Likert-style survey response. You can put those in order, but "the average disturbance class was 2.4" isn't meaningful the way "the average pH was 6.8" is. These rank-based tests only need the ordering, which ordinal data already has, so they apply directly where a *t*-test can't.

7.2 Nonparametric alternative to one-sample & paired t-test: the Wilcoxon signed-rank test

Chapter 6 built the paired *t*-test around this same idea, wetland restoration and soil moisture, with a full 20-plot dataset. Let's run a smaller version of that same design, and this time actually check whether we should trust the result before reporting it.

\begin{table}

\caption{Soil moisture (%) before and after restoration, 12 plots}

Table 7.1: Paired t-test: soil moisture before vs. after restoration

Mean difference	t	df	p-value
2.583333	2.57	11	0.026

Table 7.2: Shapiro-Wilk normality test on the differences

Statistic	p-value
0.85	0.039

Plot	Before	After	Difference
1	20	26	6
2	15	21	6
3	22	21	-1
4	18	22	4
5	16	23	7
6	24	23	-1
7	19	18	-1
8	14	22	8
9	21	21	0
10	17	17	0
11	25	25	0
12	20	23	3

\end{table}

Step 1: run the paired t-test, the way Chapter 6 taught it.

$t(11) = 2.57$, $p = .026$. Significant! Restoration looks like it increased soil moisture.

Step 2: wait, did we check assumptions first? Let's look at the differences before trusting that p-value.

The Shapiro-Wilk test agrees with the picture: $p = .039$, a real violation, not a hypothetical one. Three plots didn't change at all, three dropped slightly, and the rest jumped up by quite a lot; that's not symmetric, let alone normal. The t-test's "significant" result was built on an assumption the data don't meet. Time for the **Wilcoxon signed-rank test**.

The question: did soil moisture systematically change after restoration, or is any change we see just plot-to-plot noise?

- H_0: no systematic shift; the differences are symmetric around a median of 0 (positive and negative ranks are balanced).
- H_A: there is a systematic shift; the median difference is not 0.

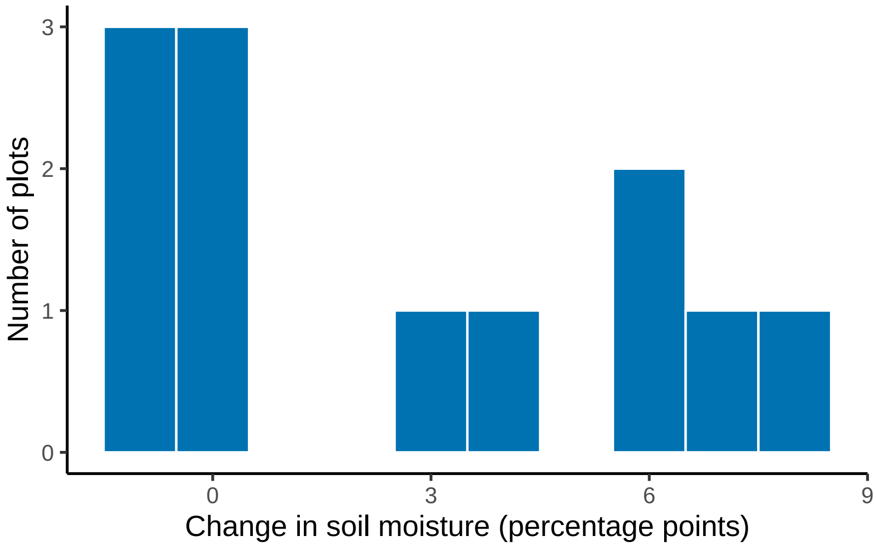

Figure 7.1: Change in soil moisture per plot. Three plots showed no change at all, and the positive changes stretch out much further than the negative ones: not the symmetric, bell-shaped pattern a paired t-test assumes.

Table 7.3: Wilcoxon signed-rank test on restoration soil-moisture differences

Statistic (V)	p-value	Method	Tails
39	0.057	Wilcoxon signed rank test with continuity correction	two.sided

How it works, conceptually. Instead of using the actual sizes of the differences, the test converts them to ranks: it ranks the *absolute* differences from smallest to largest, then checks whether the positive-signed ranks and negative-signed ranks are roughly balanced. Balanced ranks are the null case: it would mean a plot's rank (how extreme its change was) has nothing to do with whether that change was an increase or a decrease. If soil moisture really increased at most plots, the larger ranks should belong mostly to positive differences instead. Because it works with ranks rather than raw values, one wildly extreme plot can't dominate the result the way it could with a mean.

Its assumption. The Wilcoxon signed-rank test does not require normality, but it assumes the differences come from a **symmetric distribution around their median**. That's a milder assumption than normality (symmetric-but-not-normal distributions are common), but our histogram above is pushing it: the positive side stretches to 8, the negative side barely reaches -1. We'll run the test anyway, since it's still the better-justified choice here, and note the strain honestly when we interpret the result.

"Two.sided" in the Tails column just means we tested for a shift in either direction, the same one-vs-two-tailed distinction from Chapter 6.

Table 7.4: Ranking the nonzero restoration differences to calculate V

Difference	Abs. difference	Rank	Sign
6	6	6.5	+
6	6	6.5	+
-1	1	2.0	-
4	4	5.0	+
7	7	8.0	+
-1	1	2.0	-
-1	1	2.0	-
8	8	9.0	+
3	3	4.0	+

R falls back to a normal approximation twice over here: once for the tied differences (two plots at $+6$, three at -1), and once because three plots showed *no* change at all. Zero differences can't be ranked as positive or negative, so the Wilcoxon test quietly drops them and runs on the remaining 9 plots instead of all 12. That's worth thinking about: the rank test isn't just less sensitive to magnitude than the t-test, it can end up with less data entirely.

How V is actually calculated. Drop the zero differences, take the absolute value of what's left, rank those absolute values from smallest to largest (tied absolute values share the average of their ranks), then sum the ranks belonging to the *positive* differences. That sum is V.

V is the sum of the ranks where Sign is "+": $6.5 + 6.5 + 5 + 8 + 9 + 4 = 39$, matching the table R gave us above.

How the reference distribution is actually built. Under H_0, each plot's rank has an independent 50/50 chance of ending up positive or negative, purely by chance. That gives you 2^n equally likely sign patterns for n nonzero differences (here, $n = 9$, not 12, since the zeros already dropped out), and summing the positive ranks for every one of those patterns builds a discrete probability distribution, not just a raw count. It has a known mean and variance:

$$E[V] = \frac{n(n+1)}{4}, \qquad \mathrm{Var}(V) = \frac{n(n+1)(2n+1)}{24}$$

Plugging in our $n = 9$ nonzero plots:

$$E[V] = \frac{9 \times 10}{4} = 22.5, \qquad \mathrm{Var}(V) = \frac{9 \times 10 \times 19}{24} = 71.25$$

That's an SD of about 8.4. Our observed $V = 39$ sits about 1.96 SDs above that expectation ($\frac{39-22.5}{8.4} \approx 1.96$), which is where R's $p = .057$ comes from: a normal approximation to this exact distribution (with a ties and zeros adjustment). For small, tie-free samples with no zeros, R can instead read the exact enumerated distribution directly.

Where do the critical values actually come from? The same place they always do in this book: we build the null distribution ourselves, by simulating 20,000 experiments where each of the 9 nonzero ranks independently gets a random + or − sign, exactly what H_0 says should happen, and adding up the positive ranks each time.

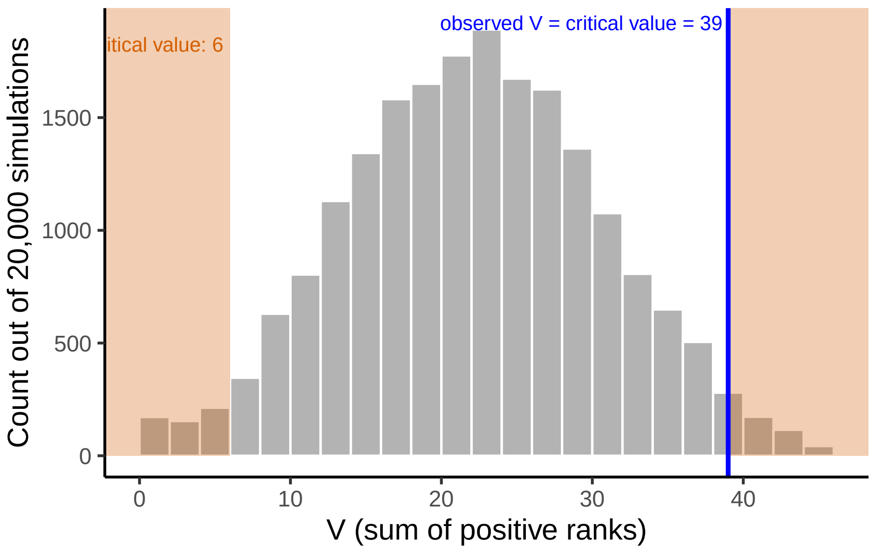

Figure 7.2: Simulated null distribution of the Wilcoxon signed-rank statistic V, built from 20,000 random sign-flips of the 9 nonzero ranked differences. Shaded regions mark the two-tailed 5% rejection region; our observed V = 39 lands right on the boundary.

Our observed $V = 39$ isn't just close to the critical value; it *is* the critical value, 39, the exact boundary where the two-tailed 5% rejection region begins. That's why $p = .057$ sits so close to $\alpha = .05$ without crossing it. Not significant, but only barely.

So far: t-test $p = .026$ (significant), Wilcoxon $p = .057$ (not quite). Same 12 plots, two different verdicts, because the t-test's numerator and denominator both used the exact magnitudes the Wilcoxon test deliberately ignores.

The sign test. There's an even simpler nonparametric option, the **sign test**, which looks only at whether each difference is positive or negative. It discards the magnitude entirely and asks whether pluses outnumber minuses more than chance would suggest. It has essentially no distributional assumption beyond the universal one (independence of observations, from a sound study design; see Chapter 6), not even symmetry. That's

Table 7.5: Sign test on restoration soil-moisture differences

Plots increased	Plots total	p-value
6	9	0.508

Table 7.6: Plant species richness at restored and unrestored wetland plots

Plot	Restored	Unrestored
1	21	17
2	16	18
3	22	18
4	16	12
5	17	11
6	16	11
7	17	11
8	34	10

genuinely useful here: we already flagged the symmetry assumption as shaky for this data (the positive side stretching to 8, the negative side barely reaching -1), so the sign test sidesteps the one assumption the Wilcoxon test still needed. That robustness comes at a big cost in power, though, so it's usually a last resort.

Six of the 9 nonzero plots increased. $p = .508$: nowhere close to significant.

The full picture. t-test: $p = .026$, significant. Wilcoxon signed-rank: $p = .057$, not quite. Sign test: $p = .508$, not close. Same 12 plots, three tests, three different verdicts. Each test gives up a little more than the last one, the t-test uses the full magnitude of every difference, Wilcoxon keeps the ranks but drops the zeros, and the sign test keeps only direction, and the weaker the information, the weaker the evidence.

That's not an argument that nonparametric tests are worse. The t-test's answer looked strongest here specifically because it leaned hardest on the assumption that turned out not to hold. Check your assumptions before you report a p-value, not after you've already seen one you like.

7.3 Nonparametric alternative to the independent-samples t-test: the Mann-Whitney-Wilcoxon test

Now let's run through the same kind of example for the independent-samples case. Researchers surveyed **plant species richness** (the number of distinct plant species found) at 8 restored wetland plots and 8 unrestored reference plots, to see whether restoration increased plant diversity.

Table 7.7: Independent-samples t-test: restored vs. unrestored richness

Mean difference	t	df	p-value
6.375	2.54	14	0.024

Table 7.8: Shapiro-Wilk normality test, by group

Group	Statistic	p-value
Restored	0.70	0.002
Unrestored	0.77	0.015

Step 1: run the t-test, the way Chapter 6 taught it.

$t(14) = 2.54$, $p = .024$. Significant! Restoration looks like it increased richness.

Step 2: wait, did we check assumptions first? We didn't, and Chapter 6 was explicit that you're supposed to. Let's actually look at the data before trusting that p-value.

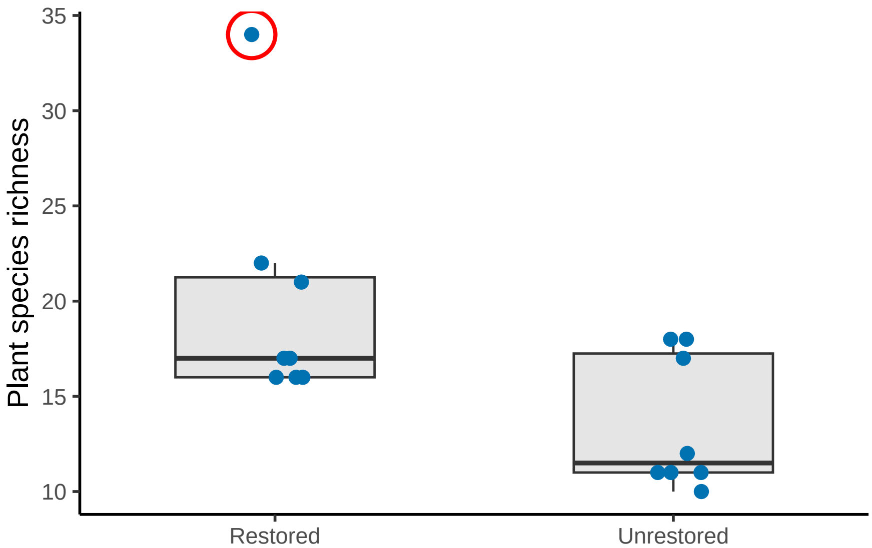

Figure 7.3: Plant species richness by group. The circled restored plot has an unusually high richness count, visibly pulling that group's distribution away from a normal, bell-shaped one.

The circled plot has 34 species, nearly double any other plot in either group. We don't just throw away outliers; maybe that plot sits in an unusually favorable microhabitat. Either way, it's a real violation of the normality assumption, not a hypothetical one:

Both groups fail the Shapiro-Wilk test, the restored group badly ($p = .002$), driven almost entirely by that one plot. The t-test's "significant" result was built on top of an assumption

Table 7.9: Mann-Whitney-Wilcoxon test: restored vs. unrestored richness

Statistic (U)	p-value	Method	Tails
50	0.063	Wilcoxon rank sum test with continuity correction	two.sided

the data don't actually meet. This is exactly the situation the **Mann-Whitney-Wilcoxon test** was built for.

The question: did restoration increase plant species richness, or is the difference we see consistent with random plot-to-plot variation?

- H_0: no difference in richness between restored and unrestored plots; ranks are well-mixed between the groups.
- H_A: richness tends to rank systematically higher in one group than the other.

How it works, conceptually. Pool all the observations from both groups together, rank them from smallest to largest, then compare the sum of ranks in Group A to the sum of ranks in Group B. If the groups truly don't differ, ranks should be well mixed between them; if one group tends to have higher values, its ranks should be systematically higher too. Critically, that outlier plot still gets *a* rank, but only one rank, the highest one, no matter how far above the pack it sits. Its magnitude, which is what distorted the *t*-test, can't distort the Mann-Whitney-Wilcoxon test the same way.

Its assumption. The Mann-Whitney-Wilcoxon test assumes both groups' parent distributions have the **same shape**: it does *not* require normality, and the assumption is **not** "equal variance," even though same-shape distributions happen to have equal variance as a side effect. If the two groups have differently-shaped distributions (say, one skewed and one symmetric), the test still runs, but interpreting a significant result purely as "the medians differ" becomes murkier, since it could be a difference in location, shape, or both. Our two groups are both right-skewed by roughly the same amount, so that's a reasonable assumption here.

R falls back to the normal approximation again, because both groups share some tied richness values (two plots at 16, two at 11, and so on).

How the reference distribution is actually built. Same idea as the Wilcoxon signed-rank case above, just built from group membership instead of signs. Under H_0, both groups are draws from the same population, so once you pool and rank all $N = n_A + n_B = 16$ values, which ranks end up in the restored group is just a matter of chance: any of the $\binom{16}{8}$ ways of splitting those 16 ranks into two groups of 8 is equally likely. That's a real, discrete probability distribution, and its mean and variance are known:

$$E[U] = \frac{n_A n_B}{2}, \qquad \mathrm{Var}(U) = \frac{n_A n_B(n_A + n_B + 1)}{12}$$

Plugging in our two groups of 8:

$$E[U] = \frac{8 \times 8}{2} = 32, \qquad \mathrm{Var}(U) = \frac{8 \times 8 \times 17}{12} \approx 90.7$$

That's an SD of about 9.5. Our observed $U = 50$ sits about 1.9 SDs above that expectation ($\frac{50-32}{9.5} \approx 1.9$), which is where R's $p = .063$ comes from: a normal approximation to this exact distribution (with a small ties adjustment), same mechanism as the Wilcoxon case. For small samples without ties, R can instead read the exact permutation distribution directly, no approximation needed.

Same question as before: where do the critical values come from? We build them the same way, by simulation. Under H_0, group membership is random with respect to rank, so we pool the richness values, then repeatedly shuffle which 8 ranks land in the restored group:

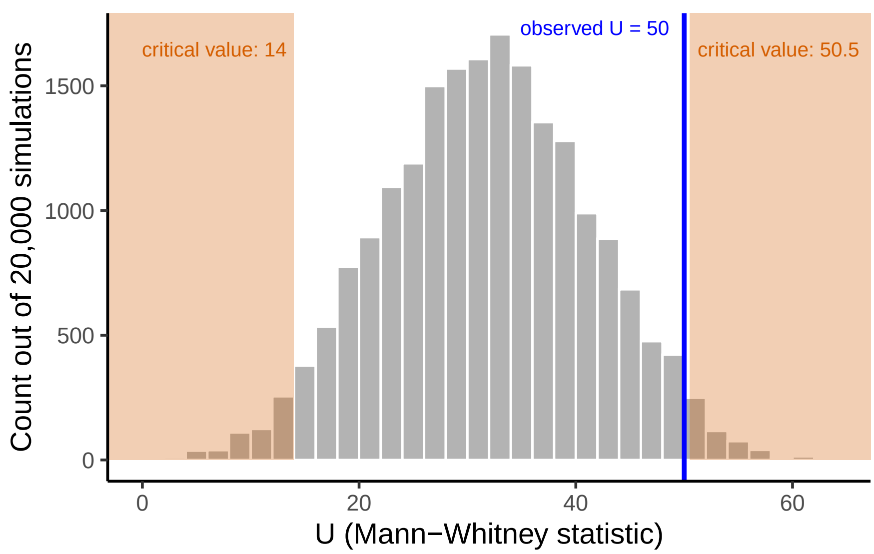

Figure 7.4: Simulated null distribution of the Mann-Whitney U statistic, built from 20,000 random reshuffles of group membership among the pooled, ranked richness values. Shaded regions mark the two-tailed 5% rejection region; our observed U = 50 falls just inside the boundary.

Our observed $U = 50$ lands just inside the critical value of 50.5, matching R's $p = .063$: close to significant, but not quite.

The full picture. The t-test said $p = .024$: significant, restoration works. The Mann-Whitney-Wilcoxon test on the exact same data says $p = .063$: not significant at $\alpha = .05$, right on the boundary. Same 16 plots, same question, two different answers. One outlier plot was doing a lot of the work behind that "significant" t-test result, and that's exactly the kind of influence the normality assumption is supposed to rule out. That's not a reason to always prefer the nonparametric test; if the data really had been roughly normal, the t-test's extra power would have been the right tool. It's why you check assumptions *before* reporting a p-value, not after.

7.4 Categorical data: the chi-square tests

Every test so far (the t-tests in Chapter 6, and the Wilcoxon and Mann-Whitney-Wilcoxon tests above) needs a **continuous** outcome variable. But a lot of environmental data isn't continuous at all: a site either has an invasive species or it doesn't; a stream sample either exceeds a safety threshold or it doesn't; a plot falls into one land-cover category, not a number line. For **categorical** data, none of the tests above apply: you need a different tool entirely, the **chi-square (χ^2) test**.

Chapter 3 introduced the shape of the chi-square distribution: always non-negative, right-skewed, naturally one-tailed, with its exact shape depending on degrees of freedom. Here's what that rejection region looks like in practice, using the goodness-of-fit example below (one of the two chi-square tests you'll meet in this section; more on what "goodness-of-fit" means right before we run it), which has $df = 3$. The critical value at $\alpha = .05$ is 7.81; any statistic beyond that point (the shaded region) occurs less than 5% of the time by chance.

If you're wondering how a "nonparametric" test can still have its own named distribution, that's worth clearing up. "Nonparametric" is an assumption about your **data**, not the test statistic: you don't need to assume the data came from a specific family like the normal distribution. Every test still needs a known reference distribution for its **test statistic** to produce a p-value; what differs is how that reference distribution gets built. The chi-square distribution here, like the Wilcoxon and Mann-Whitney-Wilcoxon distributions earlier in this chapter, is built without any assumption about the data's shape: chi-square from squared deviations under H_0, the rank tests from all possible rank arrangements. That's what "distribution-free" actually means: the statistic's distribution is known, but building it never required assuming anything about the data.

There are two versions of the chi-square test, depending on the question you're asking.

7.4.1 Chi-square goodness-of-fit: does one categorical variable match an expected pattern?

The name comes straight from what the test does: it checks how well ("goodness") your observed counts match ("fit") a specific pattern you'd expect, whether that expectation comes from a historical baseline, a theory, or an assumption of "equally likely." A small

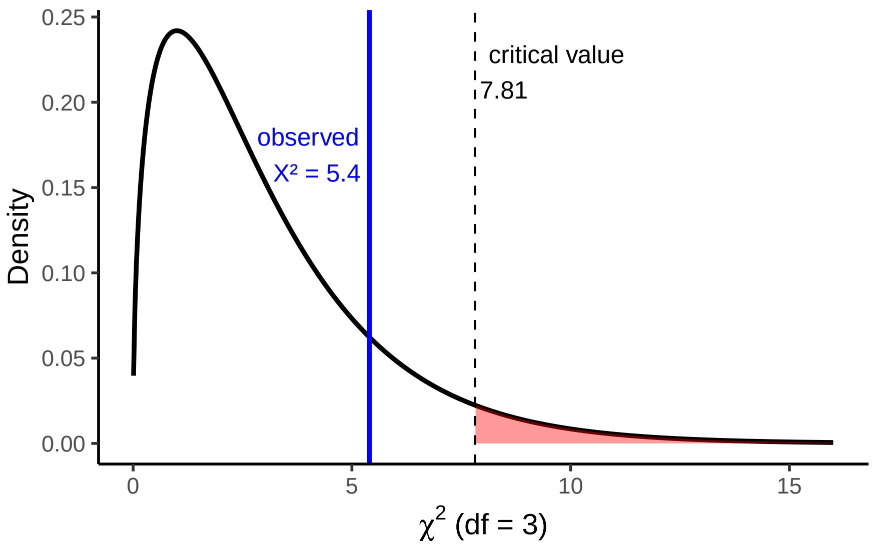

Figure 7.5: The chi-square distribution with df = 3. The shaded region marks the rejection region at alpha = .05 (critical value 7.81). Our goodness-of-fit statistic, X-squared = 5.4, falls short of it.

Table 7.10: Observed fish counts by lake zone (n = 200)

Zone	Observed
A	38
B	61
C	52
D	49

chi-square statistic means a good fit (observed counts close to expected); a large one means a poor fit (observed counts far from expected).

A state wildlife agency has historically found fish evenly distributed across four zones of a lake: 25% of fish in each zone. This year's survey of 200 fish gives a different picture:

The question: does this year's distribution across zones differ from the historical 25%-each pattern, or is this the kind of unevenness you'd expect from random sampling alone?

- H_0: fish are still distributed 25% per zone; the pattern matches the historical baseline.
- H_A: fish are not evenly distributed across zones; the pattern has changed.

Assumption. The goodness-of-fit test doesn't assume anything about the shape of your data (categories aren't numbers, so there's no shape to assume). It does have a practical requirement instead: **expected counts should generally be 5 or more** in each category.

Table 7.11: Chi-square goodness-of-fit test

Statistic (^{2})	df	p-value	Method
5.4	3	0.145	Chi-squared test for given probabilities

Table 7.12: Chi-square contribution by zone

Zone	Observed	Expected	Contribution
A	38	50	2.88
B	61	50	2.42
C	52	50	0.08
D	49	50	0.02

That "5" is a conventional rule of thumb, not something calculated from this data or compared to a critical value; it's a check on whether the chi-square curve is a trustworthy approximation, done *before* the test itself runs. Here, expected count per zone is $200 \times 0.25 = 50$ (total fish times the 25%-per-zone expectation under H_0), comfortably above that minimum. When expected counts are too small, the chi-square approximation breaks down and a different method (like Fisher's exact test) is more appropriate.

Where χ^2 actually comes from. For each category, take the squared difference between observed and expected, scaled by expected: $\chi^2 = \sum \frac{(O-E)^2}{E}$.

Summing the last column: $2.88 + 2.42 + 0.08 + 0.02 = 5.4$, matching R's statistic above. With $df = 3$, the critical value at $\alpha = .05$ is 7.81 (the same one marked on the distribution figure earlier in this chapter); our 5.4 falls well short of it.

Interpretation. $\chi^2(3) = 5.4$, $p = .145$: not significant. This year's zone counts are consistent with the historical 25%-each pattern; the unevenness we see is within the range chance alone would produce.

7.4.2 Chi-square test of independence: are two categorical variables related?

Now suppose researchers classify 150 stream sites by **land use** (Forested, Agricultural, Urban) and whether each site's *E. coli* level **exceeds** a recreational safety threshold:

Table 7.13: Stream sites by land use and E. coli exceedance

	No	Yes
Forested	42	8
Agricultural	25	25
Urban	20	30

Table 7.14: Chi-square test of independence

Statistic (^{2})	df	p-value	Method
21.84	2	$< .001$	Pearson's Chi-squared test

Table 7.15: Expected counts under H0 (independence)

	No	Yes
Forested	29	21
Agricultural	29	21
Urban	29	21

The question: is *E. coli* exceedance **independent** of land use, or are they related?

- H_0: land use and exceedance are independent; knowing one tells you nothing about the other.
- H_A: land use and exceedance are related; they are not independent.

Assumption. Same rule as goodness-of-fit: expected counts should generally be 5 or more in every cell of the table. Unlike the goodness-of-fit test, where expected counts came from a stated proportion (25% per zone), here they come from the table's own row and column totals: the expected count for a cell is $\frac{\text{row total}\times\text{column total}}{\text{grand total}}$, the count you'd see in that cell if land use and exceedance really were unrelated. For Forested/No, that's $\frac{50\times 87}{150} = 29$ (50 forested sites total, 87 sites total with no exceedance, 150 sites overall).

All six expected counts are between 21 and 29, well above the 5-count minimum from the assumption check above; that's a separate question from whether the test is significant, just whether the chi-square approximation is trustworthy here.

Where χ^2 actually comes from. Same formula as the goodness-of-fit test, applied cell by cell instead of category by category: $\chi^2 = \sum \frac{(O-E)^2}{E}$, using the observed table and the expected table above.

Summing all six cells: $5.83 + 8.05 + 0.55 + 0.76 + 2.79 + 3.86 = 21.84$, matching R's statistic above. With $df = 2$, the critical value at $\alpha = .05$ is 5.99; our 21.84 is more than three times that, which is where $p < .001$ comes from.

Table 7.16: Chi-square contribution by cell

	No	Yes
Forested	5.83	8.05
Agricultural	0.55	0.76
Urban	2.79	3.86

Interpretation. $\chi^2(2) = 21.84$, $p < .001$: significant. *E. coli* exceedance is not independent of land use: forested sites exceed the threshold far less often (8 of 50) than agricultural (25 of 50) or urban (30 of 50) sites. Note what this test does *not* tell you: it says the two variables are related, but, just like correlation, it doesn't by itself establish that land use *causes* the exceedance.

7.5 Chapter Summary

Why it matters. Real environmental data often breaks the rules a t-test needs: small samples, skewed distributions, or outcomes that are counts and categories rather than measurements. This chapter is the toolkit for those cases, and for recognizing when you are in one.

Core ideas

- **Rank-based tests trade power for weaker assumptions.** The **Wilcoxon signed-rank** and **Mann-Whitney-Wilcoxon** tests ask the same questions as the paired and independent-samples t-tests, but work from ranks instead of raw magnitudes. That buys a milder distributional assumption at the cost of some sensitivity.
- **The trade-off is a spectrum.** The **sign test** goes further still, using only the direction of each difference. On the same restoration data, the t-test, Wilcoxon, and sign test gave progressively weaker verdicts, because each one throws away more information than the last.
- **Nonparametric does not mean assumption-free.** Wilcoxon assumes the differences are symmetric about their median. Mann-Whitney-Wilcoxon assumes the two distributions have the same *shape*, which is not the same as equal variance. State the assumption you are actually making.
- **Check assumptions before you trust a p-value, not after.** Both worked examples in this chapter ran the parametric test first, then went back and found the violation. That order is deliberate, because it is the mistake most people actually make.
- **Categorical data needs a different tool entirely.** The **chi-square tests** handle counts: goodness-of-fit asks whether one variable's counts match an expected pattern, and the test of independence asks whether two categorical variables are related.
- **The testing logic never changes.** State H_0 and H_A, compute a statistic, and ask how surprising that statistic would be if H_0 were true. Only the statistic and its reference distribution differ from test to test.

8 ANOVA

A fun bit of stats history (Salsburg 2001). Sir Ronald Fisher invented the ANOVA, which we learn about in this section. He wanted to publish his new test in the journal Biometrika. The editor at the time was Karl Pearson (remember Pearson's r for correlation?). Pearson and Fisher were apparently not on good terms, they didn't like each other. Pearson refused to publish Fisher's new test. So, Fisher eventually published his work in the Journal of Agricultural Science. Funnily enough, the feud continued onto the next generation. Years after Fisher published his ANOVA, Karl Pearson's son Egon Pearson, and Jersey Neyman revamped Fisher's ideas, and re-cast them into what is commonly known as null vs. alternative hypothesis testing. Fisher didn't like this very much.

We present the ANOVA in the Fisherian sense, and at the end describe the Neyman-Pearson approach that invokes the concept of null vs. alternative hypotheses.

8.1 ANOVA is Analysis of Variance

ANOVA stands for Analysis Of Variance. It tests whether differences among several group means are larger than we'd expect from chance alone.

With two groups, a t-test already answers that question. ANOVA extends the same logic to **three or more** groups, which comes up constantly in environmental science: multiple sites, multiple treatments, multiple time periods.

8.1.1 Why not just run multiple t-tests?

Suppose we're comparing three restoration treatments for vegetation cover: **compost**, **mulch**, and a bare-soil **control**. With three groups, there are three possible pairwise comparisons: compost vs. mulch, compost vs. control, and mulch vs. control.

We could run a separate t-test for each. The problem is that every test carries its own 5% chance of a false positive, and those chances accumulate across tests. If we run m independent tests at $\alpha = .05$, the probability of at least one false positive across all of them is

$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^m.$$

With three groups, that's three tests and a familywise error rate of 14%, not 5%:

$$1 - (1 - .05)^3 = .14.$$

Number of groups (k)	Pairwise tests (m)	Familywise error rate
3	3	$1 - .95^3 = .14$
4	6	$1 - .95^6 = .26$
5	10	$1 - .95^{10} = .40$

Even with only five groups, there's a 40% chance of turning up at least one "significant" difference somewhere purely by chance, even if every population mean is identical. Figure 8.1 shows this inflation directly: as the number of independent t-tests grows, so does the chance that at least one is a false positive.

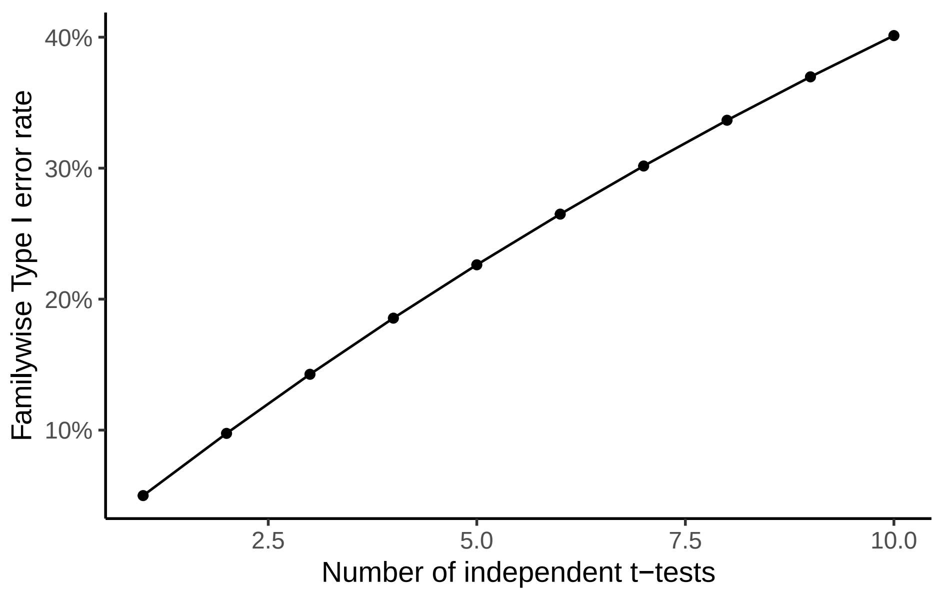

Figure 8.1: Familywise Type I error rate as the number of independent t-tests increases, at alpha = .05 per test.

ANOVA solves this by replacing many separate t-tests with **one** test that controls the overall error rate. It asks a single question:

$$H_0 : \mu_1 = \mu_2 = \mu_3 \qquad H_a : \text{at least one mean differs}$$

Just like the t-test, different research designs call for different kinds of ANOVA: a one-factor (between-subjects) ANOVA when every unit is measured once, and a repeated-measures ANOVA when the same units are measured under multiple conditions. We start with the one-factor case.

8.2 One-factor ANOVA

The one-factor ANOVA (also called a between-subjects, independent-factor, or one-way ANOVA) is the simplest version of the test: one independent variable with two or more levels, and every unit measured once.

With exactly two groups, a t-test and a one-factor ANOVA give the same answer: $t^2 = F$. Both compare the size of an effect to the amount of variation you'd expect from sampling error alone; the t-test expresses that comparison as t, and the ANOVA expresses the same underlying ratio as F (named for Fisher, its inventor).

$$F = \frac{\text{measure of effect}}{\text{measure of error}}$$

The difference is that F uses **variances** for both the numerator and the denominator, so F is a ratio of two variances: the variance you can explain (differences between group means) over the variance you can't (leftover noise within each group).

$$F = \frac{\text{Can explain}}{\text{Can't explain}}$$

When the two are about the same size, $F \approx 1$: group differences are no bigger than ordinary sampling noise. When explained variance is much larger than unexplained variance, F climbs well above 1, evidence that the groups really differ; when it's smaller, F drops below 1. An F of 5, for example, means the groups explain five times more variance than they leave unexplained, a substantial effect; an F of 0.6 means the opposite, mostly unexplained noise.

8.2.1 Computing the F-value

ANOVA splits the total variation in the data into two sources: variation caused by group membership (the manipulation), and variation left over as sampling error. If we can measure those two parts separately, we can turn them into a ratio and get an F-value.

$$\text{Total variation} = \text{Variation between groups} + \text{Variation within groups}$$

To make that precise we need an actual measure of variation. As in Chapter 1, that's the **sums of squares**:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}$$

8.2.2 SS Total

SS_{Total} captures all of the variation in a dataset: the difference between each score and the grand mean, squared and summed.

Let's imagine we had some data in three groups, A, B, and C. For example, we might have 3 scores in each group. The data could look like this:

groups	scores	diff	diff_squared
A	20	13	169
A	11	4	16
A	2	-5	25
B	6	-1	1
B	2	-5	25
B	7	0	0
C	2	-5	25
C	11	4	16
C	2	-5	25
Sums	63	0	302
Means	7	0	33.5555555555556

The data are in long format, so each row is a single score. The mean of all of the scores is the **Grand Mean**, calculated in the table as 7.

The `diff` column holds every score's distance from the grand mean; `diff_squared` squares those distances. Squaring matters because the raw differences sum to zero (the mean is the balancing point of the data), so working with them directly wouldn't tell us anything. Squaring turns every deviation positive and, as a side effect, weights larger deviations more heavily.

Adding up all of the squared deviations gives the sums of squares, which is where the name comes from. That's the first piece of our answer:

$$SS_{\text{Total}} = SS_{\text{Between}} + SS_{\text{Within}}, \qquad 302 = SS_{\text{Between}} + SS_{\text{Within}}.$$

We only need to compute one of the two remaining pieces directly; the other follows by subtraction.

8.2.3 SS Between

SS_{Total} captured all of the variation in the data. SS_{Between} isolates the piece of that variation caused by group membership: how far each group's mean sits from the grand mean.

To find it, replace every score with its own group's mean, then square and sum each group mean's distance from the grand mean.

groups	scores	means	diff	diff_squared
A	20	11	4	16
A	11	11	4	16
A	2	11	4	16
B	6	5	-2	4
B	2	5	-2	4
B	7	5	-2	4
C	2	5	-2	4
C	11	5	-2	4
C	2	5	-2	4
Sums	63	63	0	72
Means	7	7	0	8

Notice the new `means` column: the mean for group A was 11, so there are three 11s, one per observation in row A; groups B and C both average to 5, so the rest of the column is 5s.

Each score has been replaced by its own group's mean. From the mean's point of view, every score in the group looks identical: a mean has no memory of how far off it was from any individual data point, only that everything in the group is centered on it.

Finding the differences between each of those group-mean values and the grand mean, squaring them, and summing gives $SS_{\text{Between}} = 72$: the amount of variation caused by differences between the group means, i.e., the variation we can explain by group membership.

SS_{Between} can never be larger than SS_{Total}, and indeed 72 is smaller than 302.

8.2.4 SS Within

We already have $SS_{\text{Total}} = 302$ and $SS_{\text{Between}} = 72$, so we can solve for SS_{Within} by subtraction:

$$SS_{\text{Within}} = SS_{\text{Total}} - SS_{\text{Between}} = 302 - 72 = 230.$$

It's worth calculating SS_{Within} directly from the data too, both to see where it comes from and to double-check that the subtraction was right.

groups	scores	means	diff	diff_squared
A	20	11	-9	81
A	11	11	0	0
A	2	11	9	81
B	6	5	-1	1
B	2	5	3	9
B	7	5	-2	4
C	2	5	3	9
C	11	5	-6	36
C	2	5	3	9
Sums	63	63	0	230
Means	7	7	0	25.5555555555556

This time, for each score we found the group mean first, then found the leftover error in that estimate: the `diff` column is the difference between each score and its own group mean, and `diff_squared` squares those differences. Summing them gives another sum of squares, SS_{Within}: the deviations the group means can't explain.

8.2.5 Degrees of freedom

Degrees of freedom work the same way here as for the t-test: every time we estimate something from the data, we use up a degree of freedom.

$df_{\text{Between}} = \text{groups} - 1$. With 3 groups, $df_{\text{Between}} = 3 - 1 = 2$: once we fix the grand mean, two of the three group means are free to be anything, but the third is determined by the other two.

$df_{\text{Within}} = \text{scores} - \text{groups}$. With 9 scores and 3 groups, $df_{\text{Within}} = 9 - 3 = 6$: estimating each of the 3 group means uses up 3 degrees of freedom, so only 6 of the 9 individual deviations are free to vary.

8.2.6 Mean Squares

We're building toward $F = \frac{\text{measure of effect}}{\text{measure of error}}$, and it's tempting to just divide SS_{Between} by SS_{Within} directly. The trouble is that SS_{Within} sums over all 9 individual scores while SS_{Between} sums over only 3 group means, so SS_{Within} is almost always the much larger number, and the raw ratio wouldn't be interpretable the way t or z are.

The fix is to convert each sum of squares into an average, or **mean square**, by dividing by its degrees of freedom (rather than its raw count, since these are estimated quantities):

$$MS_{\text{Between}} = \frac{SS_{\text{Between}}}{df_{\text{Between}}} = \frac{72}{2} = 36$$

$$MS_{\text{Within}} = \frac{SS_{\text{Within}}}{df_{\text{Within}}} = \frac{230}{6} = 38.33$$

Both are variances: MS_{Between} measures how much the group means vary, and MS_{Within} measures how much individual scores vary around their own group's mean.

8.2.7 Calculate F

With MS_{Between} and MS_{Within} in hand, F is their ratio:

$$F = \frac{MS_{\text{Between}}}{MS_{\text{Within}}} = \frac{36}{38.33} = 0.94$$

8.2.8 The ANOVA table

All of these pieces, the dfs, SSes, MSes, and finally F, are conveniently organized into an **ANOVA table**:

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
groups	2	72	36.00000	0.9391304	0.4417359
Residuals	6	230	38.33333	NA	NA

R labels the two rows `groups` (the between-groups row: what the means can explain) and `Residuals` (the within-groups row: what they can't). Other software uses slightly different labels for the same two rows. The table doesn't add new information beyond what we already computed; it just organizes it, with MS_{Between} (36) divided by MS_{Within} (38.33) giving the F-value.

8.3 What does F mean?

Every number in the ANOVA table so far was computed directly from the data, except one: the p-value. Its meaning is the same as for any other test we've covered: the probability of observing an F this large or larger if the null hypothesis (no real group differences) were true.

Since F is a sample statistic, we can find out how it behaves the same way we found out how t behaves: simulate it many times under the null and look at the resulting distribution.

To build intuition for the F distribution, consider a simulated monitoring study with three site types (A, B, C). Each site type has 10 sampling plots, for 30 total. We measure species abundance at each plot. Critically, the **null scenario** applies: all 30 plots are drawn from the same distribution (mean = 100, SD = 10), so site type has no real effect on abundance. Any differences between group means arise from sampling error alone.

We run this simulation 10,000 times. Each time we compute F from the sample data. The histogram of those 10,000 F-values, shown in Figure 8.2, is the null distribution of F for this design.

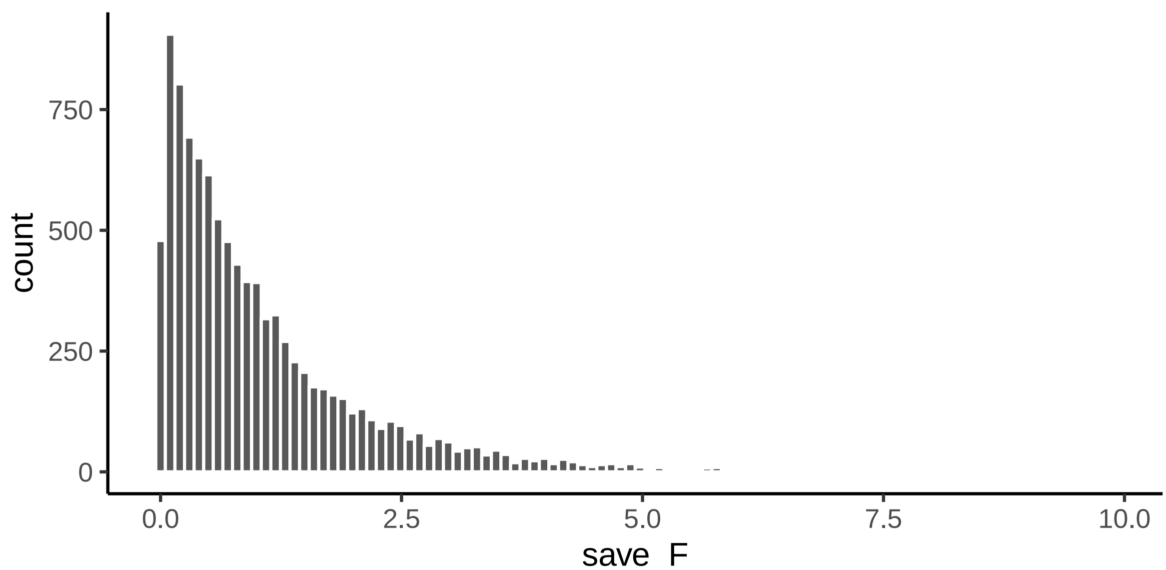

Figure 8.2: A simulation of 10,000 experiments from a null distribution where there is no differences. The histogram shows 10,000 F-values, one for each simulation. These are values that F can take in this situation. All of these F-values were produced by random sampling error.

A couple of things are worth noting about this F distribution. First, it never goes below 0: F is a ratio of two variances, and variances can't be negative, so neither can F. Second, it isn't normal or symmetric; it's right-skewed, and its exact shape depends on the numerator and denominator degrees of freedom.

Most simulated F-values here cluster below 1, which makes sense: all 30 plots were drawn from the same distribution, so on average the explained and unexplained variance should be about the same size. A smaller number of F-values reach up toward 5, an F that large would mean the groups explain five times more variance than they leave unexplained, which is a lot, and the histogram shows that values that large are rare. Every F in this histogram was produced purely by random sampling error, since we built the simulation with no real group differences at all.

8.3.1 Making decisions

We can use this null distribution of F to decide when a real experiment's result is too large to be explained by chance:

1. Set an alpha criterion of $p = 0.05$
2. Find the critical value of F for our design's numerator and denominator dfs.

Figure 8.3 marks that critical value on the histogram:

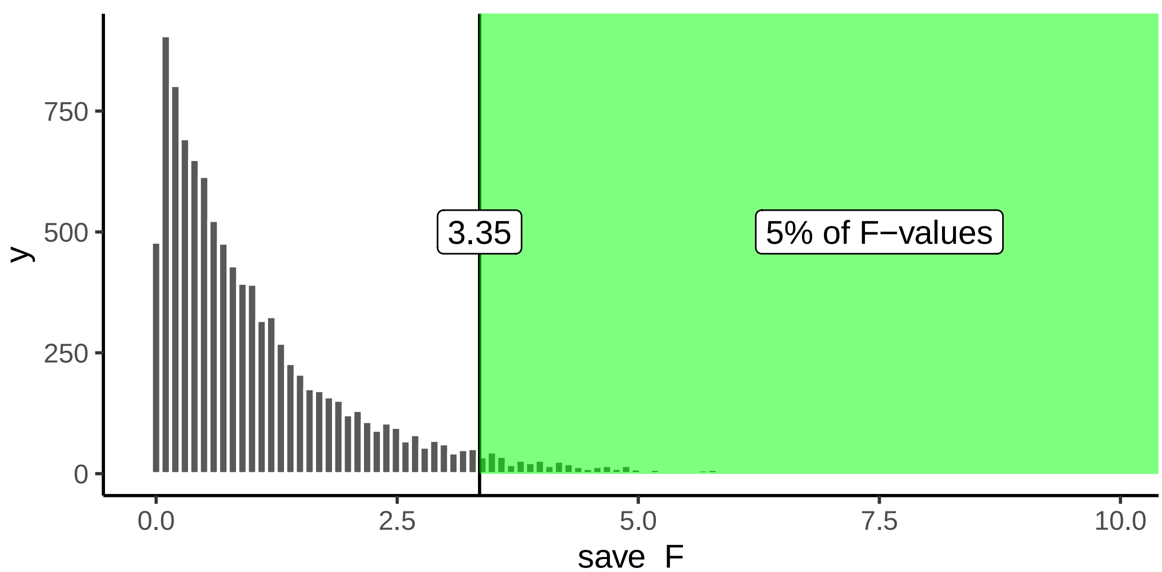

Figure 8.3: The critical value for F where 5% of all F-values lie beyond this point

Only 5% of F-values from this null distribution are 3.35 or larger. That gives us a decision rule: if we ran the same design for real (30 plots, 10 per site type) and site type genuinely affected abundance, we could treat $F \geq 3.35$ as evidence the differences aren't just chance.

An observed F of 3.4, for instance, would occur less than 5% of the time under the null, so seeing it would give us reasonably good grounds to conclude that site type actually mattered.

8.3.2 F and the means

An F-value and its p-value don't, by themselves, tell us what the group means actually were. If a study reports $F(2, 27) = 6.0$, $p = .001$, we know only that an F this large would happen by chance about 0.1% of the time, strong evidence against the null. We still don't know which means differed, or by how much, without looking at the group means directly (the ANOVA table itself doesn't report them).

That said, a large F with a small p-value does tell us one thing for certain: the group means **must** differ somewhere. If they didn't, the variance explained by group membership (the numerator of F) wouldn't be large enough to produce that result. We know a difference exists; identifying it requires a separate step.

8.3.2.1 ANOVA is an omnibus test

This is why ANOVA is called an **omnibus test**: omnibus meaning "comprising several items." It answers one overall question that stands in for several pairwise ones at once.

With three groups A, B, and C, there are three possible pairwise differences: A vs. B, B vs. C, and A vs. C. We could test each with a separate t-test, or ask the more general omnibus question with a single ANOVA:

Hypothesis of no differences anywhere: $A = B = C$

Any differences anywhere:

a. $A \neq B = C$
b. $A = B \neq C$
c. $A \neq C = B$

A small F with a large p-value means we don't reject the hypothesis of no differences: we have no evidence the three group means differ, and whatever gaps exist could easily be chance. A large F with a small p-value (below our alpha criterion) means we reject that hypothesis and conclude at least one group mean differs from the others. That's the omnibus test: it tells us *whether* the means differ, not yet *which* ones.

8.3.2.2 Looking at group means directly

The 10,000 simulated experiments above never once showed us the actual group means behind each F-value. Figure 8.4 shows the group means (with standard-error bars) from 10 of those null simulations.

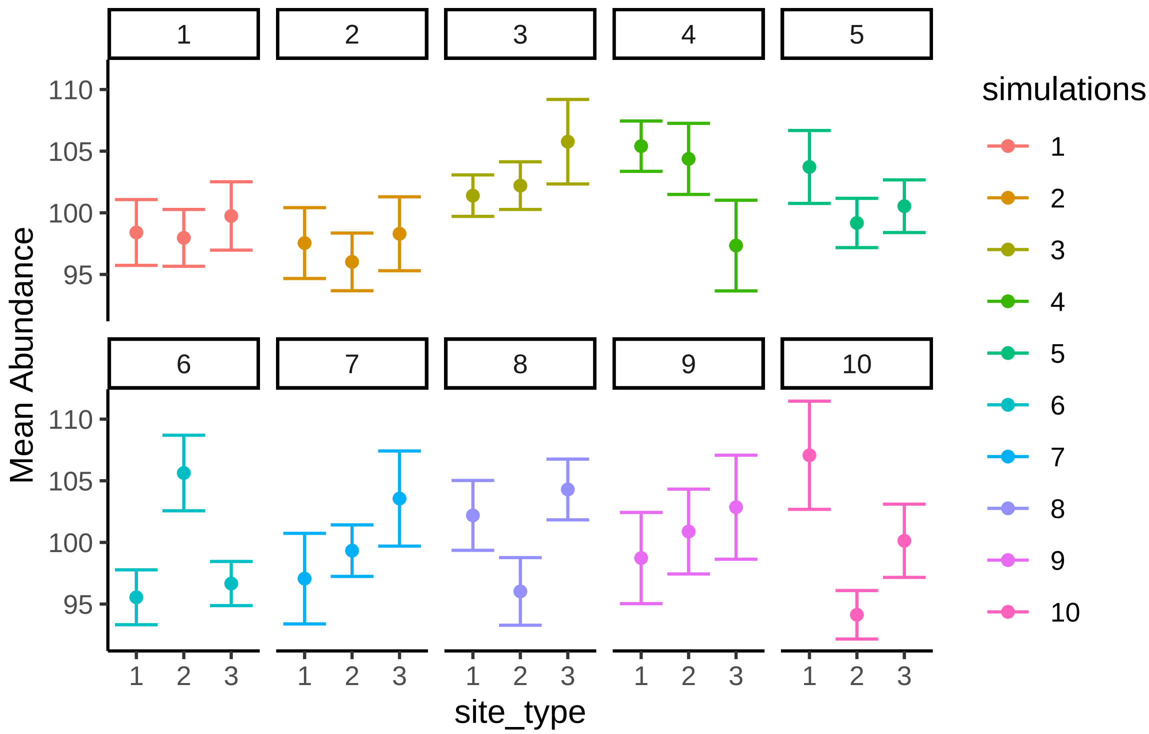

Figure 8.4: Different patterns of group means under the null (all scores for each group sampled from the same distribution).

Each panel is one simulated experiment; the dots are the group means for site types A, B, and C, with error bars showing standard error. All 30 plots in every panel were drawn from the same distribution (mean = 100, sd = 10), so the group means are only ever different because of sampling error, not any real effect of site type.

In most panels, the error bars overlap heavily, visually suggesting no real difference. In a few, they overlap less, which would suggest a difference that isn't actually there. Concluding a real effect from one of those panels would be a Type I error: we know there's no true difference here because we built the simulation that way, but in a real study you don't get to know that in advance.

8.3.2.3 What group means look like when F is less than 1

Figure 8.5 shows 10 simulated experiments, this time filtered to only the ones where $F < 1$, the cases where we'd clearly fail to reject the null.

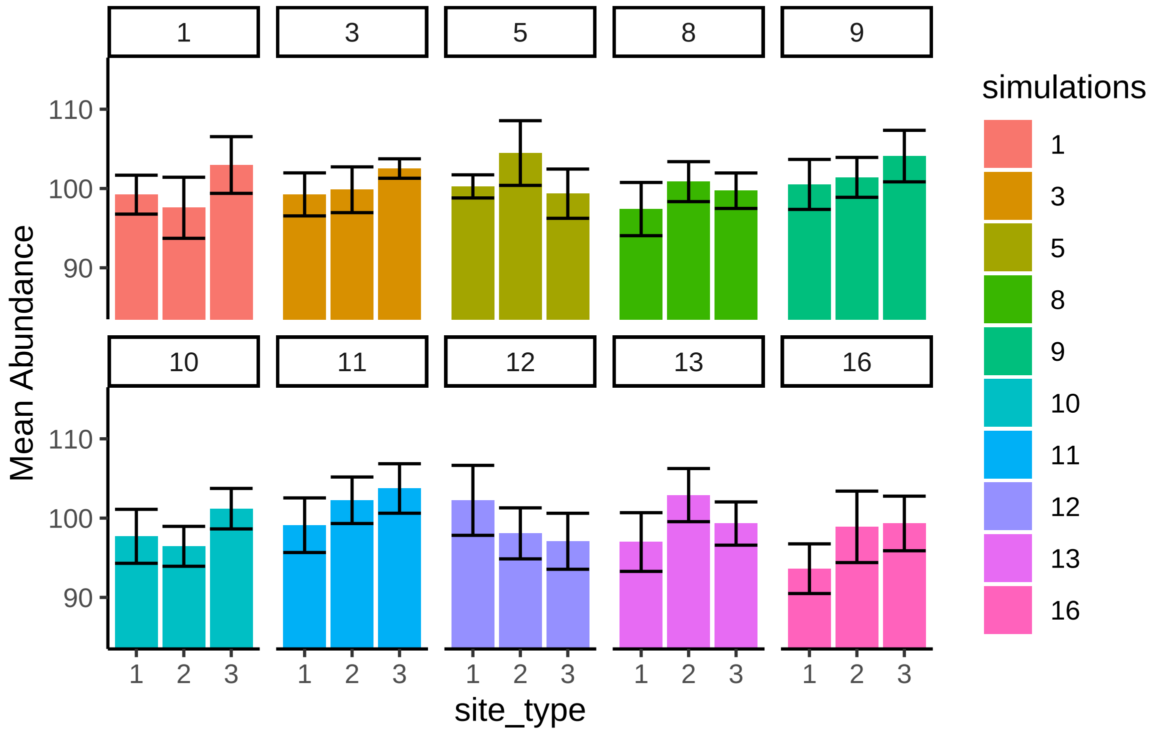

Figure 8.5: Different patterns of group means under the null (sampled from same distribution) when F is less than 1.

The panel numbers show which simulations actually produced $F < 1$. The bars aren't perfectly flat, but what matters is that within each panel, the error bars for all three means overlap heavily: our estimate of the mean is about the same across groups, and we wouldn't be making a Type I error here.

8.3.2.4 What group means look like when F exceeds the critical value

Earlier we found a critical value of 3.35: only 5% of F-values from this null distribution exceed it. Figure 8.6 shows 10 simulations filtered to exactly that rare case, $F > 3.35$, even though every one of them is still drawn from the same null distribution with no real group differences. Rejecting the null whenever F clears 3.35 is the right long-run decision rule, but every one of these particular simulations would be a Type I error if we did.

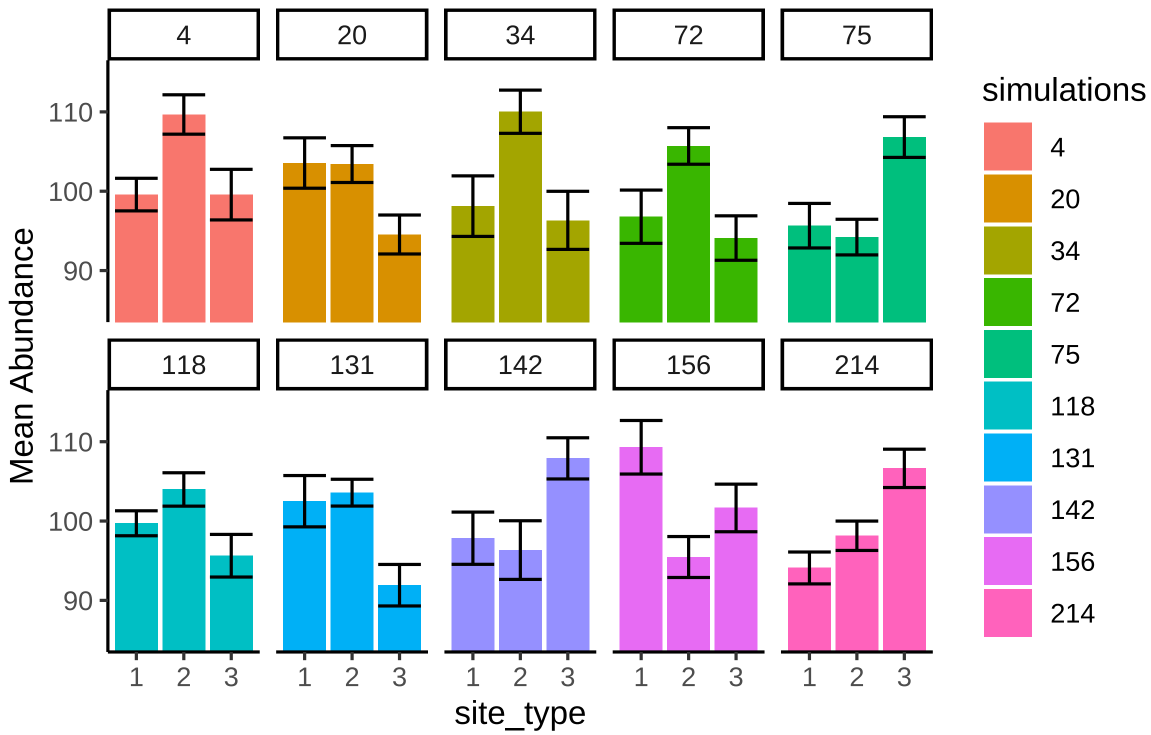

Figure 8.6: Different patterns of group means under the null when F is above critical value (these are all type I Errors).

Every panel of Figure 8.6 now shows at least one group mean whose error bars don't overlap with another's: a visually convincing "real" difference. All ten are Type I errors. That's the deceptive part: when this happens by chance, the data really does look like it shows a pattern, the F-value really is large, and the p-value really is small. A genuine Type I error looks exactly like a genuine effect, which is exactly why it's easy to be convinced by one.

8.4 ANOVA on Real Data

We've covered the fundamentals of ANOVA: how to compute an F-statistic from the data, and how to interpret it along with its p-value. In practice, software handles these calculations for you, but understanding what the numbers mean is what lets you interpret them correctly. Computing at least one ANOVA by hand, as we just did, is worth the effort: it's the fastest way to build real intuition for what the software is doing under the hood.

The following example applies the full ANOVA workflow, from descriptive summaries through the omnibus F test to post-hoc comparisons, to a real ecological dataset. This is the same dataset used in Lab 10.

8.4.1 Salamander abundance across forest types

Plethodontid salamanders are sensitive indicators of forest condition: they depend on moist microhabitats, abundant coarse woody debris, and stable soil temperatures, all features that decline with forest disturbance. A field ecologist surveys salamander abundance at 24 plots distributed across three forest types: **old-growth** (8 plots), **secondary** forest (8 plots), and **managed plantation** (8 plots). The research question is whether mean salamander abundance differs among forest types.

Let's look at the data and findings. Note that you will work through these steps yourself in Lab 10.

The bar graph shows a clear gradient: old-growth plots have the highest mean abundance, managed plantations the lowest, with secondary forest in between. The individual plot counts (dots) show considerable within-group variability, a reminder that even where a real difference exists, individual plots can vary substantially.

We now conduct the one-way ANOVA to test whether these differences are larger than expected by chance alone.

We report the result as:

> There was a significant effect of forest type on salamander abundance, $F(2, 21) = 14.60$, $MSE = 10.65$, $p < 0.001$.

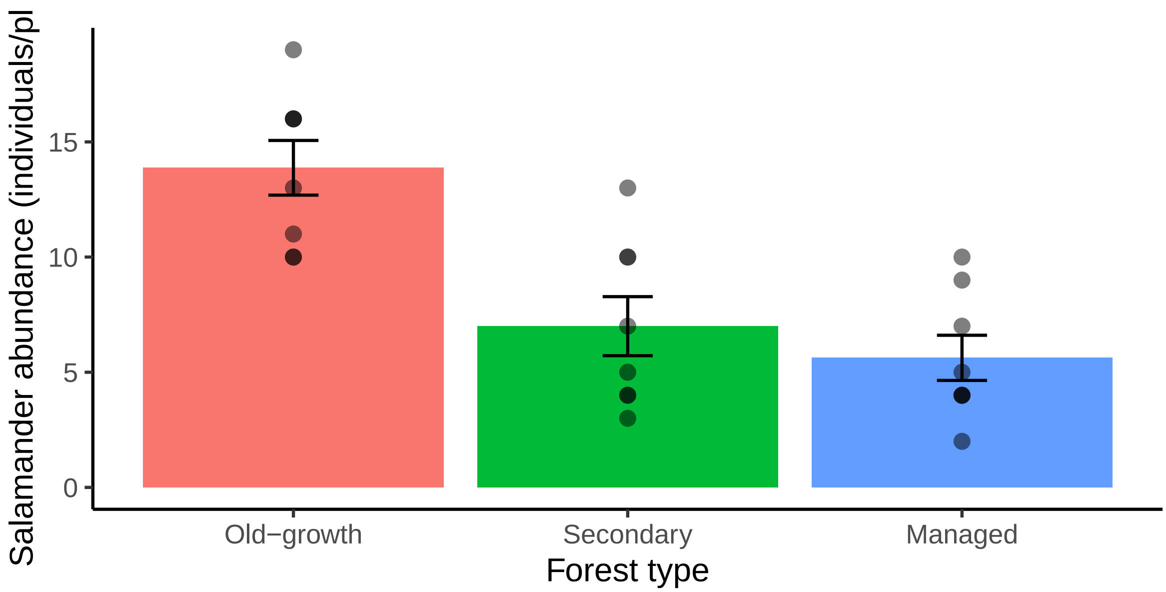

Figure 8.7: Mean salamander abundance (individuals per plot) across three forest types. Points show individual plot counts; error bars show ±1 SE.

Table 8.2: One-way ANOVA table for salamander abundance by forest type.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
forest_type	2	312.5833	156.29167	14.60345	0.000106
Residuals	21	224.7500	10.70238	NA	NA

Table 8.3: Levene's test for homogeneity of variance

Statistic (F)	p-value	df	df residual
0.52	0.599	2	21

Table 8.4: Shapiro-Wilk test on ANOVA residuals

Statistic (W)	p-value	Method
0.93	0.078	Shapiro-Wilk normality test

The F-value of 14.60 falls far into the tail of the null F distribution for df(2, 21): values this large arise by chance less than 0.1% of the time. We reject the null hypothesis that all three group means are equal.

8.4.2 Checking ANOVA's assumptions

Before trusting this result and moving on to post-hoc comparisons, it's worth pausing to check whether ANOVA's assumptions actually hold. (Recall from Chapter 6 that independence is the universal assumption that applies to every test we cover and won't be re-listed here; what follows are ANOVA's *test-specific* assumptions.)

ANOVA has two: **homogeneity of variance** (the groups' spread around their own means should be roughly comparable) and **approximately normal residuals** (the leftover variation after accounting for group membership should be roughly bell-shaped). Both matter for the same underlying reason: the F-test pools information across groups into single numbers (a common error variance, a reference F-distribution), and that pooling is only trustworthy if the groups are behaving similarly enough to be pooled.

Levene's test gives $F(2, 21) = 0.52$, $p = .60$: no evidence of unequal variance across forest types. The Shapiro-Wilk test on the model's residuals gives $W = 0.93$, $p = .078$: no strong evidence against normality either. Both checks are consistent with the boxplot and dot plot we already looked at in Figure 8.7: similar spread across groups, no glaring outliers.

A word of caution about formal assumption tests. It's tempting to treat Levene's test as a strict gate, where significant means "stop" and not significant means "proceed." Resist that. These tests have an awkward property: with a small sample (exactly when a violated assumption matters most), they have low power and will often say "no problem" even when there is one; with a very large sample (where ANOVA is fairly robust to mild violations anyway), they have high power and will flag trivial differences as "significant." A formal test is one piece of evidence, not the whole verdict, so look at the diagnostic plots too and use judgment. In practice, one-way ANOVA is moderately robust to mild departures from these assumptions, especially with balanced group sizes like we have here (8 plots per forest type). The situation to genuinely worry about is the combination of small, unbalanced samples with

Table 8.5: Tukey HSD post-hoc comparisons for salamander abundance by forest type.

	diff	lwr	upr	p adj
Secondary-Old-growth	-6.875	-10.998	-2.752	0.0011
Managed-Old-growth	-8.250	-12.373	-4.127	0.0002
Managed-Secondary	-1.375	-5.498	2.748	0.6825

visibly skewed data or wildly different spreads, and that's exactly when the nonparametric alternative below becomes the better choice.

8.4.3 Post-hoc comparisons with Tukey HSD

The omnibus F test tells us that at least one group mean differs, but not which pairs. To identify which forest types differ, we use **Tukey's Honestly Significant Difference (HSD)** test. Tukey HSD controls the family-wise error rate across all pairwise comparisons, making it the appropriate choice after a significant ANOVA.

The Tukey results show:

- **Old-growth vs. Managed**: mean difference = 8.25 individuals/plot, p = 0.0002. Old-growth plots support significantly higher salamander abundance than managed plantations.
- **Old-growth vs. Secondary**: mean difference = 6.88 individuals/plot, p = 0.0011. Old-growth plots also differ significantly from secondary forest.
- **Secondary vs. Managed**: mean difference = 1.38 individuals/plot, p = 0.68. Secondary forest and managed plantation plots are **not** significantly different from each other.

The pattern is ecologically interpretable: old-growth forest supports the highest salamander abundance, consistent with its structural complexity and stable microclimate. Secondary forest and managed plantation are statistically indistinguishable, suggesting that the early stages of forest recovery may not yet restore the conditions salamanders require.

8.4.4 Nonparametric alternative: the Kruskal-Wallis test

Our assumption checks above came back clean, so the ANOVA result stands as-is. But suppose they hadn't: visibly unequal spread across groups, or clearly skewed residuals, especially with small or unbalanced samples. The **Kruskal-Wallis test** is the nonparametric alternative to one-way ANOVA, in exactly the same relationship that the Mann-Whitney-Wilcoxon test has to the independent-samples t-test (in fact, Kruskal-Wallis *is* the Mann-Whitney-Wilcoxon logic extended to more than two groups).

Table 8.6: Kruskal-Wallis test on salamander abundance by forest type

Statistic (^{2})	p-value	df	Method
13.09	0.001	2	Kruskal-Wallis rank sum test

Table 8.7: Pairwise Mann-Whitney-Wilcoxon comparisons (Holm-adjusted)

Group 1	Group 2	p-value
Secondary	Old-growth	0.011
Managed	Old-growth	0.004
Managed	Secondary	0.486

How it works, conceptually. Rank every observation across all groups together, from smallest to largest, ignoring group membership. Then check whether the average rank differs more across groups than chance alone would produce. If forest type genuinely matters, one group's observations should occupy systematically higher or lower ranks than the others.

Its assumption. Kruskal-Wallis does not require normality. Like the Mann-Whitney-Wilcoxon test, it's cleanest to interpret as a test of differences in **location** (median) when the groups' distributions have similar shape; if the shapes differ a lot, a significant result tells you the groups differ somehow, but not cleanly whether that's a difference in location, spread, or shape.

$\chi^2(2) = 13.09$, $p = .001$: significant, the same conclusion as the ANOVA ($F(2, 21) = 14.60$, $p < .001$). That agreement is exactly what you'd hope for when the parametric assumptions were actually fine to begin with, as ours were here.

Post-hoc comparisons. Just as a significant ANOVA needs Tukey HSD to identify *which* groups differ, a significant Kruskal-Wallis result needs its own follow-up: pairwise rank-sum tests (Mann-Whitney-Wilcoxon, applied to each pair) with a correction for running multiple comparisons.

The pattern matches Tukey HSD exactly: Old-growth differs significantly from both Secondary ($p = .011$) and Managed ($p = .004$), while Secondary and Managed do not differ from each other ($p = .49$). Two different sets of assumptions, two different sets of machinery, same ecological story: that's the point of having both tools available, not a coincidence to be suspicious of.

8.5 Chapter Summary

Why it matters. Comparing more than two groups with a stack of t-tests inflates your false-positive rate fast. ANOVA solves that by asking one omnibus question first, and its variance-partitioning logic is the same idea that reappears in regression, factorial designs, and ANCOVA.

Core ideas

- **Why not just run more t-tests.** Each comparison carries its own chance of a false positive, and they accumulate: with m comparisons the familywise error rate is $1-(1-\alpha)^m$, which is already 14% for three groups.
- **Partition the variation.** Total variation splits into a piece explained by group membership (SS_{Between}) and unexplained residual variation (SS_{Within}). That split is the whole idea.
- **F is a ratio of the two.** After adjusting each sum of squares for its degrees of freedom, $F = MS_{\text{Between}}/MS_{\text{Within}}$. A large F means between-group differences are large relative to within-group noise.
- **Two steps, always in order.** First the omnibus test: do *any* group means differ? Only if that is significant do you run post-hoc comparisons (Tukey HSD) to find out *which* pairs differ.
- **Check the assumptions.** ANOVA needs approximately equal variances across groups (Levene's test) and roughly normal residuals. Treat formal tests as evidence rather than a gate, since they have least power exactly when small samples make violations matter most.
- **When assumptions fail.** The **Kruskal-Wallis** test is the rank-based alternative, with `pairwise.wilcox.test()` as its post-hoc counterpart.

9 Simple Linear Regression

Back in the correlation chapter, we fit a best-fit line through a scatterplot of chocolate and happiness, and learned how to compute its slope and intercept from the data. That was a description: given this sample, here is the line that summarizes it. What we could not yet do was ask the inference question that every other test in this book has asked. Is the slope we found in this sample close to what we'd expect from a real population relationship, or is it the kind of thing chance alone could produce? Now that we have hypothesis testing, sampling distributions, and the logic of comparing an observed statistic to a null distribution, we can put the regression line through the same machinery.

9.1 From description to inference

A sample slope is a statistic, and like any statistic, it varies from sample to sample. If we drew a new sample from the same population and fit a new line, we would get a slightly different slope, just by chance. Simple linear regression asks whether the slope we observed is large enough, relative to that sampling variability, to conclude that a real linear relationship exists in the population, rather than one produced by chance alone.

Example: stream temperature and dissolved oxygen. Warmer water holds less dissolved gas, so ecologists expect dissolved oxygen (DO) to decline as stream temperature rises. This matters for stream health: low DO can stress or kill fish and invertebrates. Suppose we sample 30 stream sites, measuring water temperature (°C) and dissolved oxygen (mg/L) at each.

The scatterplot in Figure 9.1 shows a clear downward trend. The question this chapter answers: is that trend real, or could a sample of 30 sites show a downward-sloping line like this even if temperature and DO were unrelated in the population?

9.2 The regression model

We model dissolved oxygen as a linear function of stream temperature, plus error:

$$DO = \beta_0 + \beta_1(\text{Temperature}) + \varepsilon$$

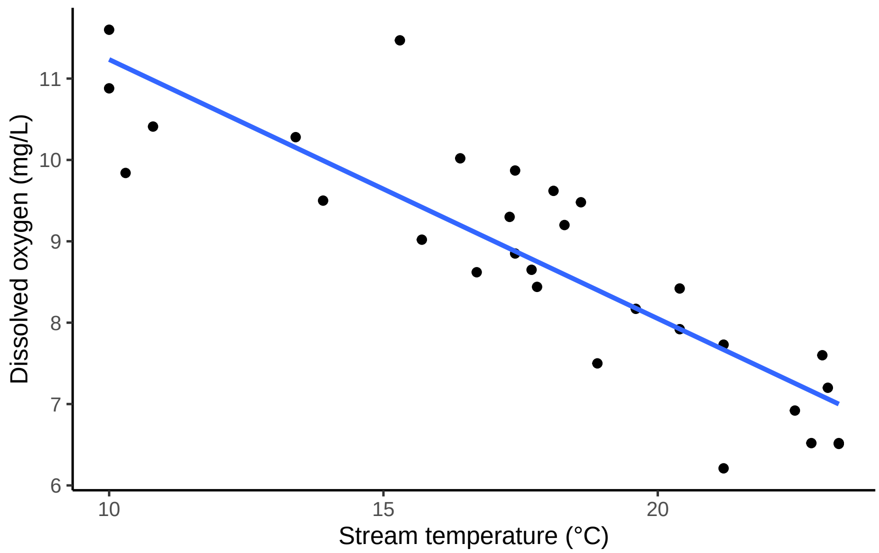

Figure 9.1: Stream temperature and dissolved oxygen at 30 sample sites, with the fitted regression line.

Table 9.1: Coefficient table for DO   stream temperature

Term	Estimate	Std. Error	t	p	CI low	CI high
(Intercept)	14.418	0.615	23.430	$< .001$	13.158	15.679
stream_temp	-0.318	0.034	-9.453	$< .001$	-0.387	-0.249

Here β_0 is the population intercept, β_1 is the population slope, and ε is the error, the part of DO that temperature alone doesn't explain. We don't know β_0 and β_1; we estimate them from the sample as b_0 and b_1, using the same least-squares logic from the correlation chapter (the line that minimizes the sum of squared residuals).

In R, this is a single line:

```
fit <- lm(DO ~ stream_temp, data = stream_df)
```

Reading the table: the fitted line is $\widehat{DO} = 14.42 - 0.32 \times \text{Temperature}$. The intercept ($b_0 = 14.42$) is the model's predicted DO at 0°C, technically part of the model but outside the range of temperatures we actually sampled (10 to 24°C), so we won't lean on it for interpretation. The slope ($b_1 = -0.32$) is the part we care about: for every 1°C increase in stream temperature, predicted dissolved oxygen drops by about 0.32 mg/L.

9.3 Testing the slope

The slope b_1 is a sample estimate. The hypothesis test asks whether the population slope β_1 could plausibly be zero, that is, whether temperature and DO are linearly unrelated.

- $H_0 : \beta_1 = 0$ (no linear relationship between stream temperature and dissolved oxygen)
- $H_A : \beta_1 \neq 0$ (there is a linear relationship)

The test statistic compares the estimated slope to its standard error:

$$t = \frac{b_1}{SE(b_1)} = \frac{-0.318}{0.034} = -9.45$$

with $df = n - 2 = 28$ (one degree of freedom spent estimating the intercept, one for the slope). From the coefficient table above, $t = -9.45$, $p < .001$. This crosses the conventional $\alpha = .05$ threshold, so we reject H_0: stream temperature is linearly related to dissolved oxygen.

The 95% confidence interval for the slope, also in the table, runs from -0.387 to -0.249 mg/L per °C. Because this interval does not contain 0, it agrees with the hypothesis test, as it always will.

> **Note**
>
> **The same test in different clothes.** This is the correlation test from Chapter 2, just written differently. Testing $H_0 : \beta_1 = 0$ for the slope and testing $H_0 : \rho = 0$ for the correlation coefficient give the identical t-statistic and identical p-value. Regression and correlation are two views of the same underlying relationship: correlation asks how strongly two variables move together, and regression additionally lets us write down an equation and make predictions.

9.4 R-squared: variance explained

Just as one-way ANOVA in the last chapter partitioned total variability into a between-groups piece and a within-groups piece, regression partitions the total variability in Y into a piece explained by the line and a piece left over as residual error:

$$SS_{Total} = SS_{Regression} + SS_{Residual}$$

R^2 is the proportion of that total variability the line accounts for:

Table 9.2: Model fit for DO stream temperature

	R-squared	F	df1	df2	p
value	0.761	89.36	1	28	$<$ .001

Table 9.3: Shapiro-Wilk test on the regression residuals

Statistic	p-value
0.97	0.658

$$R^2 = \frac{SS_{Regression}}{SS_{Total}}$$

Here $R^2 = 0.76$: stream temperature accounts for about 76% of the variation we observed in dissolved oxygen across sites. The remaining variation comes from everything else the model doesn't capture (discharge, shading, groundwater input, and so on), folded into ε.

R^2 is regression's effect size. A relationship can be tested (is β_1 significantly different from 0?) and separately sized (how much of the variation does it actually explain?), the same distinction Chapter 5 introduced for Cohen's d and that eta-squared makes for ANOVA. A small sample can produce a significant slope that explains very little variance; R^2 is what tells you whether the relationship is worth acting on, not just whether it's real.

9.5 Checking assumptions

Simple linear regression assumes:

- **Linearity**: the true relationship between X and Y is a straight line.
- **Independence**: observations don't influence each other (the universal assumption from Chapter 6, still not something you list when asked to enumerate a test's specific assumptions, but always worth checking your study design against).
- **Constant variance (homoscedasticity)**: the spread of residuals is roughly the same across the whole range of X.
- **Normality of residuals**: the residuals are approximately normally distributed.

The most useful diagnostic is a residuals-versus-fitted plot:

Figure 9.2 shows what we want to see: residuals scattered evenly above and below zero, with no curve (which would suggest the true relationship isn't linear) and no funnel shape (which would suggest variance changes across the range of X). For normality, we check the residuals directly:

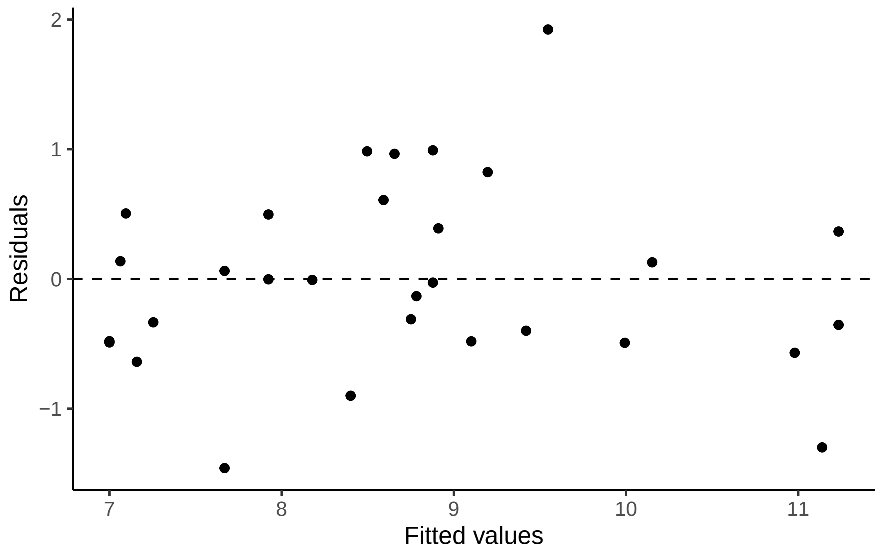

Figure 9.2: Residuals versus fitted values for the stream temperature model. A random scatter around zero, with no funnel shape or curve, supports linearity and constant variance.

The residuals don't depart from normality here (p0.658). As with any formal normality test, treat this as one piece of evidence alongside the residual plot, not a strict gate. Shapiro-Wilk has the least power to detect real departures exactly when the sample is small, which is exactly when a visual check matters most.

9.6 Using the model: prediction

Once we've confirmed a real linear relationship, the fitted equation lets us predict DO at any stream temperature. For example, at 20°C:

\begin{table}

\caption{Predicted dissolved oxygen at 20°C, with 95% confidence interval}

Stream temperature	Predicted	CI low	CI high
20	8.05	7.74	8.36

\end{table}

At 20°C, the model predicts DO of 8.05 mg/L, with a 95% confidence interval from 7.74 to 8.36. Notice that 20°C falls inside our sampled range of 10 to 24°C, this is interpolation, and it's what regression is good for. Predicting DO at, say, 35°C would be extrapolation: nothing in the data tells us the relationship stays linear (or even stays a relationship at all) beyond the range we actually observed, and the further outside that range, the less trustworthy the prediction.

9.7 Chapter Summary

Why it matters. Regression is where description becomes inference. Chapter 2 drew a best-fit line through a scatterplot; this chapter turns that line into something you can test, quantify, and make predictions from.

Core ideas

- **The slope is a hypothesis.** $H_0 : \beta_1 = 0$ says there is no linear relationship. Testing it uses a t-test on the slope divided by its standard error, with $df = n - 2$, and it is mechanically identical to testing a correlation coefficient.
- R^2 **is the effect size.** It partitions variance exactly as ANOVA does, reporting the proportion of variation in y the line accounts for. A relationship can be real (p small) and still explain very little, which is why you report both.
- **Assumptions are not bookkeeping.** Linearity, independence, constant variance, and roughly normal residuals are what license the test and the predictions. A residuals-versus-fitted plot catches most problems faster than any formal test.
- **Predict inside your data, not outside it.** Interpolating within the observed range of x is what regression is for. Extrapolating beyond it assumes a linear relationship continues where you have no evidence at all.
- **This model keeps growing.** Add a categorical group alongside the continuous predictor and the same machinery becomes ANCOVA, which closes out the course in Chapter 11.

10 Factorial ANOVA

In environmental science, outcomes rarely depend on just one thing. Climate, soil type, and water availability all act together, shaping ecosystems in ways a single-factor design can't capture. Factorial designs let us study several factors, and how they interact with each other, at once.

This chapter covers both halves of that: **how to compute** a factorial ANOVA, including a full by-hand walkthrough of the sums-of-squares arithmetic, and **how to interpret** the result, which is where factorial designs actually go wrong. Interpretation is not a step you do after the analysis. When an interaction is present, it changes what every other number in the table means.

Imagine investigating the growth rate of a plant species, with independent variables like soil pH or sunlight exposure. Each added IV multiplies the number of conditions you can observe, and with it, the questions you can ask about how those factors work together.

Let's look at a few factorial design examples in an environmental setting:

1. **1 IV (two levels)**

A t-test would suffice here, as we are dealing with just two levels of our independent variable.

a. Soil pH (Acidic vs. Neutral): How does soil pH affect plant growth? We have one IV (soil pH), with two levels (Acidic vs. Neutral).

b. Sunlight Exposure (Full Sun vs. Partial Shade): Does the amount of sunlight influence plant growth rates? Here, there's one IV (sunlight exposure), with two levels (Full Sun vs. Partial Shade).

2. **1 IV (three levels):**

An ANOVA is appropriate here, given the variable has more than two levels.

a. Water Availability (Low, Medium, High): How does varying water availability impact plant growth? Our single IV (water availability) has three levels (Low, Medium, High).

b. Fertilizer Type (No fertilizer, Organic, Synthetic): What's the effect of different fertilizer types on plant growth? One IV (fertilizer type) has three levels.

3. **2 IVs: IV1 (two levels), IV2 (two levels)**

We would employ a factorial ANOVA for this design, which we haven't discussed yet.

a. IV1 (Soil pH: Acidic vs. Neutral); IV2 (Water Availability: Low vs. High): How do soil pH and water availability together affect plant growth? Each IV has two levels, leading to a **2x2** Factorial Design. This type of design is fully crossed: each level of one IV is paired with each level of the other IV, so the combined effect on growth rates can be observed.

The first two designs above both had one IV. The third has two (soil pH and water availability), each with two levels, giving a **2x2 Factorial Design**. It's called a **factorial** design because the levels of each independent variable are fully crossed: every level of one IV occurs alongside every level of the other.

10.1 Factorial basics

Why This Matters: Factorial designs let us study how multiple environmental drivers act together. This is often a more realistic picture of environmental systems.

10.1.1 2x2 Designs

We've just started talking about a **2x2 Factorial design**. We said this means the IVs are crossed. To illustrate this, take a look at Figure 10.1. We show an abstract version and a concrete version using soil pH and water availability as the two IVs, each with two levels in the design:

2x2 Design		IV 1	
		IV1: Level 1	IV1: Level 2
IV 2	IV2: Level 1	dv	dv
	IV2: Level 2	dv	dv

2x2 Design		Time of Day	
		Morning	Afternoon
Caffeine	Some Caffeine	dv	dv
	No Caffeine	dv	dv

Figure 10.1: Structure of 2x2 factorial designs

- IV1: Soil Type (Clay vs. Loam)
- IV2: Drought Condition (Yes vs. No)

Each combination of soil type and drought condition creates a unique environment for the plants, giving four distinct scenarios to measure growth rates: 2 IVs, each with 2 levels, for 2x2 = 4 conditions total. The notation directly tells you the number of conditions.

10.1.2 Factorial Notation

Anytime **all of the levels** of each IV in a design are fully crossed, so that they all occur for each level of every other IV, we can say the design is a **fully factorial** design.

Our notation system succinctly encapsulates the structure of factorial designs. Each IV gets a number representing its levels. Here are a few examples:

- 2x2: Two IVs, each with two levels, yielding four unique conditions.
- 2x3: Two IVs, with the first having two levels and the second three, resulting in six conditions.
- 3x2: Similar to the 2x3, but the first IV has three levels, and the second has two, also giving us six conditions.
- 4x4: Two IVs, each with four levels, resulting in sixteen conditions.

Expanding on our designs, a 2x3 factorial design could investigate the impact of two soil types across three different levels of water availability, equating to six unique conditions to analyze plant growth.

10.2 The running example: pollution, flow, and stream life

The rest of this chapter uses one example throughout, so it's worth setting it up carefully before any arithmetic starts.

The question. Streams are affected by pollution, but they are also affected by how fast the water moves. Fast-moving water flushes contaminants downstream and re-oxygenates the channel; slow water lets them sit. So we might reasonably suspect that pollution does more damage in a sluggish stream than in a fast one. A factorial design is what lets us test that suspicion directly.

The setup. We run an experiment in artificial stream channels. Each channel is assigned one of two pollution conditions and one of two flow rates, fully crossed, giving four conditions. Figure 10.2 lays out the design.

The dependent variable is **macroinvertebrate species count**: how many distinct kinds of aquatic insects, snails, worms, and crustaceans we find in a channel. This is a standard measure of stream health. These animals sit in the middle of the food web, many of them are pollution-sensitive, and they can't quickly flee a degraded reach the way fish can, so the number of species present is a good integrative signal of how habitable the water actually is. More species means a healthier stream.

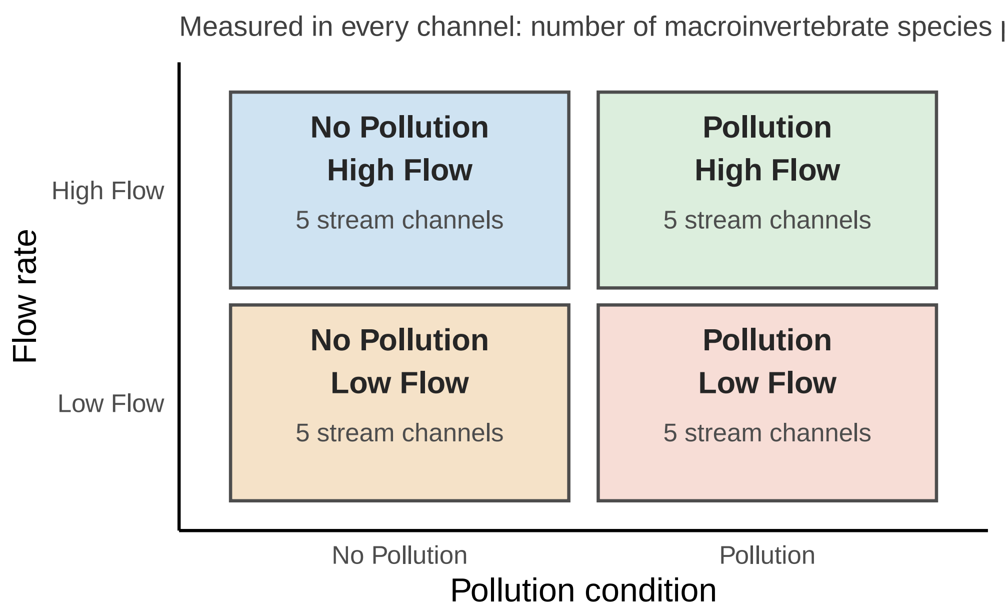

Figure 10.2: The 2x2 experimental design. Two pollution conditions crossed with two flow rates gives four stream-channel conditions. In every condition we count how many macroinvertebrate species are present.

10.3 Why factorial designs matter

Factorial designs let us ask more nuanced questions than a single-factor study can. Suppose we ignored flow entirely and just compared polluted to unpolluted channels. That gives us a **Pollution effect**: the difference in species count between polluted and unpolluted water, shown in Figure 10.3.

A factorial design lets us ask the follow-up question a one-factor design can't: does the Pollution effect itself depend on flow? With flow rate added as a second IV, there are four condition means instead of two, so we can compare the Pollution effect *within* each flow condition, shown in Figure 10.4.

Under low flow, species count drops sharply with pollution (35 to 15, a difference of 20). Under high flow, the drop is much smaller (30 to 25, a difference of 5). The Pollution effect isn't constant: it depends on flow rate, exactly as we suspected it might.

That's the core payoff of a factorial design. It doesn't just tell you whether a manipulation has an effect, it tells you what changes that effect's size.

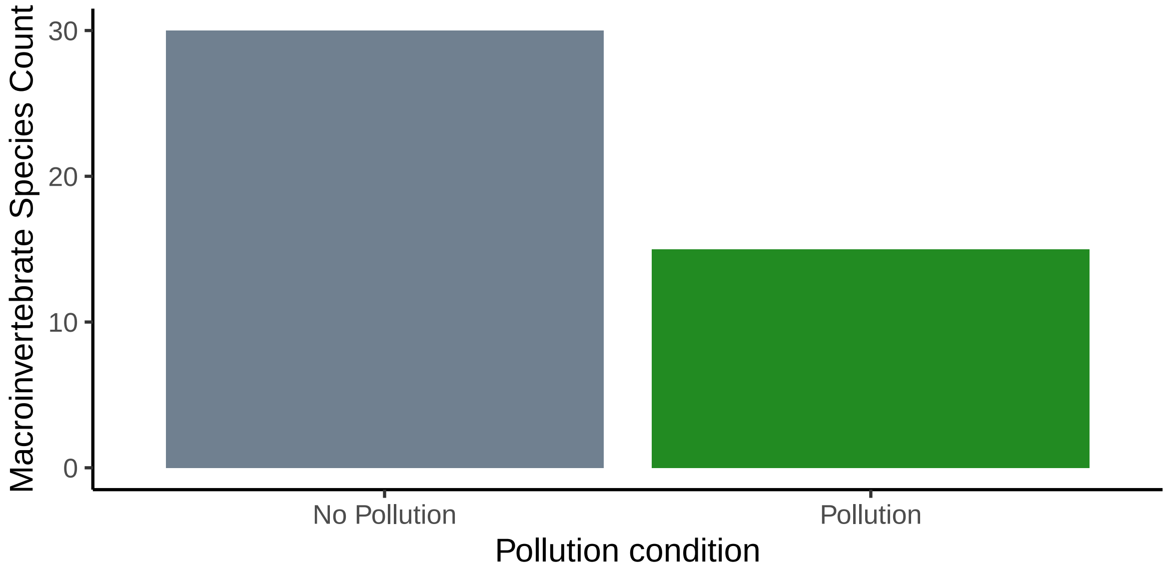

Figure 10.3: Macroinvertebrate species count under pollution vs. no pollution, ignoring flow.

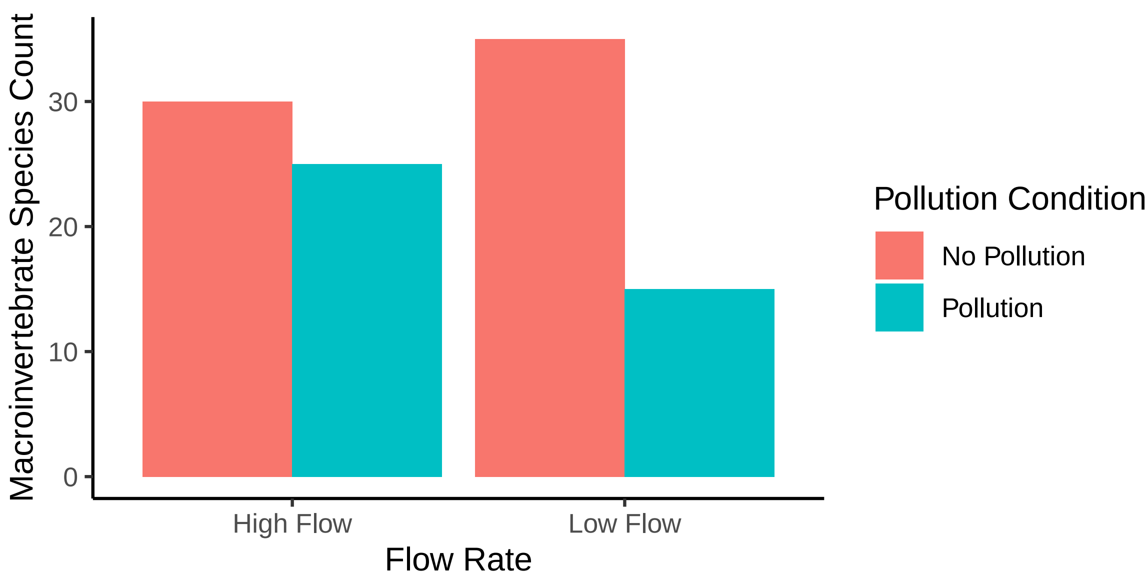

Figure 10.4: The Pollution effect (the gap between the two bars within a flow condition) is larger under low flow than high flow.

10.3.1 Main effects and interactions

Those two things, **main effects** and **interactions**, are what any factorial design gives you. How many of each depends on the number of IVs.

A **main effect** is the average difference associated with a single IV, one per IV. A 2x2 design has two IVs, so two main effects: one for Pollution and one for Flow. A 2x2x2 design has three main effects, one per IV; a 3x3x3 design also has three, since the number of main effects tracks the number of IVs, not their levels.

An **interaction** occurs when the effect of one IV depends on the level of another, which is exactly what Figure 10.4 shows: the size of the Pollution effect changed with Flow.

Equivalently, and this framing is worth holding onto because it's what the arithmetic actually computes, an interaction is a **difference between differences**. The Pollution effect is a difference (polluted minus unpolluted). If that difference itself changes across Flow conditions, that's an interaction.

The number of possible interactions scales with the number of IVs. A 2x2 design has exactly one interaction, the combined effect of the two IVs. A design with three IVs (A, B, C) has three two-way interactions (AB, AC, BC) and one three-way interaction (ABC).

10.4 What a 2x2 actually tests

Before computing anything, it's worth being precise about what the three tests are, because a two-way ANOVA tests **three effects** and therefore has **three null hypotheses**:

1. a main effect of pollution,
2. a main effect of flow, and
3. a pollution $\times$ flow interaction.

We can represent the four combinations of conditions and their mean species counts as:

Pollution	Flow	Mean species count
No Pollution	Low Flow	$\mu_{\text{NoPoll,Low}}$
No Pollution	High Flow	$\mu_{\text{NoPoll,High}}$
Pollution	Low Flow	$\mu_{\text{Poll,Low}}$
Pollution	High Flow	$\mu_{\text{Poll,High}}$

10.4.1 Hypotheses

Main effect of pollution

- **Null (H_0):** There is no effect of pollution on mean species count.
- **Alternative (H_1):** Pollution affects mean species count.

Main effect of flow

- **Null (H_0):** There is no effect of flow rate on mean species count.
- **Alternative (H_1):** Flow rate affects mean species count.

Pollution × flow interaction

- **Null (H_0):** The effect of pollution is the same at both flow rates. *(In symbols:* $(\mu_{\text{NoPoll,Low}} - \mu_{\text{Poll,Low}}) = (\mu_{\text{NoPoll,High}} - \mu_{\text{Poll,High}})$*.)*
- **Alternative (H_1):** The effect of pollution differs across flow rates.

  Plain English: Under the interaction null, the gap between polluted and unpolluted channels is the same whether the water is moving fast or slow. If that gap changes, for example if pollution does far more damage under low flow, you've found an interaction.

10.4.2 Template sentences (swap in your variables)

Main effect of A

- **Null:** There is no effect of A on mean [DV].
- **Alternative:** A affects mean [DV].

Main effect of B

- **Null:** There is no effect of B on mean [DV].
- **Alternative:** B affects mean [DV].

A×B interaction

- **Null:** The effect of A on [DV] is the same at all levels of B. *(In symbols:* $(\mu_{A1,B1} - \mu_{A2,B1}) = (\mu_{A1,B2} - \mu_{A2,B2})$*.)*
- **Alternative:** The effect of A on [DV] differs across levels of B.

> **Note**
>
> **General form.** You'll sometimes see this written in abstract terms, with "Factor A" and "Factor B." For example: $H_0 : (\mu_{A1,B1} - \mu_{A2,B1}) = (\mu_{A1,B2} - \mu_{A2,B2})$. It's the same

logic: just replace A and B with your own variables.

10.5 What the ANOVA is really comparing

Before running the full machinery, it's worth seeing what the interaction term is actually testing, because the answer is less mysterious than the arithmetic makes it look.

Suppose the same five stream channels were measured under all four conditions, making this a repeated-measures design. We could compute each channel's Pollution effect under low flow, then again under high flow, and simply ask whether those two numbers differ:

```
#>
#>  Paired t-test
#>
#> data:  A_B and C_D
#> t = 2.493, df = 4, p-value = 0.06727
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#>  -0.3865663  7.1865663
#> sample estimates:
#> mean difference
#>             3.4
```

The mean Pollution effect was 6 under low flow and 2.6 under high flow, a difference of differences of 3.4. A paired t-test on those two sets of difference scores gives $t(4) = 2.49$, $p = .067$: not significant at $\alpha = .05$.

That is the interaction, computed with nothing more than a paired t-test on difference scores. The same trick works for each main effect, and the resulting t values relate to the ANOVA's F values by $t^2 = F$ in this design.

Note

Going deeper on repeated measures. A full 2x2 repeated-measures design can be worked entirely through paired t-tests this way, one per main effect plus the interaction, and it produces results identical to a repeated-measures ANOVA. That walkthrough, along with the `aov()` syntax for repeated-measures designs and the error-term structure they require, lives in the out-of-sequence chapter *"Repeated Measures ANOVA"* in this book's repository. For the rest of this chapter we work with the more general **between-subjects** case, where different channels appear in each condition.

	All Conditions				Difference from Grand Mea		
	Low Flow		High Flow		Low Flow		Hig
	No Pollution	Pollution	No Pollution	Pollution	No Pollution	Pollution	No Pollutio
	A	B	C	D	A-GrandM	B-GrandM	C-GrandM
	10	5	12	9	1.05	-3.95	3.05
	8	4	13	8	-0.95	-4.95	4.05
	11	3	14	10	2.05	-5.95	5.05
	9	4	11	11	0.05	-4.95	2.05
	10	2	13	12	1.05	-6.95	4.05
Means	9.6	3.6	12.6	10			
Grand Mean	8.95						
sums							
SS Total							

10.6 Computing a factorial ANOVA by hand

In the **between-subjects** 2x2 design, different stream channels appear in each condition, so there are no per-channel difference scores to work with and we compute the ANOVA table directly. The logic is the same as the one-way ANOVA: partition the total variance into parts. For a 2x2 design, there's now one more part, the interaction:

$$SS_{\text{Total}} = SS_{\text{Effect IV1}} + SS_{\text{Effect IV2}} + SS_{\text{Effect IV1xIV2}} + SS_{\text{Error}}$$

The sections below compute each SS in turn, using the same Pollution × Flow example, now as a between-subjects design: 5 different subjects per condition, 20 subjects total, so there's no subjects column.

10.6.1 SS Total

SS_{Total} works exactly as it did for the one-way ANOVA: find the grand mean, subtract it from each score, square, and sum, reported in the bottom yellow row.

10.6.2 SS Pollution

$SS_{\text{Pollution}}$ isolates the variation due to the Pollution main effect: replace every score with its own Pollution-condition mean, find each of those means' distance from the grand mean, then square and sum, reported in the bottom yellow row.

	All Conditions				Pollution Mean - G		
	Low Flow		High Flow		Low Flow		
	No Pollution	Pollution	No Pollution	Pollution	No Pollution	Pollution	No Pollt
	A	B	C	D	NDM-GM A	DM-GM B	NDM-G
	10	5	12	9	2.15	-2.15	2.15
	8	4	13	8	2.15	-2.15	2.15
	11	3	14	10	2.15	-2.15	2.15
	9	4	11	11	2.15	-2.15	2.15
	10	2	13	12	2.15	-2.15	2.15
Means	9.6	3.6	12.6	10			
Grand Mean	8.95	No Pollution	11.1	Pollution	6.8		
sums							
SS Pollution							

	All Conditions				Flow Mean - GM		
	Low Flow		High Flow		Low Flow		
	No Pollution	Pollution	No Pollution	Pollution	No Pollution	Pollution	No Pollu
	A	B	C	D	NRM-GM A	NRM-GM B	RM-GM
	10	5	12	9	-2.35	-2.35	2.35
	8	4	13	8	-2.35	-2.35	2.35
	11	3	14	10	-2.35	-2.35	2.35
	9	4	11	11	-2.35	-2.35	2.35
	10	2	13	12	-2.35	-2.35	2.35
Means	9.6	3.6	12.6	10			
Grand Mean	8.95	Low Flow	6.6	High Flow	11.3		
sums							
SS Flow							

The grand mean is 8.95. The no-Pollution mean (columns A and C) is 11.1, giving a difference score of 11.1-8.95 = 2.15 for every no-Pollution observation. The Pollution mean (columns B and D) is 6.8, giving -2.15 for every Pollution observation. Since means are the balancing point of the data, the grand mean 8.95 sits exactly between the two condition means, 2.15 from each.

10.6.3 SS Flow

SS_{Flow} works the same way: replace every score with its own Flow-condition mean, find each of those means' distance from the grand mean, then square and sum, reported in the bottom yellow row.

Every Low Flow score is replaced with the Low Flow mean (6.6), 2.35 below the grand mean

	All Conditions				Interaction Differ		
	Low Flow		High Flow		Low Flow		
	No Pollution	Pollution	No Pollution	Pollution	No Pollution	Pollution	No
	A	B	C	D	A-NP-LF+GM	B-P-LF+GM	C-NP
	10	5	12	9	0.85	-0.85	-0.85
	8	4	13	8	0.85	-0.85	-0.85
	11	3	14	10	0.85	-0.85	-0.85
	9	4	11	11	0.85	-0.85	-0.85
	10	2	13	12	0.85	-0.85	-0.85
Means	9.6	3.6	12.6	10			
Grand Mean	8.95						
sums							
SS Interaction							

(8.95); every High Flow score is replaced with the High Flow mean (11.3), 2.35 above it. Squaring and summing those distances gives SS_{Flow}.

10.6.4 SS Pollution by Flow

The interaction is the one new piece an ANOVA with more than one IV requires. The question it answers: do the four individual condition means behave differently than the two main-effect group means alone would predict?

Take the Low Flow group: its overall mean is 6.6. If Flow's effect were consistent across Pollution levels, each Pollution subgroup's Low Flow mean should also be about 6.6. It isn't: the no-Pollution/Low-Flow mean (column A) is 9.6, and the Pollution/Low-Flow mean (column B) is 3.6 (averaging to 6.6, but neither equal to it). That gap, variance the Flow mean alone can't explain, is exactly what the interaction captures.

To compute it: replace each score with its own condition mean (one of the four cells), then subtract the score's Pollution-group mean and Flow-group mean, and add back the grand mean:

$$\bar{X}_{\text{condition}} - \bar{X}_{\text{IV1}} - \bar{X}_{\text{IV2}} + \bar{X}_{\text{Grand Mean}}$$

Equivalently, subtract the grand mean from the condition mean, then add back both main-effect means. Squaring and summing these gives $SS_{\text{Interaction}}$, reported in the bottom yellow row.

10.6.5 SS Error

The last piece, SS_{Error}, follows by subtraction rather than a fresh table, since every other term in the identity is already known:

$$SS_{\text{Total}} = SS_{\text{Effect IV1}} + SS_{\text{Effect IV2}} + SS_{\text{Effect IV1xIV2}} + SS_{\text{Error}}$$

$$SS_{\text{Error}} = SS_{\text{Total}} - SS_{\text{Effect IV1}} - SS_{\text{Effect IV2}} - SS_{\text{Effect IV1xIV2}} = 242.95 - 92.45 - 110.45 - 14.45 = 25.6$$

10.6.6 Check your work

Converting each SS to an F-value works exactly as it did for the one-way ANOVA (divide by df to get MS, then take the ratio), so we skip repeating that here. Instead, the check that matters: these hand-calculated SS values should match what R computes directly.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Pollution	1	92.45	92.45	57.78125	0.0000011
Flow	1	110.45	110.45	69.03125	0.0000003
Pollution:Flow	1	14.45	14.45	9.03125	0.0083879
Residuals	16	25.60	1.60	NA	NA

The `Sum Sq` column matches the hand-calculated values above. Note these results differ from the repeated-measures calculation earlier in the chapter: a between-subjects design has no channel-to-channel variation to split out of the error, which also means more degrees of freedom in the error term. With this specific dataset, that extra power is enough to make all three effects (both main effects and the interaction) significant here, where the interaction wasn't when the same numbers were treated as repeated measures.

10.7 Reading patterns in a 2x2

Having computed one ANOVA table by hand, we can stop doing arithmetic and start doing the part that actually determines whether you get the science right: reading the result.

In any 2x2 ANOVA, there are eight general outcome patterns, different combinations of main effects and interactions. You don't need to memorize all eight, but you should learn to **read them in a line plot**:

Pattern	Main Effect: Pollution	Main Effect: Flow	Interaction
1			
2	X		
3	X		X
4	X	X	
5	X	X	X
6		X	
7		X	X
8			X

We'll create bar and line graphs to illustrate these patterns. Bar graphs are great for seeing differences in means directly, while line graphs help us spot interactions. Figure 10.5 shows the possible patterns in bar graph form. These are abstract illustrations of every combination that can occur, so a few of them describe patterns you would not expect ecologically; the point is to recognize the shapes, not to defend each one.

Here's what to look for:

- **Parallel lines** → no interaction.
- **Non-parallel or crossing lines** → interaction likely.
- **Vertical separation between group means** → main effect.

For example:

- If the **no-pollution vs. pollution** lines are consistently apart, there's a main effect of pollution.
- If the lines cross or diverge, the **pollution effect changes across flow rates**, showing an interaction.
- If one line sits higher overall (say, high flow always supports more species), that's a main effect of flow.

Figure 10.6 shows the same patterns in line graph form:

In the line graphs, interactions are easiest to see:

- **Parallel lines** mean no interaction (the pollution effect is the same at both flow rates).
- **Crossing lines** mean an interaction (the pollution effect differs by flow).
- **Vertical shifts** between the two lines reflect a main effect of flow.

Order of interpretation. When we conduct a two-way ANOVA, we always begin by testing the **interaction** term.

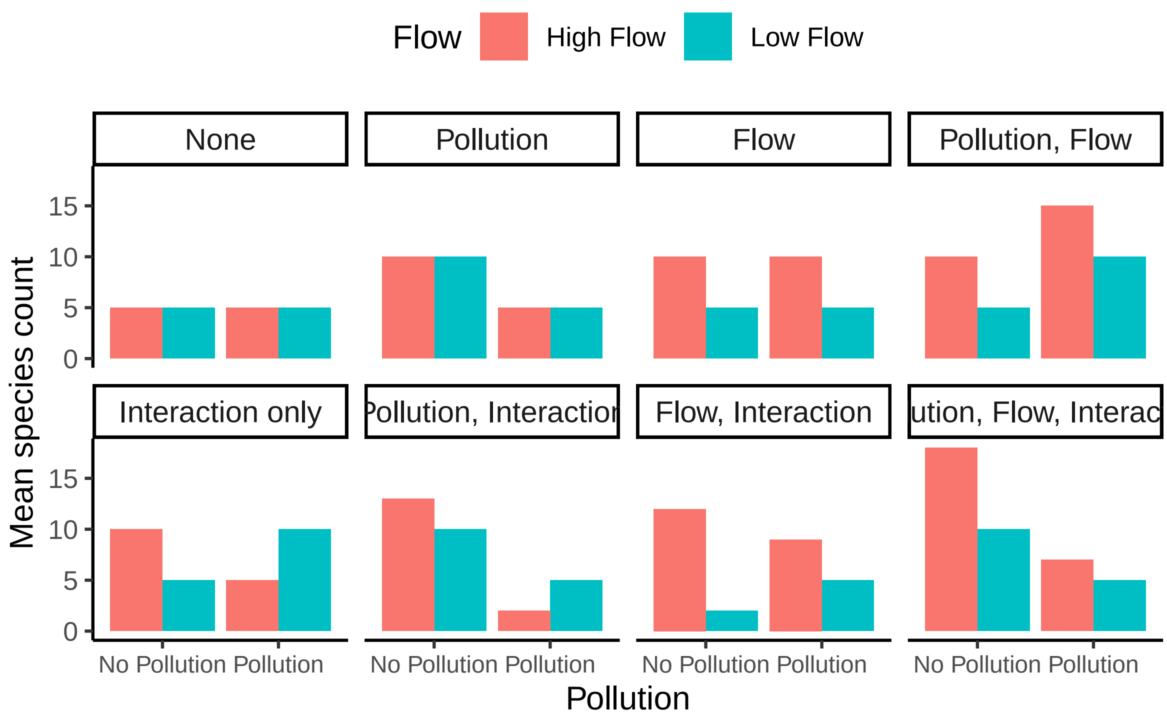

Figure 10.5: Eight example patterns of main effects and interactions in a pollution × flow design.

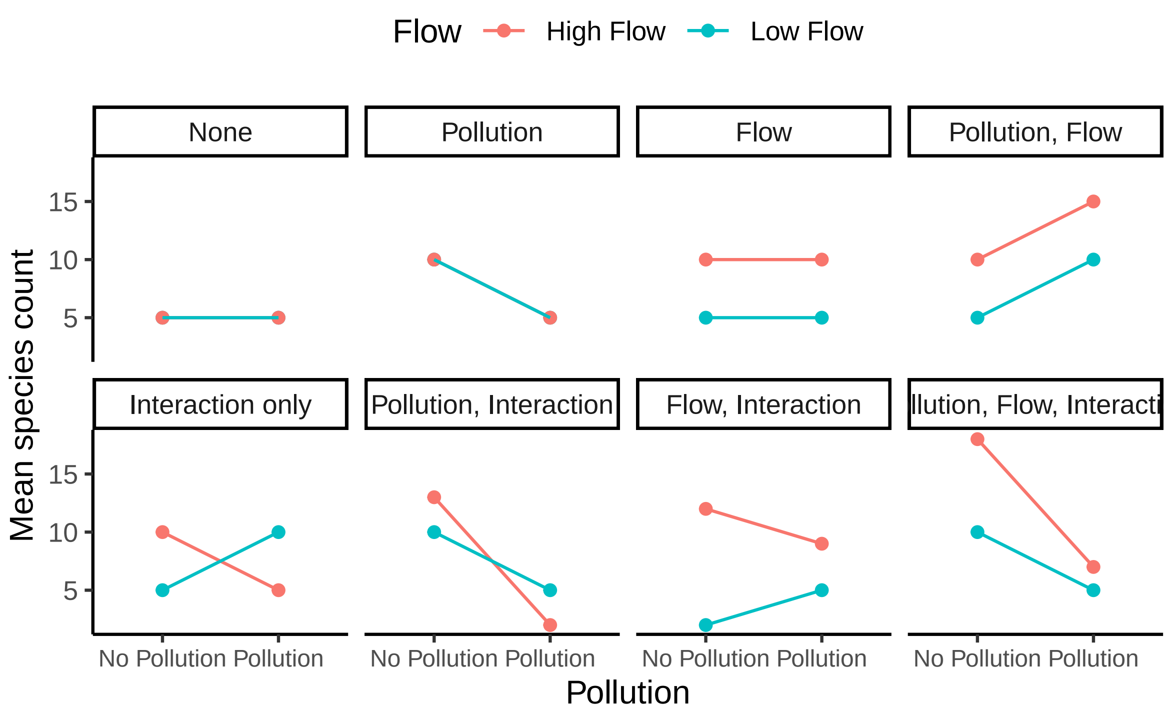

Figure 10.6: Line graphs showing eight possible outcomes for a pollution × flow design.

- If the null hypothesis of *no interaction* is **rejected**, we **do not** interpret the main effects in isolation, because they are not consistent across conditions. Instead, we focus on the simple effects that describe how one factor behaves at each level of the other. Report simple effects (A at each level of B, or vice versa) rather than standalone main effects.
- If the interaction is **not significant**, we then examine the two main effects separately, interpreting each as an overall difference averaged across the other factor.

10.8 Interpreting main effects and interactions

It's convenient to think of a main effect as one manipulation's consistent influence. An interaction complicates that picture: it means the effect of one IV depends on another, so by definition, some main effect isn't behaving consistently across conditions.

Why interactions matter. In our stream example, pollution reduces species count most under low flow and less under high flow. The pollution effect is not constant across flow rates, so the average "main effect" of pollution can mislead if read without context.

How to read it.

- Look for non-parallel or crossing lines. That signals that simple effects differ across levels of the other factor.
- Report the pattern using simple effects: e.g., "Pollution reduced species count sharply under low flow, and only slightly under high flow."
- If you must report a main effect, qualify it: "There was a main effect of pollution, but it varied by flow rate."

Here are two generalized examples to help you make sense of these issues:

10.8.1 A consistent main effect and an interaction

Figure 10.7 shows both a main effect and an interaction. Unpolluted channels sit above polluted ones at both flow rates, that's the main effect. But the size of that gap is larger under low flow (12 vs. 5) than under high flow (14 vs. 11), so there's an interaction too.

There was a main effect of pollution, qualified by a Pollution × Flow interaction: the size of the pollution effect changed with flow rate, larger under low flow and smaller under high flow. That qualification doesn't mean the main effect isn't real. Pollution is associated with lower species counts fairly consistently; flow changes how much lower, not the direction.

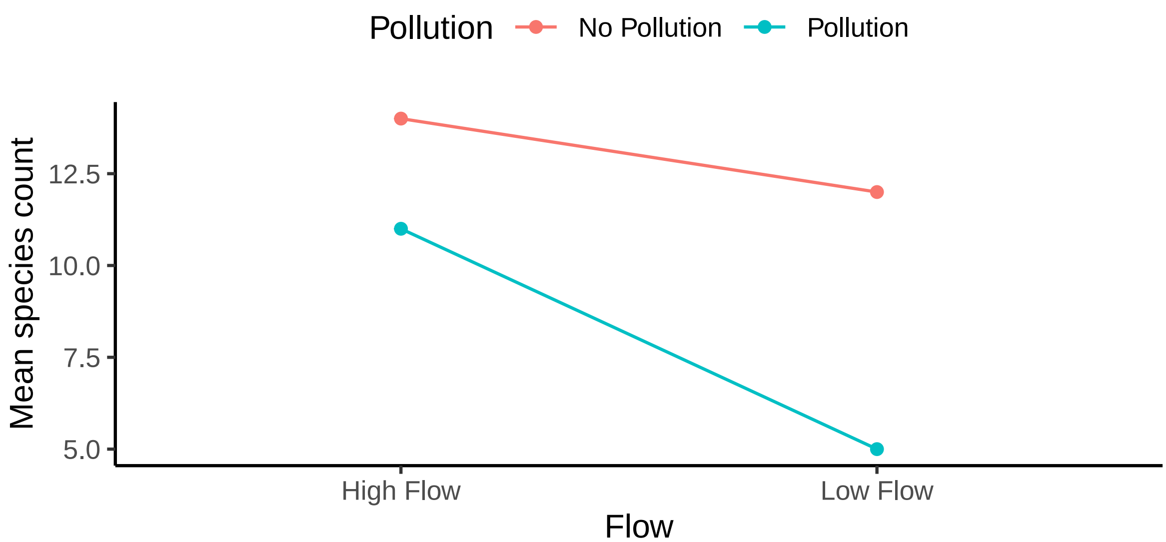

Figure 10.7: Pollution shows a general (main) effect, but the size of that effect differs by flow rate (interaction).

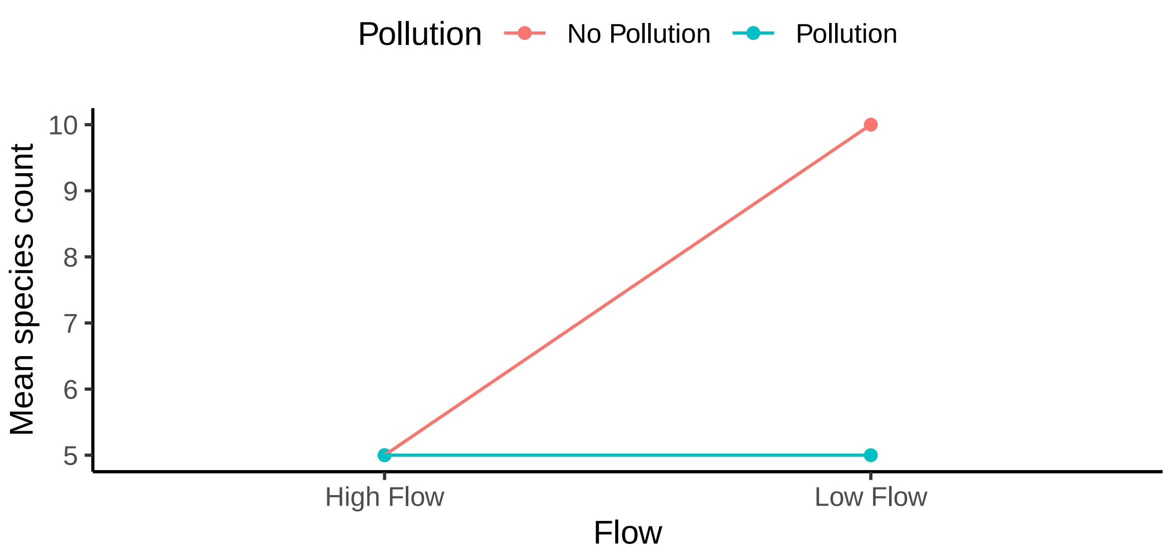

Figure 10.8: An apparent main effect of pollution arises from averaging over an interaction.

10.8.2 An "averaged" main effect that's actually driven by an interaction

Figure 10.8 shows a starker 2x2 pattern: a large difference between polluted and unpolluted channels under low flow, and no difference at all under high flow, an interaction visible at a glance.

Yet an ANOVA run on this data can still report a significant main effect of pollution. Averaged across flow rates, unpolluted channels give $(10 + 5)/2 = 7.5$ species and polluted channels give 5, a main-effect difference of 2.5. That number is real, but it doesn't describe what pollution actually does: it removes 5 species under low flow and none under high flow, not 2.5 everywhere. The "main effect" here is an artifact of averaging over a strong interaction, not a description of anything pollution is doing in general.

There was a main effect of pollution, but it was entirely qualified by a Pollution $\times$ Flow interaction: the pollution effect was large under low flow and effectively zero under high flow. In a case like this, it's fair to doubt whether pollution has a *general* effect on species count at all. It may only do damage under specific conditions, in combination with other factors, rather than on its own.

10.9 More complicated designs

Up until now we have focused on the simplest case for factorial designs, the 2x2 design, with two IVs, each with 2 levels. It is worth spending some time looking at a few more complicated designs and how to interpret them.

10.9.1 3x2 design

This design works exactly like the 2x2 you already know, it just adds one more level to one factor. The test structure is the same (two main effects and one interaction). What changes is how easy it is to interpret the interaction.

Let's expand our stream example. Suppose we examine macroinvertebrate species count (DV) under two pollution conditions (no pollution vs. pollution) across **three flow rates** (low, medium, high).

- The **main effect of pollution** tests whether, on average, polluted and unpolluted channels differ in species count across all flow rates.
- The **main effect of flow** tests whether mean species count differs among flow rates.
- The **interaction** tests whether the pollution effect (unpolluted − polluted) changes with flow rate.

In a 2x2 design, you can easily spot interactions by looking for non-parallel lines. In a 3x2, it's still the same idea; but with three points per line, patterns get trickier to read. The statistical test is identical, but interpretation requires more care.

You have three hypotheses in a 3x2 factorial design:

H : Main effect of Flow

- **Null ($H_{0,1}$)**: $\mu_{\text{Low}} = \mu_{\text{Medium}} = \mu_{\text{High}}$
- **Alternative ($H_{a,1}$)**: At least one flow rate mean differs from the others.

H : Main effect of Pollution

- **Null ($H_{0,2}$)**: $\mu_{\text{NoPoll}} = \mu_{\text{Poll}}$
- **Alternative ($H_{a,2}$)**: Mean species count differs between polluted and unpolluted channels.

H : Pollution × Flow interaction

- **Null ($H_{0,3}$)**: The pollution effect is the same across all flow rates. $[(\mu_{\text{NoPoll,Low}} - \mu_{\text{Poll,Low}}) = (\mu_{\text{NoPoll,Med}} - \mu_{\text{Poll,Med}}) = (\mu_{\text{NoPoll,High}} - \mu_{\text{Poll,High}})]$
- **Alternative ($H_{a,3}$)**: The pollution effect differs across flow rates (i.e., at least one difference of differences is not equal).

Remember, this is far better than running two separate single factor ANOVAs that contrast pollution effects at each flow rate, because you have more statistical power (higher degrees of freedom) for the tests of interest, and you get a formal test of the interaction between factors which is often scientifically interesting.

We might expect data like shown in Figure 10.9:

The figure shows some pretend means in all conditions. Let's talk about the main effects and interaction.

1. **Main Effect of Pollution**: The main effect of pollution is evident. Unpolluted channels (red line) generally support higher species counts than polluted ones (aqua line).
2. **Main Effect of Flow**: There is a main effect of flow as well. Averaged over pollution condition, species counts are lowest at low flow and higher at medium and high flow.
3. **Interaction Between Pollution and Flow**: Is there an interaction? Yes. Remember, an interaction occurs when the effect of one IV depends on the levels of another. The gap between polluted and unpolluted channels is much larger at low flow than at high flow. So the size of the pollution effect changes with flow rate. There is evidence in the means for an interaction. You would have to conduct an inferential test on the interaction term to see if these differences were likely or unlikely to be due to sampling error.

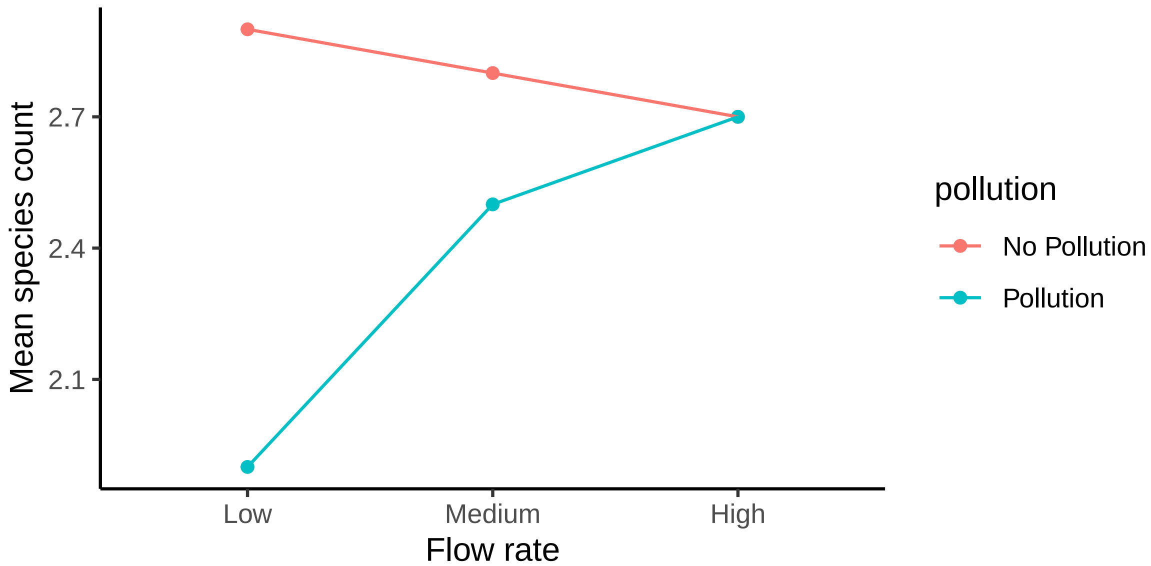

Figure 10.9: Example means for a 3x2 factorial design in environmental science

If there was no interaction and no main effect of flow, we would see something like the pattern in Figure 10.10.

What would you say about the interaction if you saw the pattern in Figure 10.11?

The correct answer is that there is evidence in the means for an interaction. Remember, we are measuring the pollution effect three times, once at each flow rate. The pollution effect is the same at low and medium flow, but it is much smaller at high flow. The size of the pollution effect depends on the level of the flow IV, so here again there is an interaction.

10.10 Designs Beyond This Course

Once you understand the 2x2 logic, everything else in factorial ANOVA builds from the same foundation. But complexity grows quickly, both in the number of effects and in how hard they are to interpret.

10.10.1 2×2×2 (Three-Factor) Designs

A 2×2×2 design includes **three independent variables**, each with two levels. That creates:

- three **main effects** (one per variable)
- three **two-way interactions** (A×B, A×C, B×C)

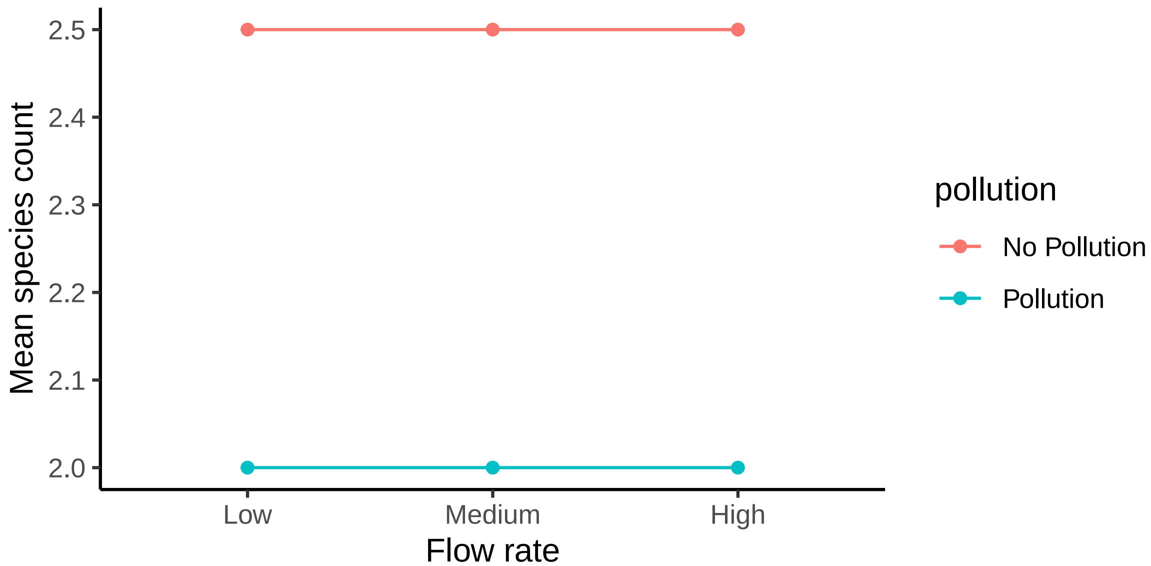

Figure 10.10: Example means for a 3x2 design with only one main effect

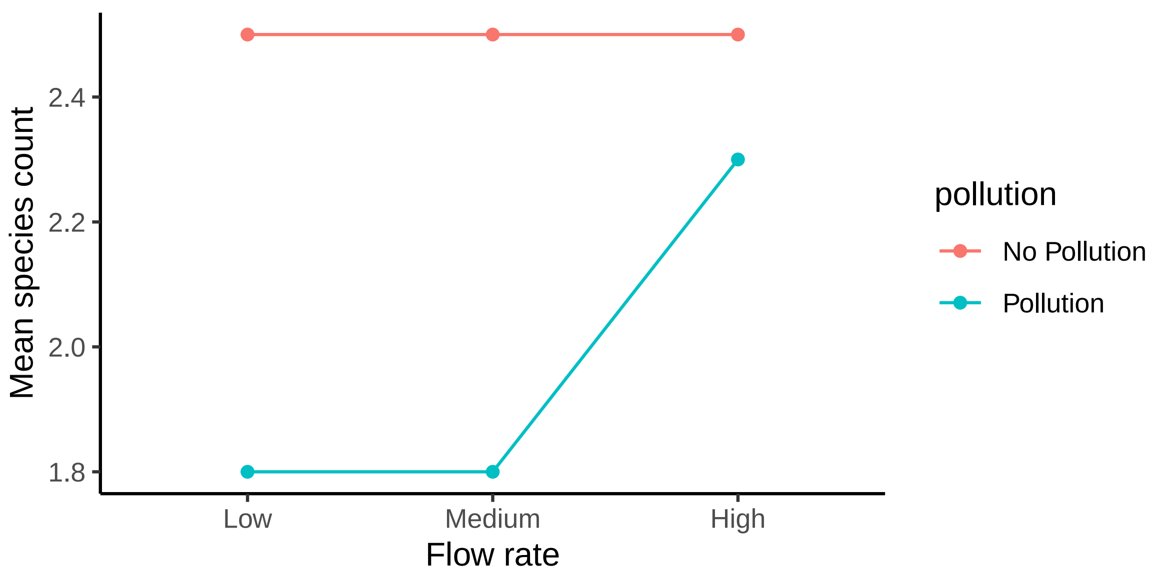

Figure 10.11: Example means for a 3x2 design showing a different interaction pattern

- one **three-way interaction** (A×B×C)

You can think of it as two 2×2 plots side-by-side; one for each level of the third variable.

A **three-way interaction** means that one of the two-way interactions changes across the third variable. For example, the pollution × flow interaction might look different in summer than in winter.

That's all you really need to know: interactions can themselves interact. The math is identical in structure, but the number of effects grows fast, and the graphs become difficult to interpret. In applied research, most studies stop at **two factors** unless there's a strong reason to add more.

10.10.2 Mixed Designs

Sometimes one factor is **between-subjects** (e.g., different sites or treatments) and another is **within-subjects** (e.g., repeated measures over time). This is called a **mixed design**.

The interpretation of main effects and interactions is the same, but the error terms used in the F-tests differ, and the data structure must account for repeated measures. These models are important in longitudinal and environmental time-series studies, but they're beyond our current scope.

Takeaway: Factorial ANOVA can handle almost any experimental structure, but interpretability drops as factors multiply. Start simple: use only the number of factors needed to answer your scientific question clearly.

10.11 Chapter Summary

Why it matters. Single-factor designs can tell you whether something has an effect. Factorial designs tell you what *changes* that effect, which is usually the more interesting scientific question, and often the more realistic one for environmental systems where drivers act together. Getting the numbers is only half the job; an interaction changes what every other term in the table means, so interpretation is not a step you do after the analysis, it is the analysis.

Core ideas

- **Fully crossed means every combination.** A factorial design pairs every level of each IV with every level of the others. The notation says the structure: 2×2 is four conditions, 3×2 is six.
- **A 2×2 tests three things.** One main effect per factor, plus their interaction. That means three null hypotheses, and you should be able to state all three in plain language before running anything.

- **An interaction is a difference between differences.** It asks whether the effect of one factor changes across levels of another. That framing is what the arithmetic is actually computing, and you can compute it directly with a paired t-test on difference scores.
- **The partition extends the one-way logic.** SS_{Total} now splits into each main effect, the interaction, and error. Working one table through by hand shows that the interaction term is the only genuinely new piece.
- **Read the interaction first.** If it's significant, do not interpret the main effects on their own, because they are not consistent across conditions. Report simple effects instead: what factor A does at each level of B. Only when the interaction is not significant do the main effects stand on their own.
- **Learn to read the picture.** In a line plot, parallel lines mean no interaction, crossing or diverging lines mean an interaction, and vertical separation between lines means a main effect. There are eight possible patterns in a 2×2, and recognizing them by eye is faster than parsing the table.
- **An averaged main effect can be an artifact.** A factor that matters a lot in one condition and not at all in another will still produce a "significant main effect" once you average across conditions. That number is real arithmetic, but it describes something that never happens in either condition.
- **Bigger designs, same tests, harder reading.** A 3×2 has the same three tests as a 2×2, but three points per line are harder to read than two. Beyond that, a three-factor design brings three main effects, three two-way interactions, and a three-way interaction, and interpretability falls off fast. Use only as many factors as your question actually needs.
- **You could do this by hand, but you won't.** Having built one ANOVA table from scratch, the rest of the course lets R do the arithmetic while you concentrate on what the output means.

11 Analysis of Covariance (ANCOVA)

ANCOVA uses the same linear model ideas from regression, with one continuous covariate and one categorical factor. We first test whether groups have different **slopes** with respect to the covariate, then, only if slopes are parallel, we test whether groups differ in **adjusted means**.

A quick refresher, since regression was a few weeks ago. A simple linear model is

$$y = \beta_0 + \beta_1 x + \varepsilon$$

one intercept, one slope, one line drawn through the data. ANOVA has the same shape, with a categorical predictor standing in for the continuous one. ANCOVA puts both kinds of predictor in the same equation, and that raises the question this whole chapter turns on: do our groups share that single line, or does each group need its own?

> **Note**
>
> **Going deeper.** Every test in this book, t-tests, ANOVA, regression, and ANCOVA, is the same underlying model with different kinds of predictors. Chasing that unification all the way, through reference coding across many categories, planned contrasts, and several continuous predictors at once, is what a second course in regression is for. Here we need exactly one piece of it: the two-category indicator variable.

11.1 Moving Beyond Regression and ANOVA to ANCOVA

As we've explored statistical modeling, we've understood the power of both regression and ANOVA. Now, let's merge them. ANCOVA allows us to examine relationships across different categories.

Consider these questions that may arise in environmental research:

- Are dbh (Diameter at Breast Height) and height related similarly for tulip poplars and oaks?
- Are biomass and BTUs (British Thermal Units) related similarly for corn stover and Miscanthus?
- Does the exposure to PFAS correlate with the lifetime incidence of cancer uniformly across low- and high-income American populations?

This specific subsection of general linear models is known as **Analysis of Covariance – ANCOVA**. ANCOVA is a linear model with one continuous covariate and one categorical factor; it tests group differences **after** adjusting for the covariate and, first, whether slopes differ across groups.

11.1.1 Conceptual Background for ANCOVA

At its core, ANCOVA checks whether groups share the same line, or whether each group needs its own slope and intercept. That's exactly what the interaction test is doing. Figure 11.1 shows the three possibilities, and it's worth keeping this picture in mind for the rest of the chapter.

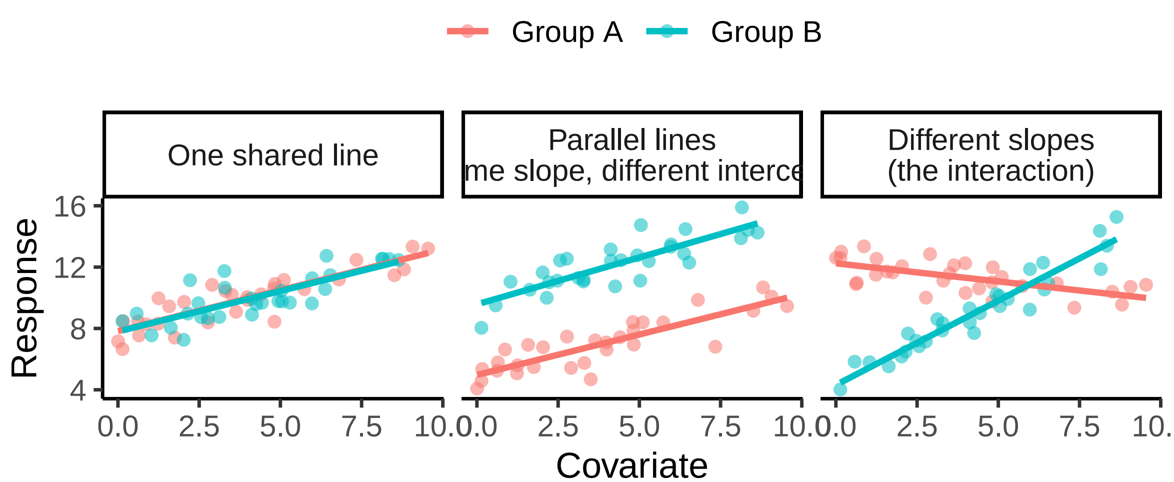

Figure 11.1: The question ANCOVA answers. Left: one line describes both groups. Middle: the groups need different intercepts but share a slope, so the lines are parallel. Right: the groups need different slopes, so the lines are not parallel and the gap between them depends on where you look.

ANCOVA works right to left across that figure: first it asks whether the third panel is what we have (do the slopes differ?), and only if the answer is no does it ask whether we have the second panel or the first (do the intercepts differ?).

11.1.2 Terminology in LMs

A few key terms help clarify how variables play different roles in linear models:

- **Fixed factor:** A categorical variable with specific, intentional levels chosen by the researcher.

  Example: comparing plant growth across seasons (spring, summer, autumn, winter). Each season is a fixed, predefined category of interest.

- **Random factor:** A categorical variable whose levels represent a random sample from a larger population.

  Example: if tree plots are randomly selected from many possible sites, "plot" is a random factor because it reflects sampling variation rather than a treatment.

- **Covariate:** A continuous variable that changes alongside the response variable and helps explain part of its variation.

  In environmental studies, common covariates include temperature, rainfall, soil moisture, or pollution concentration, all factors that may influence the outcome but are not the main treatment of interest.

  The purpose of including a covariate is to **adjust for its influence**, allowing a clearer test of the categorical effects once that continuous background variation is accounted for.

11.2 What ANCOVA Is

Analysis of Covariance (ANCOVA) extends ANOVA by adding a continuous predictor. It fits a linear model with both a categorical factor and a covariate, allowing us to compare groups **and** account for how the response changes with the continuous variable.

The questions that opened this chapter all ask the same kind of thing: does a *relationship* hold up across groups? That framing fits environmental work particularly well, because you usually can't experimentally strip out a covariate like temperature, rainfall, or year anyway. Often you wouldn't want to. It may be the reason you designed the study in the first place. **The covariate is the point, and the interaction is the finding.**

Tip

The other way you'll see ANCOVA. Look this test up elsewhere and the covariate is usually a **nuisance**: something quietly influencing the outcome that you add to the model to control for, so a group comparison comes out fairer. There the interaction is a box to check rather than a result. Most "which test should I use" flowcharts assume that version, which is why ANCOVA rarely gets its own branch on them. Both are the same test; the difference is only which term you came to interpret, so decide that before you fit anything.

For ANCOVA, you need:

1) A continuous dependent variable (the main effect you're studying)
2) At least one continuous independent variable (covariate)
3) At least one categorical independent variable (which can be either a fixed or random factor)

11.2.1 Indicator (dummy) variables and how they work

Categorical predictors get into a linear model by being coded as **indicator variables**. For a factor with k groups you include $k-1$ indicators; the omitted group becomes the **reference**, and every coefficient is interpreted relative to it. With two groups, one indicator does the whole job:

$$x_c = \begin{cases} 0 & \text{group 1 (the reference group)} \\ 1 & \text{group 2} \end{cases}$$

In R, `lm(y ~ x + Group)` creates these indicators for you automatically.

Indicator variables in action. When an indicator variable is turned on (set to 1), it activates the terms multiplied by that variable. Depending on where the indicator appears in the model, that can change the intercept, the slope, or both.

Consider two predictors: a continuous variable x_l and an indicator x_c (0 or 1). Use this model:

$$y = 2x_l + 1(x_l x_c) - 7x_c + 3$$

When $x_c = 0$ (indicator off), only the baseline terms are active:

$$y = 2x_l + 3 \qquad \text{slope} = 2,\ \text{intercept} = 3$$

When $x_c = 1$ (indicator on), the new terms activate:

$$y = (2x_l + 3) + (1x_l - 7) = 3x_l - 4 \qquad \text{slope} = 3,\ \text{intercept} = -4$$

Turning on the indicator adds **+1 to the slope** and **−7 to the intercept**. One equation, two different lines, with the indicator deciding which one applies.

This is how linear models represent different groups: the indicator (x_c) and its interaction with x_l let a single equation shift the intercept and the slope for each group. ANCOVA uses exactly this logic to test, in order, whether the **slopes differ** between groups (the $x_l x_c$ term) and, if they don't, whether the **adjusted means** differ (the x_c term).

Naming the pieces. Write that same equation in the general form and each coefficient now has a job:

$$y = \underbrace{\beta_0}_{\text{reference intercept}} + \underbrace{\beta_1 x_l}_{\text{reference slope}} + \underbrace{\beta_2 x_c}_{\text{intercept shift}} + \underbrace{\beta_3 x_l x_c}_{\text{slope shift}} + \varepsilon$$

- β_0 and β_1 describe the line for the **reference group** ($x_c = 0$).
- β_2 is how much the **intercept** moves for the other group.
- β_3 is how much the **slope** changes for the other group. This is the interaction term.

If we only include β_2 and leave out β_3, we get a restricted version: the intercept can shift but the slope cannot, so the two lines are forced to be **parallel**. That is the parallel-lines model, and it is the one we fall back to when the slope difference turns out not to be significant.

> **Note**
>
> **Why this is worth the detour.** Every coefficient you will read out of an ANCOVA table is one of these four. Without the indicator, β_2 and β_3 are just numbers R hands you. With it, β_3 has a plain-language meaning you can check against a graph: it is the amount by which one group's line tilts away from the other's.

11.2.2 The Null Hypothesis in ANCOVA

Why this matters. ANCOVA asks two questions in order. First, do groups have the **same slope** for the covariate–response relation. Only if slopes are the same do we ask whether groups differ in **adjusted means** after controlling for the covariate.

Model. For one covariate (x) and a two-level group indicator $g \in 0, 1$,

$$y = \beta_0 + \beta_1 x + \beta_2 g + \beta_3 (x \times g) + \varepsilon.$$

Step 1: Slopes equal across groups? This tests whether the covariate effect is the same in each group.

- Null: $H_{0,\text{int}} : \beta_3 = 0$ (parallel lines, same slope)
- Alternative: $H_{A,\text{int}} : \beta_3 \neq 0$ (non-parallel lines, different slopes)

Interpretation if rejected. Report the two fitted lines and describe how the slope differs by group. Do not test adjusted means, because the idea of a single adjusted mean per group assumes parallel slopes.

Step 2: Adjusted means equal? *(only if Step 1 retained parallel slopes)* Fit the model without the interaction,

$$y = \beta_0 + \beta_1 x + \beta_2 g + \varepsilon.$$

- Null: $H_{0,\text{group}} : \beta_2 = 0$ (no difference in adjusted means)
- Alternative: $H_{A,\text{group}} : \beta_2 \neq 0$

What the parameters mean.

- β_1: common slope of y on x when slopes are parallel.
- β_2: difference in group means **after** adjusting for x (tested only when $\beta_3 = 0$).
- β_3: difference in slopes between groups.

Intercepts and centering. Intercepts are evaluated at $x = 0$. Compare intercepts only if you center the covariate, for example $\tilde{x} = x - \bar{x}$, so that the intercepts represent group means at the average covariate value.

What to report.

- If $x \times g$ is significant: "The effect of x differs by group," then provide the two line equations and a figure with separate fitted lines.
- If $x \times g$ is not significant: report the common slope β_1 and the adjusted group difference β_2, with a figure showing parallel lines.

Common pitfalls.

- Interpreting main effects when the interaction is significant.

- Comparing raw intercepts when x is not centered.
- Treating adjusted means as meaningful when slopes are not parallel.

11.2.3 Conducting the Test

When we run an ANCOVA, the output includes an **omnibus F-test**, a single test of whether there are any differences in the adjusted group means after accounting for the covariate.

If this test is **significant**, we reject the overall null hypothesis that all adjusted means are equal.

What happens next: A significant omnibus test tells us that at least one group differs, but it doesn't tell us *which* groups differ or *why*.

To interpret the result, we must **decompose the omnibus effect** into its components:

1. **Interaction (slope difference):** Are the slopes of the covariate–response relationship the same across groups?
2. **Adjusted means (intercept difference):** If the slopes are equal, do the groups still differ in their adjusted means?

This two-step logic, testing slope equality first and then adjusted means, is central to ANCOVA and parallels how we follow a significant ANOVA with post-hoc comparisons.

Interpreting the Findings:

Rejecting the omnibus null indicates that the categorical factor affects the dependent variable *after controlling for the covariate*.

However, as always, statistical significance does not guarantee ecological or practical importance. The results should be interpreted in light of the study's design, measurement precision, and the size of the observed effects.

> **Note**
>
> **ANCOVA ASSUMPTIONS**
>
> - Linearity within each group (covariate–response relation is linear).
> - **Homogeneity of slopes** (test via interaction).
> - Independent observations.
> - Approximately normal residuals and similar variances across groups.
> - Covariate measured without substantial error and not affected by treatment

11.3 Full ANCOVA Example

11.3.1 Question

Imagine we're environmental scientists studying the impact of **air pollution** (a continuous variable) on **bird species density** (our dependent variable).

We want to know whether this relationship differs between **urban** and **rural** areas (our categorical variable).

For this purpose, we've created a synthetic dataset to model the scenario.

11.3.2 Model

We'll fit an ANCOVA model to test whether location type influences bird density **after accounting for** air pollution.

The categorical variable (urban vs. rural) is represented as a dummy variable, and air pollution enters as a covariate.

- **Dataset Overview:** 200 observations of bird density across urban and rural sites, each with a measured level of air pollution.

11.3.3 Why We're Running the ANCOVA

- **Research Question:** Does the effect of air pollution on bird density differ between urban and rural areas?
- **Expectation:** Air pollution will have a stronger negative effect in urban areas than in rural areas.

11.3.4 Null and Alternative Hypotheses in our example ANCOVA

0. **Omnibus (adjusted means) hypothesis** After adjusting for air pollution, do urban and rural sites differ in mean bird density?

   $H_0 : \mu_{\text{Urban, adj}} = \mu_{\text{Rural, adj}}$

   $H_A : \mu_{\text{Urban, adj}} \neq \mu_{\text{Rural, adj}}$

   Here, $\mu_{\text{Urban, adj}}$ and $\mu_{\text{Rural, adj}}$ are the *adjusted* group means once the covariate (air pollution) is accounted for.

1. **Slope (interaction) hypothesis**

   Does air pollution affect bird density in the same way across urban and rural sites?

   $H_{0,1} : \beta_{\text{interaction}} = 0$

   $H_{A,1} : \beta_{\text{interaction}} \neq 0$

   A significant interaction means the slopes differ, and air pollution's relationship with bird density depends on area type.

2. **Intercept (baseline difference) hypothesis**

   If slopes are parallel (interaction not significant), do the adjusted means differ?

   $H_{0,2} : \beta_{0,\text{Urban}} = \beta_{0,\text{Rural}}$

   $H_{A,2} : \beta_{0,\text{Urban}} \neq \beta_{0,\text{Rural}}$

   Rejecting $H_{0,2}$ indicates an inherent group difference in bird density that is independent of pollution.

11.3.5 Analysis

Before we look at the model results, it's always good practice to **look at the data first**.

Visual inspection helps us check for broad patterns and see whether the model output later "makes sense."

Let's start with the response variable, **bird density**, in Figure 11.2. This gives us a quick visual sense of whether urban and rural areas differ in their average values.

```
ggplot(data, aes(AreaType, BirdDensity)) +
  geom_boxplot() + theme_classic(base_size = 12) +
  labs(x = "Area Type", y = "Bird Density")
```

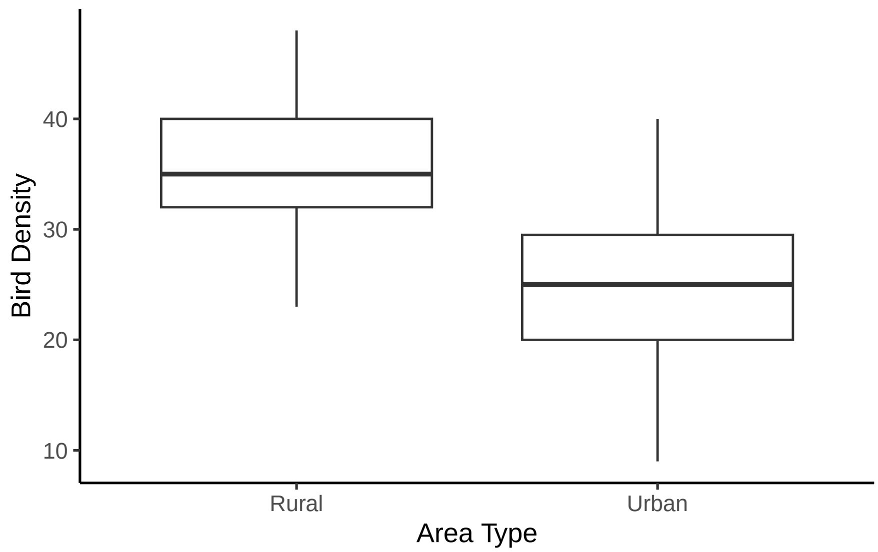

Figure 11.2: Bird density by area type. Rural sites sit higher than urban ones.

At first glance, bird density appears **higher in rural areas** than in urban ones, consistent with what we might expect ecologically.

Next, check the **covariate**, air pollution.

If the covariate differs sharply between groups, ANCOVA will "adjust" for that difference.

If it doesn't, group comparisons will look more like a simple ANOVA.

```
ggplot(data, aes(AreaType, AirPollution)) +
  geom_boxplot() + theme_classic(base_size = 12) +
  labs(x = "Area Type", y = "Air Pollution")
```

Table 11.1: ANCOVA Table

term	df	sumsq	meansq	statistic	p.value
AirPollution	1	2183.89	2183.89	88.57	0.0000
AreaType	1	6450.54	6450.54	261.60	0.0000
AirPollution:AreaType	1	269.14	269.14	10.91	0.0011
Residuals	196	4833.03	24.66	NA	NA

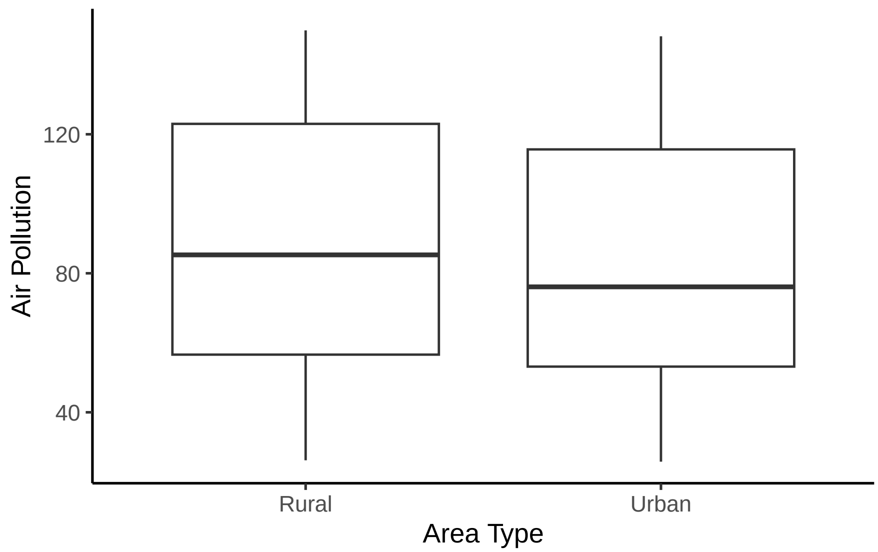

Figure 11.3: Air pollution by area type. The two groups look broadly similar, so there is no large covariate imbalance for the model to adjust away.

Air pollution looks fairly similar between urban and rural sites, with no obvious systematic difference.

This isn't diagnostic, but it helps us **anticipate what the model might show**: we may find that group differences in bird density aren't explained by pollution alone.

11.3.6 Results of the ANCOVA Model

Now that we've explored the data visually, we can test our hypotheses formally using ANCOVA.

Step 1: Fit the ANCOVA model.

```
ancova <- aov(BirdDensity ~ AirPollution * AreaType, data = data)
```

Step 2: Interpret each term in light of the hypotheses

0. Omnibus (adjusted means) :

$$H_0 : \mu_{\text{Urban, adj}} = \mu_{\text{Rural, adj}}$$

The overall ANCOVA test examines whether adjusted means differ between groups after accounting for the covariate.

Because we included an interaction term, this omnibus test is **decomposed** into two parts:

- the **interaction** (slope difference), and
- the **main effects** (covariate and group).

So we interpret the omnibus test through these individual terms rather than as a separate p-value.

1. Interaction (slope difference):

$$H_{0,1} : \beta_{\text{interaction}} = 0$$

The interaction between air pollution and area type is **significant** ($p = .001$).

Reject $H_{0,1}$ → our slopes are not parallel. Air pollution's relationship with bird density depends on area type.

2. Covariate (main effect of Air Pollution)

Although not a formal ANCOVA hypothesis, this term tests whether air pollution predicts bird density *on average*, across both groups.

It is **significant** ($p < .001$): bird density falls as pollution rises.

3. Group (main effect of Area Type) – $H_{0,2} : \mu_{\text{Urban, adj}} = \mu_{\text{Rural, adj}}$

Area type is **significant** ($p < .001$), suggesting lower average bird density in urban areas.

However, because the interaction is significant, this group difference changes with air pollution level, so it should be interpreted conditionally, not as a single overall difference.

Reject $H_{0,2}$ with caution.

Step 3: Examine regression coefficients

We can also fit the same model using `lm()` to view the **regression coefficients** directly.

This shows how each term contributes to predicting bird density and corresponds to the hypothesis tests we just discussed.

```
ancova_lm <- lm(BirdDensity ~ AirPollution * AreaType, data = data)
```

R automatically converts the categorical `AreaType` variable into a 0/1 indicator behind the scenes. It picks one level (here, "Rural," alphabetically first) as the **reference**, and every

Table 11.2: Linear Model Coefficients

term	estimate	std.error	statistic	p.value
(Intercept)	41.784	1.310	31.90	0.0000
AirPollution	-0.069	0.014	-5.07	0.0000
AreaTypeUrban	-5.909	1.802	-3.28	0.0012
AirPollution:AreaTypeUrban	-0.064	0.019	-3.30	0.0011

coefficient involving `AreaType` is interpreted relative to that reference. That's why you see a coefficient called `AreaTypeUrban` but no separate `AreaTypeRural` term.

These coefficients give the *direction* and *magnitude* of each effect:

- **(Intercept):** estimated mean bird density for rural sites (the reference group) when air pollution is zero.
- **AirPollution:** slope of the relationship between pollution and bird density in rural sites.
- **AreaTypeUrban:** difference in intercept between urban and rural areas.
- **AirPollution:AreaTypeUrban:** difference in slope between urban and rural areas (the interaction).

Step 4: Visualize the interaction

As always, plotting helps confirm what the model tells us.

A scatter plot with fitted regression lines (Figure 11.4) lets us see how air pollution relates to bird density for each area type.

Here, the slopes differ by area type, illustrating the **significant interaction** we detected statistically. Both lines slope downward, so pollution is associated with fewer birds in both settings, but the urban line drops roughly twice as fast. The two lines start close together at low pollution and pull apart as pollution rises, which is exactly what a slope difference looks like.

You'll also notice real scatter around each line, which makes sense: our overall R^2 is **0.65**.

Remember, R^2 represents the **proportion of variance in bird density explained by the model**.

So about **65% of the variation** in bird density is explained by air pollution, area type, and their interaction.

The remaining 35% reflects other factors, measurement noise, or natural ecological variability not captured by this simple model. That is a strong fit by the standards of field data, and it is strong here partly because this is a synthetic dataset built to illustrate the method. Real environmental data are usually messier.

Step 5: Centering the covariate

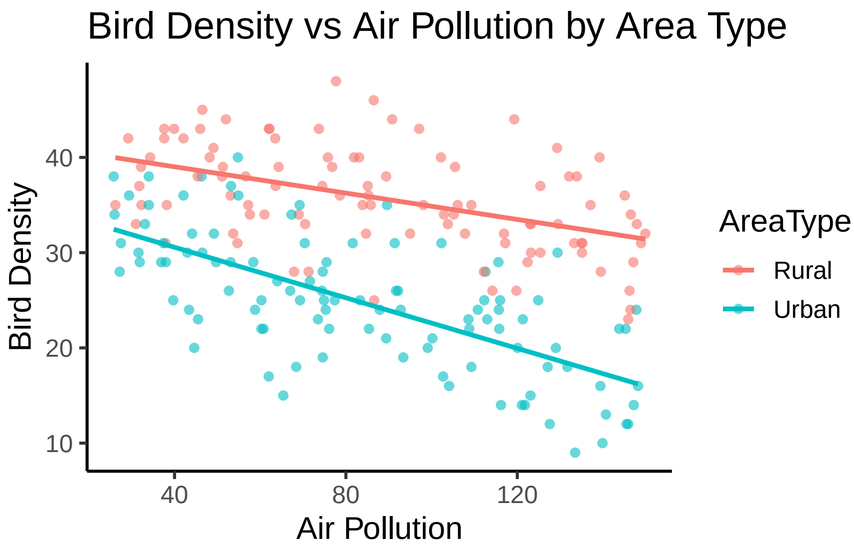

Figure 11.4: Bird density against air pollution, with a separate fitted line for each area type. Both lines slope down, but the urban line falls about twice as steeply, which is what the interaction term tests.

Table 11.3: Centered ANCOVA Model Coefficients

term	estimate	p.value
(Intercept)	35.828	0.0000
AirPollution_c	-0.069	0.0000
AreaTypeUrban	-11.385	0.0000
AirPollution_c:AreaTypeUrban	-0.064	0.0011

Interpreting intercepts at $x = 0$ can be misleading when "zero pollution" isn't meaningful.

By centering the covariate (subtracting its mean), we make the intercept represent **average bird density at mean air pollution** instead.

This re-centering doesn't change the slopes or p-values. It just shifts the intercepts, making them easier to interpret.

Interpreting the Centered Model

Centering does **not** change the slopes or their p-values.

Notice that the coefficients for `AirPollution_c` (–0.069) and `AirPollution_c:AreaTypeUrban` (–0.064) are **identical** to the slopes we estimated before centering.

Only the **intercept** and the **AreaTypeUrban** (group difference) term have changed. They are now interpreted at the *mean pollution level* (about 86 units) instead of pollution = 0.

- **Intercept (35.8):** Predicted bird density for **rural sites** at *average air pollution.*
- **AreaTypeUrban (–11.4):** Urban sites have, on average, **about 11 fewer birds** than rural sites *at the same mean pollution level.*
- **AirPollution_c (–0.069):** Slope for rural sites: bird density falls by about 0.07 birds for each additional unit of pollution.
- **Interaction (–0.064):** Difference in slopes between groups. The urban slope = $-0.069 + (-0.064) \approx$ **–0.13**, roughly twice as steep a decline as in rural sites.

Note how much the group difference changed when we centered: –5.9 before, –11.4 after. That is not a contradiction. Uncentered, the term compares groups at pollution = 0, a value we never actually observed. Centered, it compares them at average pollution, which is a real place in the data. When slopes differ, "the group difference" is not one number; it depends on where along the covariate you look.

Step 6: Summarize and Interpret

The significant interaction shows that **air pollution's effect on bird density depends on area type**: bird density declines with pollution in both settings, but the decline is about twice as steep in urban sites.

Because the interaction is significant, we focus on these **different slopes**, not on adjusted mean differences.

If the interaction had **not** been significant, we would instead drop the interaction term and interpret the **parallel slopes model**, comparing adjusted means between groups.

Our results sentences might look something like this:

Results from the ANCOVA indicated that air pollution's effect on bird density depended on area type, $F(1, 196) = 10.91$, $p = .001$. Bird density declined with increasing pollution in both settings, but the decline was roughly twice as steep at urban sites (slope –0.13) as at rural ones (slope –0.07). These findings suggest that **urbanization intensifies the pollution–bird density relationship** and may point toward the need for area-specific conservation strategies.

Key takeaways

1. Always **check for interaction first**
2. If interaction is **significant**, interpret slope differences.
3. If not, **fit the reduced model** (no interaction) and compare adjusted means.
4. Center covariates when the intercept needs to represent a meaningful point.

11.4 Chapter Summary

In this chapter, we introduced **ANCOVA** as an extension of ANOVA that includes continuous covariates. We began by framing ANOVA within the linear modeling framework, showing how ANCOVA builds on it by combining categorical and continuous predictors. Using examples such as bird density across urban and rural environments, we developed and tested null and alternative hypotheses step by step. The main advantage of ANCOVA is its ability to adjust for continuous variables, allowing clearer interpretation of group effects.

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